Tedtronics — perfboard revision 5

7×10 cm stock · two banks · six interchangeable positions · three active capsules

ENGINEERING PROTOTYPE — NOT RELEASED FOR SOLDERING

Software checks: **pass**. Actual hardware, module envelopes, timing and electrical performance remain pending. Read the release gates before soldering.

Architecture

- ESP1 owns routing, rail enable and fault reset.
- ESP2 owns smart modes, voltage scanning and joystick ADC1 inputs.
- Each bank shares an ADS1115 voltage scanner and buffered selector.
- Every position retains independent output clamps and series-path isolation.
- Pi and its peripherals have a separate supply. RC522 uses the purchased MIFARE tags.
- Socket through-hole ICs. Carrier interfaces are labelled pigtails; their connector numbering is not IC numbering.

Construction comparison

Metric	Legacy	Revision 5
Boards	17	27
Functional parts	551	457
Adapter assemblies	47	23
Local wire segments	1511	1392
Board solder estimate	5224	4946

The legacy design used larger boards. Revision 5 also has 341 distribution posts and 279 harness conductors. Counts exclude carrier internals and adapter/socket preparation. This revision reduces adapters; it is not a claim of fewer boards or a finished minimum-work optimum.

Electrical bounds and release gates

Calculated design bounds

- Voltage input impedance: ≥1.98 MΩ with 1% divider resistors.
- Worst divided input at 5.25 V: 2.919 V.
- Static source/sink clamp maximum: 11.08 mA.
- Current shunt: 1 Ω ±1%; 10 mV shunt burden at 10 mA, plus switch and wiring resistance.
- Bank scan target: ≤20 ms. Wait for conversion-ready, discard after switching and keep timestamps.
- Watchdog expiry target: 200–350 ms; actual HC123 timing must be measured before permanent installation.
- 5 V guard static trip: 5.053–5.227 V. Approximately 23 mV margin below the contact ceiling does not establish safe fault transients.

These calculations do not establish loop stability, transient limits or measured accuracy.

Pending physical evidence

- Actual stock grid and plating
- Pogo pin diameter, compressed height and mounting fit
- DevKit pinout and body measurements
- Chosen ADS1115/INA226 carrier interfaces and envelopes
- Bench qualification of output clamps, supply guards and partial-power states
- Firmware implementation and measured acceptance

Verify the exact carrier pads, actual stock grid, socket bodies, compressed pogo heights and inter-board mounting offsets.

Current contact geometry needs **156 pogo pins**. The inventory lists 150, leaving a six-pin shortage before spares.

The 12 V adapter provides a nominal 24 W budget. Measure fan, light, controller and buck input loads. Do not include the Pi in this budget.

CP01 — removable measurement leads

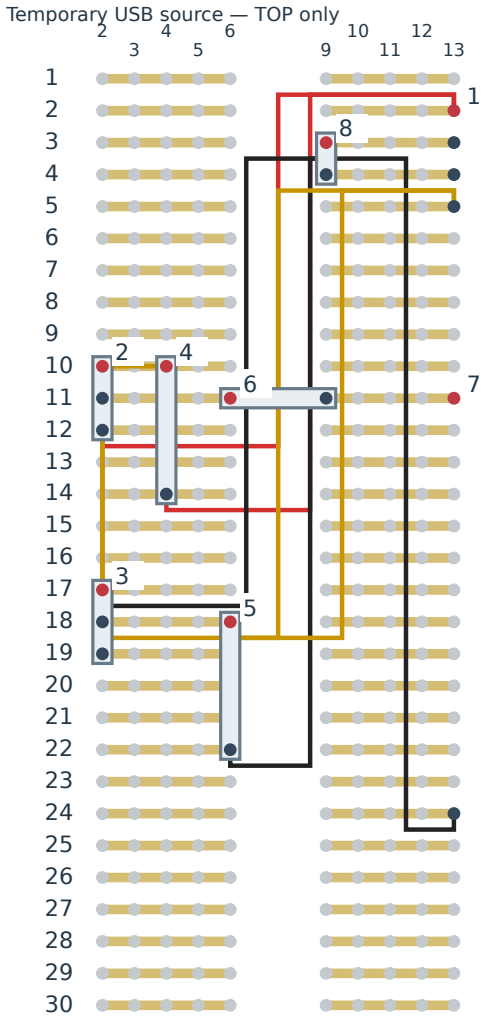
Student source positive unplugged. No matrix. Keep these leads distinct from permanent harnesses.

Purpose	From	To / condition
Temporary precision input A	measurement-a-precision / J_LINK_4.1	A1 C4 / NODE03; also J_DIAGNOSTIC.1 stimulus
Temporary precision input B	measurement-a-precision / J_LINK_5.1	A1 C1 / NODE00; also J_DIAGNOSTIC.2 return
Temporary 3.3 V supply	router-core J_FIXTURE.1	measurement-a-precision / J_LINK_3.1
Common return	router-core J_FIXTURE.2	measurement-a-precision JP_RETURN.2 / GND; permanent ground harness may supply this
Temporary ADC SDA	router-core J_FIXTURE.3 / GPIO4	measurement-a-precision J_A_ADC_MODULE.4 / SDA
Temporary ADC SCL	router-core J_FIXTURE.4 / GPIO16	measurement-a-precision J_A_ADC_MODULE.3 / SCL
Stimulus control	router-core J_FIXTURE.5 / GPIO25	measurement-a-precision / J_LINK_6.1; use the CP01 permanent harness
Test return link	measurement-a-precision JP_RETURN.1	JP_RETURN.2; fit only during unpowered continuity test

The input terminals retain their final net names even while temporary leads test C4/C1. Their permanent NODE00/NODE01 harness is absent until CP06. CD4052 selector pull-downs choose channel0; leave later ESP2 selector harnesses disconnected. At CP06 remove temporary sense/stimulus leads and JP_RETURN before installing the permanent node harness. The USB3V3 feed is removed before final regulated-rail links are fitted at CP11; temporary ESP1 bus control ends at CP15.

CP00 — temporary USB source

Solderless fixture; excluded from the 7×10 cm permanent-board inventory



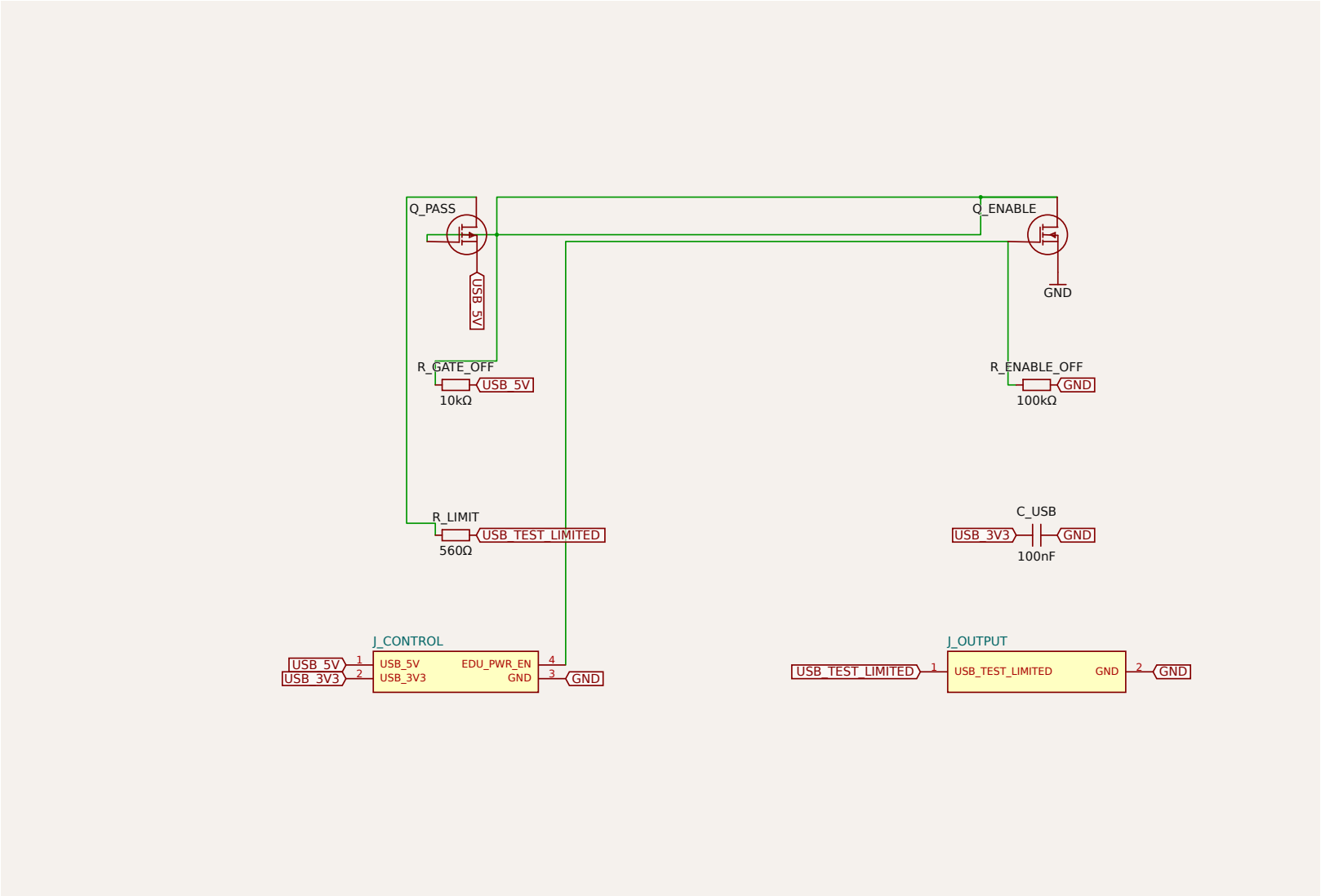
Gold: factory five-hole row ties. Check your fixture.

Coordinates follow this drawing; no underside soldering.

# / reference	Purpose
1 / J_CONTROL	Four individual flying leads from ESP1 prototype header
2 / Q_PASS	USB high-side switch
3 / Q_ENABLE	3.3V gate-control transistor
4 / R_GATE_OFF	10kΩ makes the P-channel switch OFF
5 / R_ENABLE_OFF	100kΩ holds enable LOW
6 / R_LIMIT	560Ω ±1%, 0.25W; bounds every educational-source path
7 / J_OUTPUT	Two flying leads: C2 positive / C1 return
8 / C_USB	USB3V3 fixture decoupling

J_CONTROL pins 1…4 receive router J_PROTOTYPE pins 1…4: USB5V, USB3V3, GND, GPIO32 enable. Q_PASS is NDP6020P (G/D/S); Q_ENABLE is 2N7000 (S/G/D). Confirm the purchased lead order. The five-hole groups are factory-connected; do not add wires for the listed ties.

CP00 — temporary-source schematic



CP00 — fixture jumpers and kit loop

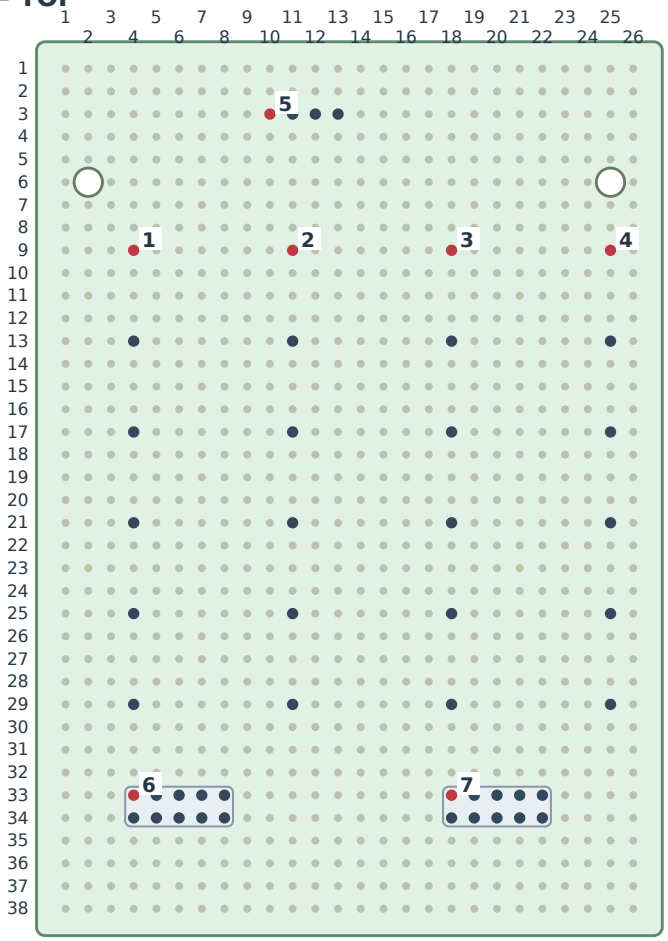
Wire / net	Endpoints	TOP holes
W1 / USB_5V	J_CONTROL.1 → Q_PASS.3	13,2 → 2,12
W2 / USB_5V	J_CONTROL.1 → R_GATE_OFF.2	13,2 → 4,14
W5 / GND	C_USB.2 → J_OUTPUT.2	9,4 → 13,24
W6 / GND	C_USB.2 → Q_ENABLE.1	9,4 → 2,17
W7 / GND	C_USB.2 → R_ENABLE_OFF.2	9,4 → 6,22
W8 / EDU_PWR_EN	J_CONTROL.4 → Q_ENABLE.2	13,5 → 2,18
W9 / EDU_PWR_EN	J_CONTROL.4 → R_ENABLE_OFF.1	13,5 → 6,18
W10 / PROTO_GATE	Q_ENABLE.3 → Q_PASS.1	2,19 → 2,10
W11 / PROTO_GATE	Q_ENABLE.3 → R_GATE_OFF.1	2,19 → 4,10

Factory ties — no added jumper C_USB.1 → J_CONTROL.2 (USB_3V3)
C_USB.2 → J_CONTROL.3 (GND)
Q_PASS.2 → R_LIMIT.1 (PROTO_DRAIN)
J_OUTPUT.1 → R_LIMIT.2 (USB_TEST_LIMITED)
Four top-side kit pieces Fake battery: C1 negative / C2 positive.
1 kΩ resistor: C2 → C3.
LED: C3 anode → C4 cathode.
Jumper: C4 → C1.
Source output pin1 → A1 J_NODE pin3 (NODE01 / C2); output pin2 → pin1 (NODE00 / C1).
No protected-power board or matrix is required. Verify reset/disable extinguishes the LED and the disconnected shorted source stays ≤9.5 mA before attaching the kit.

a1

A1 contacts

a1 — TOP

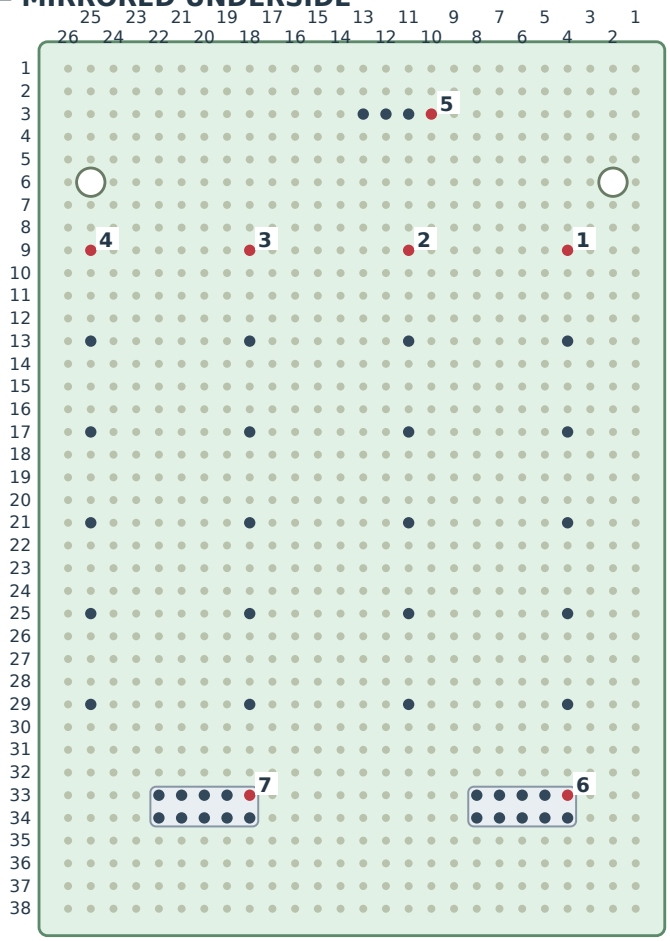


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

a1 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

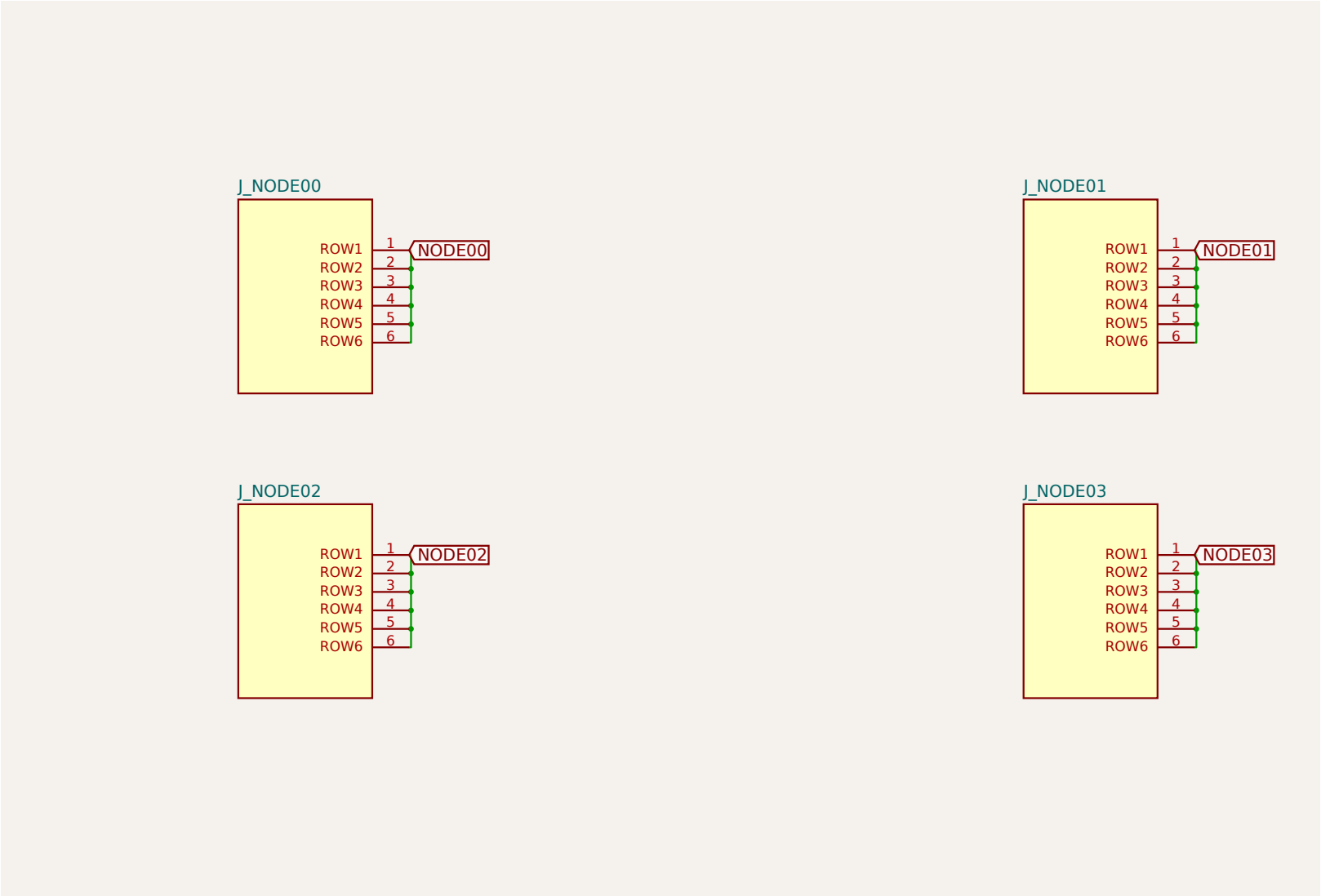
PROTOTYPE: physical fit and hardware tests pending.

a1 — numbered parts

Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

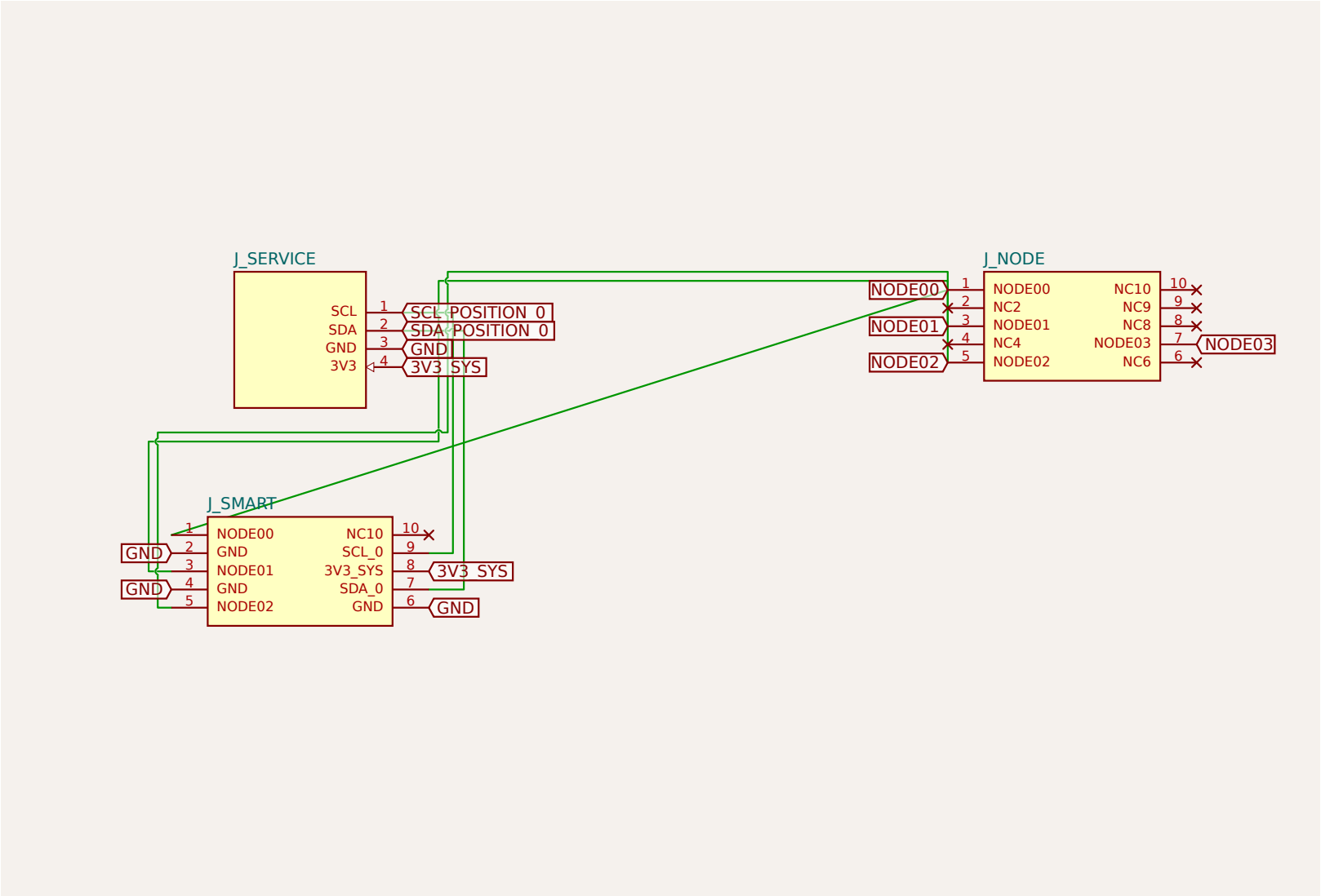
#	Reference	Purpose	Install	Adapter / carrier
1	J_NODE00	NODE00 — six contacts	CP0	
2	J_NODE01	NODE01 — six contacts	CP0	
3	J_NODE02	NODE02 — six contacts	CP0	
4	J_NODE03	NODE03 — six contacts	CP0	
5	J_SERVICE	OLED service pogo contacts	CP6	
6	J_NODE	NODE cable → matrix	CP0	
7	J_SMART	SMART_RAW v2 → channel 0	CP6	

a1 — Nodes



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

a1 — Service

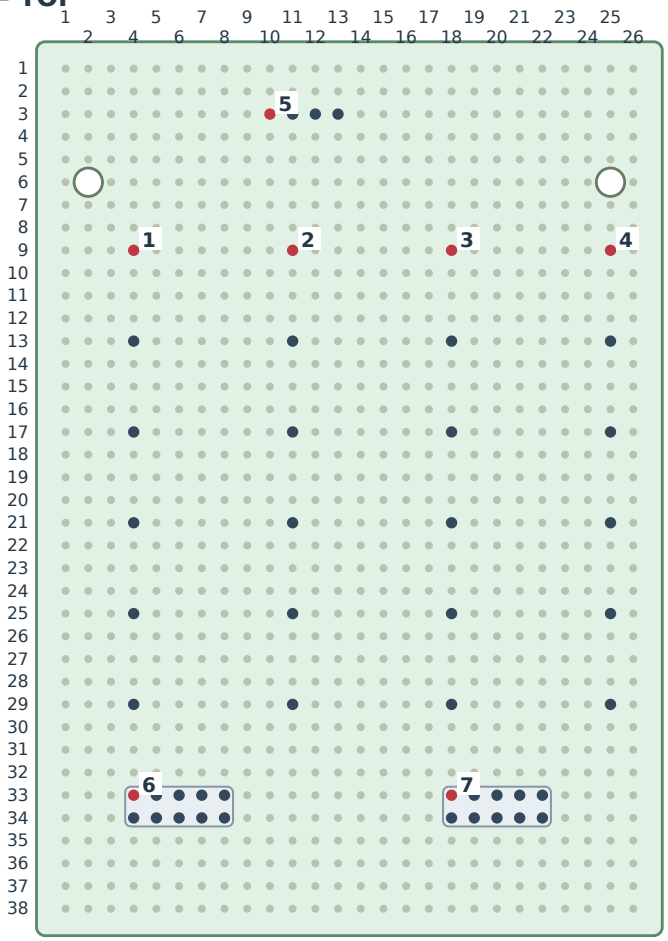


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

a2

A2 contacts

a2 — TOP

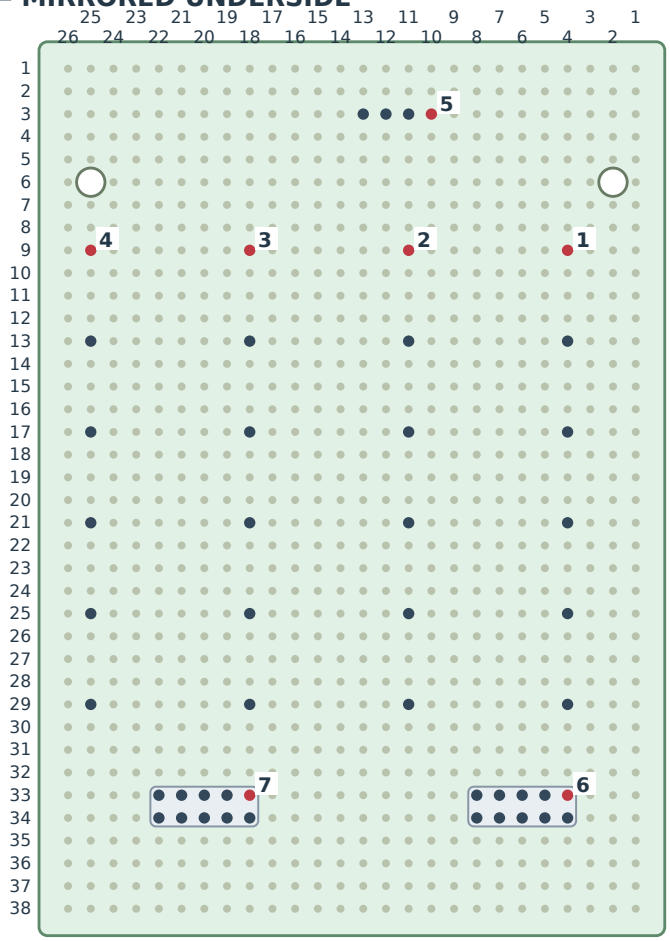


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

a2 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

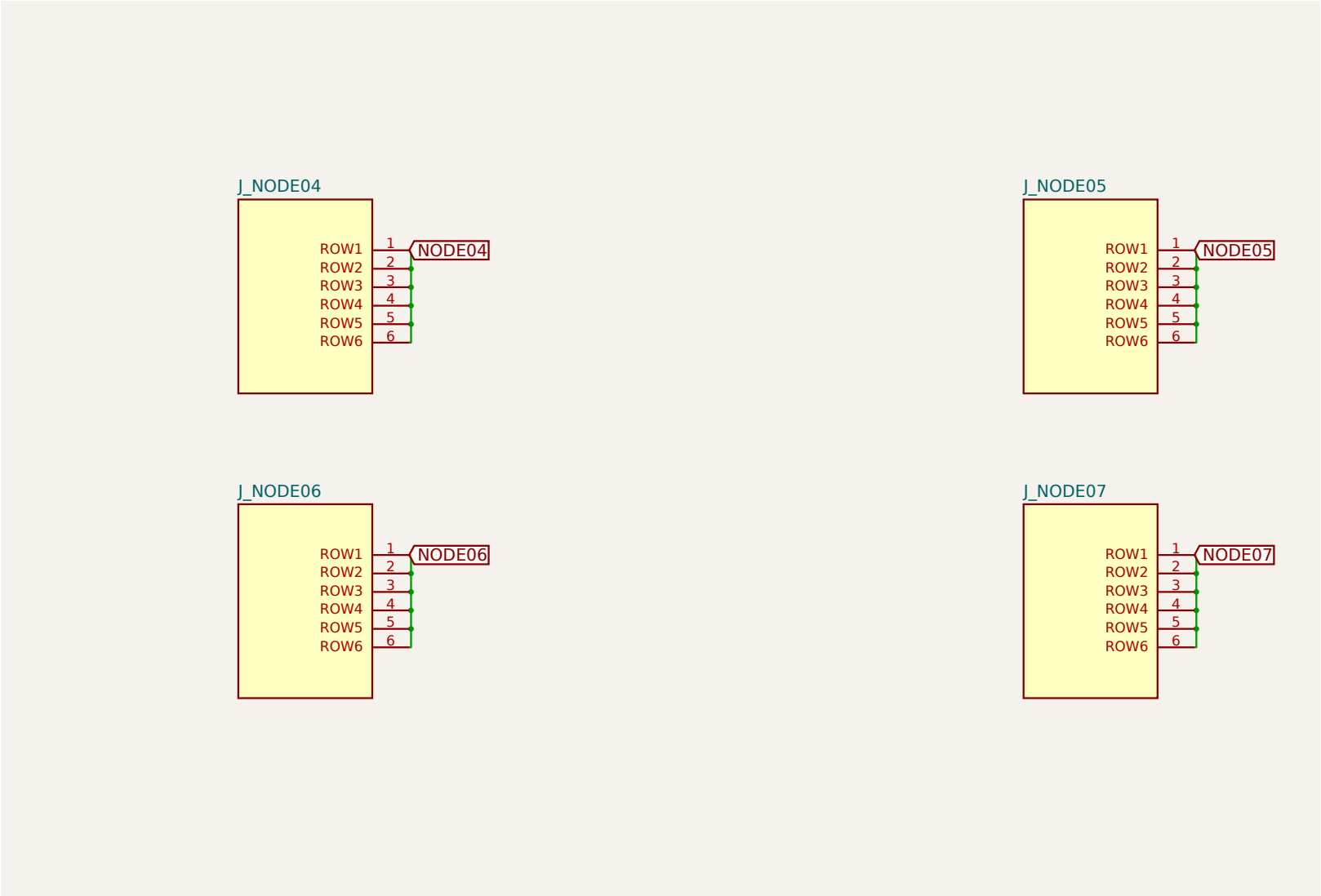
PROTOTYPE: physical fit and hardware tests pending.

a2 — numbered parts

Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

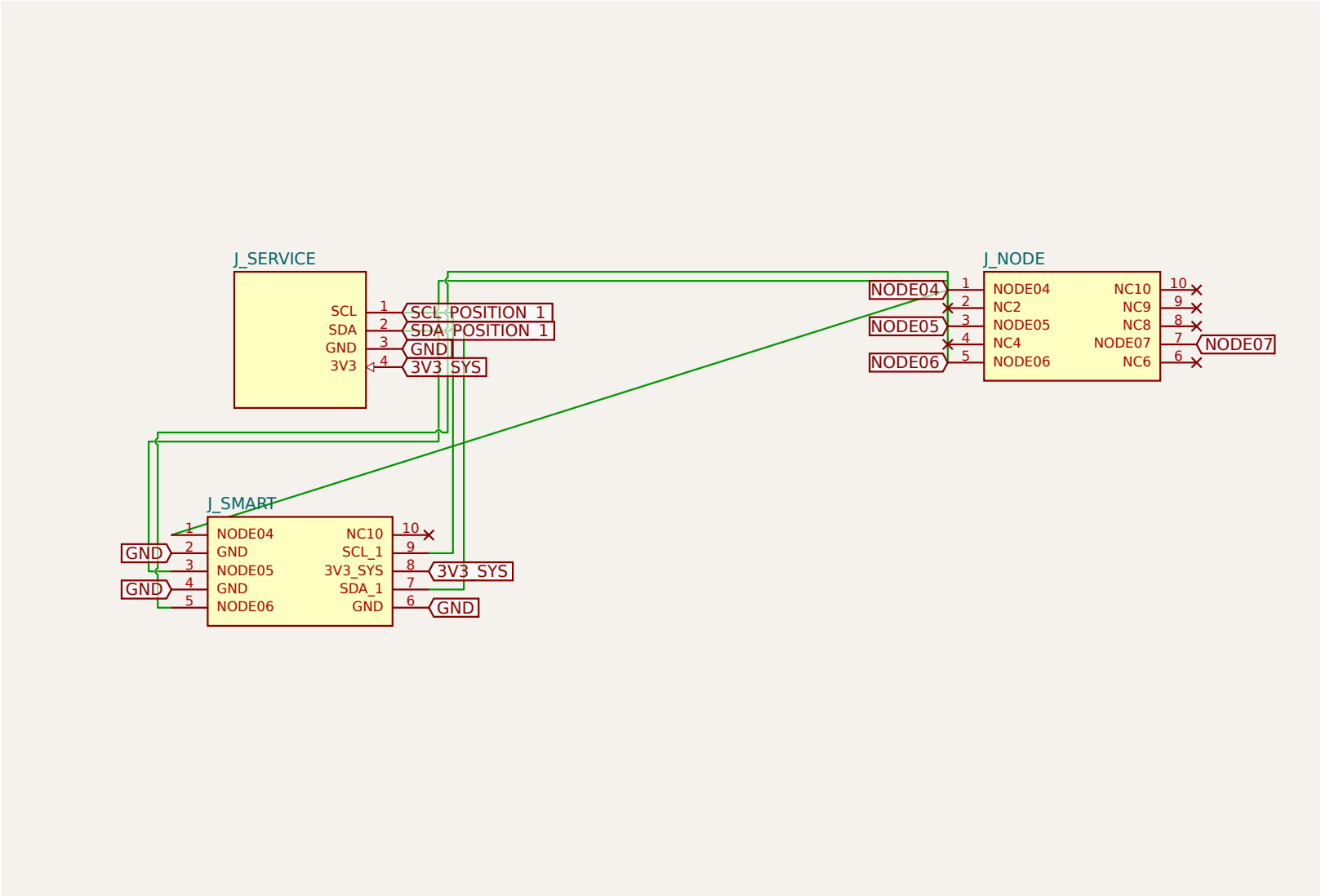
#	Reference	Purpose	Install	Adapter / carrier
1	J_NODE04	NODE04 — six contacts	CP27	
2	J_NODE05	NODE05 — six contacts	CP27	
3	J_NODE06	NODE06 — six contacts	CP27	
4	J_NODE07	NODE07 — six contacts	CP27	
5	J_SERVICE	OLED service pogo contacts	CP27	
6	J_NODE	NODE cable → matrix	CP27	
7	J_SMART	SMART_RAW v2 → channel 1	CP27	

a2 — Nodes



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

a2 — Service

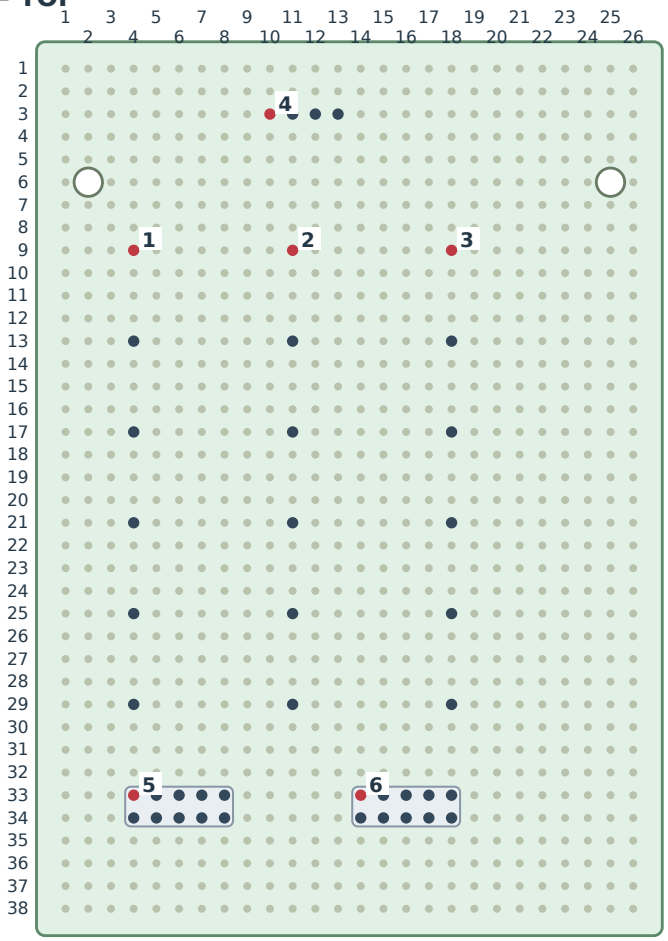


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

a3

A3 contacts

a3 — TOP

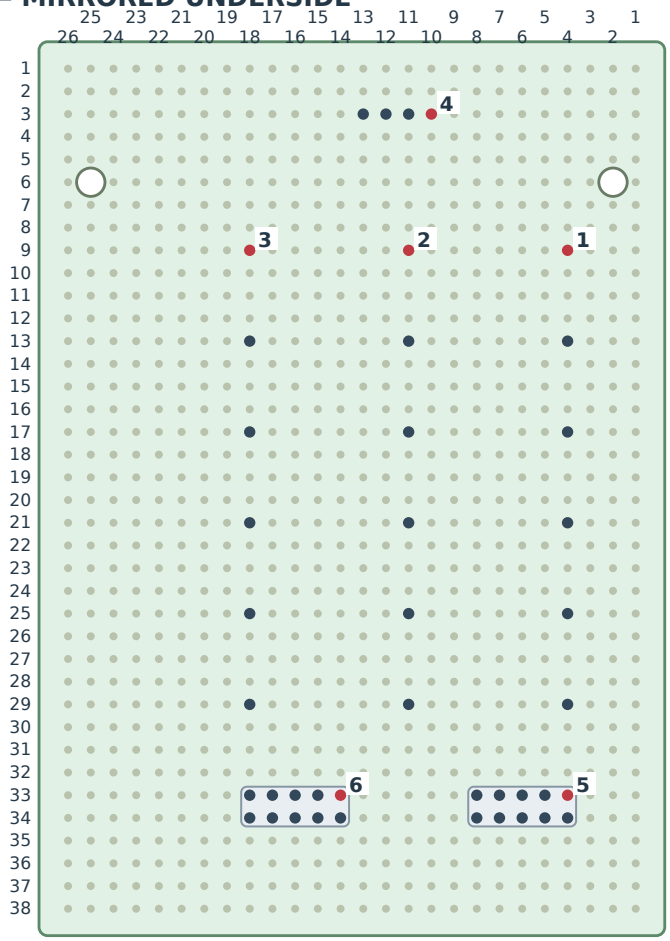


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

a3 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

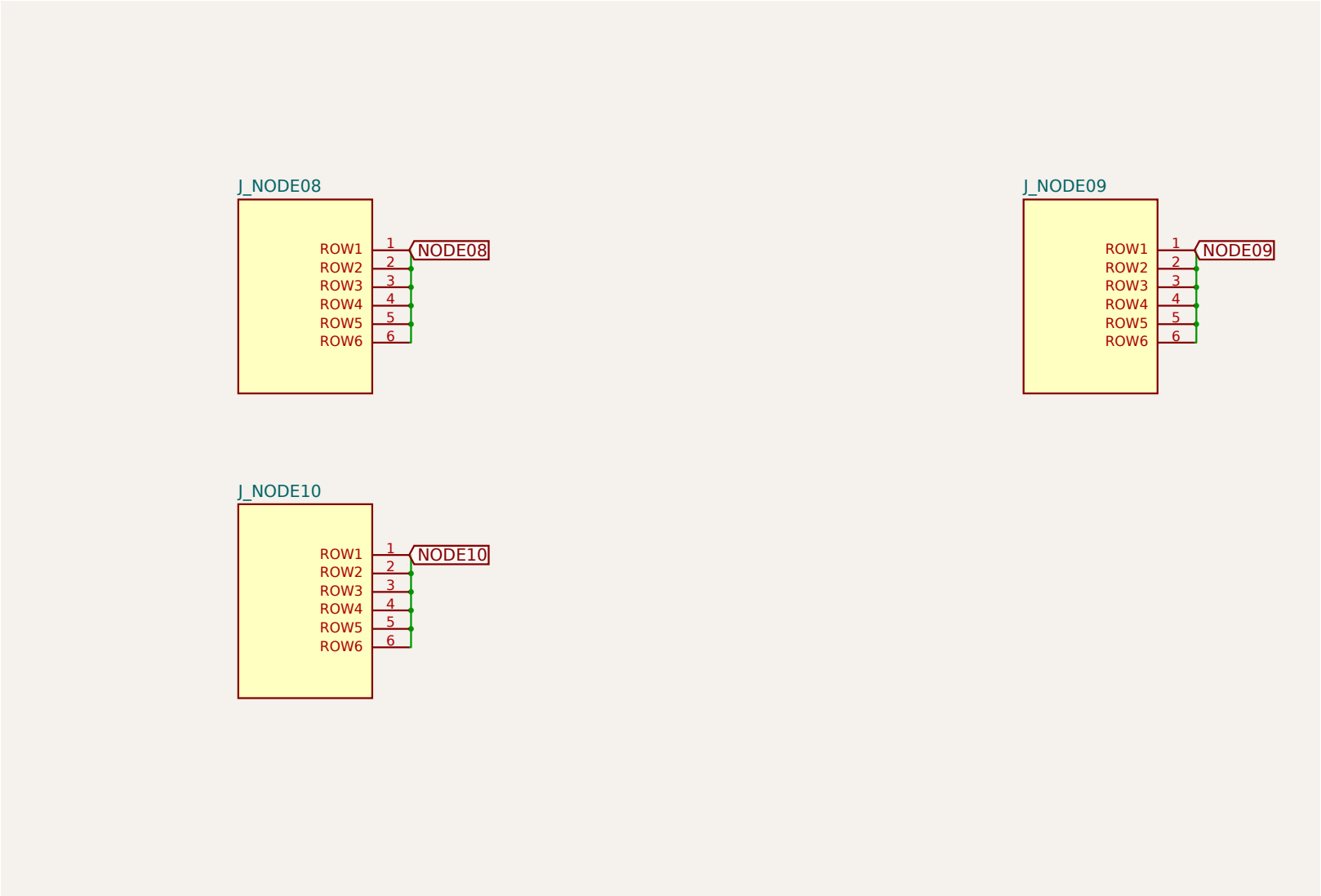
PROTOTYPE: physical fit and hardware tests pending.

a3 — numbered parts

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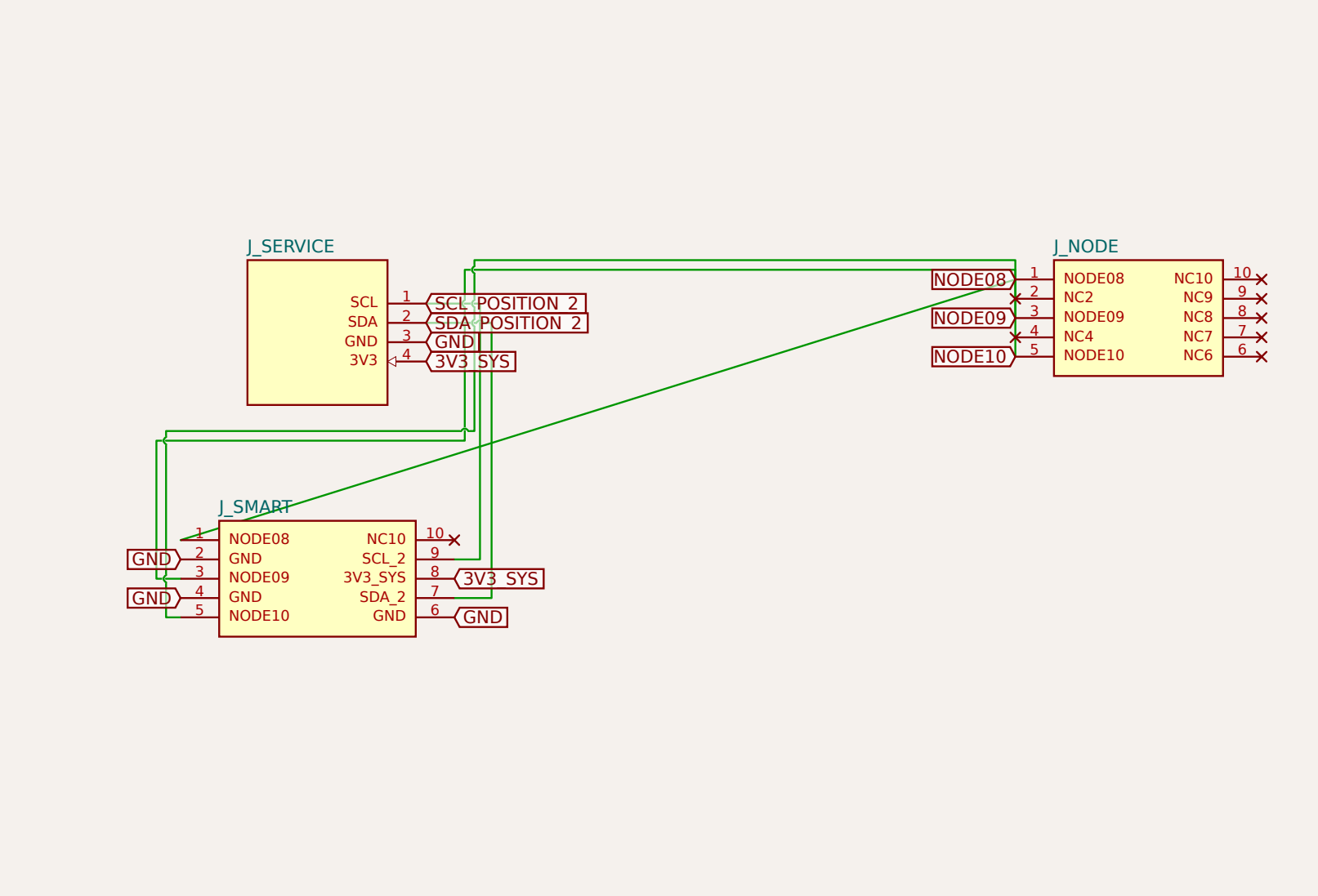
#	Reference	Purpose	Install	Adapter / carrier
1	J_NODE08	NODE08 — six contacts	CP28	
2	J_NODE09	NODE09 — six contacts	CP28	
3	J_NODE10	NODE10 — six contacts	CP28	
4	J_SERVICE	OLED service pogo contacts	CP28	
5	J_NODE	NODE cable → matrix	CP28	
6	J_SMART	SMART_RAW v2 → channel 2	CP28	

a3 — Nodes



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

a3 — Service

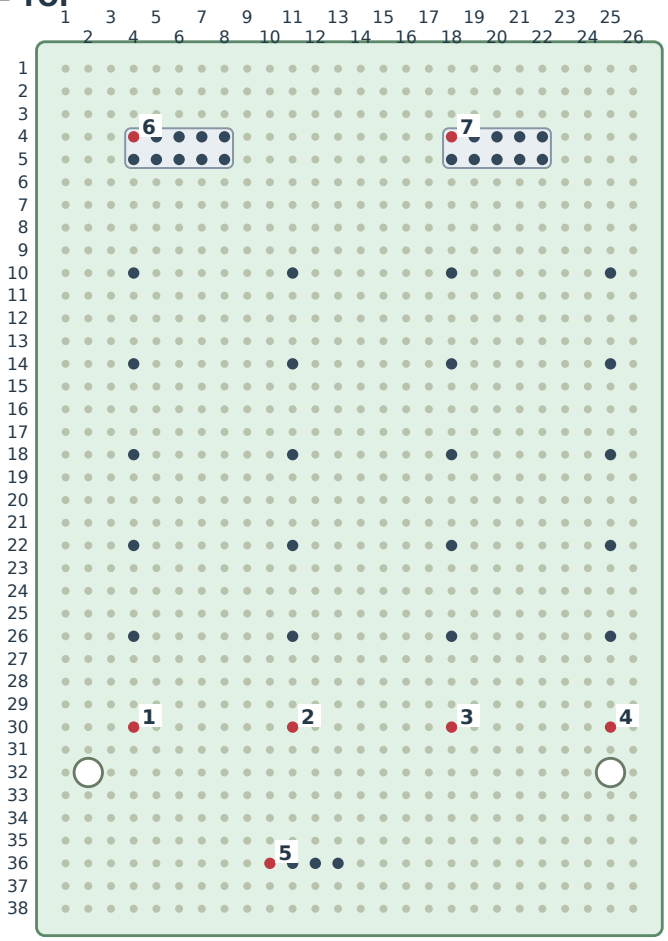


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

b1

B1 contacts

b1 — TOP

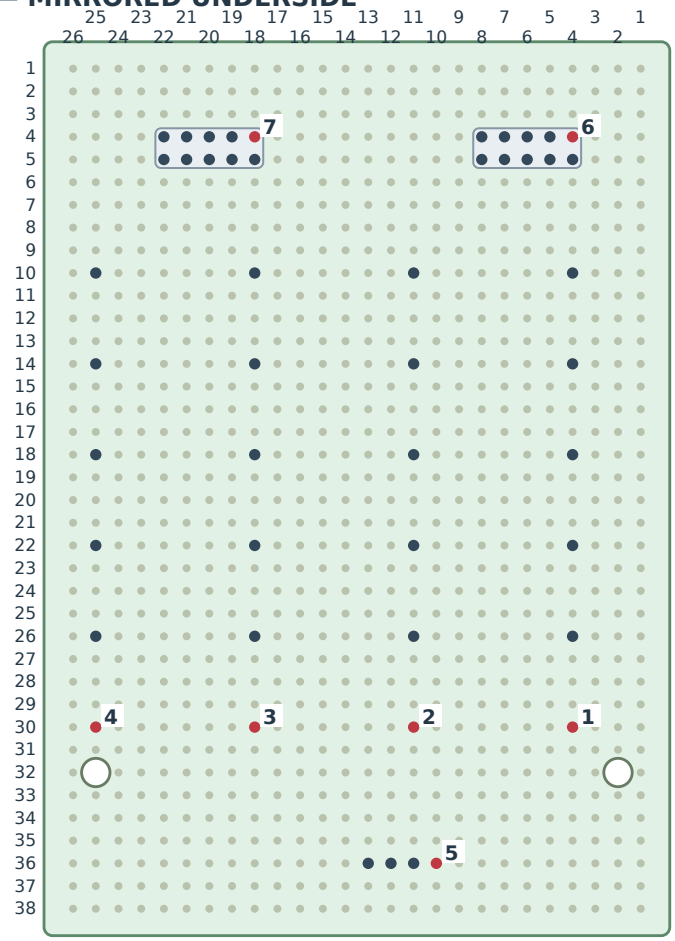


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

b1 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

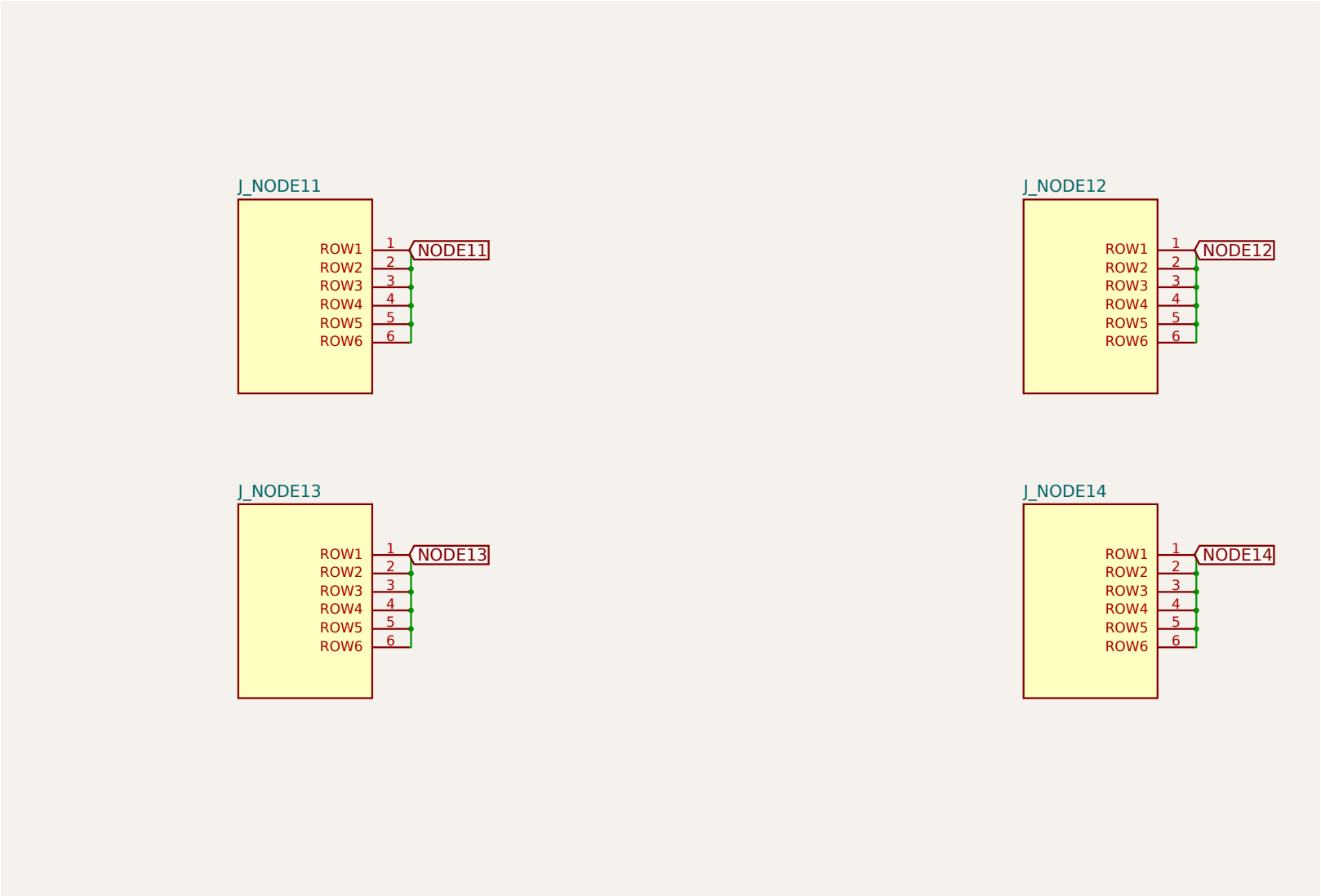
PROTOTYPE: physical fit and hardware tests pending.

b1 — numbered parts

Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

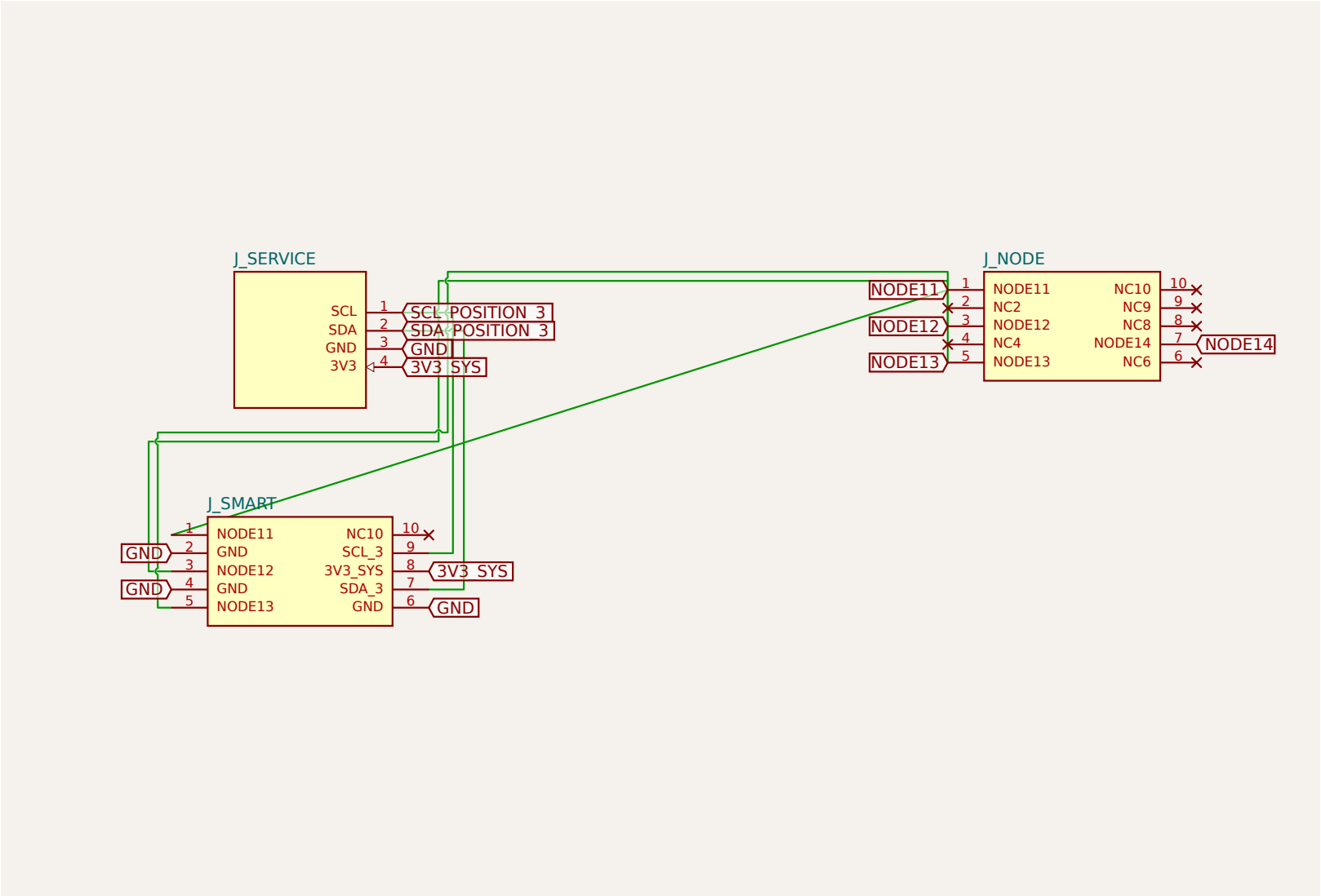
#	Reference	Purpose	Install	Adapter / carrier
1	J_NODE11	NODE11 — six contacts	CP29	
2	J_NODE12	NODE12 — six contacts	CP29	
3	J_NODE13	NODE13 — six contacts	CP29	
4	J_NODE14	NODE14 — six contacts	CP29	
5	J_SERVICE	OLED service pogo contacts	CP29	
6	J_NODE	NODE cable → matrix	CP29	
7	J_SMART	SMART_RAW v2 → channel 3	CP29	

b1 — Nodes



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

b1 — Service

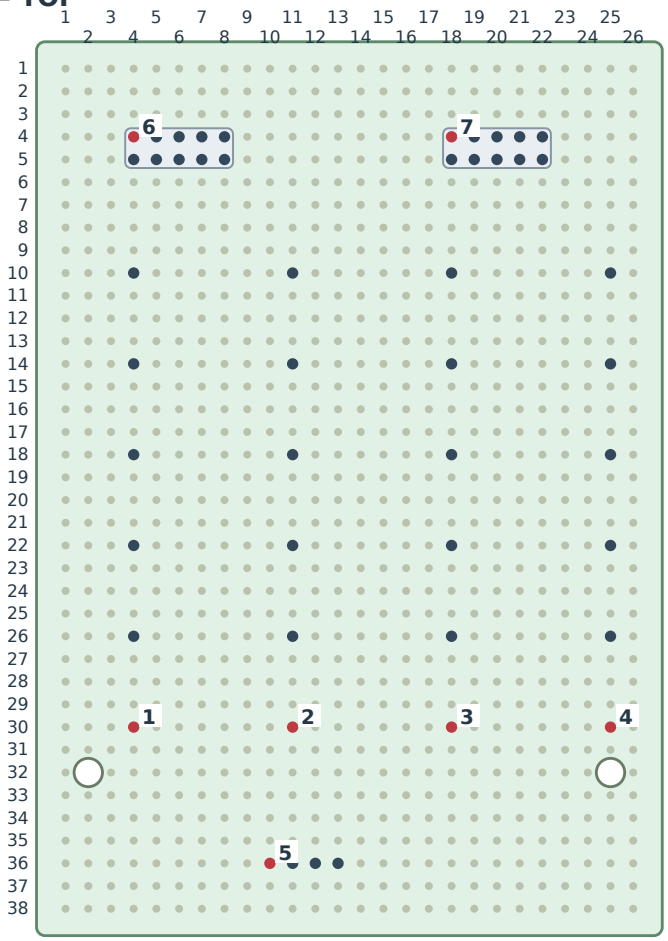


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

b2

B2 contacts

b2 — TOP

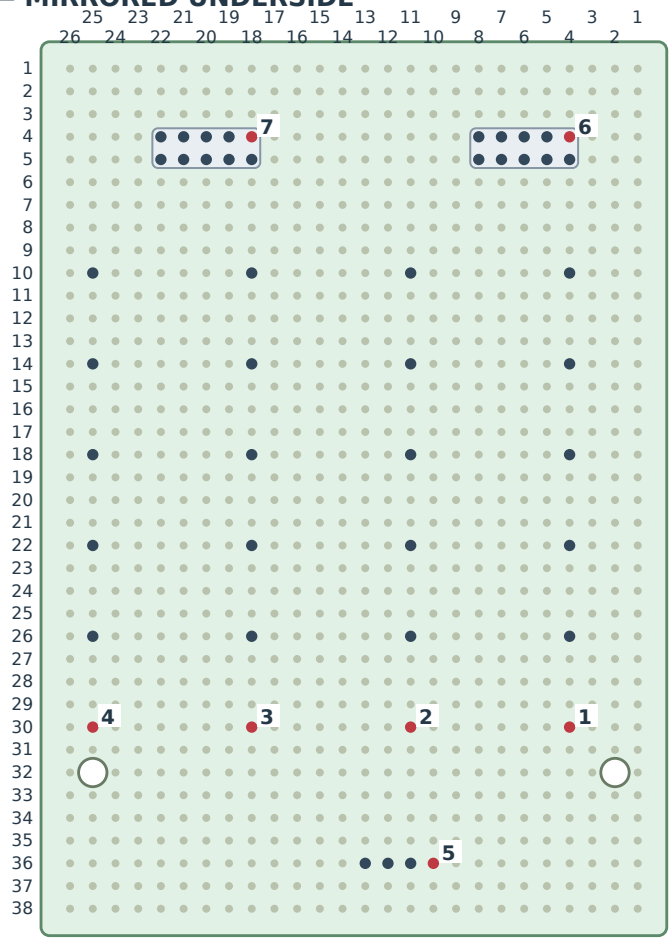


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

b2 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

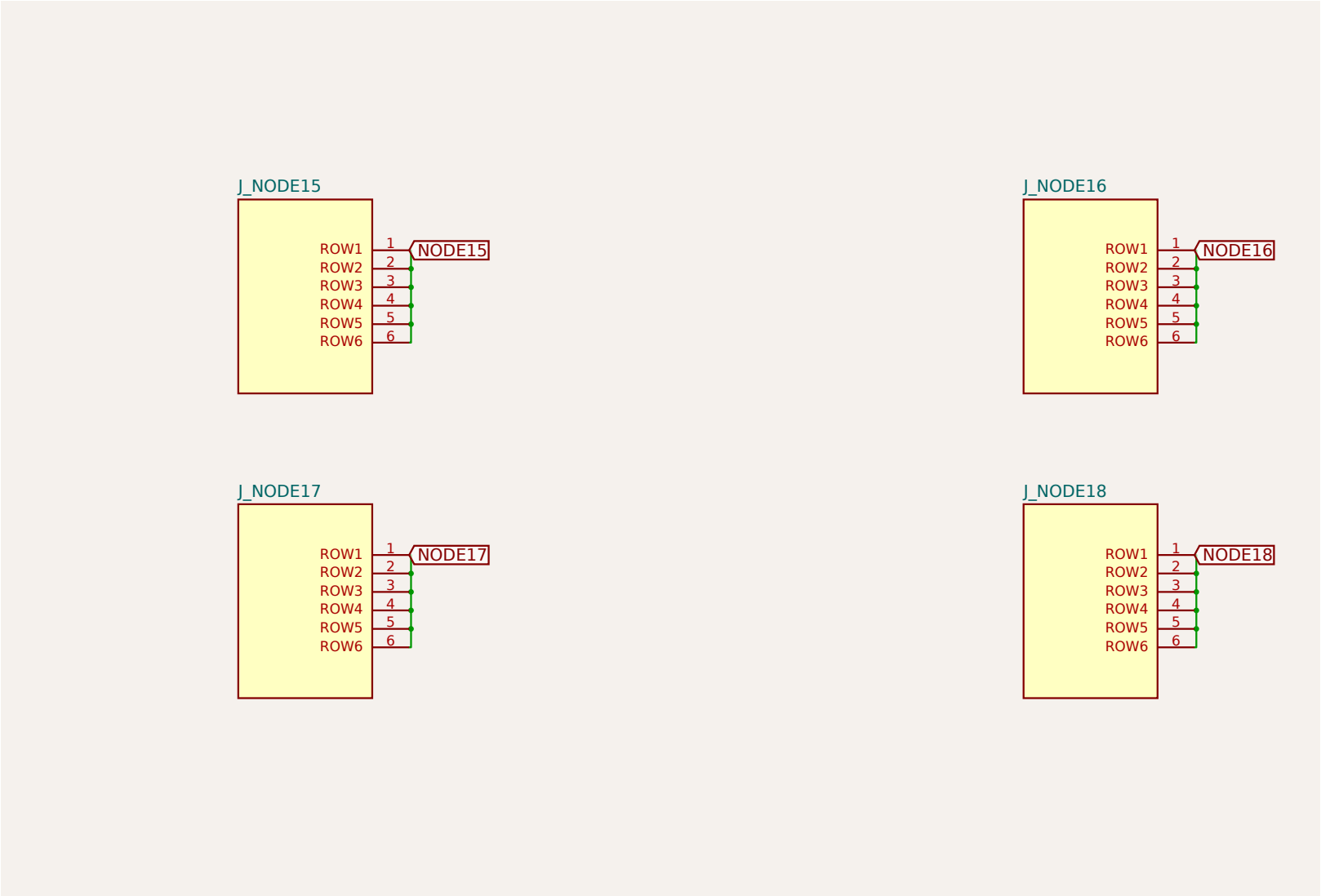
PROTOTYPE: physical fit and hardware tests pending.

b2 — numbered parts

Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

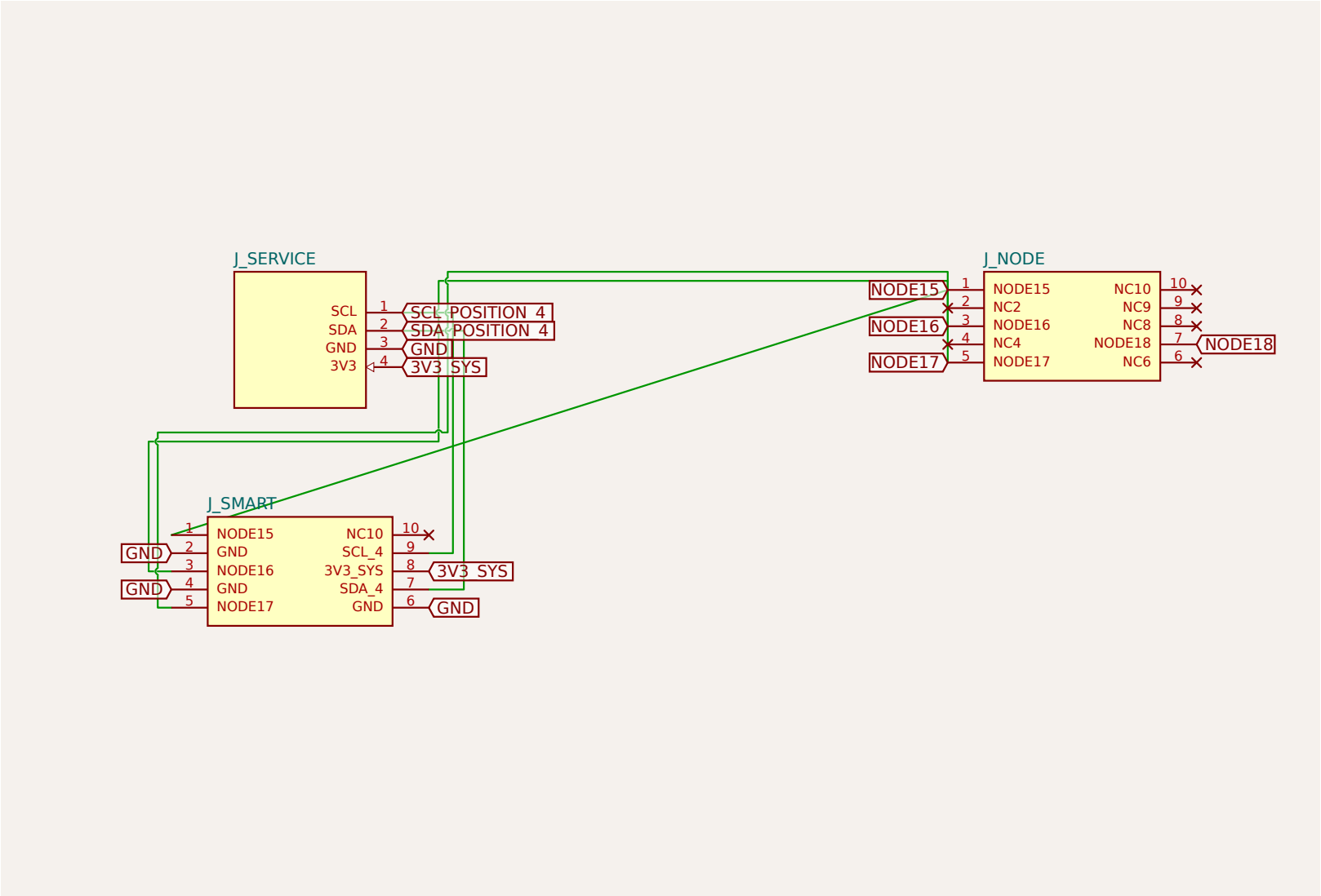
#	Reference	Purpose	Install	Adapter / carrier
1	J_NODE15	NODE15 — six contacts	CP30	
2	J_NODE16	NODE16 — six contacts	CP30	
3	J_NODE17	NODE17 — six contacts	CP30	
4	J_NODE18	NODE18 — six contacts	CP30	
5	J_SERVICE	OLED service pogo contacts	CP30	
6	J_NODE	NODE cable → matrix	CP30	
7	J_SMART	SMART_RAW v2 → channel 4	CP30	

b2 — Nodes



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

b2 — Service

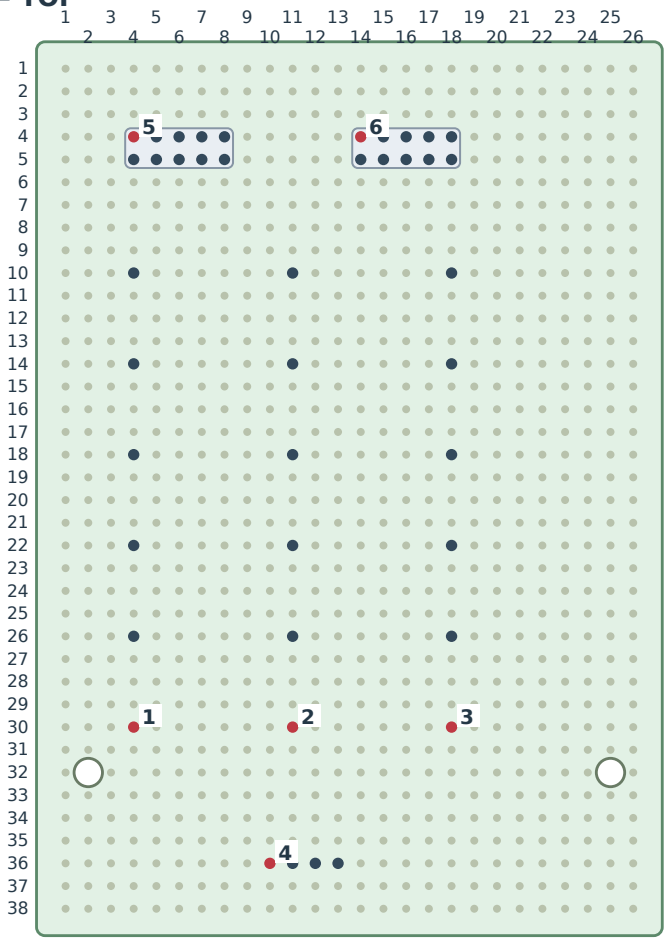


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

b3

B3 contacts

b3 — TOP

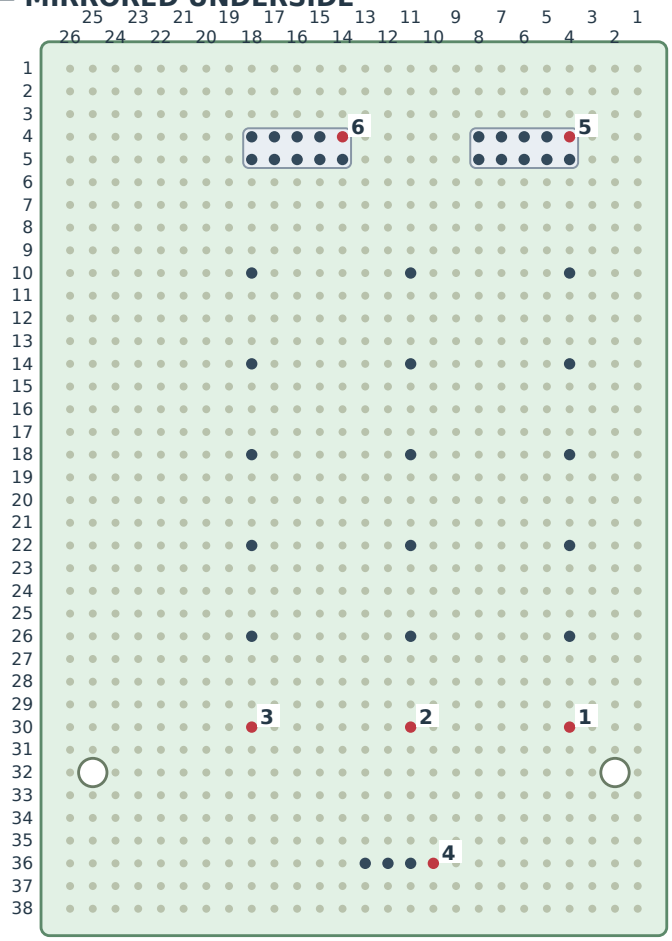


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

b3 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

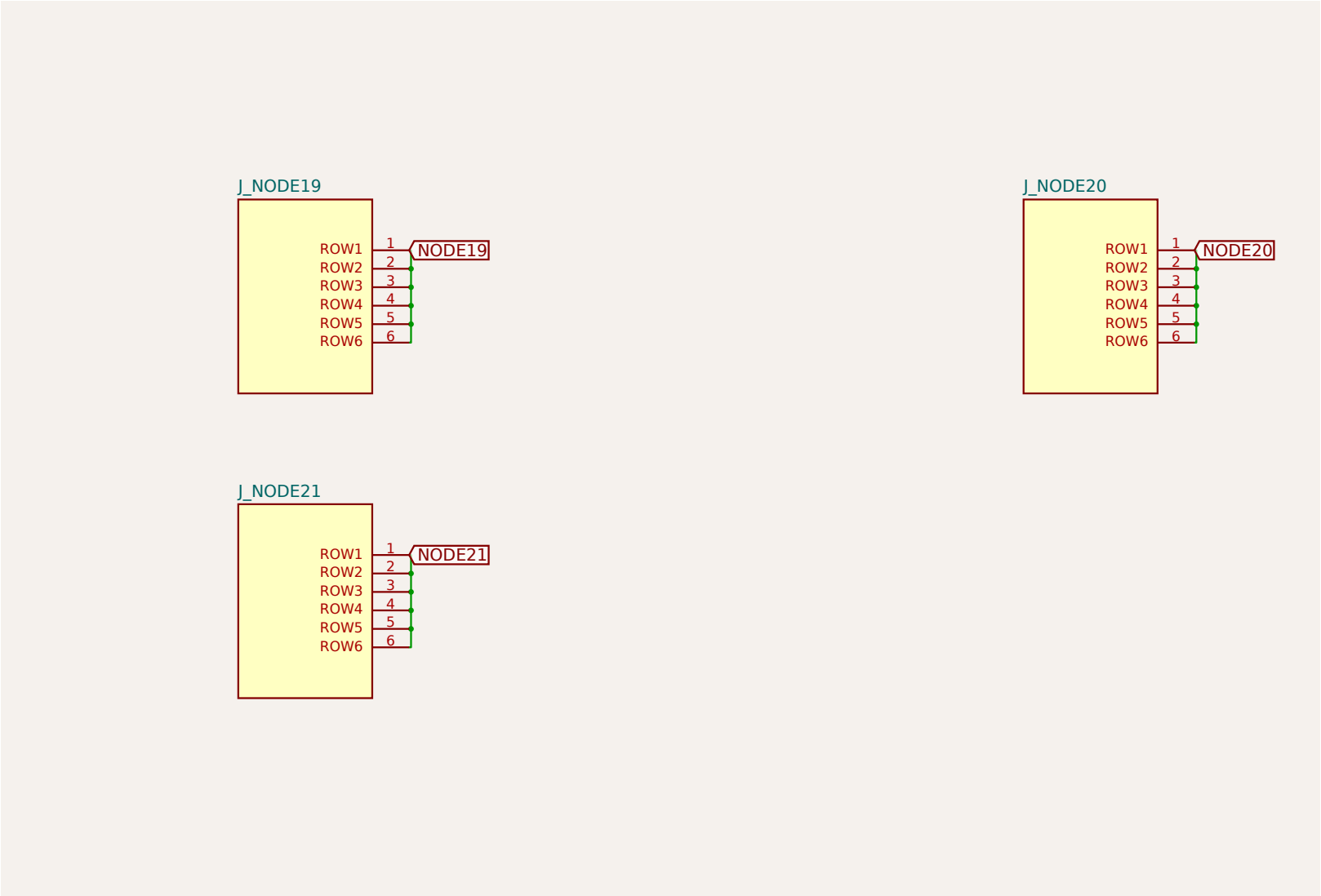
PROTOTYPE: physical fit and hardware tests pending.

b3 — numbered parts

Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

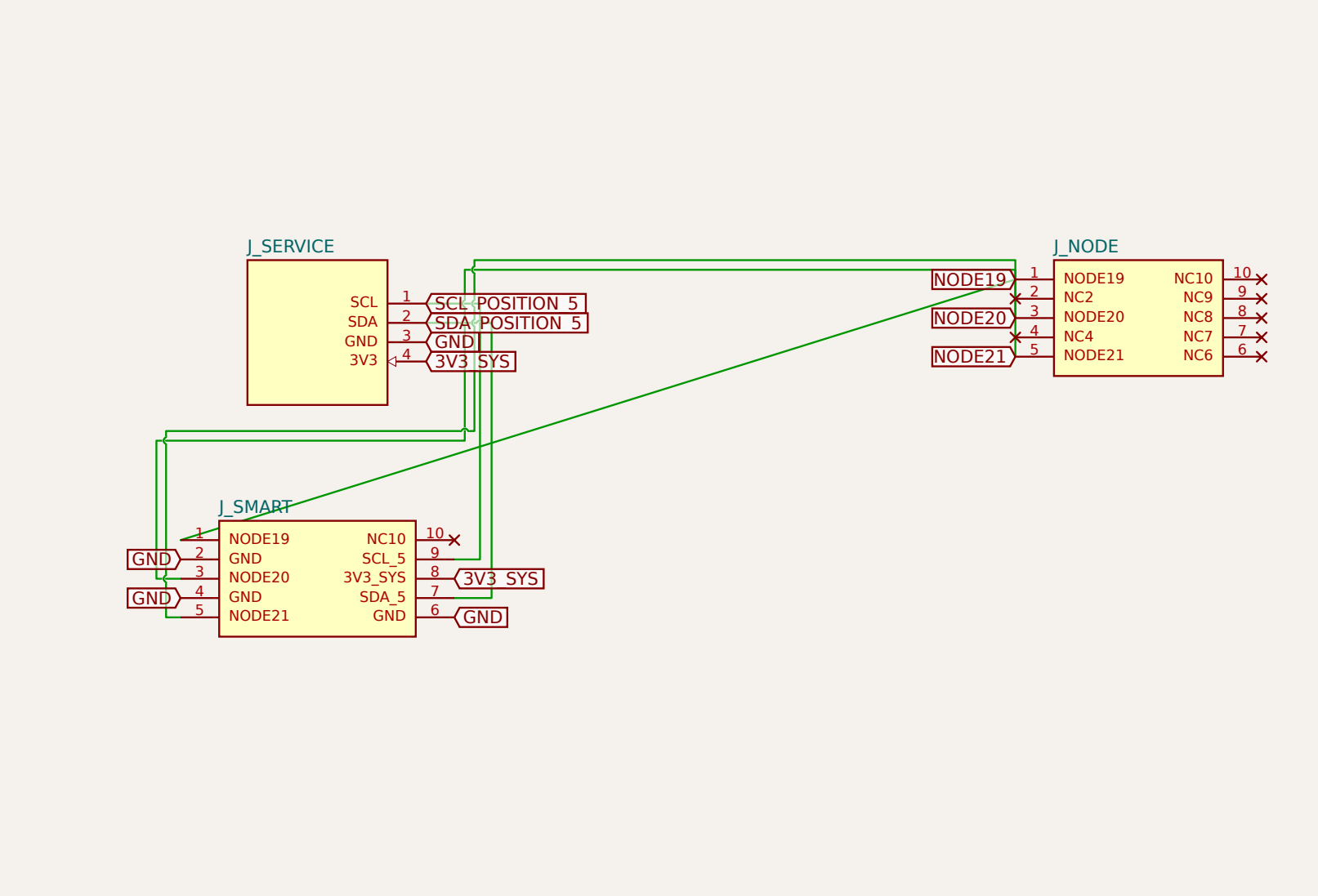
#	Reference	Purpose	Install	Adapter / carrier
1	J_NODE19	NODE19 — six contacts	CP31	
2	J_NODE20	NODE20 — six contacts	CP31	
3	J_NODE21	NODE21 — six contacts	CP31	
4	J_SERVICE	OLED service pogo contacts	CP31	
5	J_NODE	NODE cable → matrix	CP31	
6	J_SMART	SMART_RAW v2 → channel 5	CP31	

b3 — Nodes



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

b3 — Service

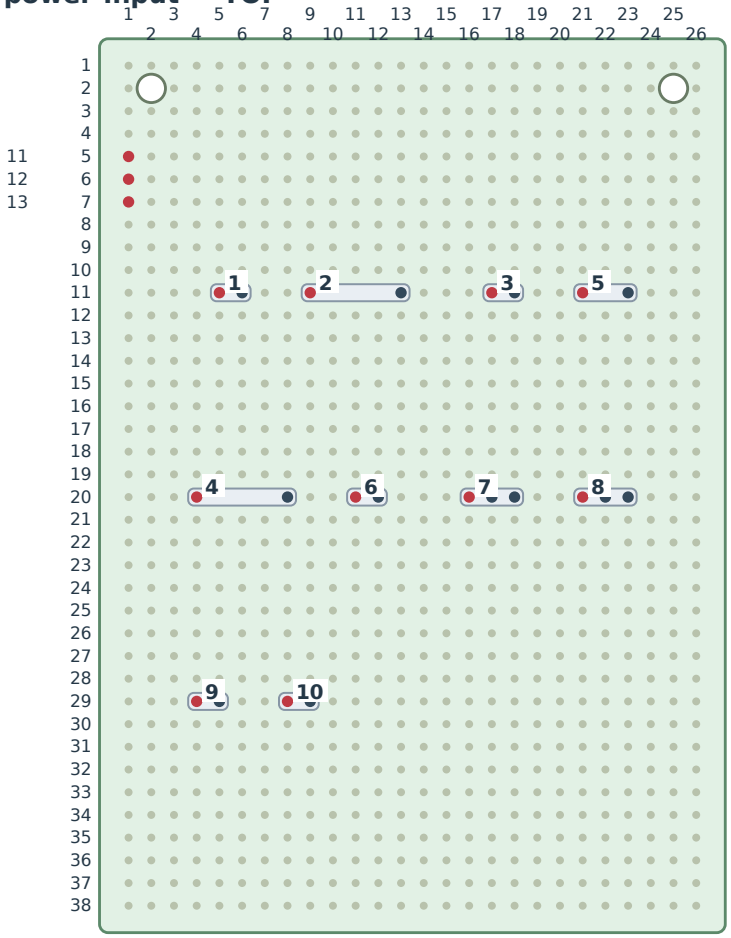


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

power-input

power input

power-input — TOP

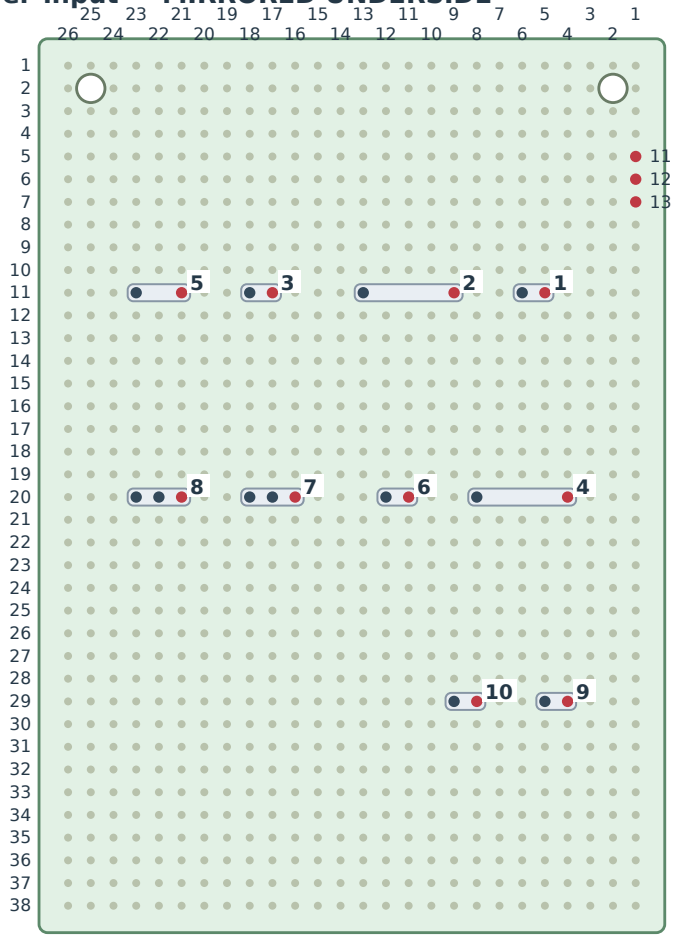


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

power-input — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

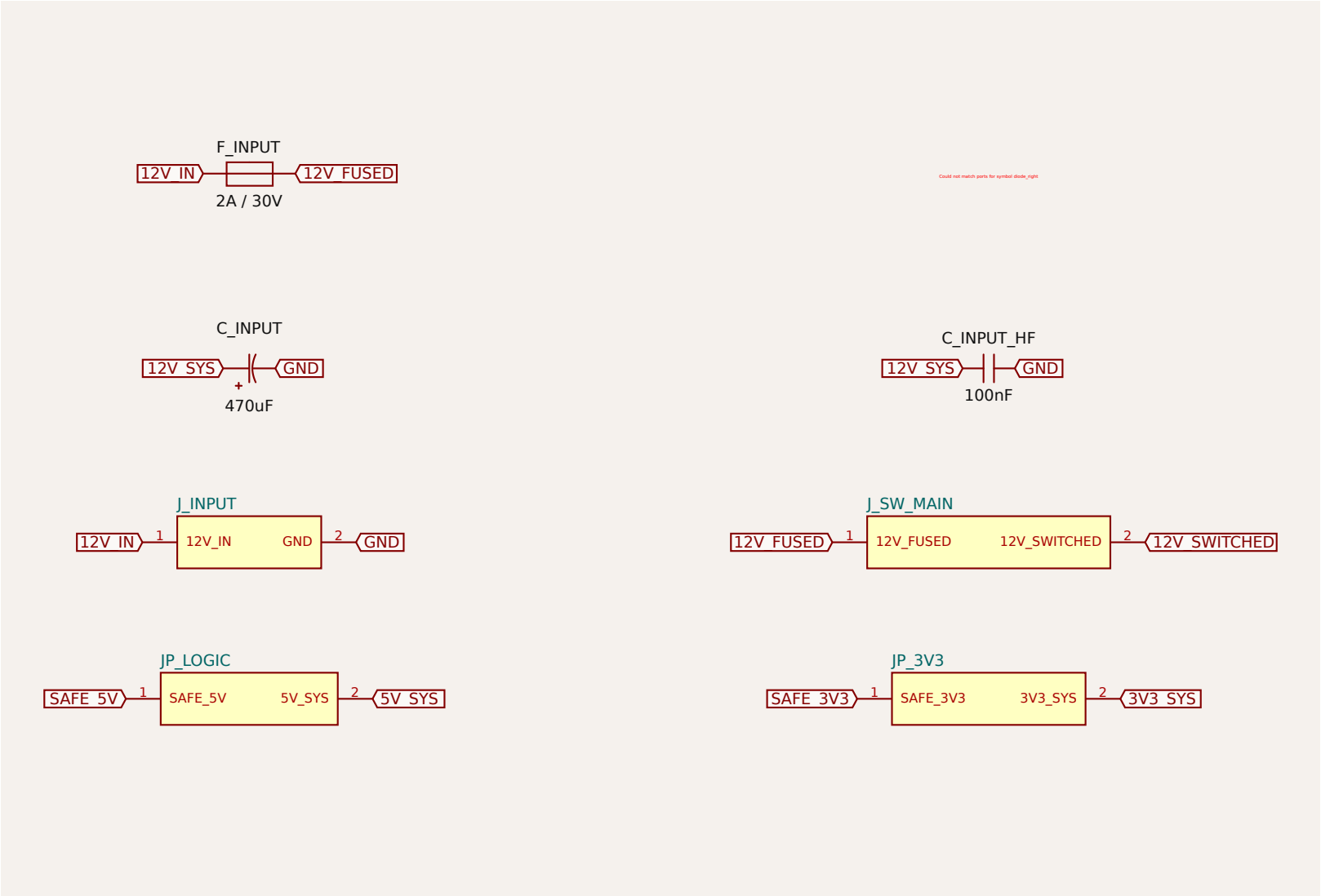
PROTOTYPE: physical fit and hardware tests pending.

power-input — numbered parts

Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

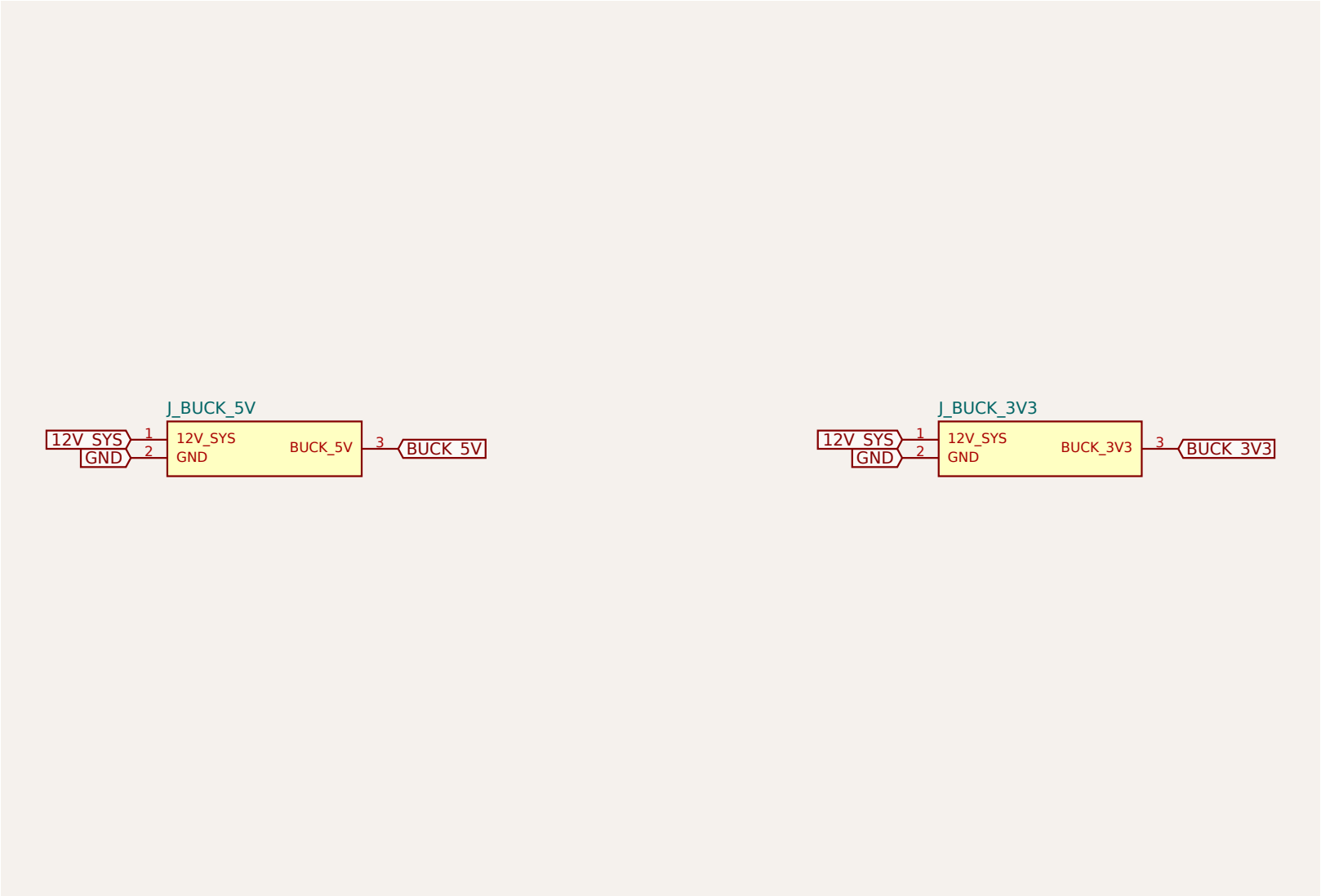
#	Reference	Purpose	Install	Adapter / carrier
1	J_INPUT	5.5x2.1 mm barrel pigtail, centre positive	CP9	
2	F_INPUT	2A slow fuse	CP9	
3	J_SW_MAIN	latching switch, 3A at 12V DC	CP9	
4	D_REVERSE	1N5822	CP9	
5	C_INPUT	470uF	CP9	
6	C_INPUT_HF	100nF	CP9	
7	J_BUCK_5V	External LM2596 set to 5V	CP9	Purchased LM2596 adjustable module
8	J_BUCK_3V3	External XL4015 set to 3.3V	CP9	Purchased XL4015 adjustable module
9	JP_LOGIC	removable 5-V consumer link	CP9	
10	JP_3V3	removable 3.3-V consumer link	CP9	
11	J_LINK_1	Harness terminal: 12V_SYS	CP9	
12	J_LINK_2	Harness terminal: BUCK_5V	CP9	
13	J_LINK_3	Harness terminal: BUCK_3V3	CP9	

power-input — Input



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

power-input — Circuit

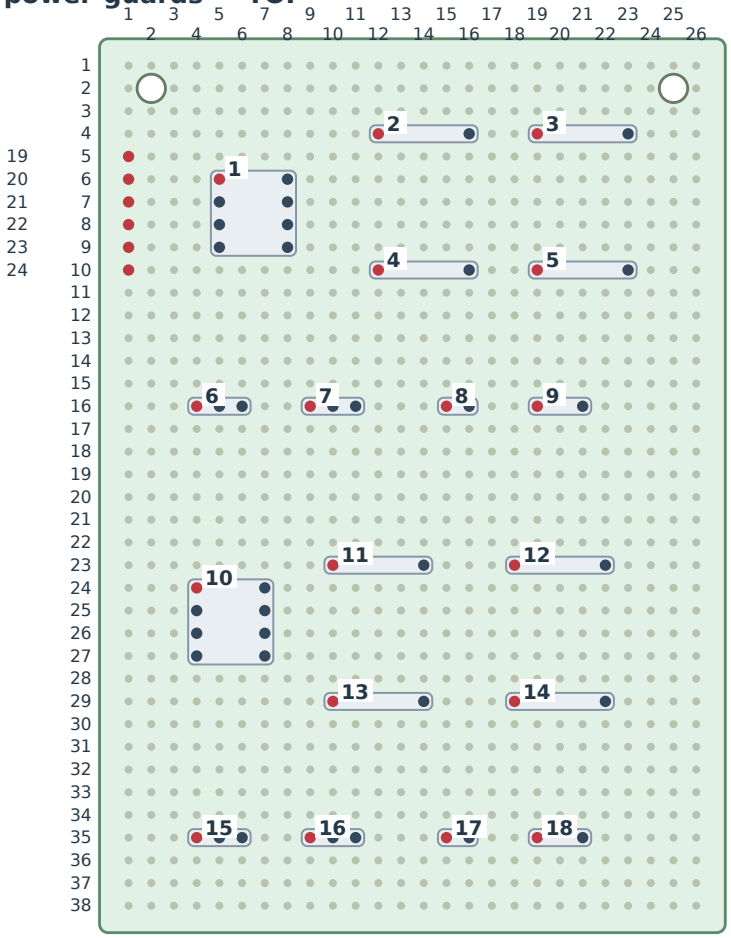


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

power-guards

power guards

power-guards — TOP

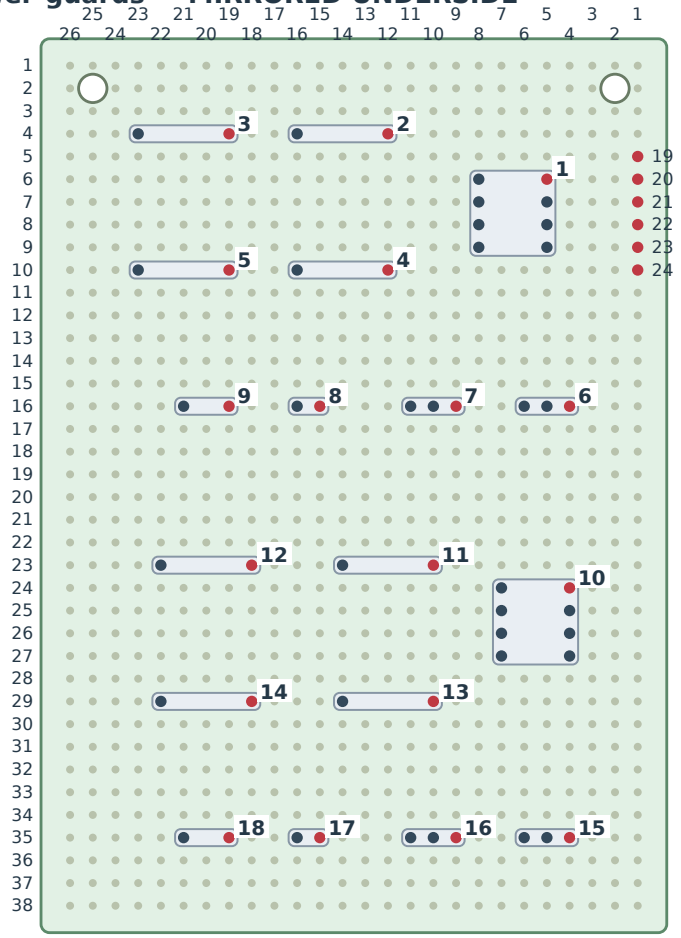


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

power-guards — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

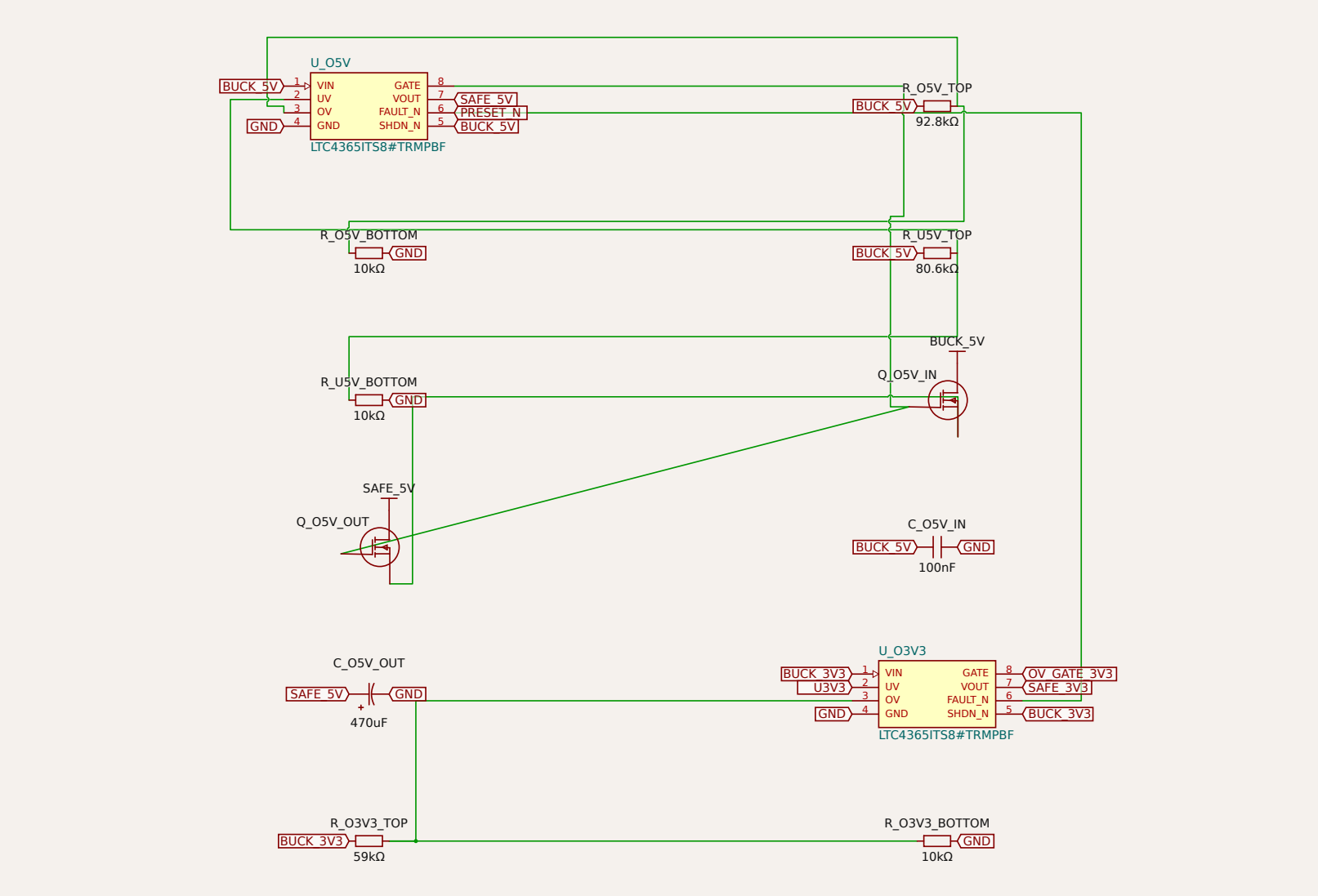
PROTOTYPE: physical fit and hardware tests pending.

power-guards — numbered parts

Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

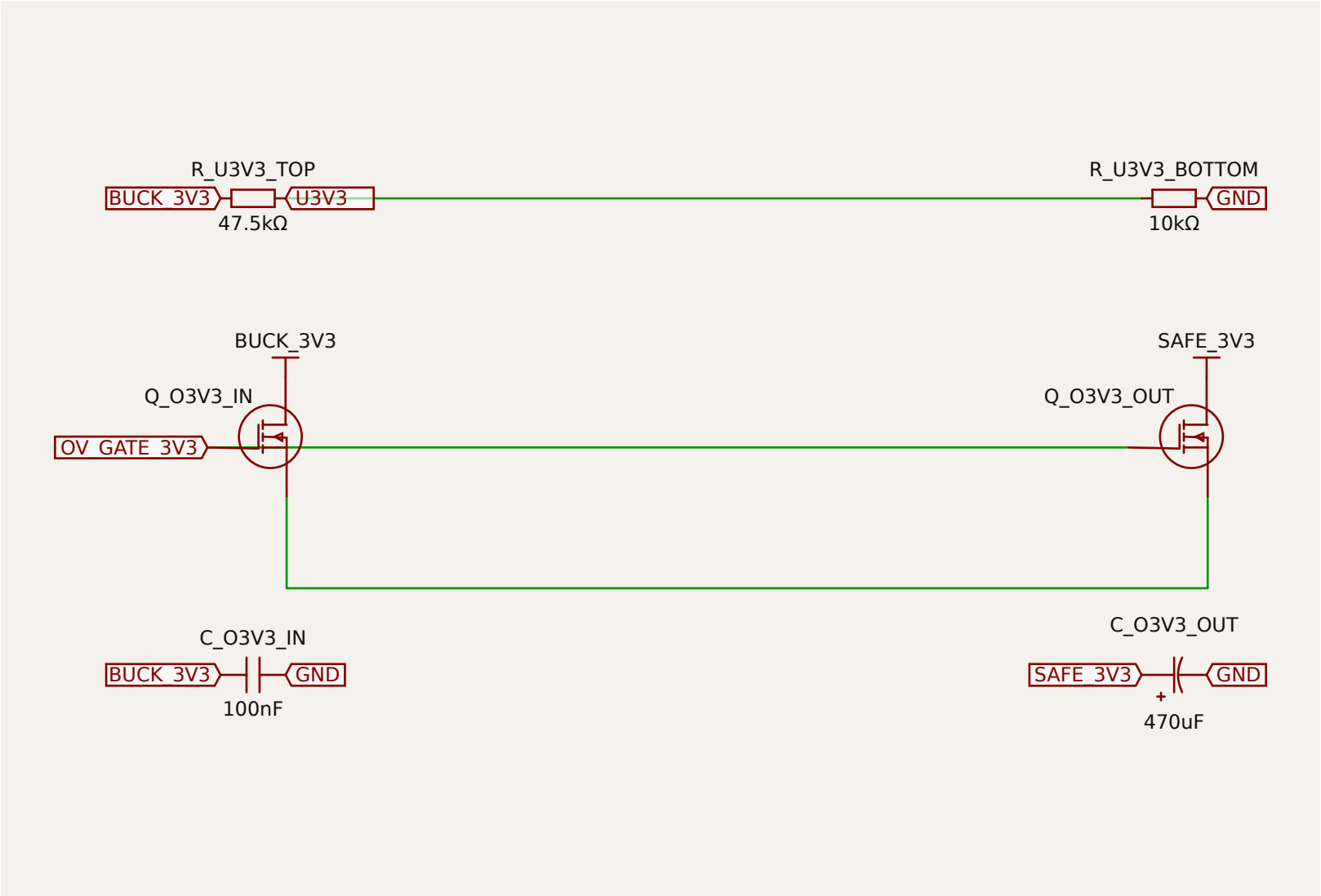
#	Reference	Purpose	Install	Adapter / carrier
1	U_OV_5V	LTC4365ITS8#TRMPBF	CP10	TSOT23-8 to numbered DIP8
2	R_OV_5V_TOP	92.8k	CP10	
3	R_OV_5V_BOTTOM	10k	CP10	
4	R_UV_5V_TOP	80.6k	CP10	
5	R_UV_5V_BOTTOM	10k	CP10	
6	Q_OV_5V_IN	IRLML6344TRPBF	CP10	SOT23 to numbered 3-pin adapter
7	Q_OV_5V_OUT	IRLML6344TRPBF	CP10	SOT23 to numbered 3-pin adapter
8	C_OV_5V_IN	100nF	CP10	
9	C_OV_5V_OUT	470uF	CP10	
10	U_OV_3V3	LTC4365ITS8#TRMPBF	CP10	TSOT23-8 to numbered DIP8
11	R_OV_3V3_TOP	59k	CP10	
12	R_OV_3V3_BOTTOM	10k	CP10	
13	R_UV_3V3_TOP	47.5k	CP10	
14	R_UV_3V3_BOTTOM	10k	CP10	
15	Q_OV_3V3_IN	IRLML6344TRPBF	CP10	SOT23 to numbered 3-pin adapter
16	Q_OV_3V3_OUT	IRLML6344TRPBF	CP10	SOT23 to numbered 3-pin adapter
17	C_OV_3V3_IN	100nF	CP10	
18	C_OV_3V3_OUT	470uF	CP10	
19	J_LINK_1	Harness terminal: GND	CP10	
20	J_LINK_2	Harness terminal: BUCK_5V	CP10	
21	J_LINK_3	Harness terminal: BUCK_3V3	CP10	
22	J_LINK_4	Harness terminal: SAFE_5V	CP10	
23	J_LINK_5	Harness terminal: SAFE_3V3	CP10	
24	J_LINK_6	Harness terminal: PRESET_N	CP10	

power-guards — VoltageGuard 1



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

power-guards — VoltageGuard 2

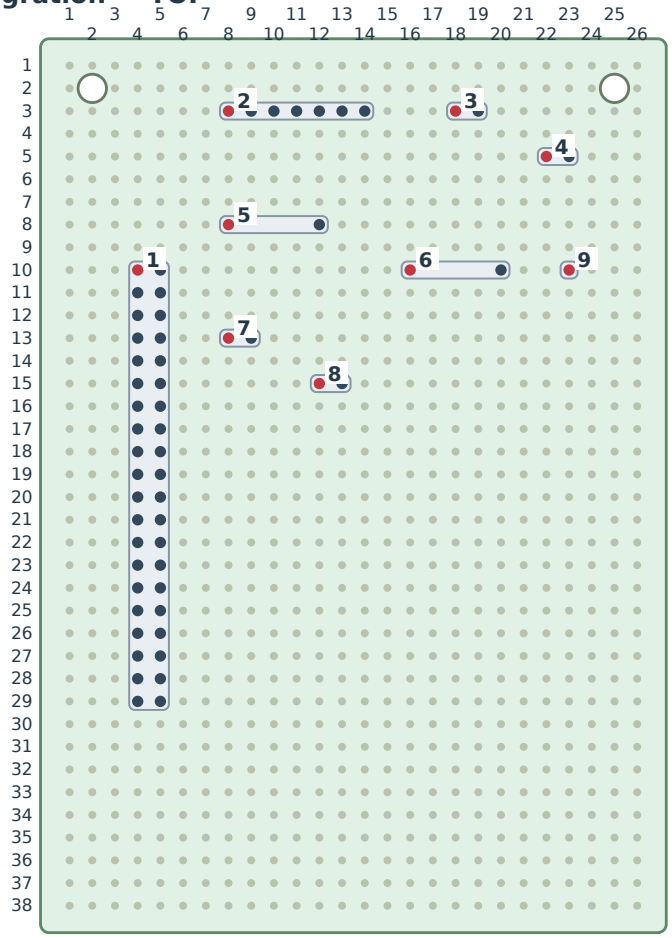


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

integration

Pi buttons / RC522 / enclosure harness

integration — TOP

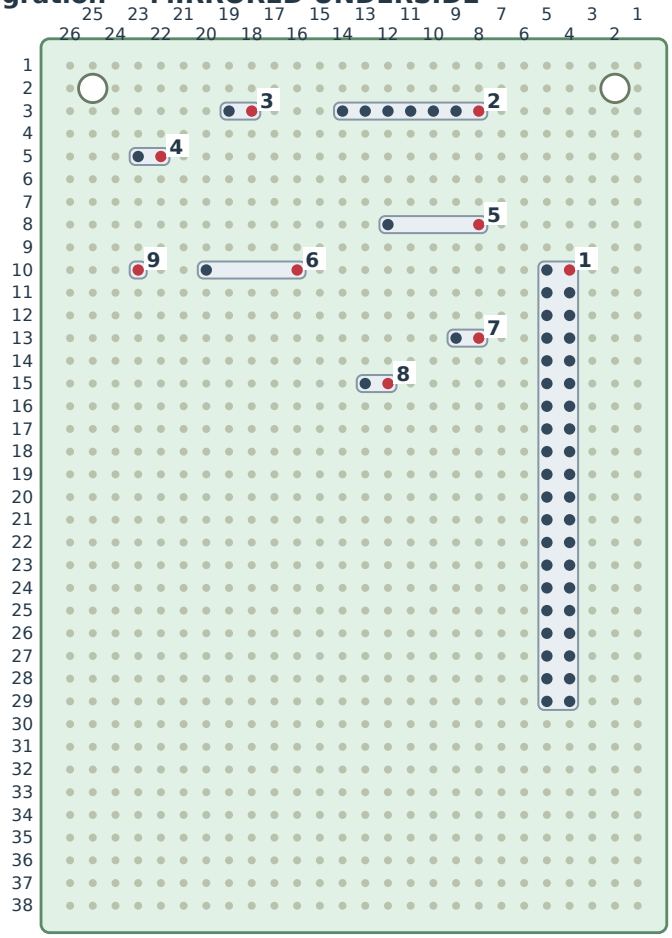


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

integration — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

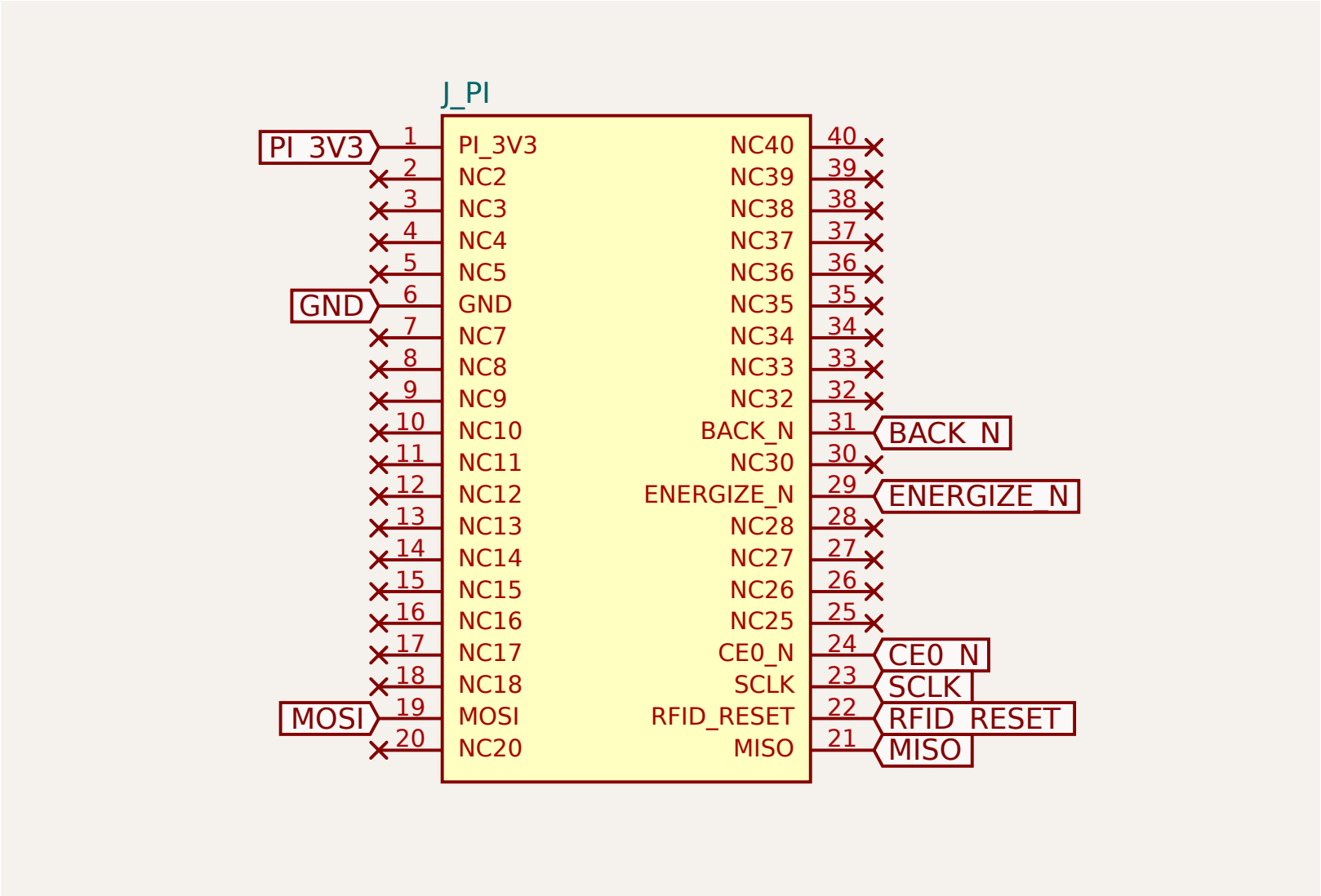
PROTOTYPE: physical fit and hardware tests pending.

integration — numbered parts

Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

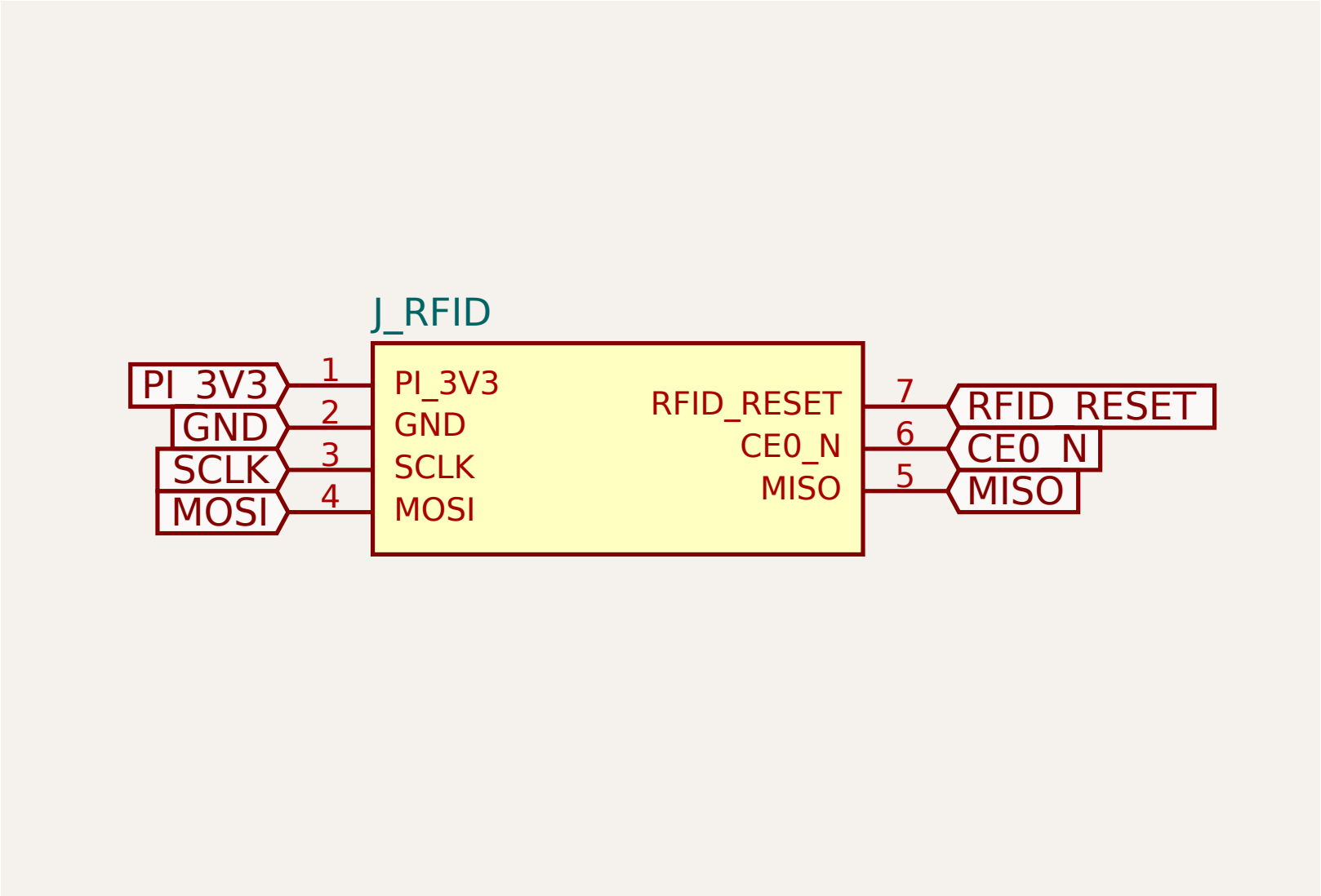
#	Reference	Purpose	Install	Adapter / carrier
1	J_PI	Pi physical header	CP32	
2	J_RFID	RC522 labelled SPI carrier interface	CP34	Purchased RC522 reader
3	J_SW_ENERGIZE	Energizar dry-contact button	CP32	
4	J_SW_BACK	Back dry-contact button	CP32	
5	R_ENERGIZE	10k	CP32	
6	R_BACK	10k	CP32	
7	C_ENERGIZE	100nF	CP32	
8	C_BACK	100nF	CP32	
9	J_GROUND	ground to electronics	CP32	

integration — Pi



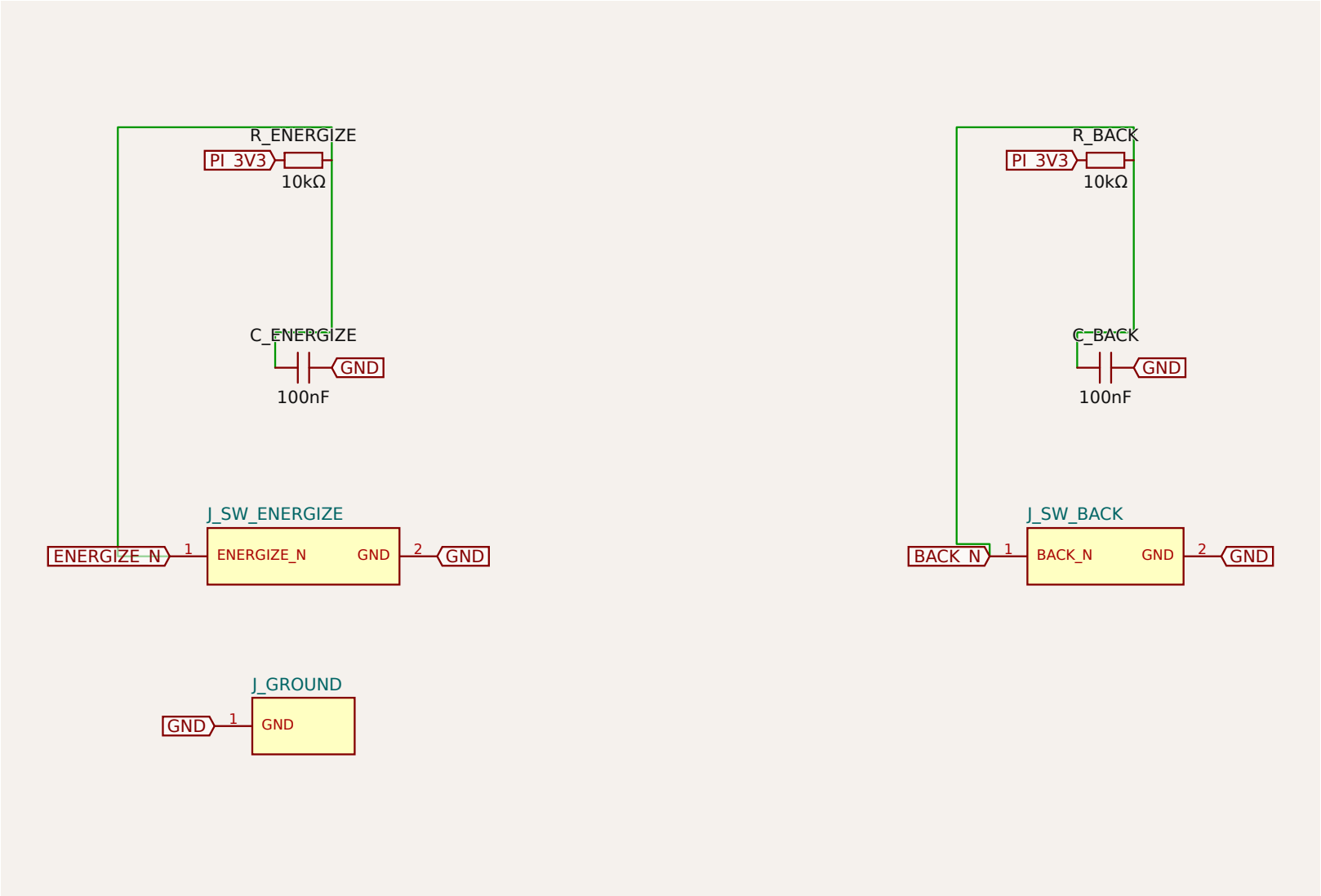
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

integration — Circuit



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

integration — Buttons

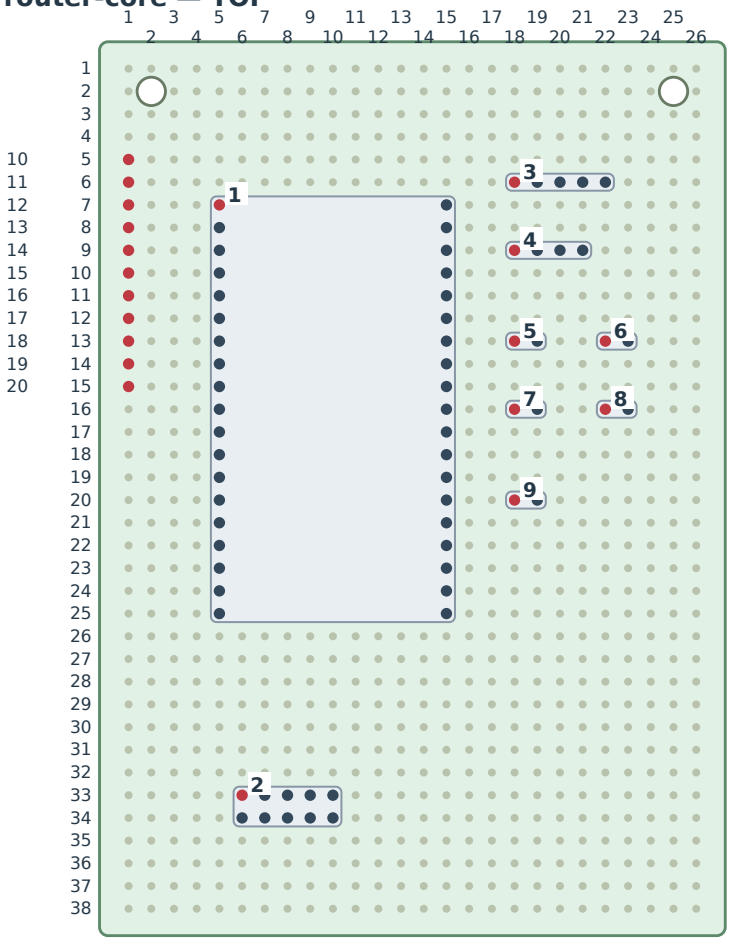


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

router-core

ESP1 carrier and enclosure I/O

router-core — TOP

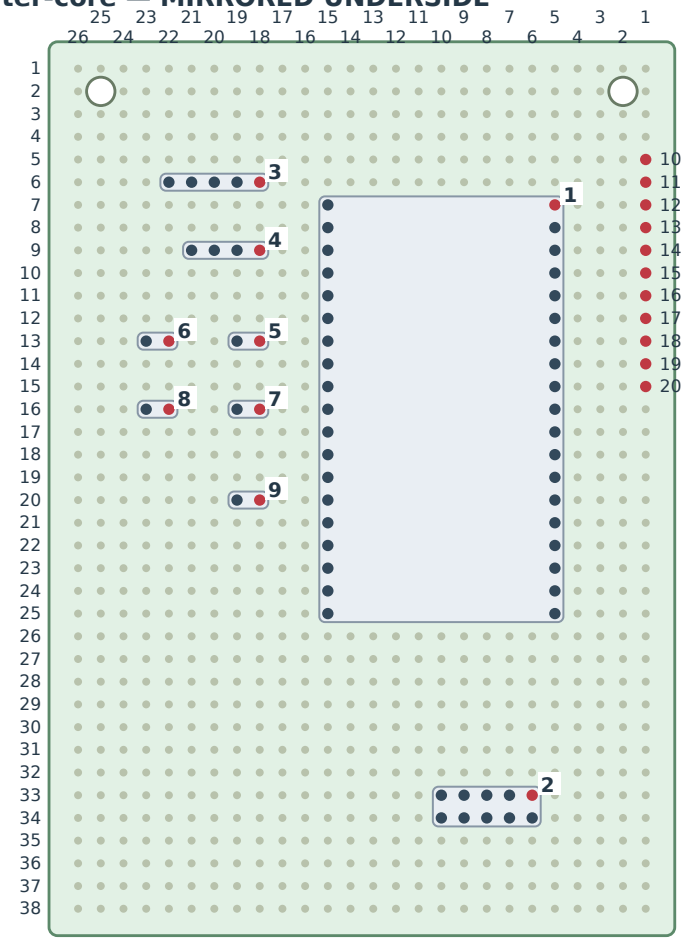


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

router-core — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

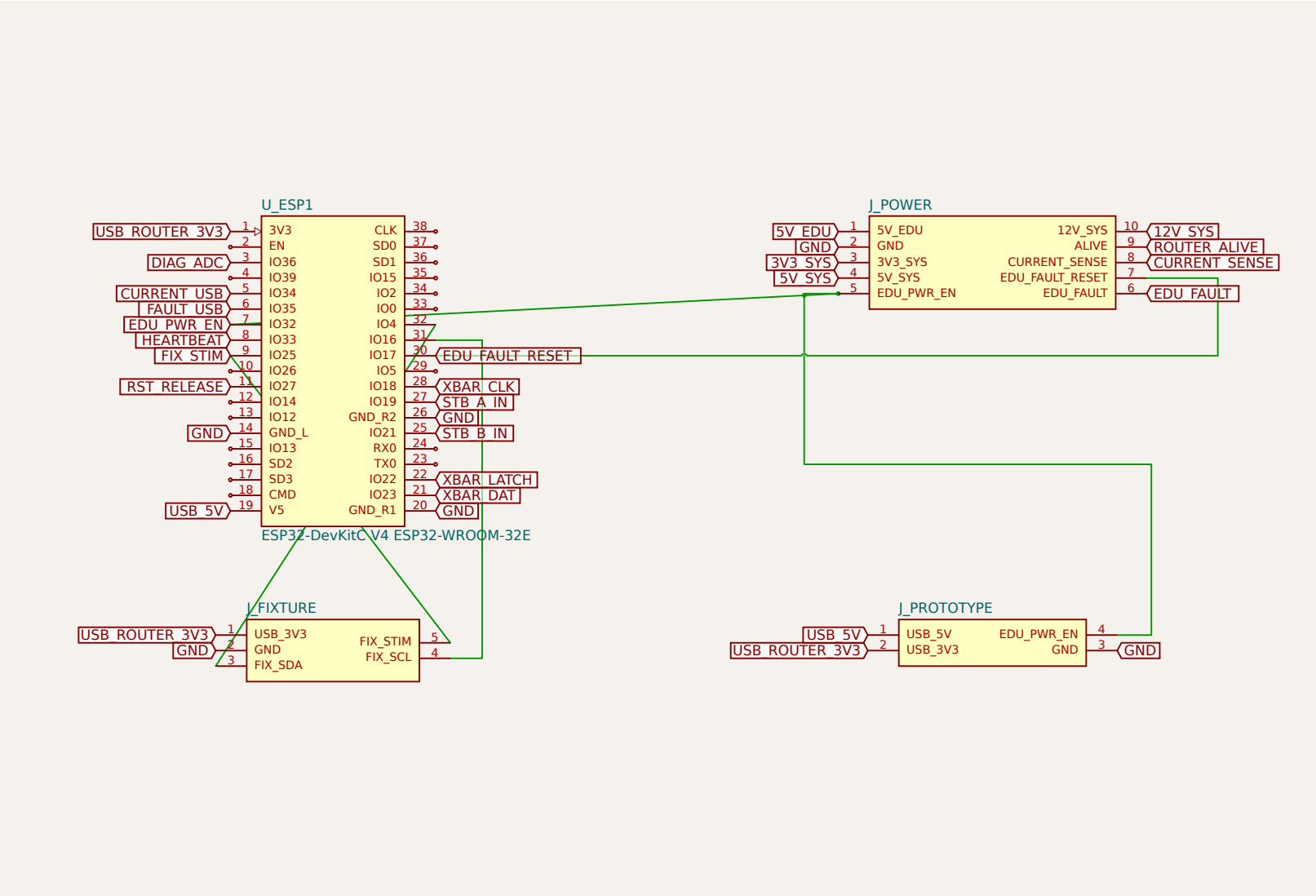
PROTOTYPE: physical fit and hardware tests pending.

router-core — numbered parts

Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

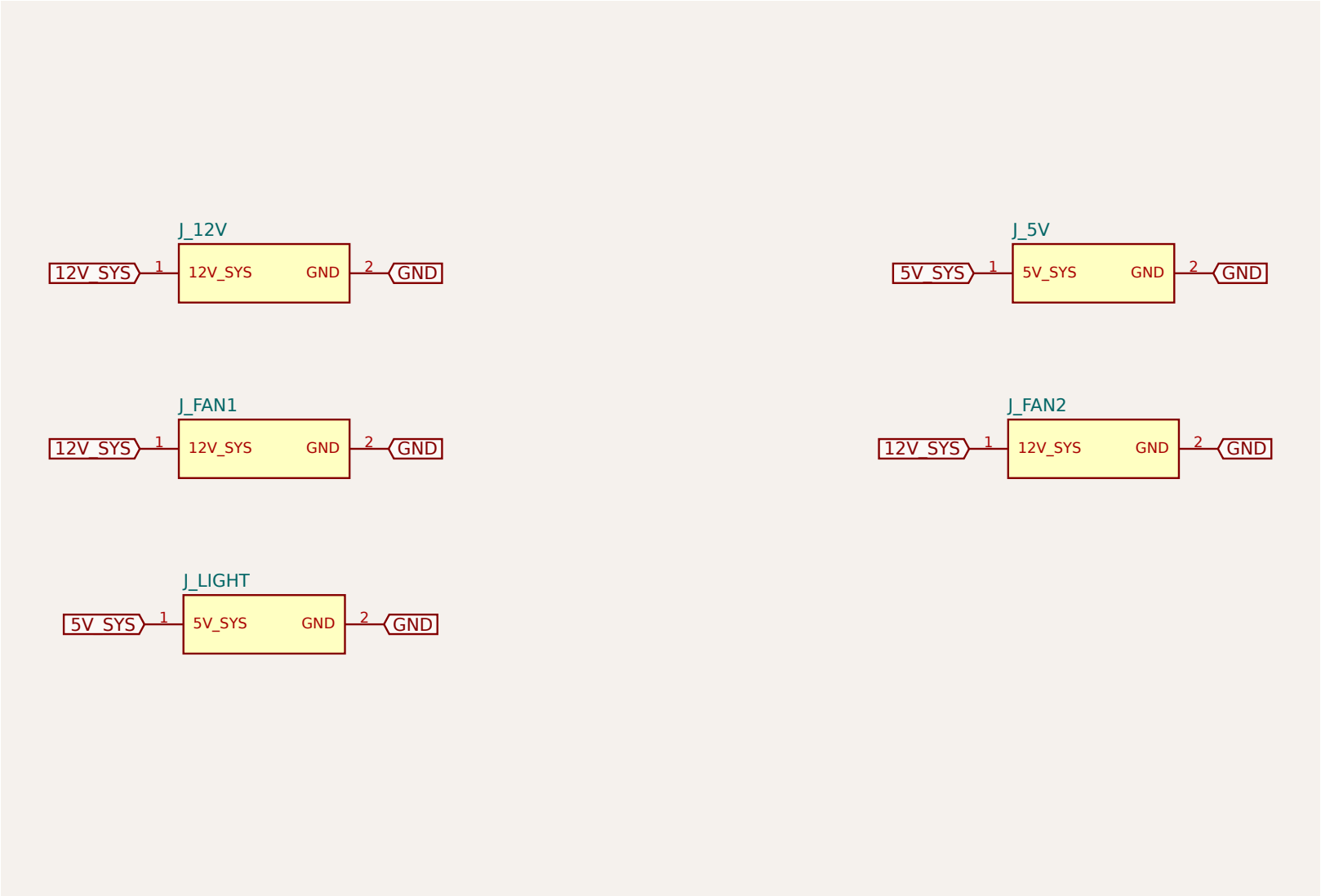
#	Reference	Purpose	Install	Adapter / carrier
1	U_ESP1	ESP32 DevKitC-38 — router	CP0	
2	J_POWER	POWER_MATRIX v2	CP11	
3	J_FIXTURE	manual continuity fixture; ESP1 GPIO4 SDA / GPIO16 SCL / GPIO25 stimulus	CP0	
4	J_PROTOTYPE	USB breadboard prototype; remove before final rail connection	CP0	
5	J_12V	Protected 12V input	CP32	
6	J_5V	System 5V input	CP32	
7	J_FAN1	12V fan 1 — fixed-speed	CP32	
8	J_FAN2	12V fan 2 — fixed-speed	CP32	
9	J_LIGHT	5V white lighting — never 5V_EDU	CP32	
10	J_LINK_1	Harness terminal: XBAR_DAT	CP0	
11	J_LINK_2	Harness terminal: XBAR_CLK	CP0	
12	J_LINK_3	Harness terminal: XBAR_LATCH	CP0	
13	J_LINK_4	Harness terminal: STB_A_IN	CP0	
14	J_LINK_5	Harness terminal: STB_B_IN	CP0	
15	J_LINK_6	Harness terminal: RST_RELEASE	CP0	
16	J_LINK_7	Harness terminal: HEARTBEAT	CP0	
17	J_LINK_8	Harness terminal: FAULT_USB	CP0	
18	J_LINK_9	Harness terminal: EDU_FAULT_RESET	CP0	
19	J_LINK_10	Harness terminal: CURRENT_USB	CP0	
20	J_LINK_11	Harness terminal: DIAG_ADC	CP0	

router-core — Controller



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

router-core — Circuit

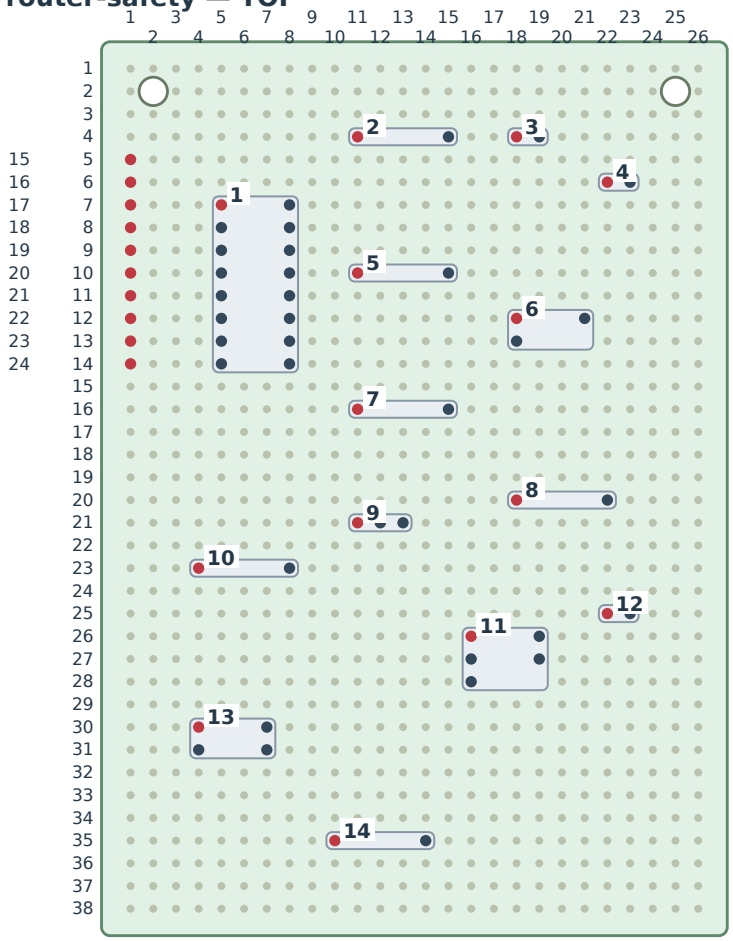


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

router-safety

ESP1 watchdog / isolated returns

router-safety — TOP

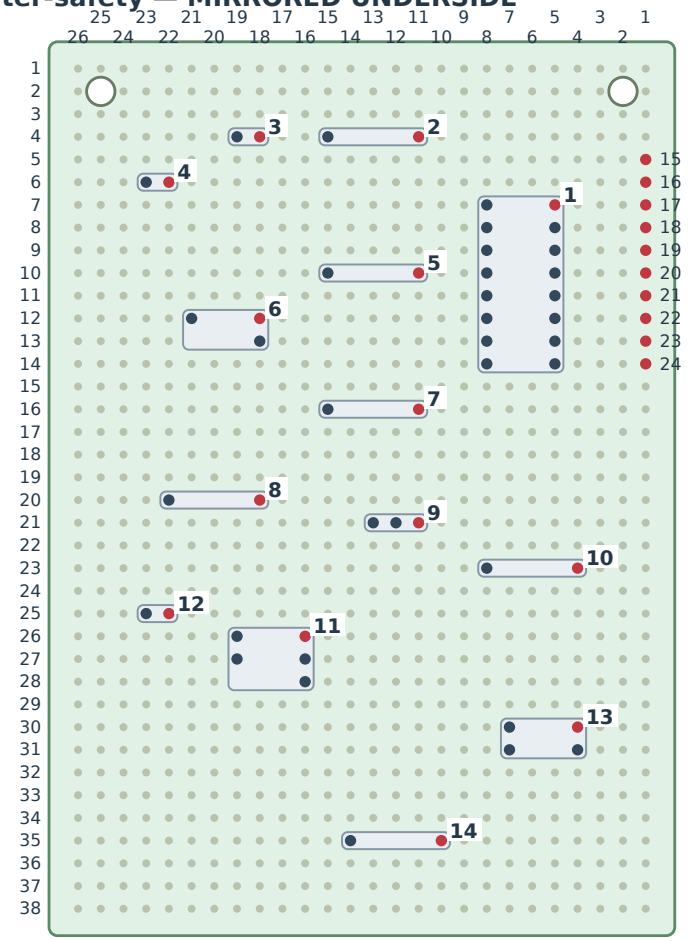


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

router-safety — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

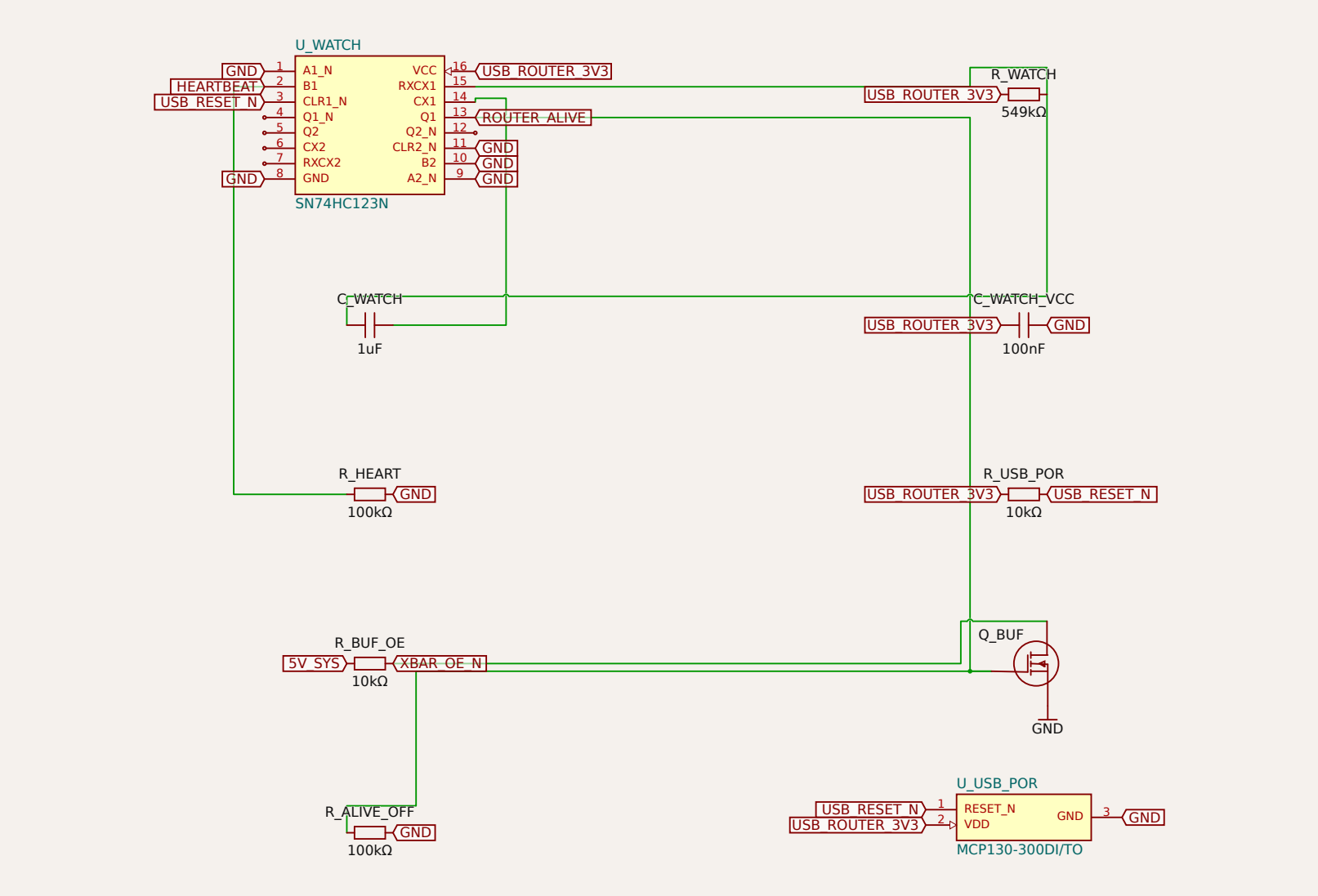
PROTOTYPE: physical fit and hardware tests pending.

router-safety — numbered parts

Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

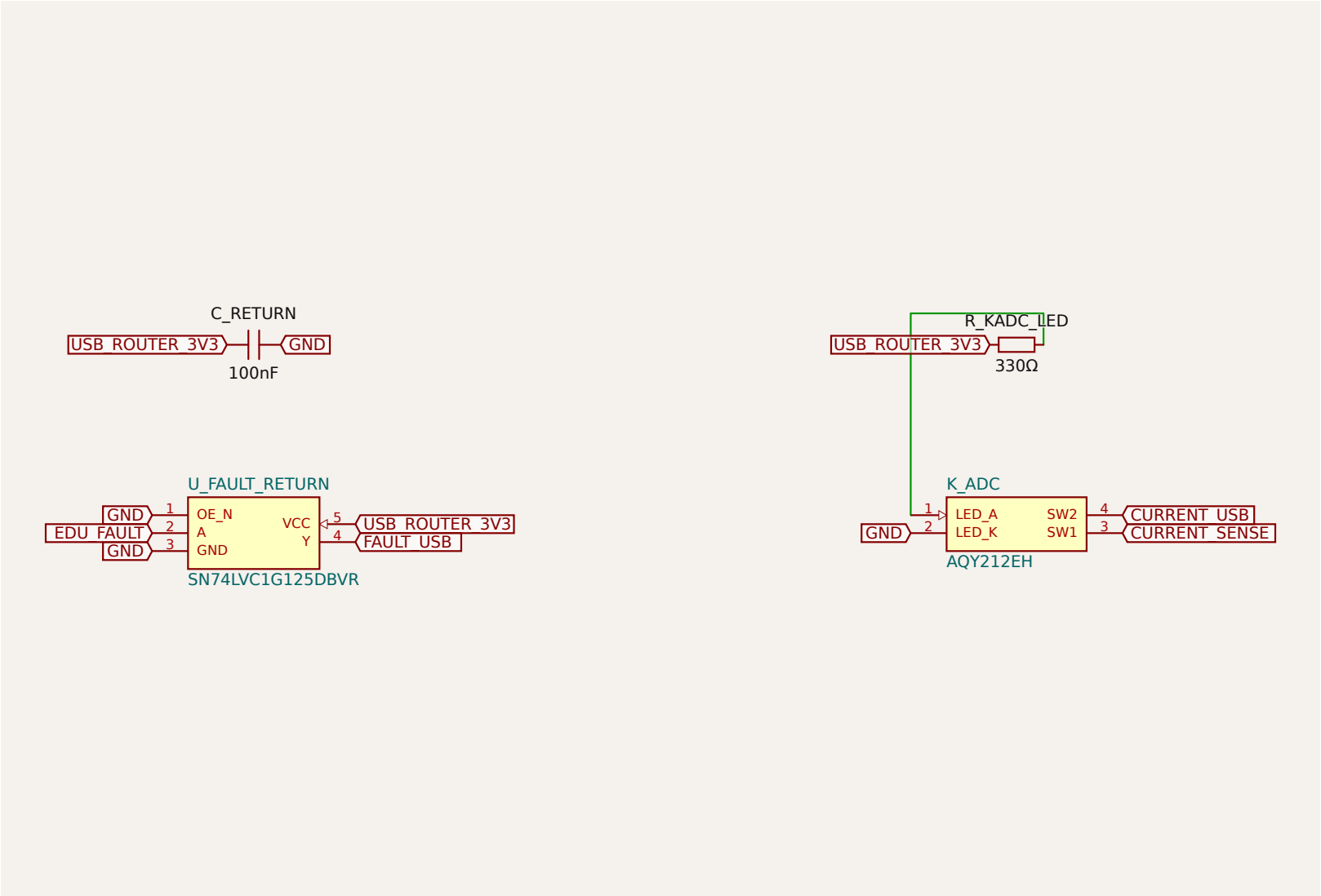
#	Reference	Purpose	Install	Adapter / carrier
1	U_WATCH	SN74HC123N	CP13	
2	R_WATCH	549kΩ ±1% watchdog timing — qualify before permanent installation	CP13	
3	C_WATCH	1µF ±5% film timing capacitor; not unverified inventory electrolytic	CP13	
4	C_WATCH_VCC	100nF	CP13	
5	R_HEART	100k	CP13	
6	U_USB_POR	MCP130-300DI/TO	CP13	
7	R_USB_POR	10k	CP13	
8	R_BUF_OE	10k	CP13	
9	Q_BUF	2N7000	CP13	
10	R_ALIVE_OFF	100k	CP13	
11	U_FAULT_RETURN	SN74LVC1G125DBVR	CP11	SOT23-5 to numbered adapter
12	C_RETURN	100nF	CP11	
13	K_ADC	AQY212EH	CP11	
14	R_KADC_LED	330	CP11	
15	J_LINK_1	Harness terminal: GND	CP11	
16	J_LINK_2	Harness terminal: 5V_SYS	CP13	
17	J_LINK_3	Harness terminal: USB_ROUTER_3V3	CP11	
18	J_LINK_4	Harness terminal: HEARTBEAT	CP13	
19	J_LINK_5	Harness terminal: FAULT_USB	CP11	
20	J_LINK_6	Harness terminal: CURRENT_USB	CP11	
21	J_LINK_7	Harness terminal: EDU_FAULT	CP11	
22	J_LINK_8	Harness terminal: CURRENT_SENSE	CP11	
23	J_LINK_9	Harness terminal: ROUTER_ALIVE	CP13	
24	J_LINK_10	Harness terminal: XBAR_OE_N	CP13	

router-safety — Watchdog



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

router-safety — Return

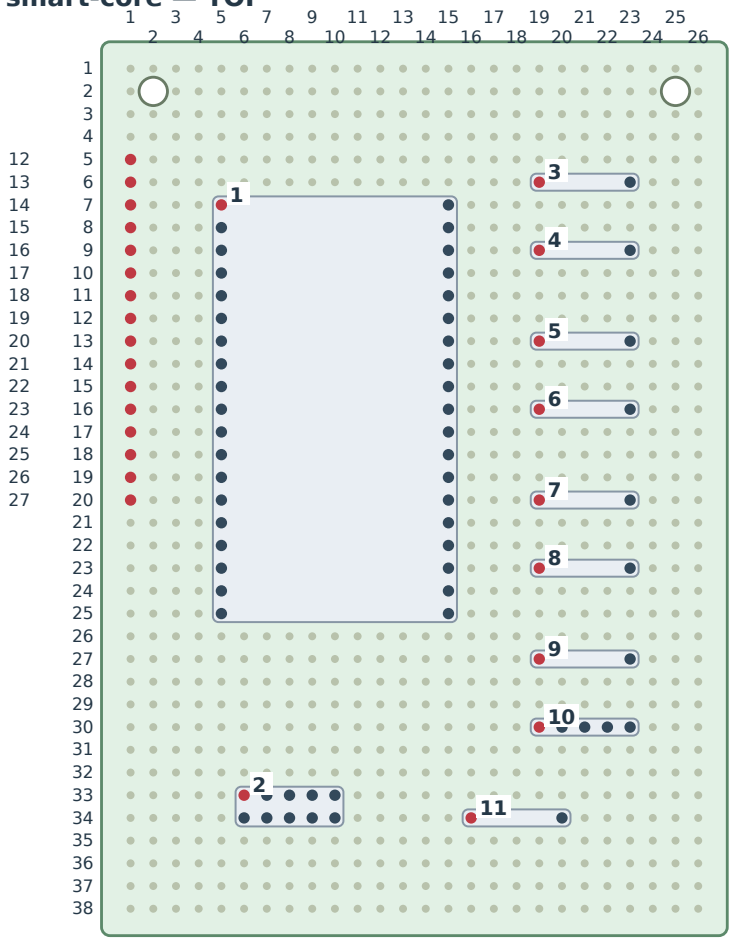


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

smart-core

ESP2 carrier and service pull-ups

smart-core — TOP

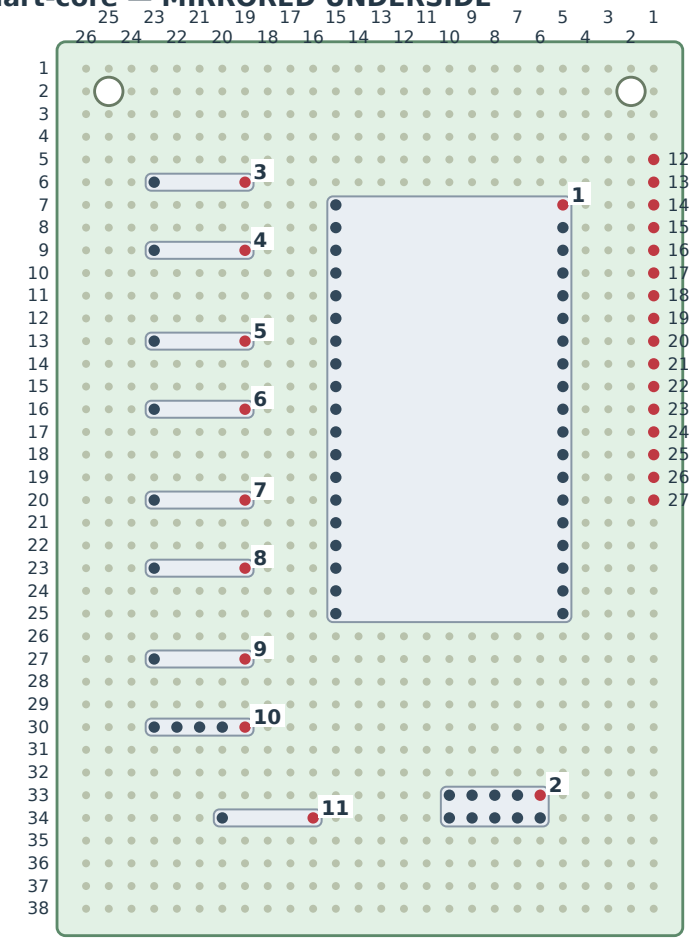


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

smart-core — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

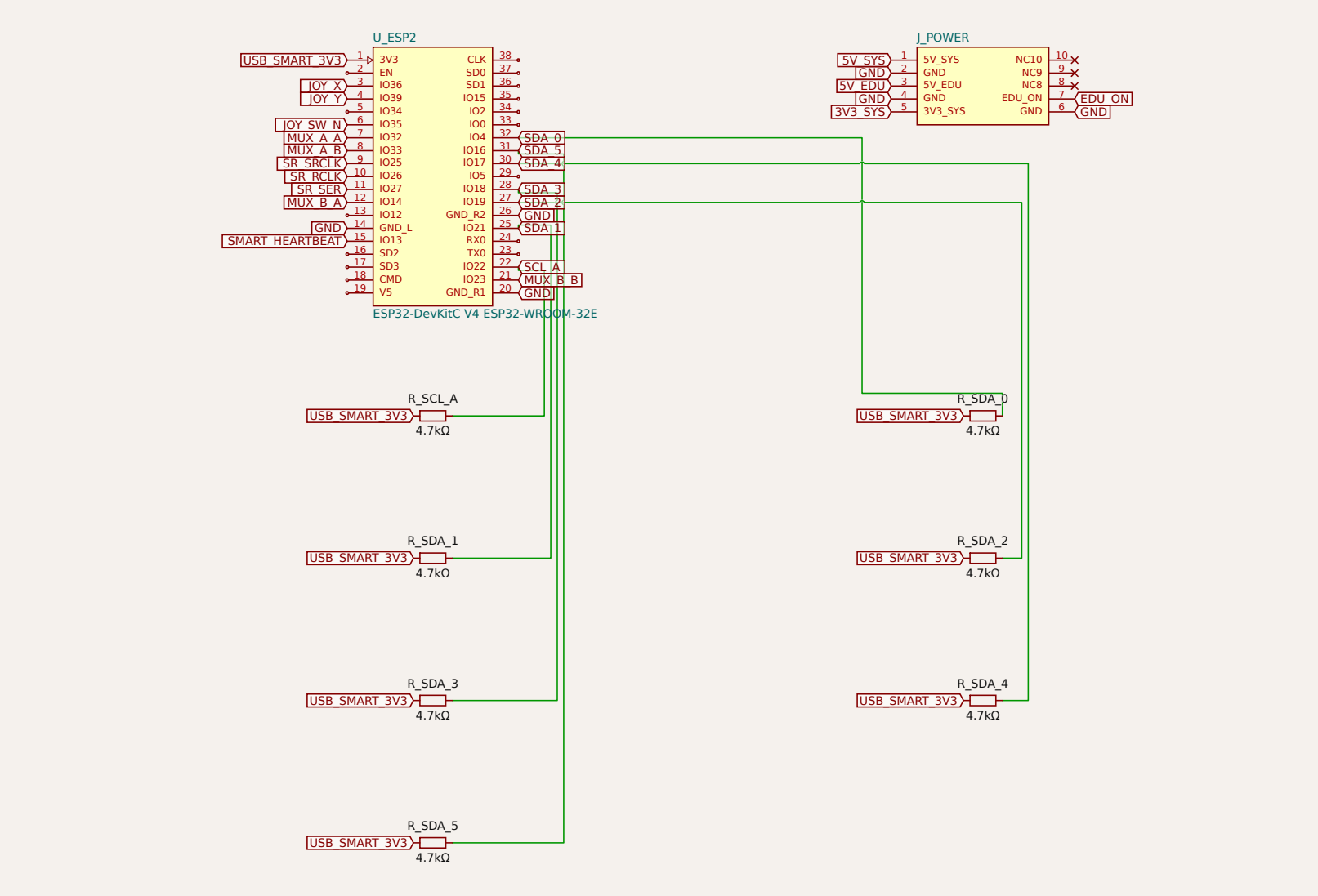
PROTOTYPE: physical fit and hardware tests pending.

smart-core — numbered parts

Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

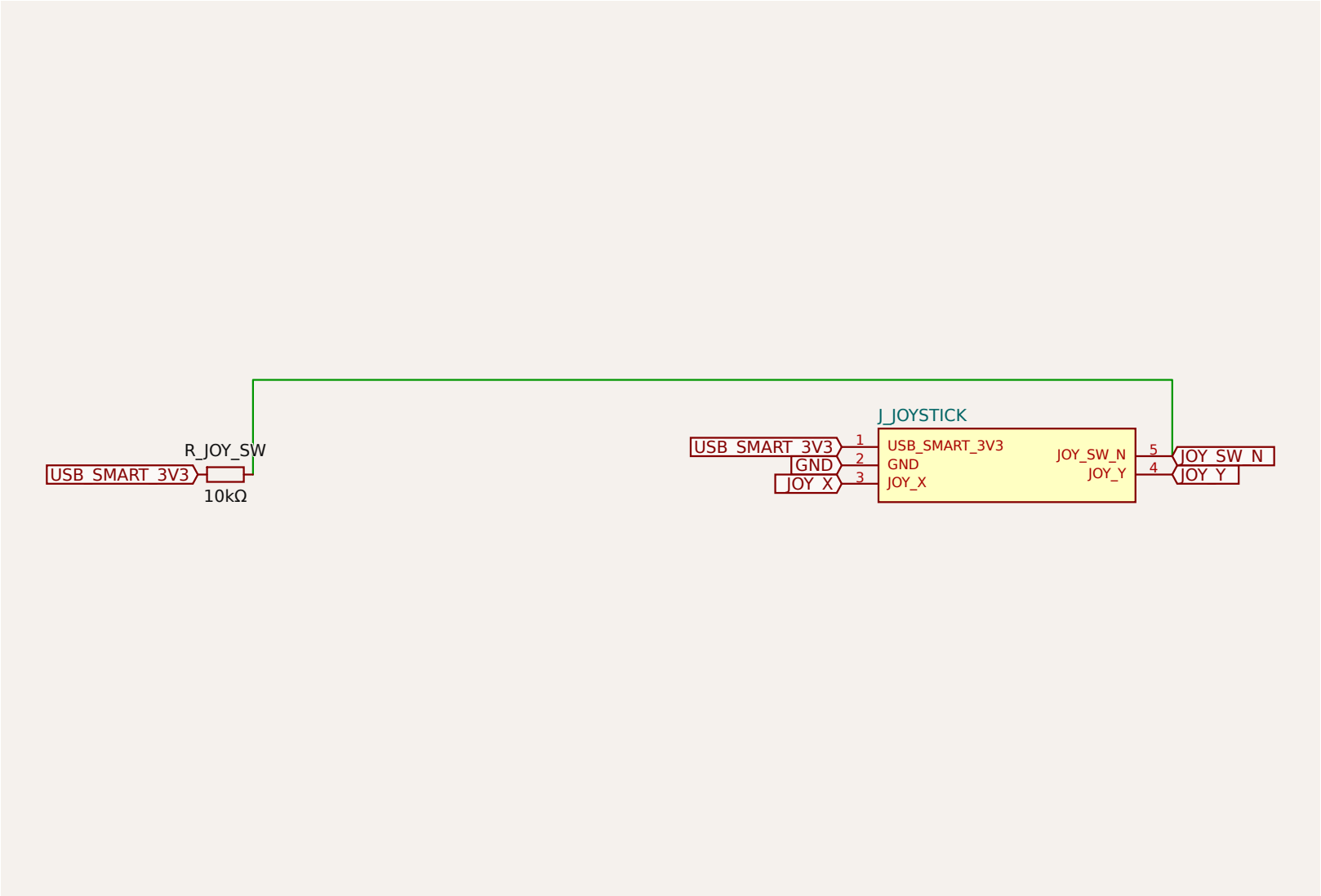
#	Reference	Purpose	Install	Adapter / carrier
1	U_ESP2	ESP32 DevKitC-38 — smart controller	CP15	
2	J_POWER	J_POWER	CP15	
3	R_SCL_A	4.7k	CP15	
4	R_SDA_0	4.7k	CP15	
5	R_SDA_1	4.7k	CP15	
6	R_SDA_2	4.7k	CP15	
7	R_SDA_3	4.7k	CP15	
8	R_SDA_4	4.7k	CP15	
9	R_SDA_5	4.7k	CP15	
10	J_JOYSTICK	Joystick powered at 3.3V	CP32	
11	R_JOY_SW	R_JOY_SW: 10kΩ	CP32	
12	J_LINK_1	Harness terminal: USB_SMART_3V3	CP15	
13	J_LINK_2	Harness terminal: SR_SRCLK	CP15	
14	J_LINK_3	Harness terminal: SR_RCLK	CP15	
15	J_LINK_4	Harness terminal: SR_SER	CP15	
16	J_LINK_5	Harness terminal: SMART_HEARTBEAT	CP15	
17	J_LINK_6	Harness terminal: SCL_A	CP15	
18	J_LINK_7	Harness terminal: SDA_0	CP15	
19	J_LINK_8	Harness terminal: SDA_1	CP15	
20	J_LINK_9	Harness terminal: SDA_2	CP15	
21	J_LINK_10	Harness terminal: SDA_3	CP15	
22	J_LINK_11	Harness terminal: SDA_4	CP15	
23	J_LINK_12	Harness terminal: SDA_5	CP15	
24	J_LINK_13	Harness terminal: MUX_A_A	CP15	
25	J_LINK_14	Harness terminal: MUX_A_B	CP15	
26	J_LINK_15	Harness terminal: MUX_B_A	CP15	
27	J_LINK_16	Harness terminal: MUX_B_B	CP15	

smart-core — Controller



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

smart-core — Circuit

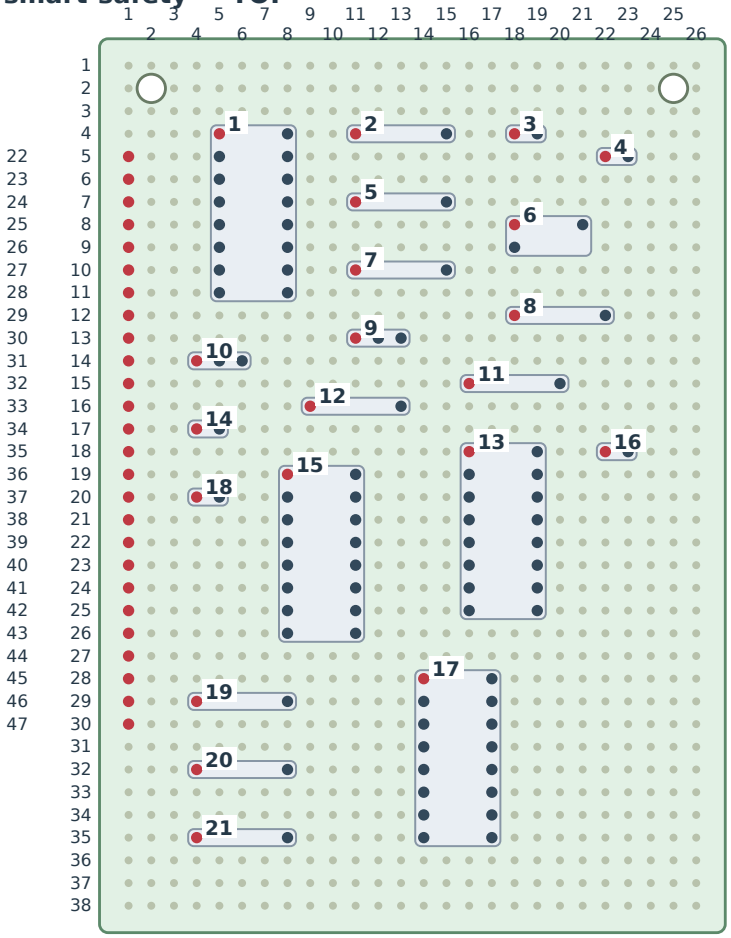


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

smart-safety

ESP2 watchdog / relay registers

smart-safety — TOP

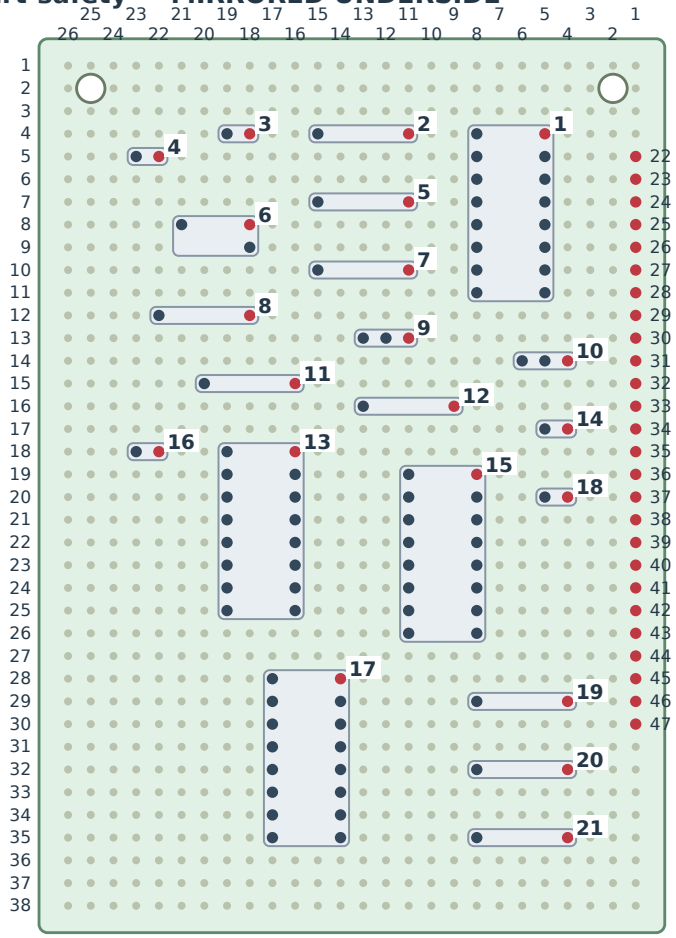


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

smart-safety — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

smart-safety — numbered parts

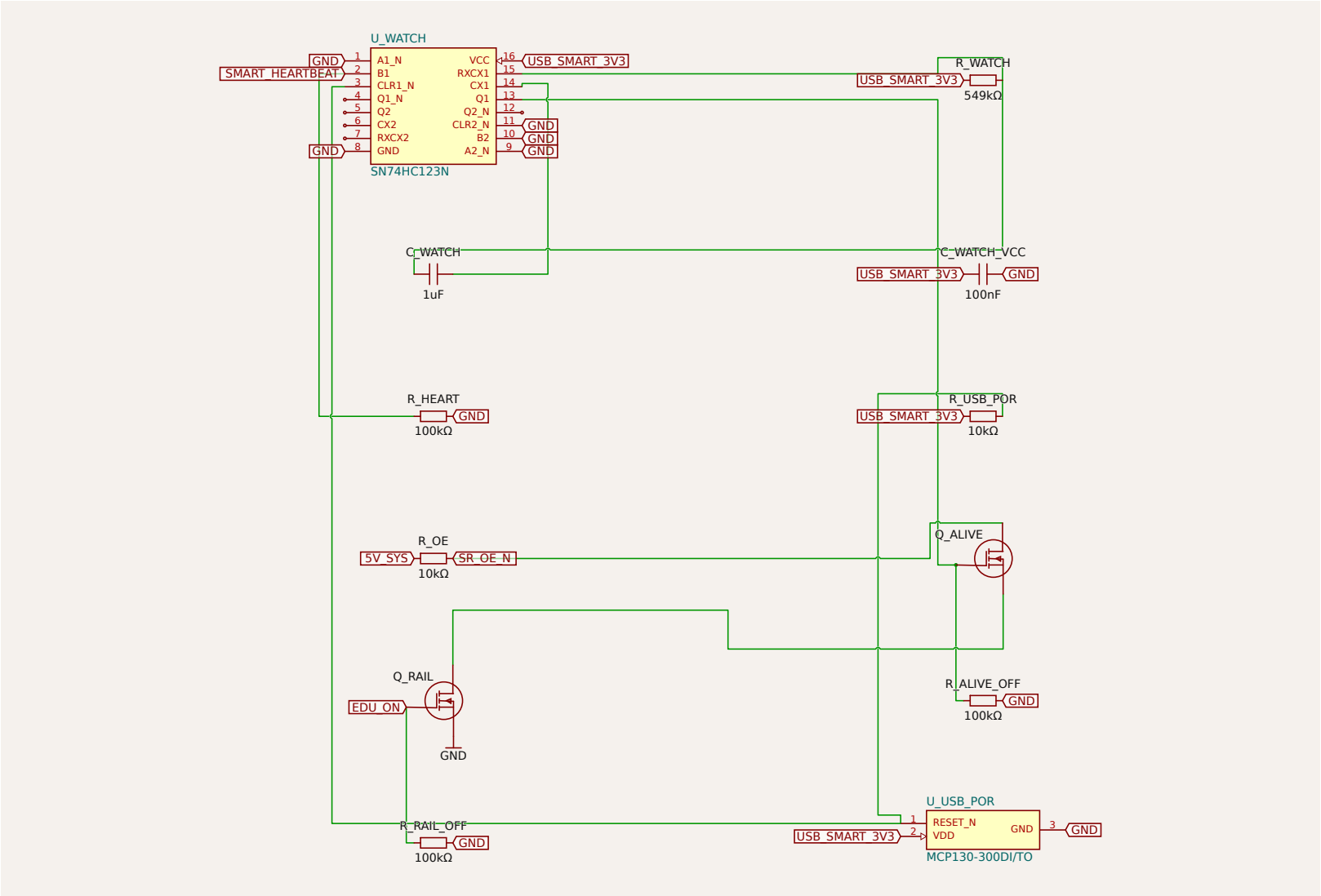
Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

#	Reference	Purpose	Install	Adapter / carrier
1	U_WATCH	SN74HC123N	CP15	
2	R_WATCH	549kΩ ±1% watchdog timing — qualify before permanent installation	CP15	
3	C_WATCH	1µF ±5% film timing capacitor; not unverified inventory electrolytic	CP15	
4	C_WATCH_VCC	100nF	CP15	
5	R_HEART	100k	CP15	
6	U_USB_POR	MCP130-300DI/TO	CP15	
7	R_USB_POR	10k	CP15	
8	R_OE	10k	CP15	
9	Q_ALIVE	2N7000	CP15	
10	Q_RAIL	2N7000	CP15	
11	R_ALIVE_OFF	100k	CP15	
12	R_RAIL_OFF	100k	CP15	
13	U_SR0	SN74HCT595N	CP15	
14	C_SR0	100nF	CP15	
15	U_SR1	SN74HCT595N	CP15	
16	C_SR1	100nF	CP15	
17	U_SR2	SN74HCT595N	CP15	
18	C_SR2	100nF	CP15	
19	R_SR_SER_OFF	100k	CP15	
20	R_SR_SRCLK_OFF	100k	CP15	
21	R_SR_RCLK_OFF	100k	CP15	
22	J_LINK_1	Harness terminal: GND	CP15	
23	J_LINK_2	Harness terminal: 5V_SYS	CP15	
24	J_LINK_3	Harness terminal: USB_SMART_3V3	CP15	
25	J_LINK_4	Harness terminal: SR_SRCLK	CP15	
26	J_LINK_5	Harness terminal: SR_RCLK	CP15	
27	J_LINK_6	Harness terminal: SR_SER	CP15	
28	J_LINK_7	Harness terminal: SMART_HEARTBEAT	CP15	

smart-safety — parts continued

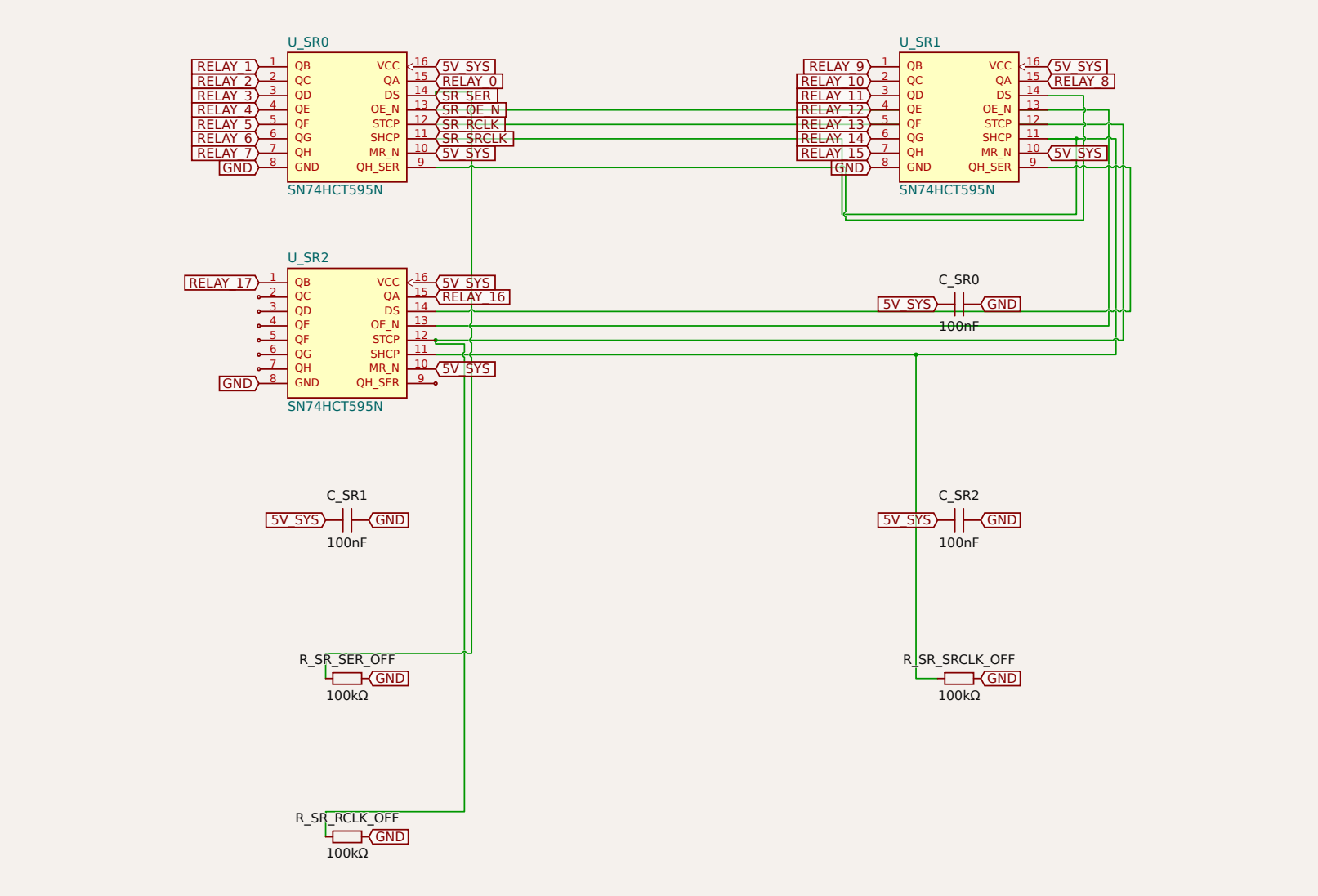
#	Reference	Purpose	Install	Adapter / carrier
29	J_LINK_8	Harness terminal: EDU_ON	CP15	
30	J_LINK_9	Harness terminal: RELAY_0	CP15	
31	J_LINK_10	Harness terminal: RELAY_1	CP15	
32	J_LINK_11	Harness terminal: RELAY_2	CP15	
33	J_LINK_12	Harness terminal: RELAY_3	CP15	
34	J_LINK_13	Harness terminal: RELAY_4	CP15	
35	J_LINK_14	Harness terminal: RELAY_5	CP15	
36	J_LINK_15	Harness terminal: RELAY_6	CP15	
37	J_LINK_16	Harness terminal: RELAY_7	CP15	
38	J_LINK_17	Harness terminal: RELAY_8	CP15	
39	J_LINK_18	Harness terminal: RELAY_9	CP15	
40	J_LINK_19	Harness terminal: RELAY_10	CP15	
41	J_LINK_20	Harness terminal: RELAY_11	CP15	
42	J_LINK_21	Harness terminal: RELAY_12	CP15	
43	J_LINK_22	Harness terminal: RELAY_13	CP15	
44	J_LINK_23	Harness terminal: RELAY_14	CP15	
45	J_LINK_24	Harness terminal: RELAY_15	CP15	
46	J_LINK_25	Harness terminal: RELAY_16	CP15	
47	J_LINK_26	Harness terminal: RELAY_17	CP15	

smart-safety — Watchdog



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

smart-safety — Registers

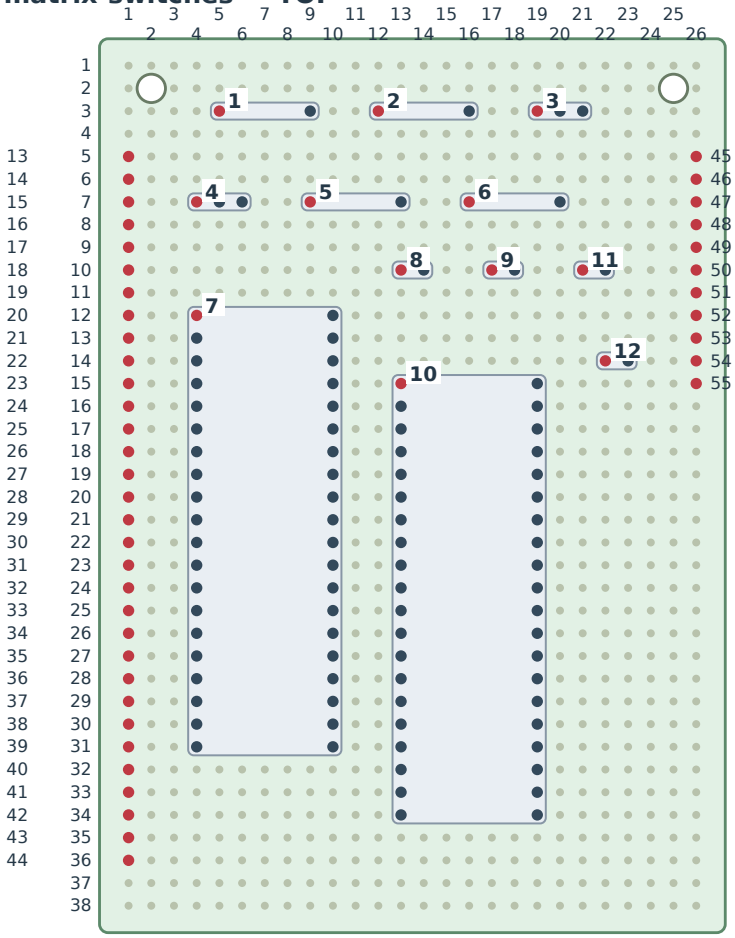


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

matrix-switches

Two MT8816 matrices / default reset

matrix-switches — TOP

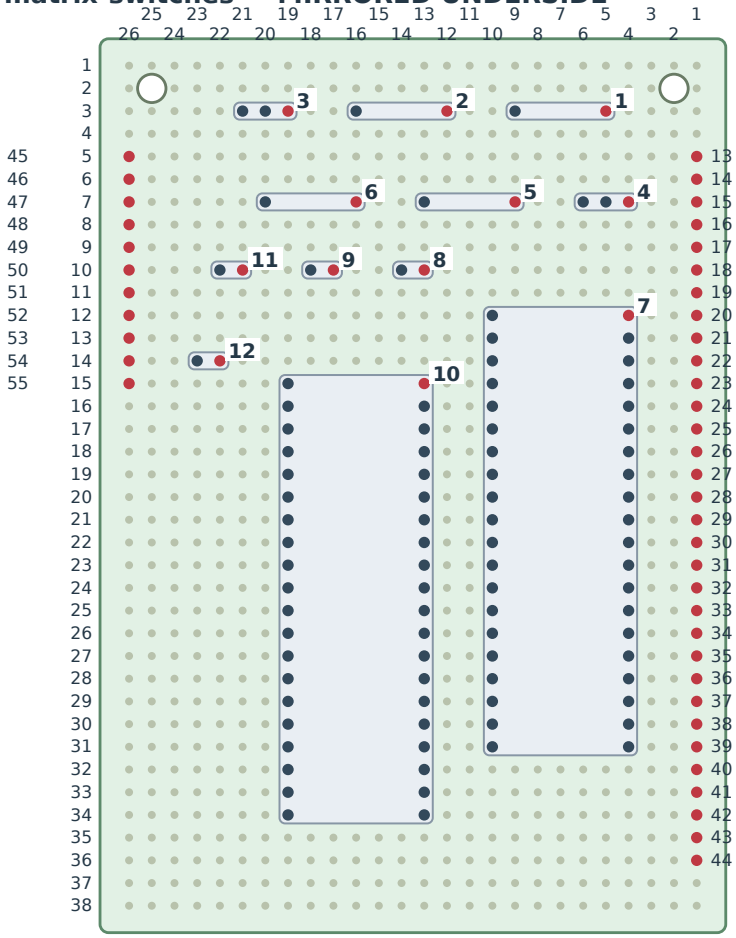


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

matrix-switches — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

matrix-switches — numbered parts

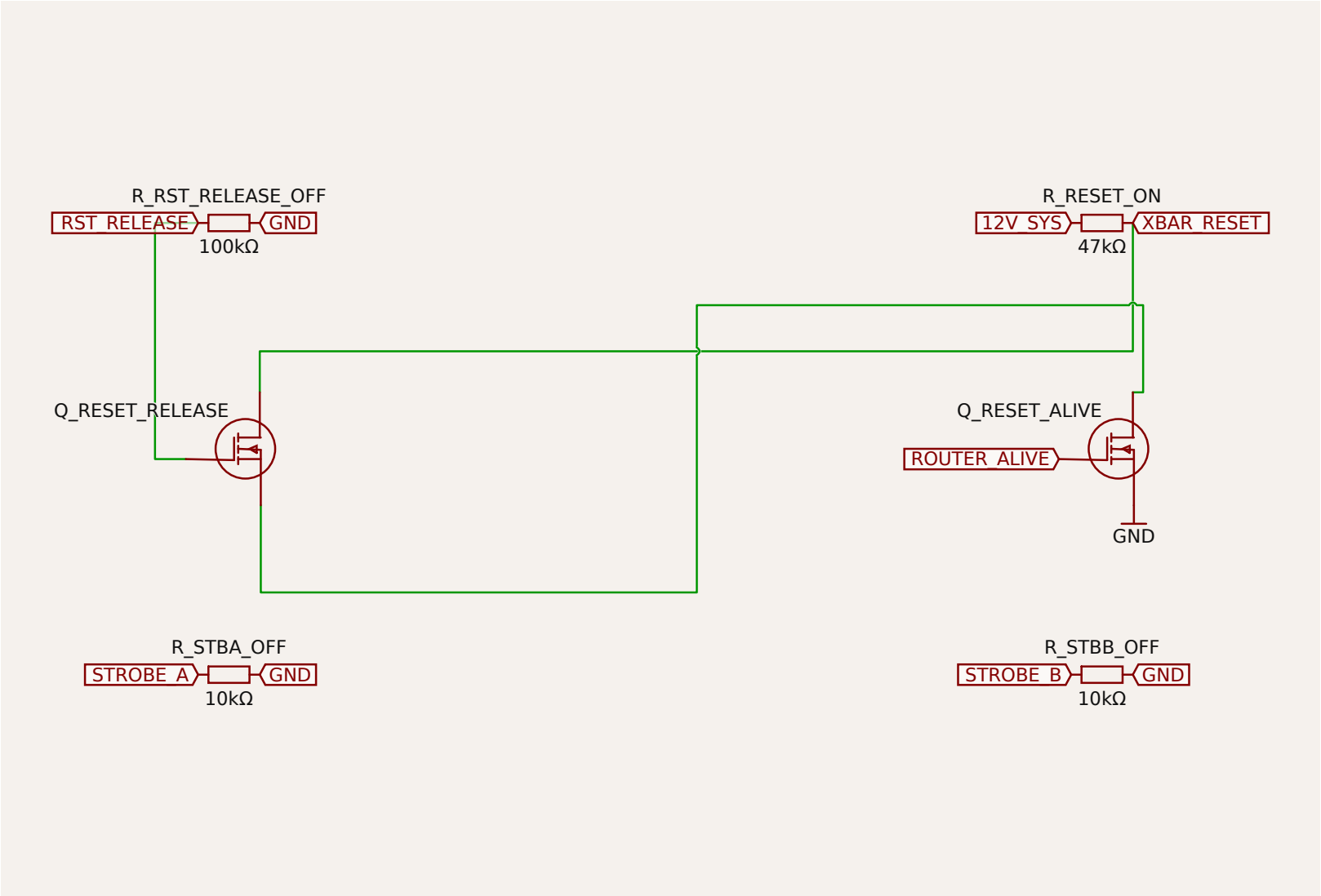
Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

#	Reference	Purpose	Install	Adapter / carrier
1	R_RST_RELEASE_OFF	100k	CP24	
2	R_RESET_ON	47k	CP24	
3	Q_RESET_RELEASE	2N7000	CP24	
4	Q_RESET_ALIVE	2N7000	CP24	
5	R_STBA_OFF	10k	CP24	
6	R_STBB_OFF	10k	CP24	
7	U_MT_A	MT8816AE1	CP24	
8	C_MT0	100nF	CP24	
9	C_MT0_BULK	10uF	CP24	
10	U_MT_B	MT8816AE1	CP24	
11	C_MT1	100nF	CP24	
12	C_MT1_BULK	10uF	CP24	
13	J_LINK_1	Harness terminal: GND	CP24	
14	J_LINK_2	Harness terminal: NODE00	CP24	
15	J_LINK_3	Harness terminal: NODE01	CP24	
16	J_LINK_4	Harness terminal: NODE02	CP24	
17	J_LINK_5	Harness terminal: NODE03	CP24	
18	J_LINK_6	Harness terminal: NODE04	CP24	
19	J_LINK_7	Harness terminal: NODE05	CP24	
20	J_LINK_8	Harness terminal: NODE06	CP24	
21	J_LINK_9	Harness terminal: NODE07	CP24	
22	J_LINK_10	Harness terminal: NODE08	CP24	
23	J_LINK_11	Harness terminal: NODE09	CP24	
24	J_LINK_12	Harness terminal: NODE10	CP24	
25	J_LINK_13	Harness terminal: NODE11	CP24	
26	J_LINK_14	Harness terminal: NODE12	CP24	
27	J_LINK_15	Harness terminal: NODE13	CP24	
28	J_LINK_16	Harness terminal: NODE14	CP24	

matrix-switches — parts continued

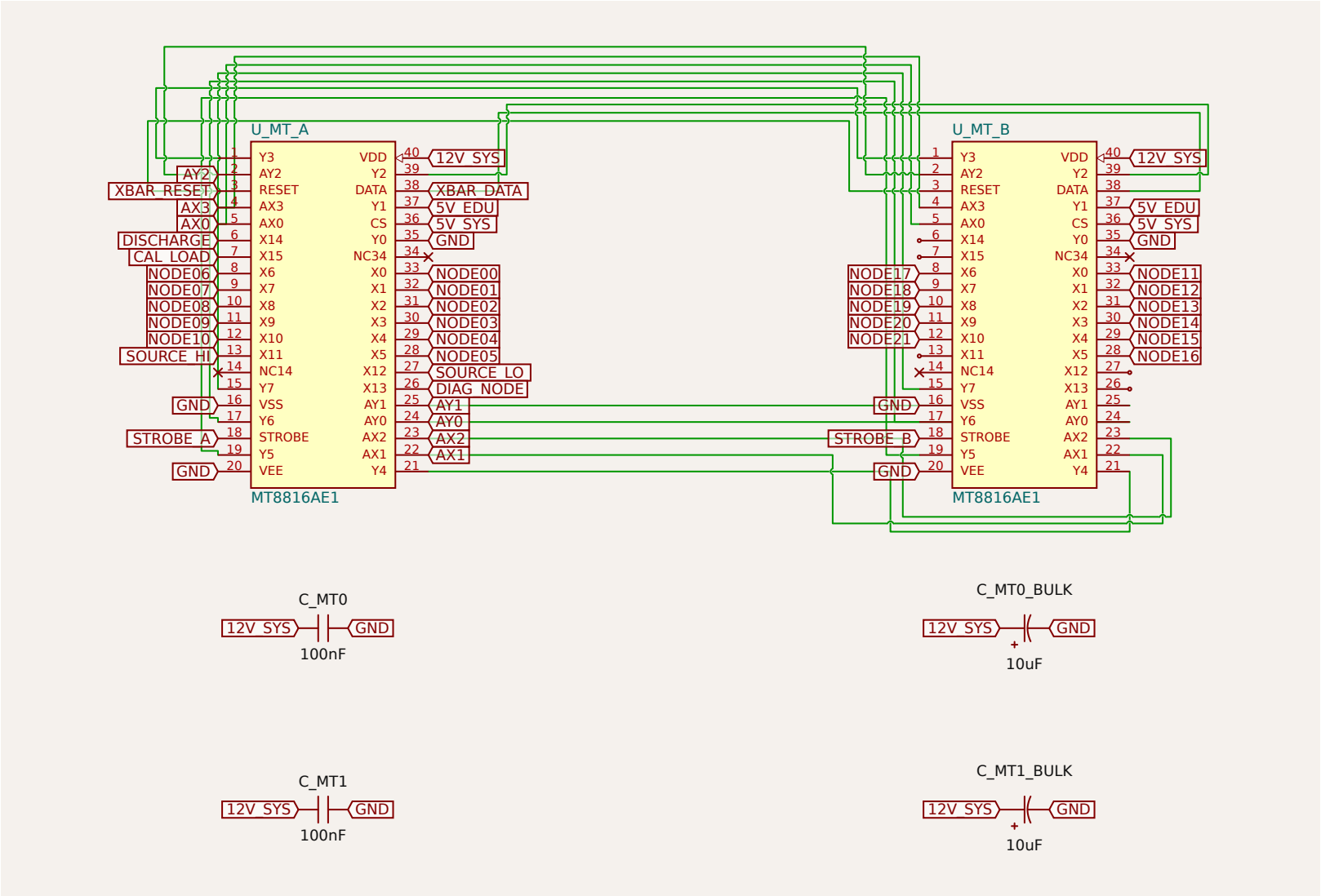
#	Reference	Purpose	Install	Adapter / carrier
29	J_LINK_17	Harness terminal: NODE15	CP24	
30	J_LINK_18	Harness terminal: NODE16	CP24	
31	J_LINK_19	Harness terminal: NODE17	CP24	
32	J_LINK_20	Harness terminal: NODE18	CP24	
33	J_LINK_21	Harness terminal: NODE19	CP24	
34	J_LINK_22	Harness terminal: NODE20	CP24	
35	J_LINK_23	Harness terminal: NODE21	CP24	
36	J_LINK_24	Harness terminal: 12V_SYS	CP24	
37	J_LINK_25	Harness terminal: 5V_SYS	CP24	
38	J_LINK_26	Harness terminal: RST_RELEASE	CP24	
39	J_LINK_27	Harness terminal: 5V_EDU	CP24	
40	J_LINK_28	Harness terminal: ROUTER_ALIVE	CP24	
41	J_LINK_29	Harness terminal: STROBE_A	CP24	
42	J_LINK_30	Harness terminal: STROBE_B	CP24	
43	J_LINK_31	Harness terminal: XBAR_DATA	CP24	
44	J_LINK_32	Harness terminal: AX0	CP24	
45	J_LINK_33	Harness terminal: AX1	CP24	
46	J_LINK_34	Harness terminal: AX2	CP24	
47	J_LINK_35	Harness terminal: AX3	CP24	
48	J_LINK_36	Harness terminal: AY0	CP24	
49	J_LINK_37	Harness terminal: AY1	CP24	
50	J_LINK_38	Harness terminal: AY2	CP24	
51	J_LINK_39	Harness terminal: SOURCE_HI	CP24	
52	J_LINK_40	Harness terminal: SOURCE_LO	CP24	
53	J_LINK_41	Harness terminal: DIAG_NODE	CP24	
54	J_LINK_42	Harness terminal: DISCHARGE	CP24	
55	J_LINK_43	Harness terminal: CAL_LOAD	CP24	

matrix-switches — Control



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

matrix-switches — Matrix

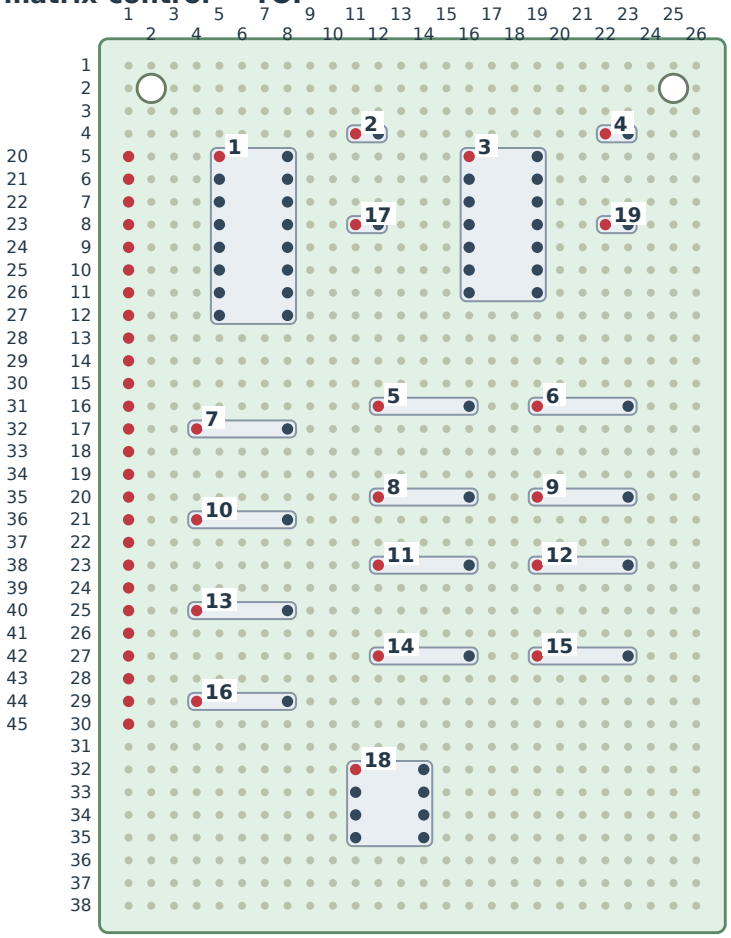


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

matrix-control

Matrix latch / local diagnostic buffer

matrix-control — TOP

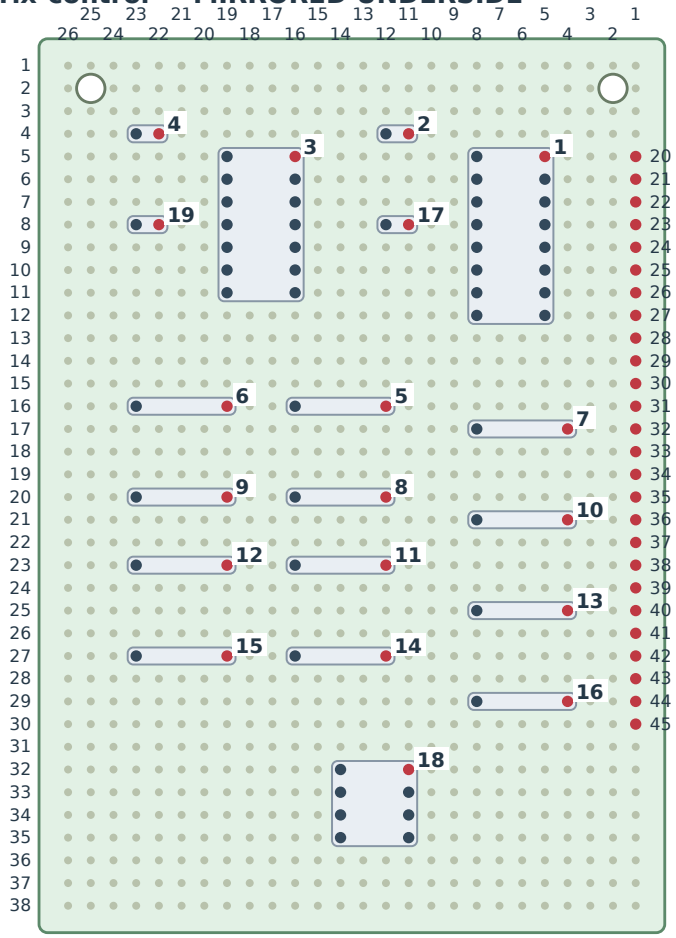


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

matrix-control — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

matrix-control — numbered parts

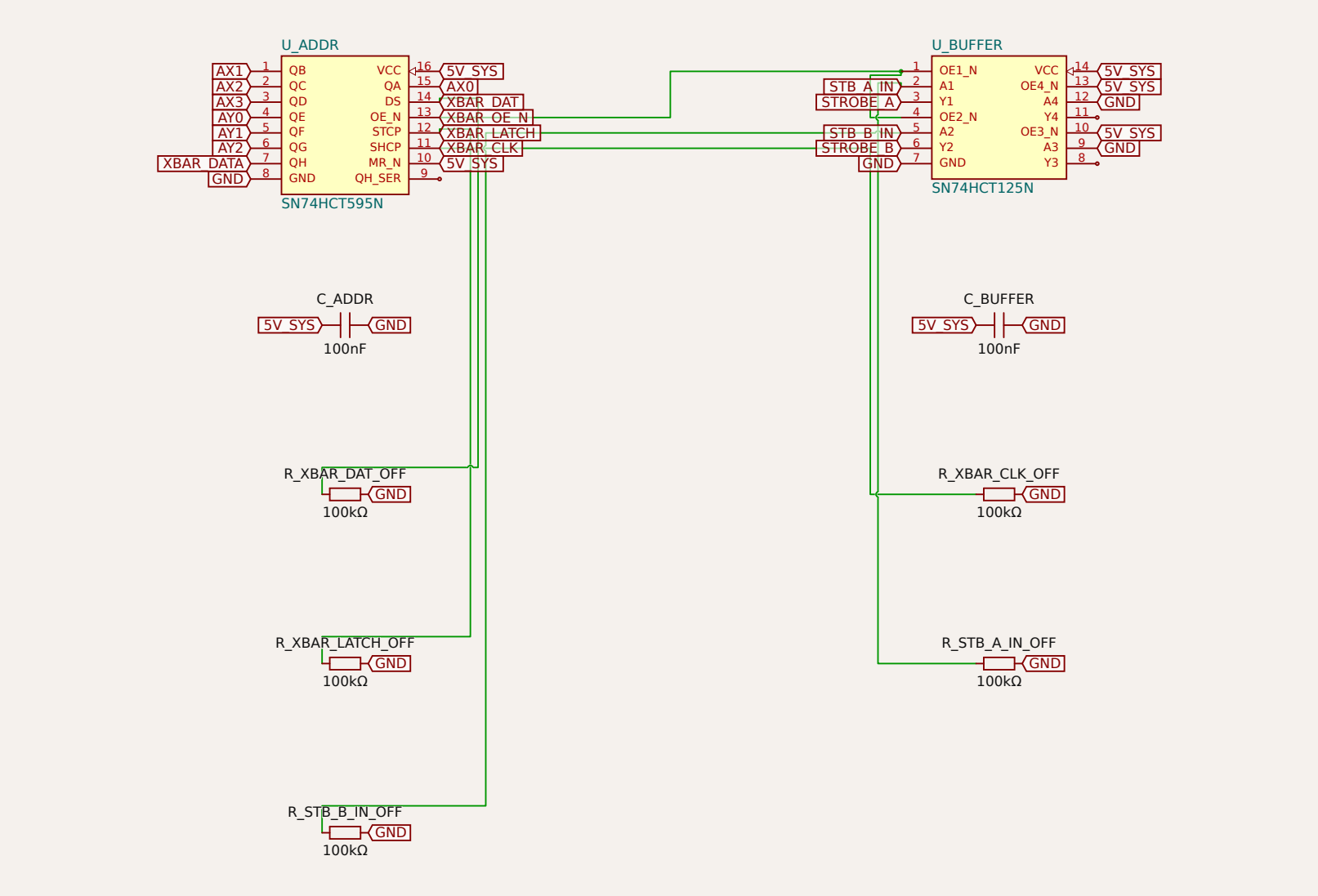
Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

#	Reference	Purpose	Install	Adapter / carrier
1	U_ADDR	SN74HCT595N	CP24	
2	C_ADDR	100nF	CP24	
3	U_BUFFER	SN74HCT125N	CP24	
4	C_BUFFER	100nF	CP24	
5	R_XBAR_DAT_OFF	100k	CP24	
6	R_XBAR_CLK_OFF	100k	CP24	
7	R_XBAR_LATCH_OFF	100k	CP24	
8	R_STB_A_IN_OFF	100k	CP24	
9	R_STB_B_IN_OFF	100k	CP24	
10	D_SOURCE	BAT85	CP24	
11	R_SOURCE_HI	10k	CP24	
12	R_SOURCE_LO	470	CP24	
13	R_DISCHARGE	470	CP24	
14	R_CAL	1k	CP24	
15	R_DIAG_TOP	1M	CP24	
16	R_DIAG_BOTTOM	1M	CP24	
17	C_DIAG	1nF	CP24	
18	U_DIAG	OPA2333AID	CP24	SOIC8 to DIP8
19	C_DIAG_VCC	100nF	CP24	
20	J_LINK_1	Harness terminal: GND	CP24	
21	J_LINK_2	Harness terminal: 3V3_SYS	CP24	
22	J_LINK_3	Harness terminal: 5V_SYS	CP24	
23	J_LINK_4	Harness terminal: USB_ROUTER_3V3	CP24	
24	J_LINK_5	Harness terminal: XBAR_DAT	CP24	
25	J_LINK_6	Harness terminal: XBAR_CLK	CP24	
26	J_LINK_7	Harness terminal: XBAR_LATCH	CP24	
27	J_LINK_8	Harness terminal: STB_A_IN	CP24	
28	J_LINK_9	Harness terminal: STB_B_IN	CP24	

matrix-control — parts continued

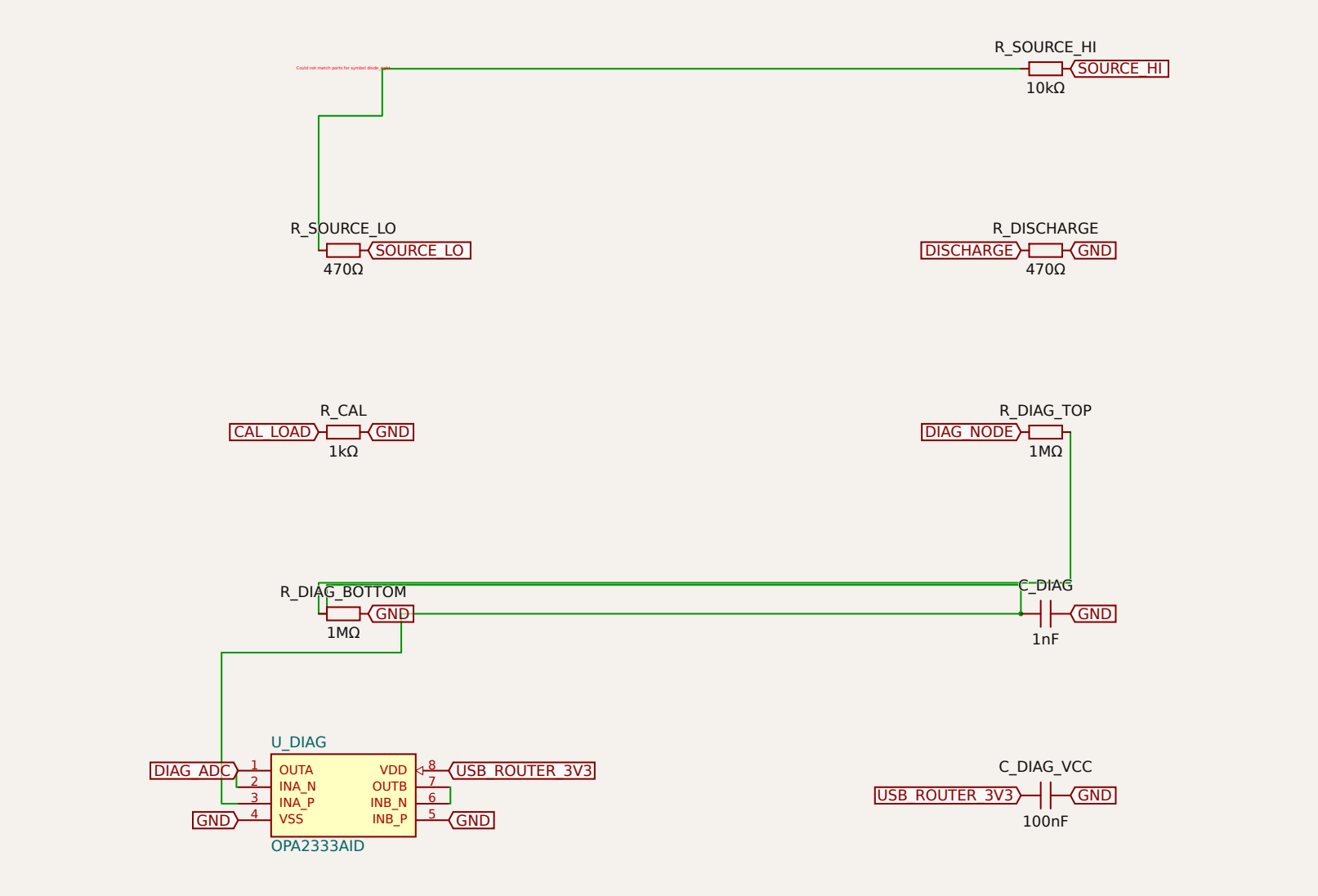
#	Reference	Purpose	Install	Adapter / carrier
29	J_LINK_10	Harness terminal: DIAG_ADC	CP24	
30	J_LINK_11	Harness terminal: XBAR_OE_N	CP24	
31	J_LINK_12	Harness terminal: STROBE_A	CP24	
32	J_LINK_13	Harness terminal: STROBE_B	CP24	
33	J_LINK_14	Harness terminal: XBAR_DATA	CP24	
34	J_LINK_15	Harness terminal: AX0	CP24	
35	J_LINK_16	Harness terminal: AX1	CP24	
36	J_LINK_17	Harness terminal: AX2	CP24	
37	J_LINK_18	Harness terminal: AX3	CP24	
38	J_LINK_19	Harness terminal: AY0	CP24	
39	J_LINK_20	Harness terminal: AY1	CP24	
40	J_LINK_21	Harness terminal: AY2	CP24	
41	J_LINK_22	Harness terminal: SOURCE_HI	CP24	
42	J_LINK_23	Harness terminal: SOURCE_LO	CP24	
43	J_LINK_24	Harness terminal: DIAG_NODE	CP24	
44	J_LINK_25	Harness terminal: DISCHARGE	CP24	
45	J_LINK_26	Harness terminal: CAL_LOAD	CP24	

matrix-control — Control



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

matrix-control — Tools

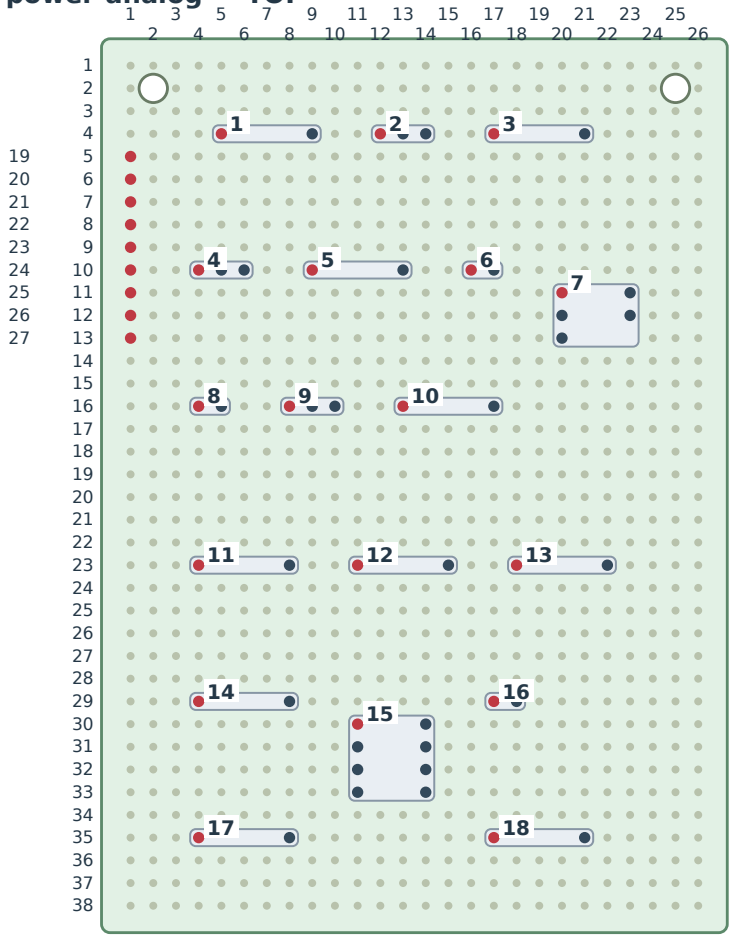


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

power-analog

Student switch / current sense / comparator

power-analog — TOP

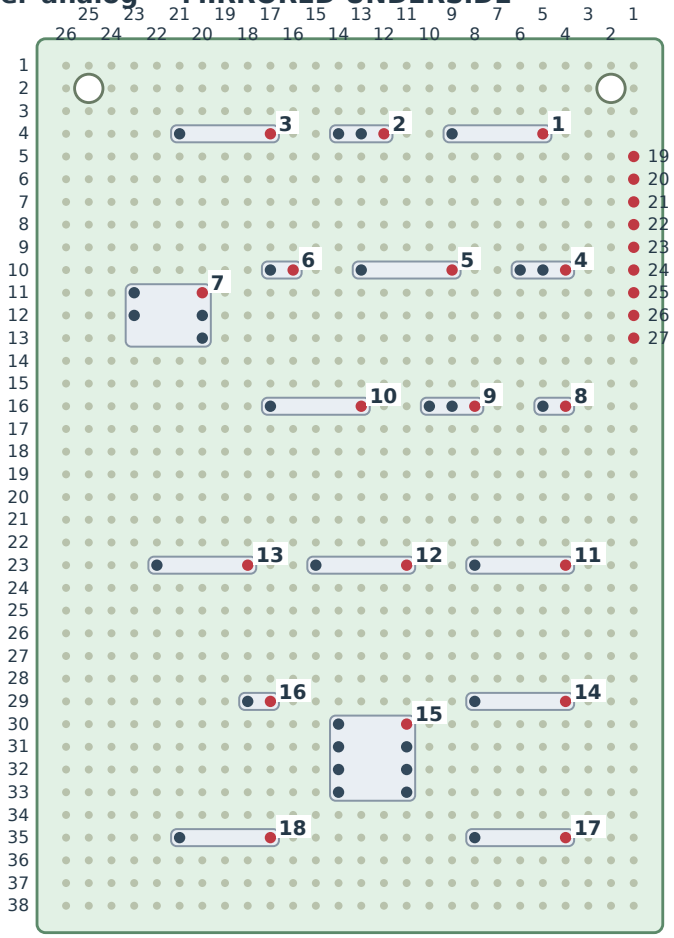


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

power-analog — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

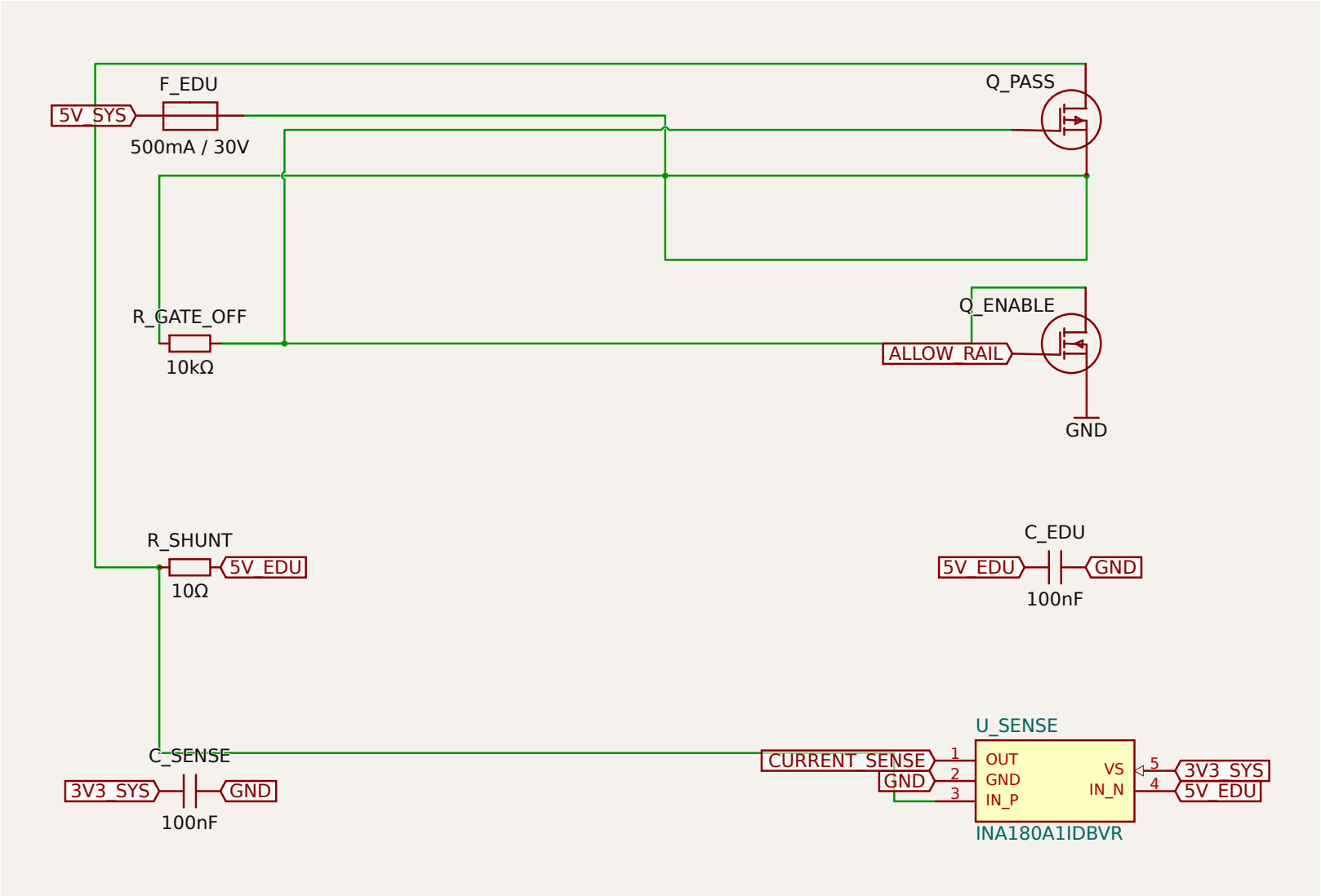
PROTOTYPE: physical fit and hardware tests pending.

power-analog — numbered parts

Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

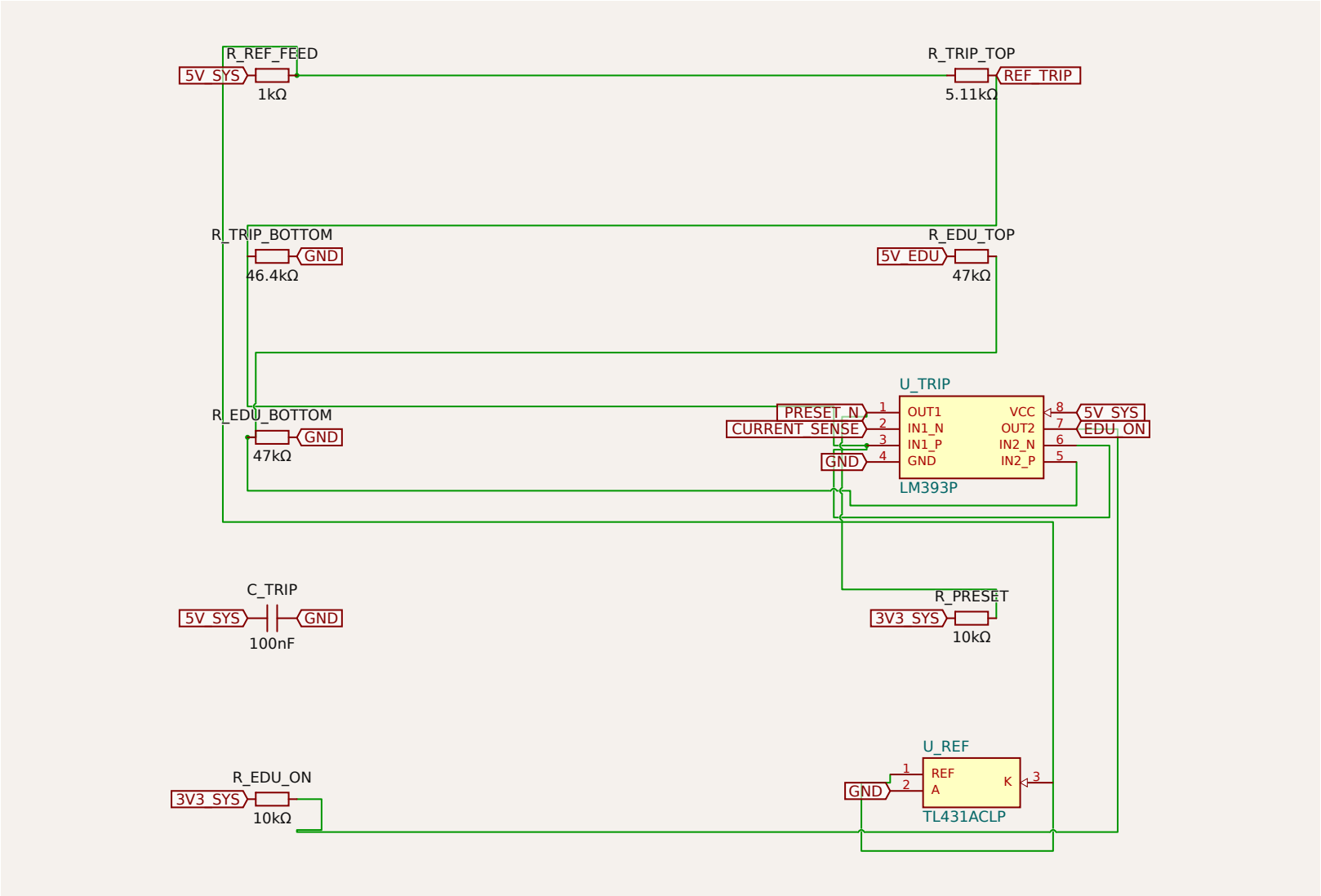
#	Reference	Purpose	Install	Adapter / carrier
1	F_EDU	MF-R050	CP11	
2	Q_PASS	NDP6020P	CP11	
3	R_GATE_OFF	10k	CP11	
4	Q_ENABLE	2N7000	CP11	
5	R_SHUNT	10	CP11	
6	C_EDU	100nF	CP11	
7	U_SENSE	INA180A1IDBVR	CP11	SOT23-5 to numbered single-row adapter
8	C_SENSE	100nF	CP11	
9	U_REF	TL431ACLP	CP11	
10	R_REF_FEED	1k	CP11	
11	R_TRIP_TOP	5.11k	CP11	
12	R_TRIP_BOTTOM	46.4k	CP11	
13	R_EDU_TOP	47k	CP11	
14	R_EDU_BOTTOM	47k	CP11	
15	U_TRIP	LM393P	CP11	
16	C_TRIP	100nF	CP11	
17	R_PRESET	10k	CP11	
18	R_EDU_ON	10k	CP11	
19	J_LINK_1	Harness terminal: GND	CP11	
20	J_LINK_2	Harness terminal: 3V3_SYS	CP11	
21	J_LINK_3	Harness terminal: 5V_SYS	CP11	
22	J_LINK_4	Harness terminal: PRESET_N	CP11	
23	J_LINK_5	Harness terminal: 5V_EDU	CP11	
24	J_LINK_6	Harness terminal: CURRENT_SENSE	CP11	
25	J_LINK_7	Harness terminal: EDU_ON	CP11	
26	J_LINK_8	Harness terminal: ALLOW_RAIL	CP11	
27	J_LINK_9	Harness terminal: REF_TRIP	CP11	

power-analog — Student



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

power-analog — Reference

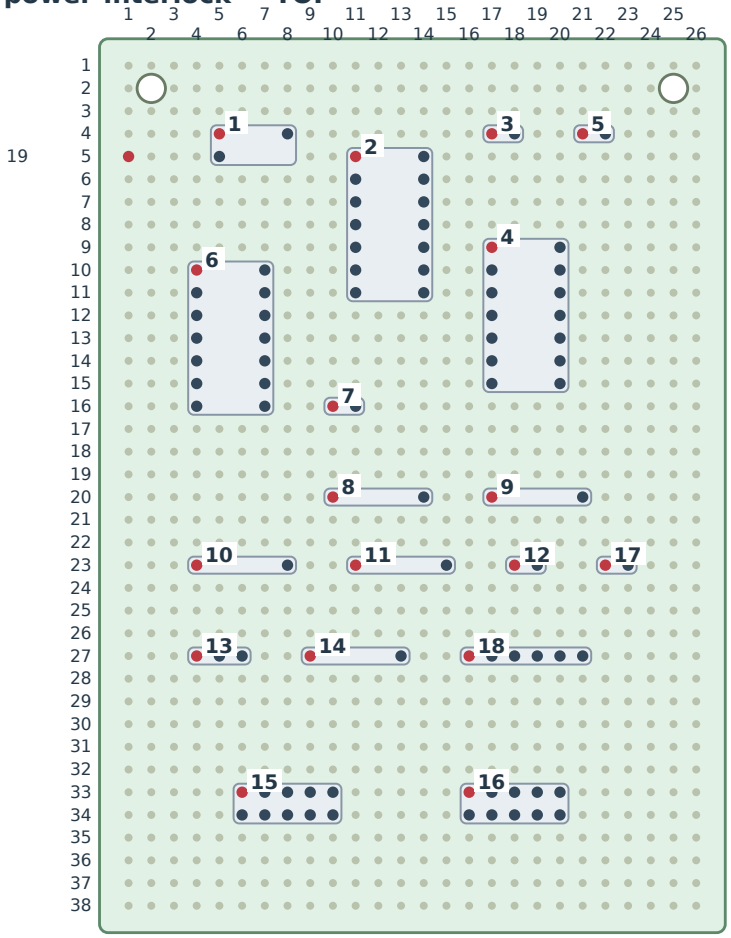


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

power-interlock

Fault latch / watchdog interlock / test headers

power-interlock — TOP

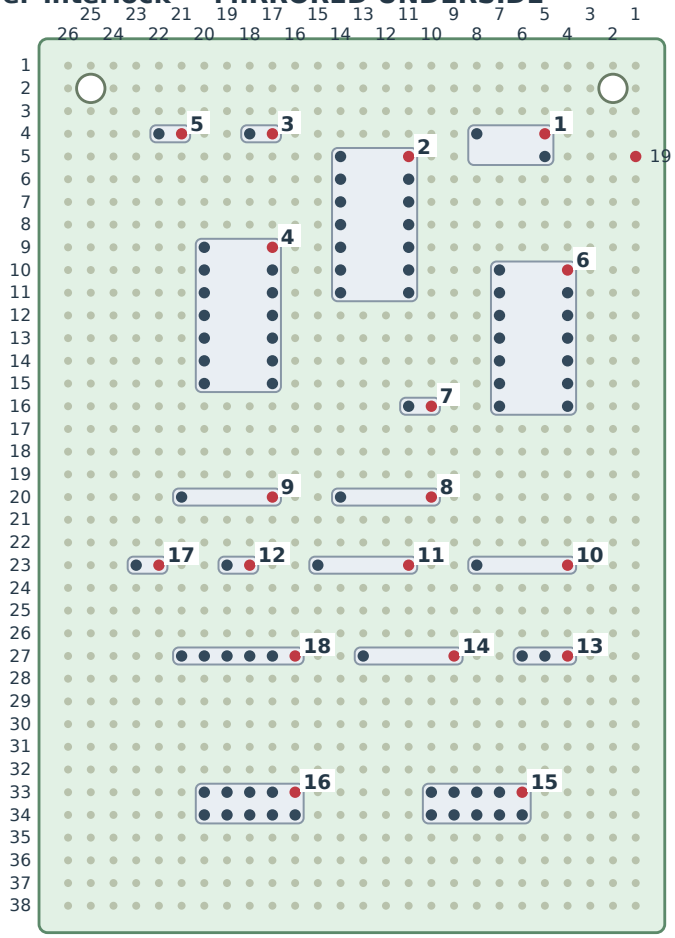


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

power-interlock — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

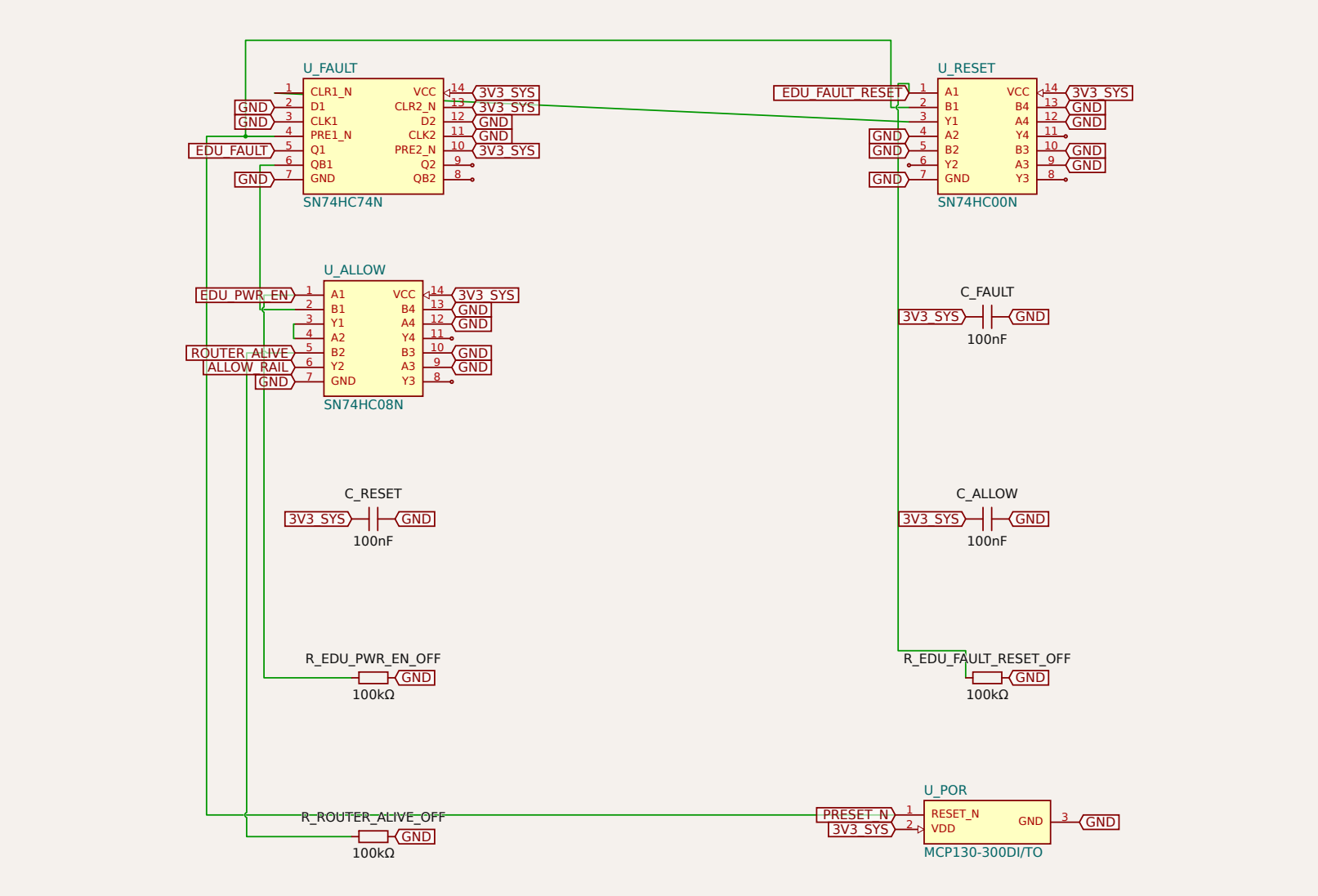
PROTOTYPE: physical fit and hardware tests pending.

power-interlock — numbered parts

Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

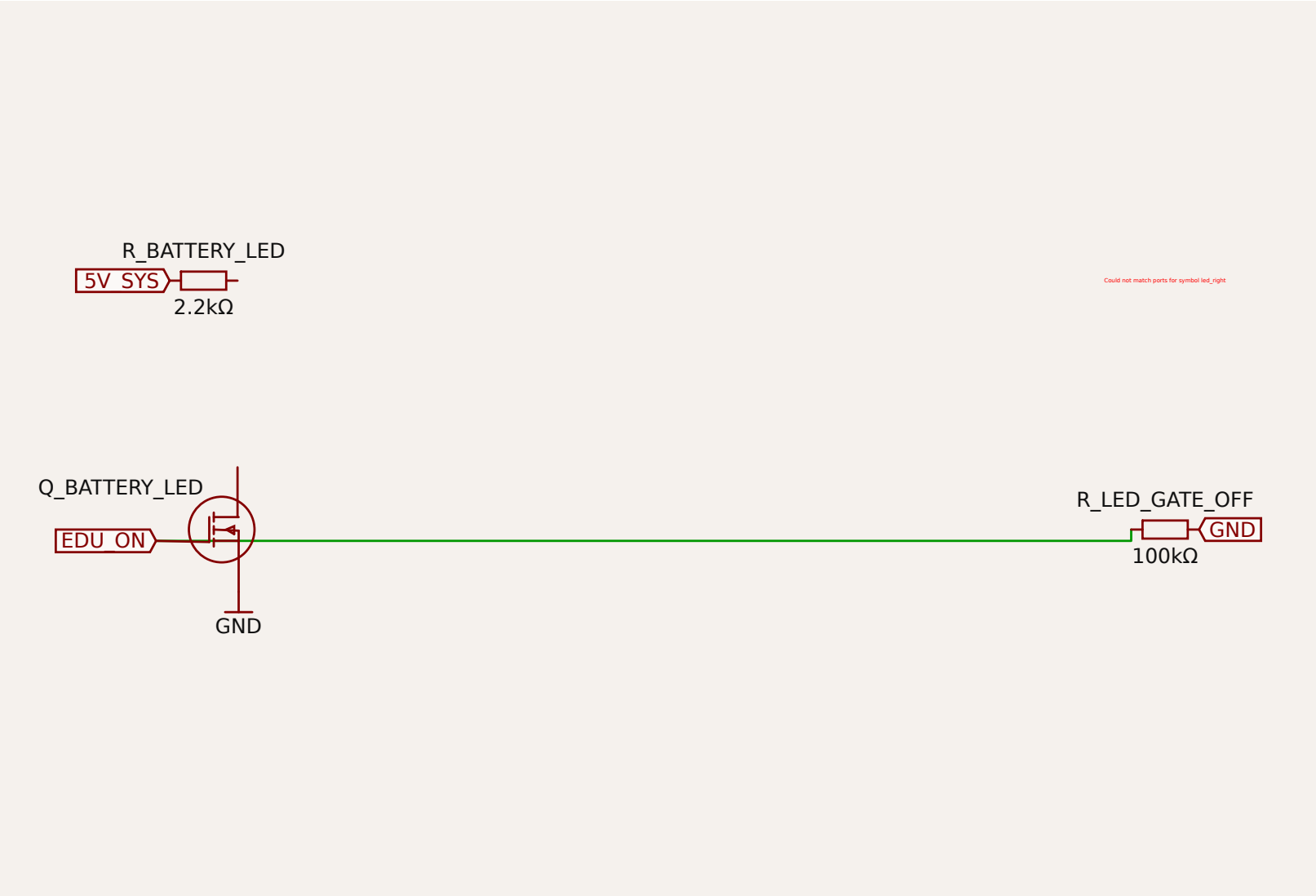
#	Reference	Purpose	Install	Adapter / carrier
1	U_POR	MCP130-300DI/TO	CP11	
2	U_FAULT	SN74HC74N	CP11	
3	C_FAULT	100nF	CP11	
4	U_RESET	SN74HC00N	CP11	
5	C_RESET	100nF	CP11	
6	U_ALLOW	SN74HC08N	CP11	
7	C_ALLOW	100nF	CP11	
8	R_EDU_PWR_EN_OFF	100k	CP11	
9	R_EDU_FAULT_RESET_OFF	100k	CP11	
10	R_ROUTER_ALIVE_OFF	100k	CP11	
11	R_BATTERY_LED	2.2k	CP11	
12	LED_BATTERY	red LED — actual student rail on	CP11	
13	Q_BATTERY_LED	2N7000	CP11	
14	R_LED_GATE_OFF	100k	CP11	
15	J_MATRIX	POWER_MATRIX v2	CP9	
16	J_SMART	POWER_SMART v2	CP9	
17	J_FIXED_BATTERY	fixed battery patch lead: C2 positive, C1 ground; unplug before matrix mode	CP9	
18	J_TEST	scope / known load fixture	CP9	
19	J_LINK_1	Harness terminal: ALLOW_RAIL	CP11	

power-interlock — Latch



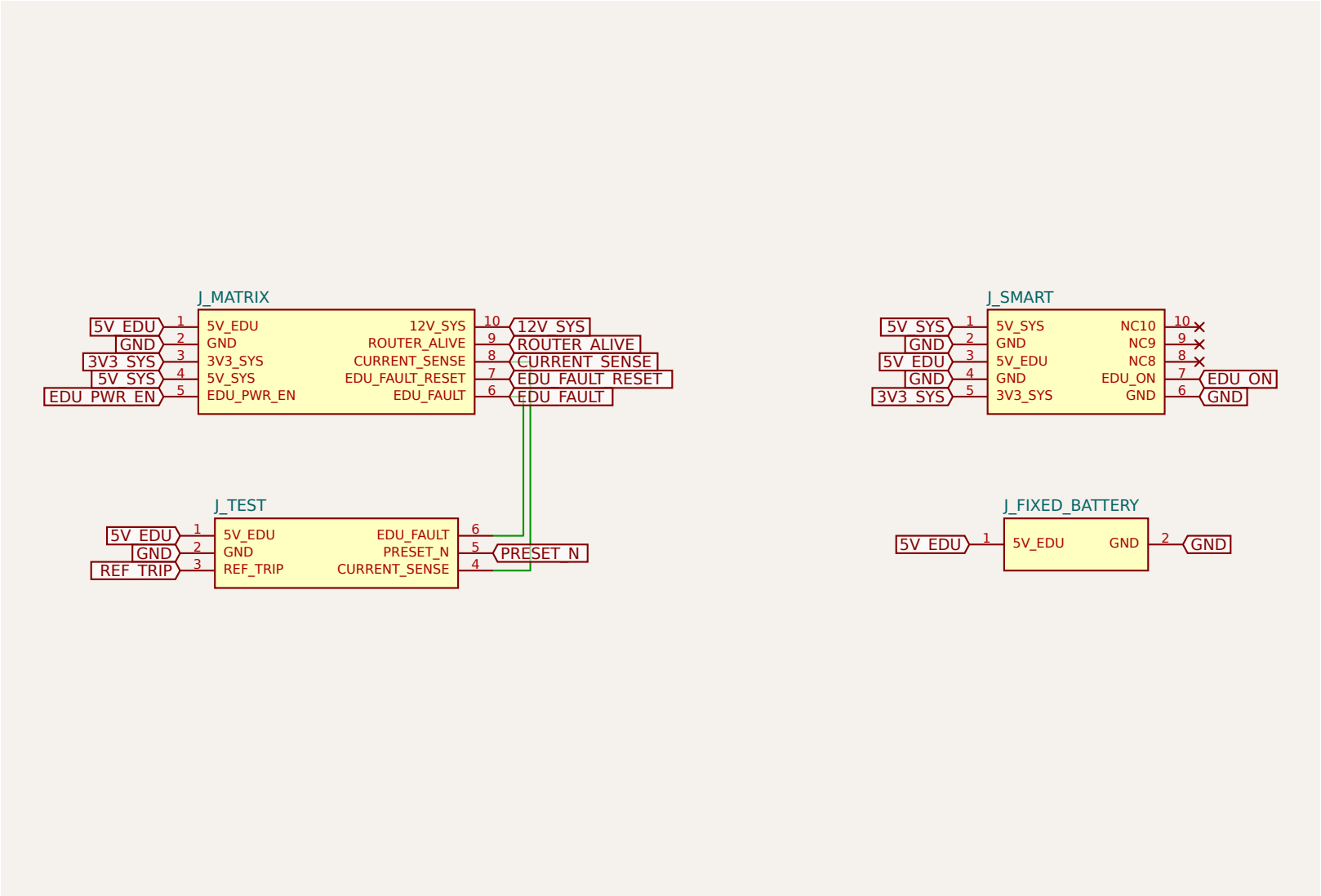
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

power-interlock — Indicator



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

power-interlock — Outputs

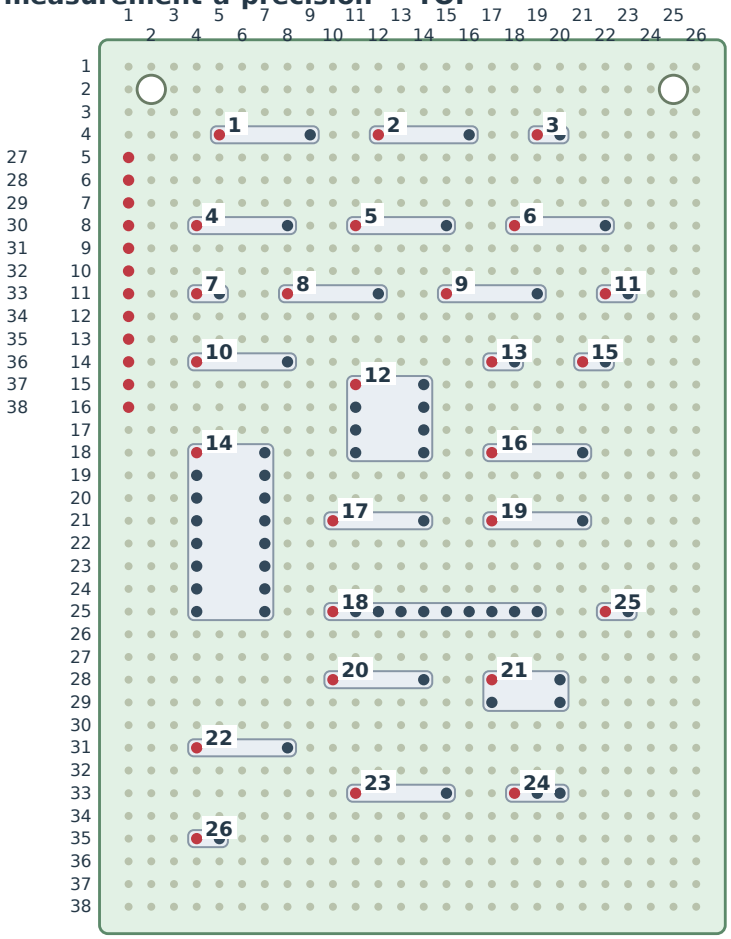


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

measurement-a-precision

Bank A: buffered voltage scanner and continuity stimulus

measurement-a-precision — TOP

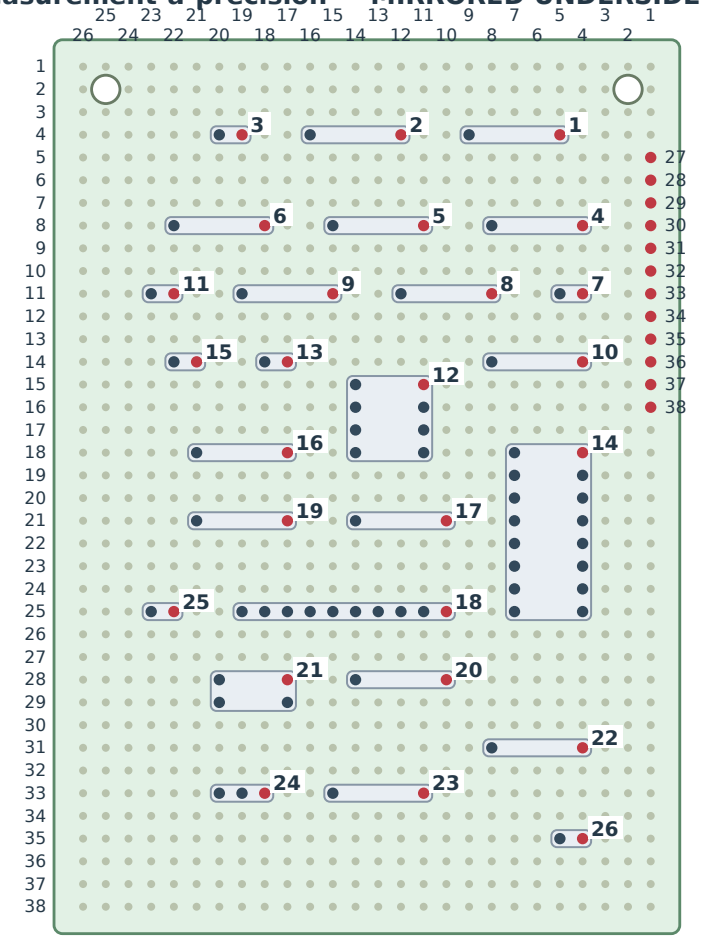


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

measurement-a-precision — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

measurement-a-precision — numbered parts

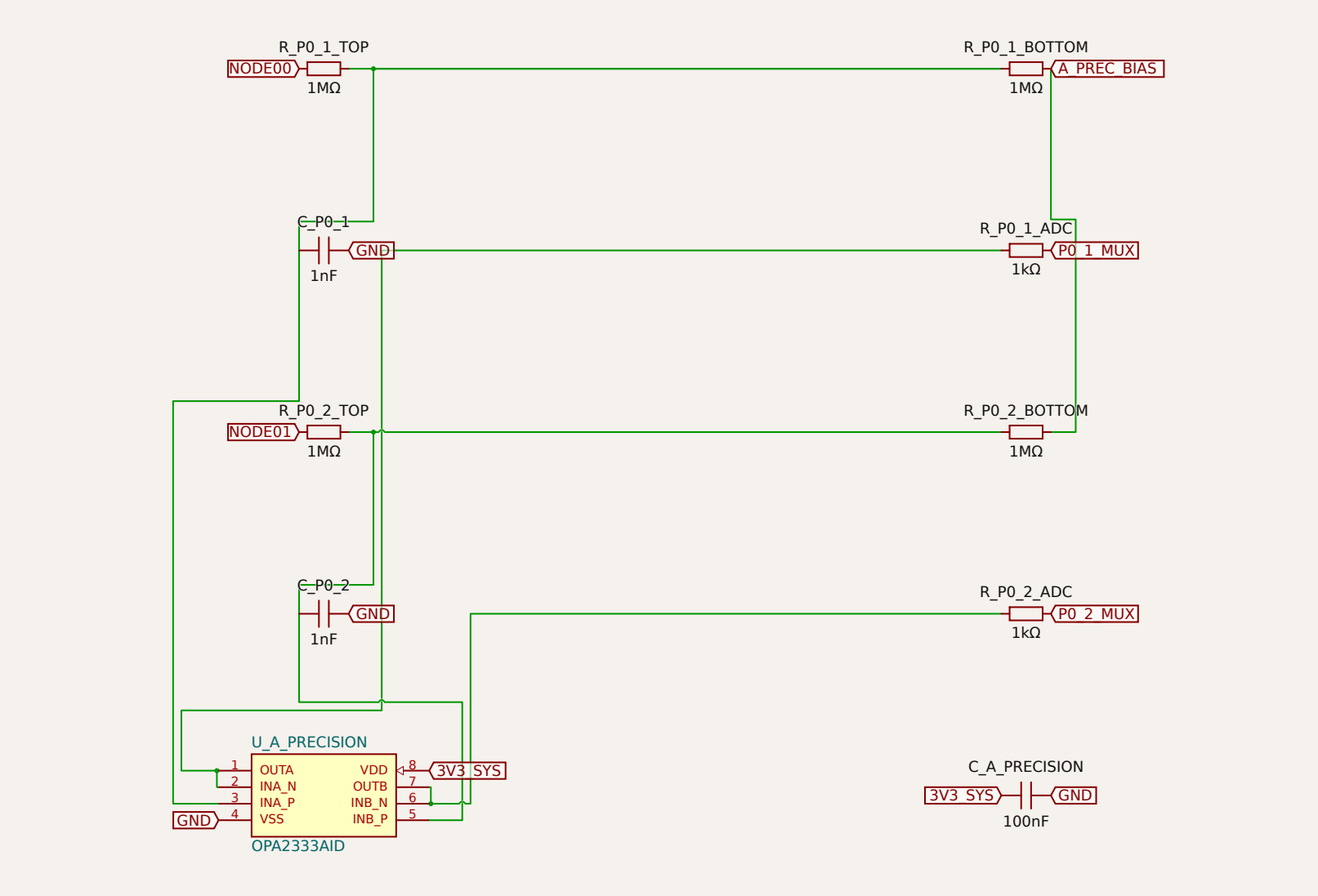
Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

#	Reference	Purpose	Install	Adapter / carrier
1	R_P0_1_TOP	R_P0_1_TOP: 1MΩ	CP1	
2	R_P0_1_BOTTOM	R_P0_1_BOTTOM: 1MΩ	CP1	
3	C_P0_1	C_P0_1: 1nFΩ	CP1	
4	R_P0_1_ADC	R_P0_1_ADC: 1kΩ	CP1	
5	R_P0_2_TOP	R_P0_2_TOP: 1MΩ	CP1	
6	R_P0_2_BOTTOM	R_P0_2_BOTTOM: 1MΩ	CP1	
7	C_P0_2	C_P0_2: 1nFΩ	CP1	
8	R_P0_2_ADC	R_P0_2_ADC: 1kΩ	CP1	
9	R_A_PREC_BIAS_TOP	R_A_PREC_BIAS_TOP: 5.62kΩ	CP1	
10	R_A_PREC_BIAS_BOTTOM	R_A_PREC_BIAS_BOTTOM: 1kΩ	CP1	
11	C_A_PREC_BIAS	C_A_PREC_BIAS: 100nFΩ	CP1	
12	U_A_PRECISION	OPA2333AID precision diagnostic/voltage buffer	CP1	SOIC8 to numbered DIP8
13	C_A_PRECISION	C_A_PRECISION: 100nFΩ	CP1	
14	U_A_MUX	CD4052BE buffered voltage selector	CP1	
15	C_A_MUX	C_A_MUX: 100nFΩ	CP1	
16	R_A_SEL_A	R_A_SEL_A: 100kΩ	CP1	
17	R_A_SEL_B	R_A_SEL_B: 100kΩ	CP1	
18	J_A_ADC_MODULE	ADS1115 carrier interface — verify labelled module pads	CP1	ADS1115 carrier
19	D_STIM	BAT85	CP1	
20	R_STIM	11kΩ ±1% independently bounds diagnostic stimulus below0.34mA	CP1	
21	K_STIM	AQY212EH	CP1	
22	R_LED	270Ω ±1% diagnostic PhotoMOS LED drive	CP1	
23	R_OFF	100k	CP1	
24	Q_STIM	2N7000	CP1	
25	J_DIAGNOSTIC	Removable diagnostic stimulus / return	CP1	
26	JP_RETURN	Fit only for unpowered continuity test	CP1	
27	J_LINK_1	Harness terminal: SCL_POSITION_0	CP1	
28	J_LINK_2	Harness terminal: SDA_POSITION_0	CP1	

measurement-a-precision — parts continued

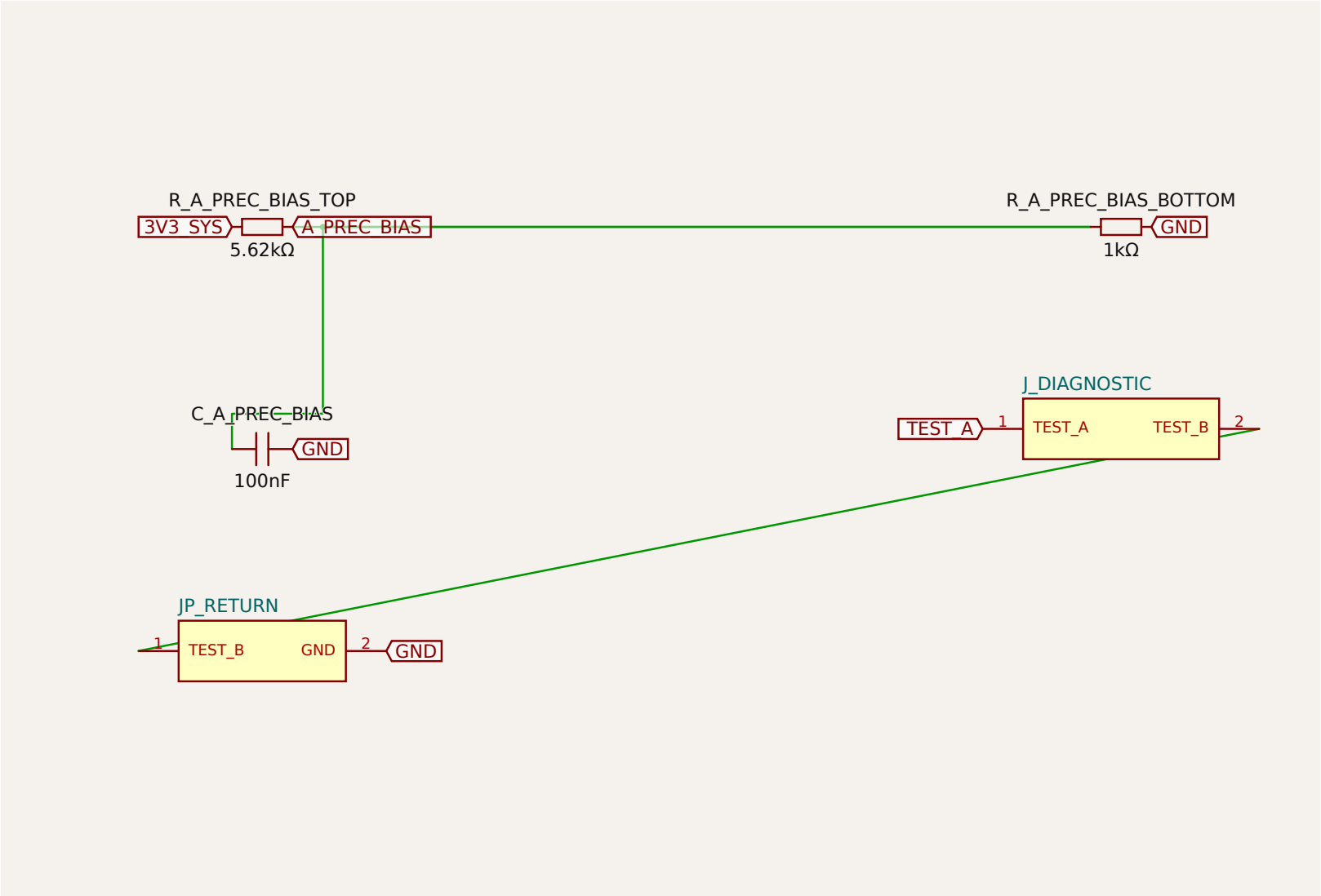
#	Reference	Purpose	Install	Adapter / carrier
29	J_LINK_3	Harness terminal: 3V3_SYS	CP1	
30	J_LINK_4	Harness terminal: NODE00	CP1	
31	J_LINK_5	Harness terminal: NODE01	CP1	
32	J_LINK_6	Harness terminal: FIX_STIM	CP1	
33	J_LINK_7	Harness terminal: MUX_A_A	CP1	
34	J_LINK_8	Harness terminal: MUX_A_B	CP1	
35	J_LINK_9	Harness terminal: P1_1_MUX	CP1	
36	J_LINK_10	Harness terminal: P1_2_MUX	CP1	
37	J_LINK_11	Harness terminal: P2_1_MUX	CP1	
38	J_LINK_12	Harness terminal: P2_2_MUX	CP1	

measurement-a-precision — Voltage inputs



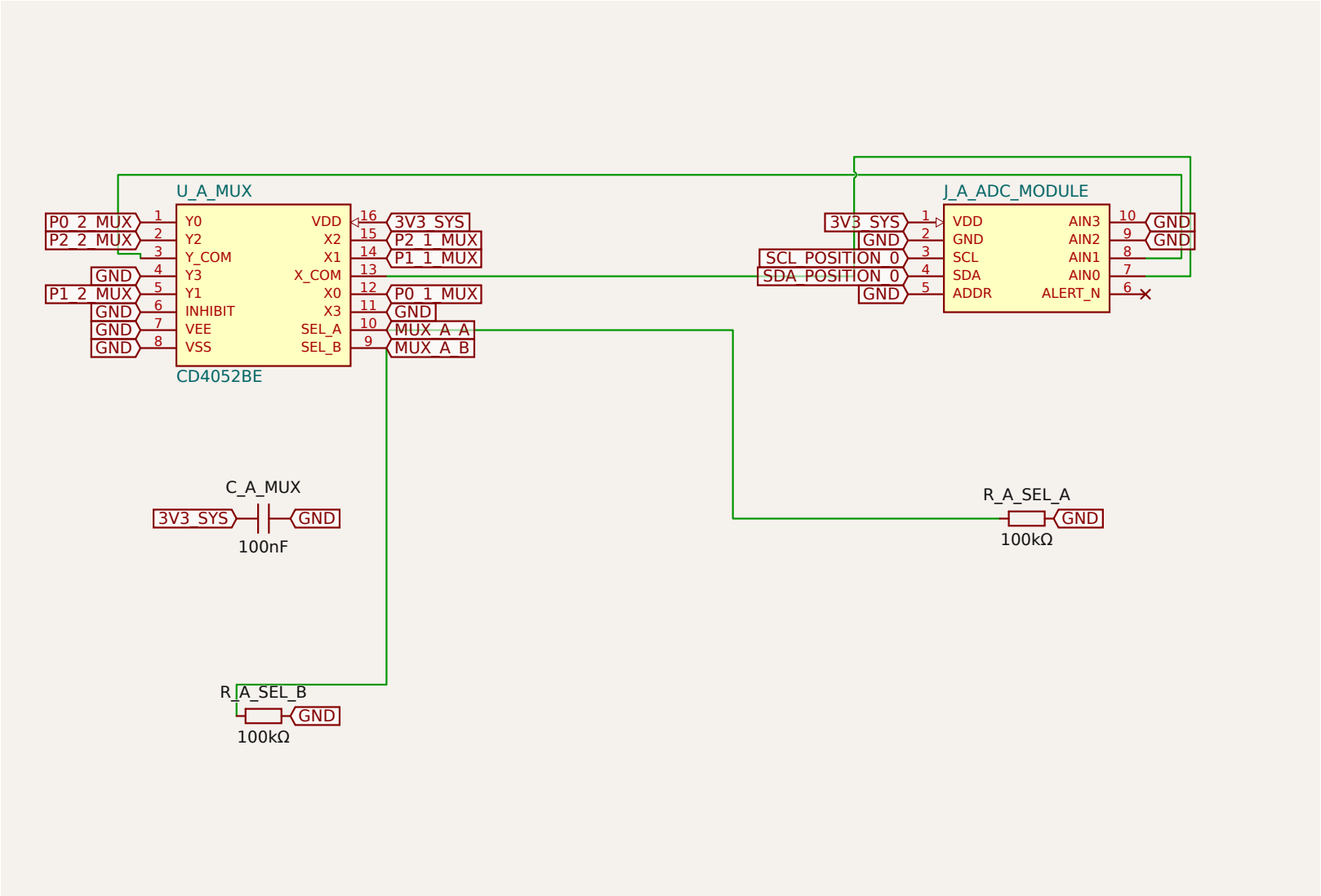
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

measurement-a-precision — Support



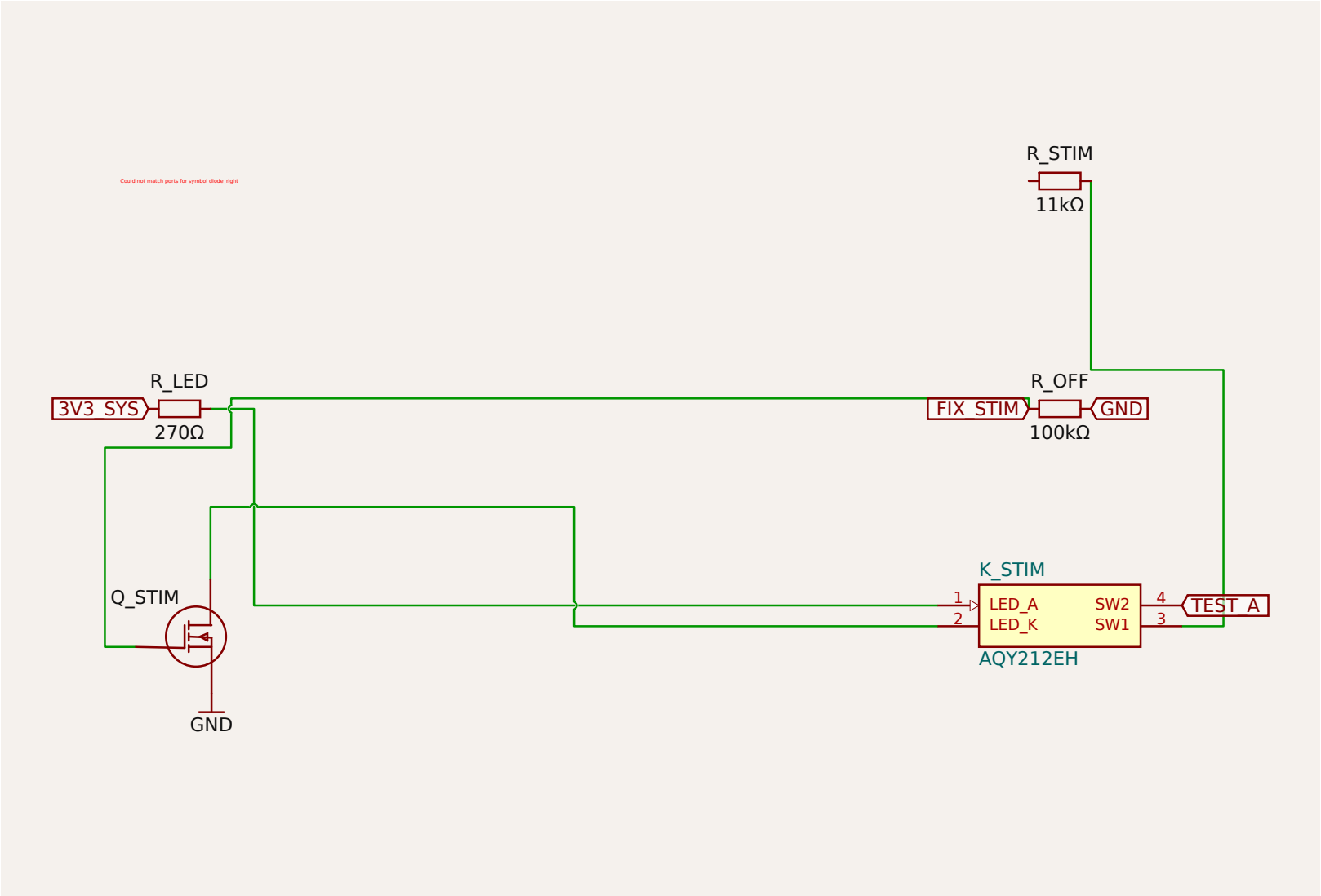
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

measurement-a-precision — Buffered selector and ADC



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

measurement-a-precision — Stimulus

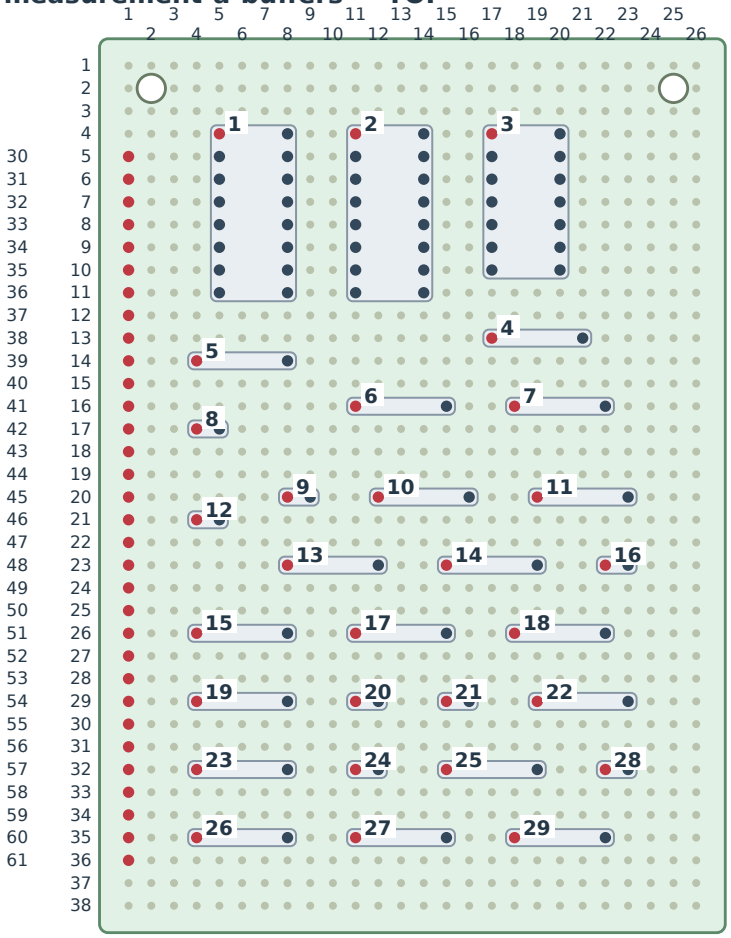


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

measurement-a-buffers

Bank A: buffered voltage scanner and continuity stimulus

measurement-a-buffers — TOP

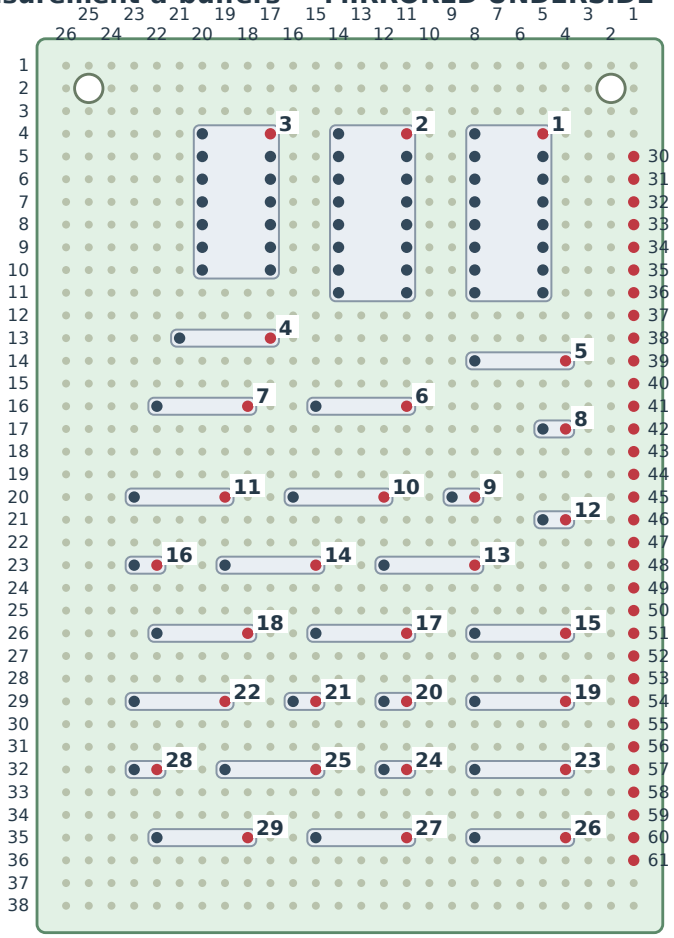


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

measurement-a-buffers — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

measurement-a-buffers — numbered parts

Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

#	Reference	Purpose	Install	Adapter / carrier
1	U_A_LED_DRIVER_0	ULN2003AN PhotoMOS LED driver array	CP15	
2	U_A_LED_DRIVER_1	ULN2003AN PhotoMOS LED driver array	CP15	
3	U_A_BUFFERS	MCP6004-I/P four voltage buffers	CP27	
4	R_A_HI_TOP	1k	CP15	
5	R_A_HI_BOTTOM	49.9k	CP15	
6	R_A_LO_TOP	49.9k	CP15	
7	R_A_LO_BOTTOM	1k	CP15	
8	C_A_REFHI	100nF	CP15	
9	C_A_REFLO	100nF	CP15	
10	R_P1_1_TOP	R_P1_1_TOP: 1MΩ	CP27	
11	R_P1_1_BOTTOM	R_P1_1_BOTTOM: 1MΩ	CP27	
12	C_P1_1	C_P1_1: 1nFΩ	CP27	
13	R_P1_1_ADC	R_P1_1_ADC: 1kΩ	CP27	
14	R_P1_2_TOP	R_P1_2_TOP: 1MΩ	CP27	
15	R_P1_2_BOTTOM	R_P1_2_BOTTOM: 1MΩ	CP27	
16	C_P1_2	C_P1_2: 1nFΩ	CP27	
17	R_P1_2_ADC	R_P1_2_ADC: 1kΩ	CP27	
18	R_A_GENERAL_BIAS_TOP	R_A_GENERAL_BIAS_TOP: 5.62kΩ	CP27	
19	R_A_GENERAL_BIAS_BOTTOM	R_A_GENERAL_BIAS_BOTTOM: 1kΩ	CP27	
20	C_A_GENERAL_BIAS	C_A_GENERAL_BIAS: 100nFΩ	CP27	
21	C_A_BUFFERS	C_A_BUFFERS: 100nFΩ	CP27	
22	R_P2_1_TOP	R_P2_1_TOP: 1MΩ	CP28	
23	R_P2_1_BOTTOM	R_P2_1_BOTTOM: 1MΩ	CP28	
24	C_P2_1	C_P2_1: 1nFΩ	CP28	
25	R_P2_1_ADC	R_P2_1_ADC: 1kΩ	CP28	
26	R_P2_2_TOP	R_P2_2_TOP: 1MΩ	CP28	
27	R_P2_2_BOTTOM	R_P2_2_BOTTOM: 1MΩ	CP28	
28	C_P2_2	C_P2_2: 1nFΩ	CP28	

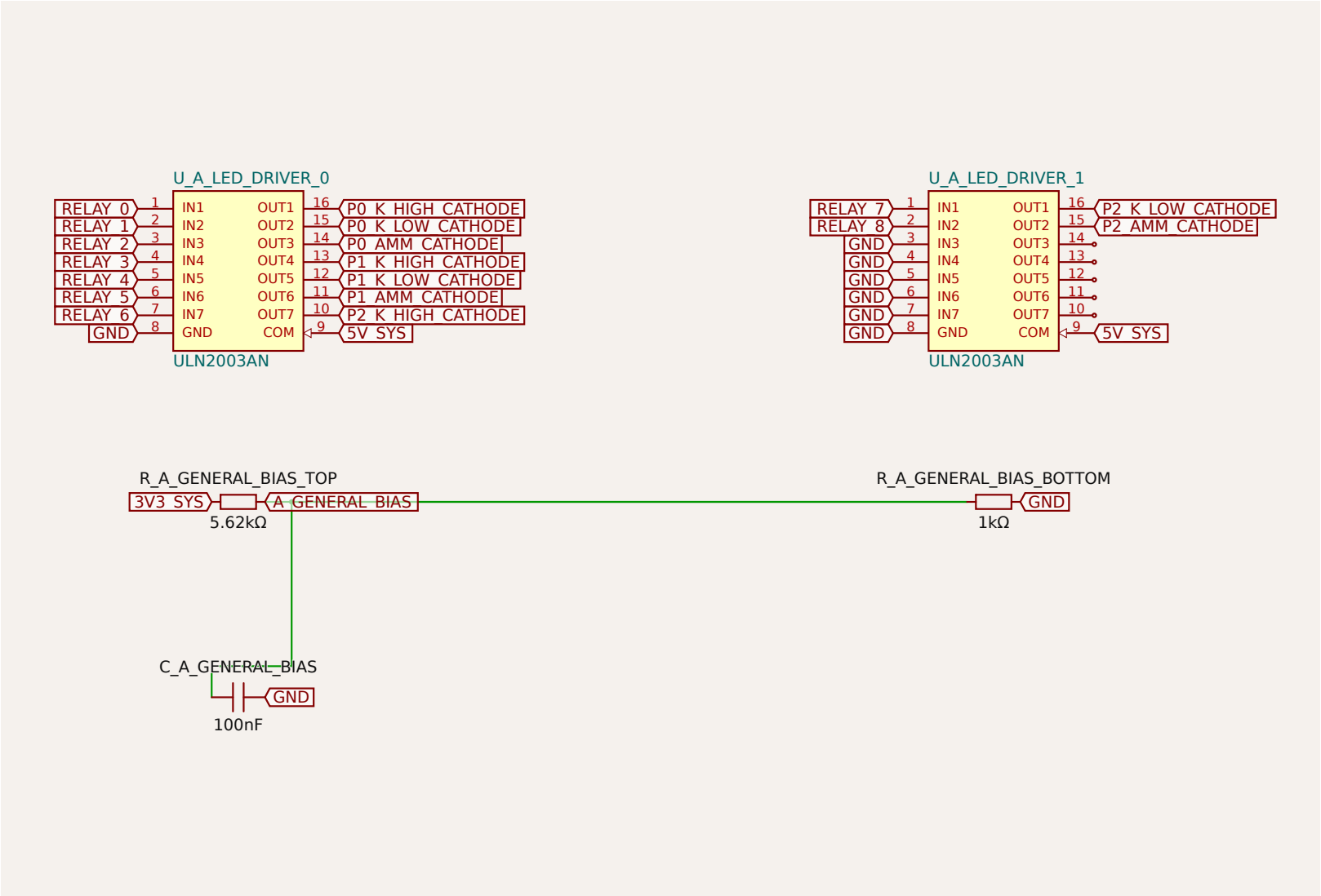
measurement-a-buffers — parts continued

#	Reference	Purpose	Install	Adapter / carrier
29	R_P2_2_ADC	R_P2_2_ADC: 1kΩ	CP28	
30	J_LINK_1	Harness terminal: GND	CP15	
31	J_LINK_2	Harness terminal: 3V3_SYS	CP27	
32	J_LINK_3	Harness terminal: NODE04	CP27	
33	J_LINK_4	Harness terminal: NODE05	CP27	
34	J_LINK_5	Harness terminal: NODE08	CP28	
35	J_LINK_6	Harness terminal: NODE09	CP28	
36	J_LINK_7	Harness terminal: 5V_SYS	CP15	
37	J_LINK_8	Harness terminal: 5V_EDU	CP15	
38	J_LINK_9	Harness terminal: RELAY_0	CP15	
39	J_LINK_10	Harness terminal: RELAY_1	CP15	
40	J_LINK_11	Harness terminal: RELAY_2	CP15	
41	J_LINK_12	Harness terminal: RELAY_3	CP15	
42	J_LINK_13	Harness terminal: RELAY_4	CP15	
43	J_LINK_14	Harness terminal: RELAY_5	CP15	
44	J_LINK_15	Harness terminal: RELAY_6	CP15	
45	J_LINK_16	Harness terminal: RELAY_7	CP15	
46	J_LINK_17	Harness terminal: RELAY_8	CP15	
47	J_LINK_18	Harness terminal: P1_1_MUX	CP27	
48	J_LINK_19	Harness terminal: P1_2_MUX	CP27	
49	J_LINK_20	Harness terminal: P2_1_MUX	CP28	
50	J_LINK_21	Harness terminal: P2_2_MUX	CP28	
51	J_LINK_22	Harness terminal: P0_K_HIGH_CATHODE	CP15	
52	J_LINK_23	Harness terminal: P0_K_LOW_CATHODE	CP15	
53	J_LINK_24	Harness terminal: P0_AMM_CATHODE	CP15	
54	J_LINK_25	Harness terminal: P1_K_HIGH_CATHODE	CP15	
55	J_LINK_26	Harness terminal: P1_K_LOW_CATHODE	CP15	
56	J_LINK_27	Harness terminal: P1_AMM_CATHODE	CP15	

measurement-a-buffers — parts continued

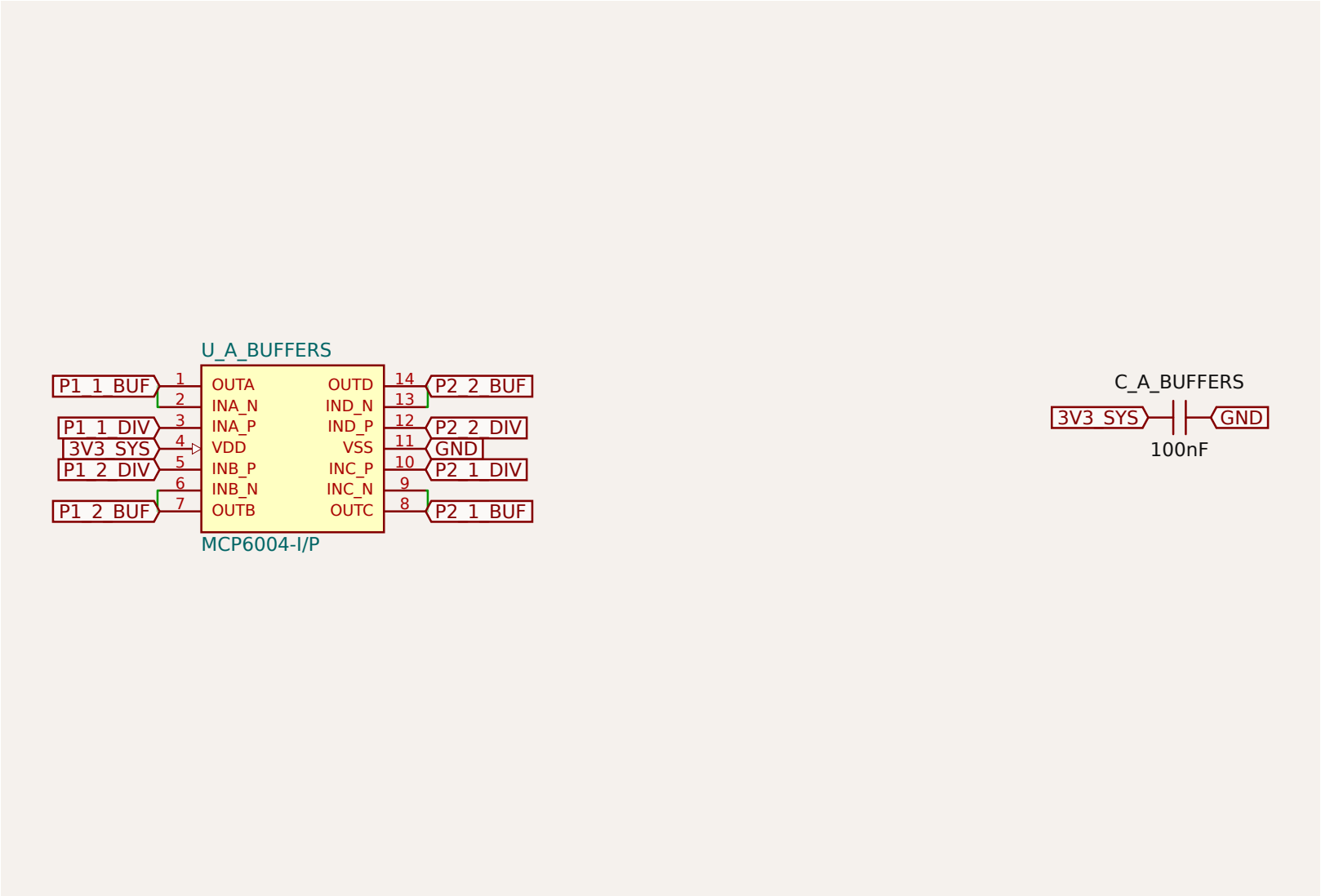
#	Reference	Purpose	Install	Adapter / carrier
57	J_LINK_28	Harness terminal: P2_K_HIGH_CATHODE	CP15	
58	J_LINK_29	Harness terminal: P2_K_LOW_CATHODE	CP15	
59	J_LINK_30	Harness terminal: P2_AMM_CATHODE	CP15	
60	J_LINK_31	Harness terminal: A_REF_HI	CP15	
61	J_LINK_32	Harness terminal: A_REF_LO	CP15	

measurement-a-buffers — Support



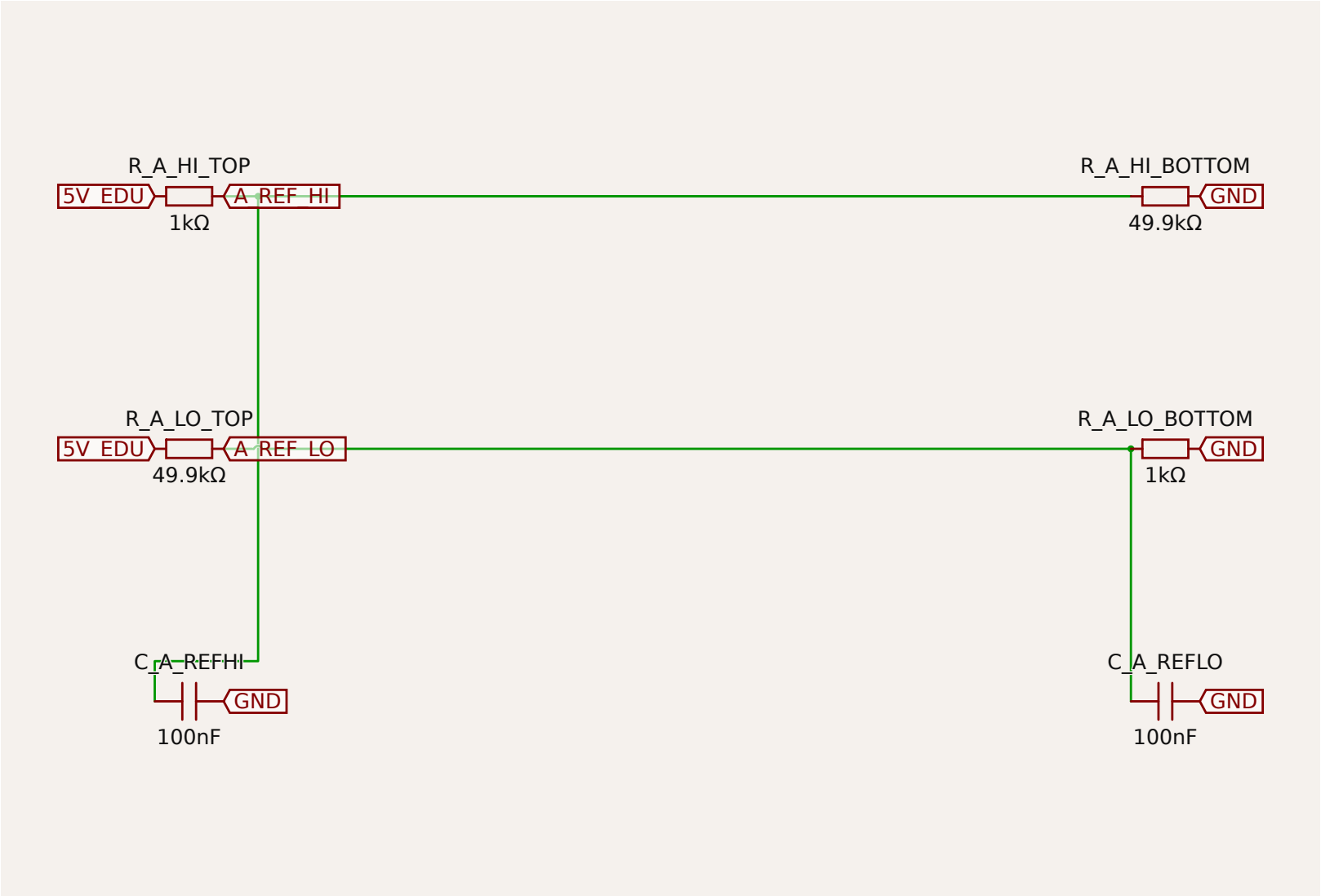
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

measurement-a-buffers — Voltage buffers



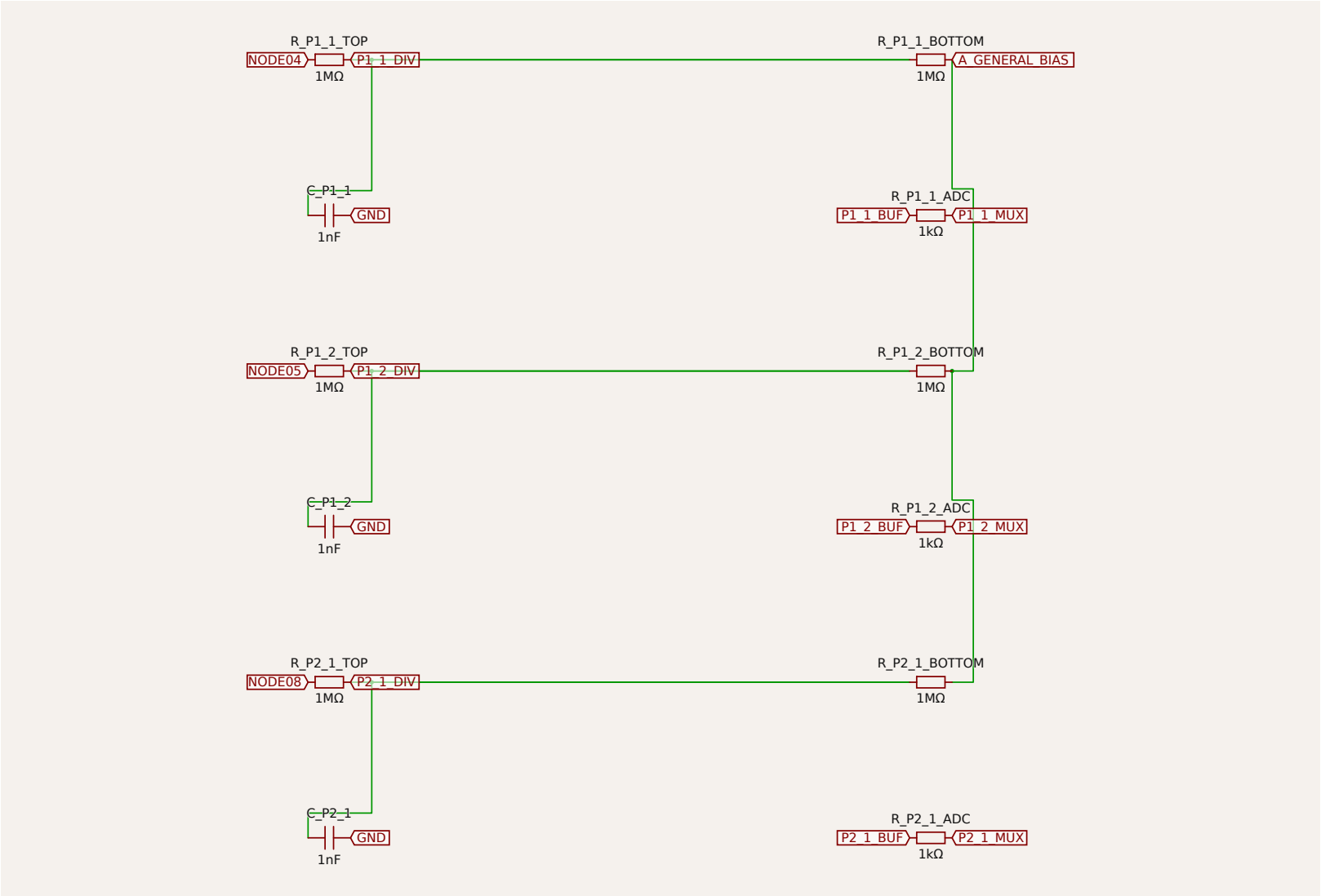
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

measurement-a-buffers — Limits



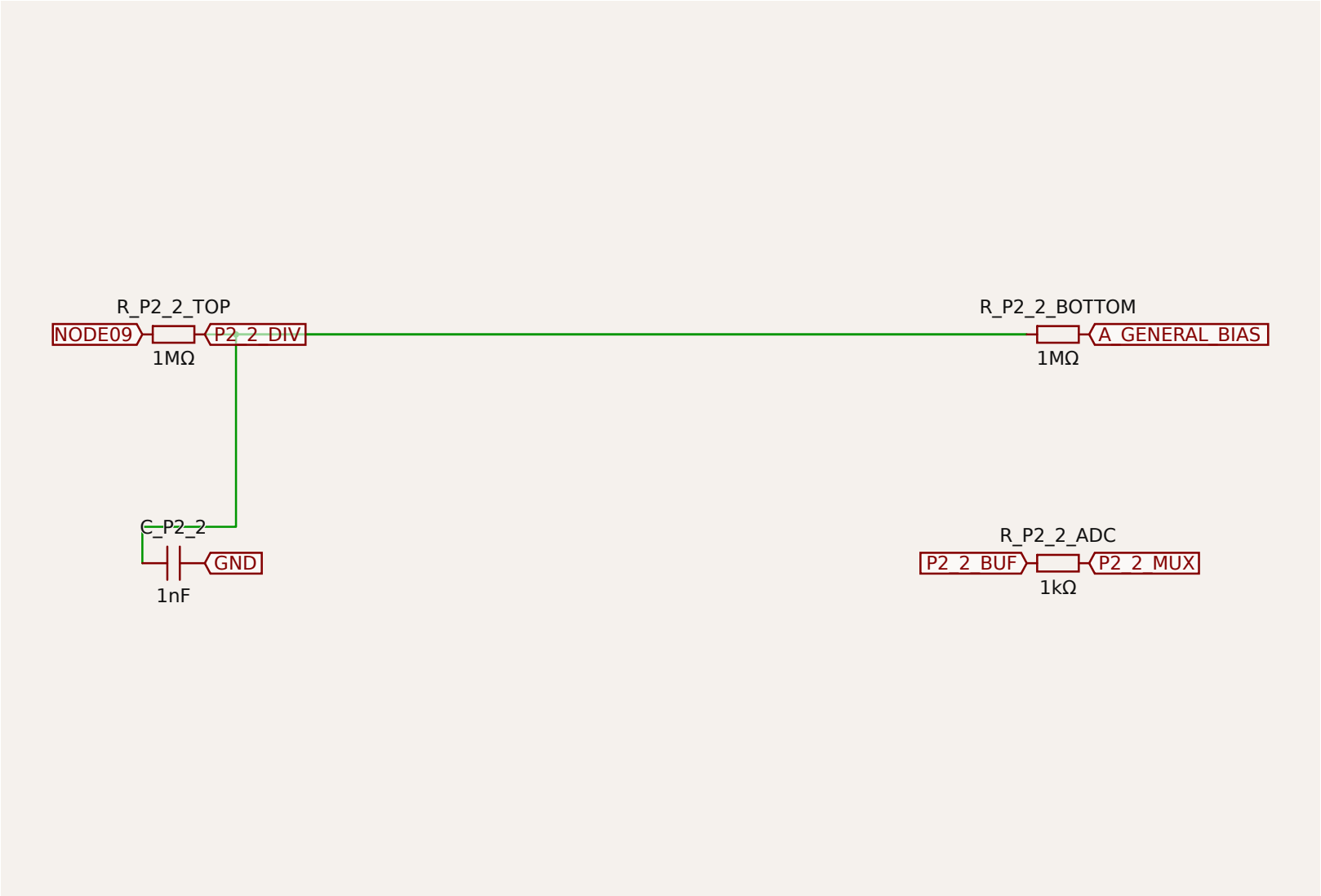
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

measurement-a-buffers — Voltage inputs 1



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

measurement-a-buffers — Voltage inputs 2

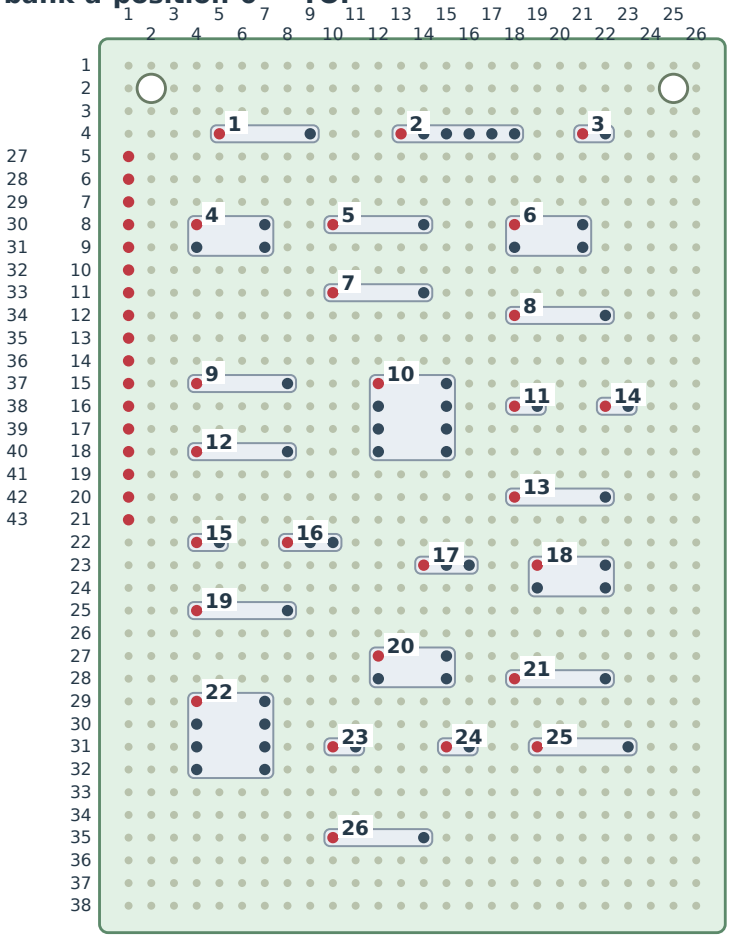


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-a-position-0

Bank A: position 0 complete isolated channel

bank-a-position-0 — TOP

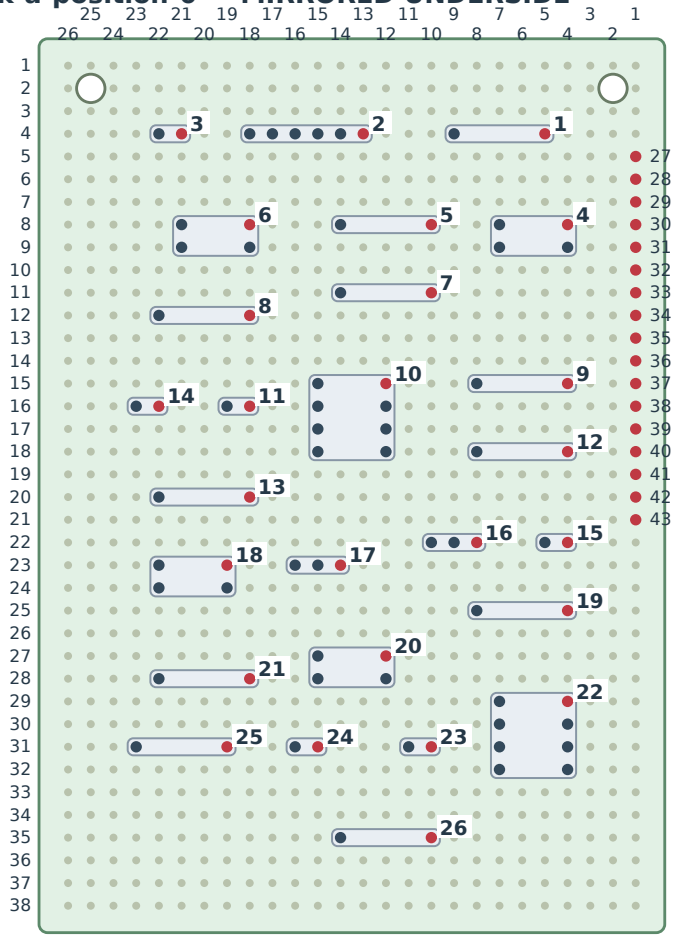


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

bank-a-position-0 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

bank-a-position-0 — numbered parts

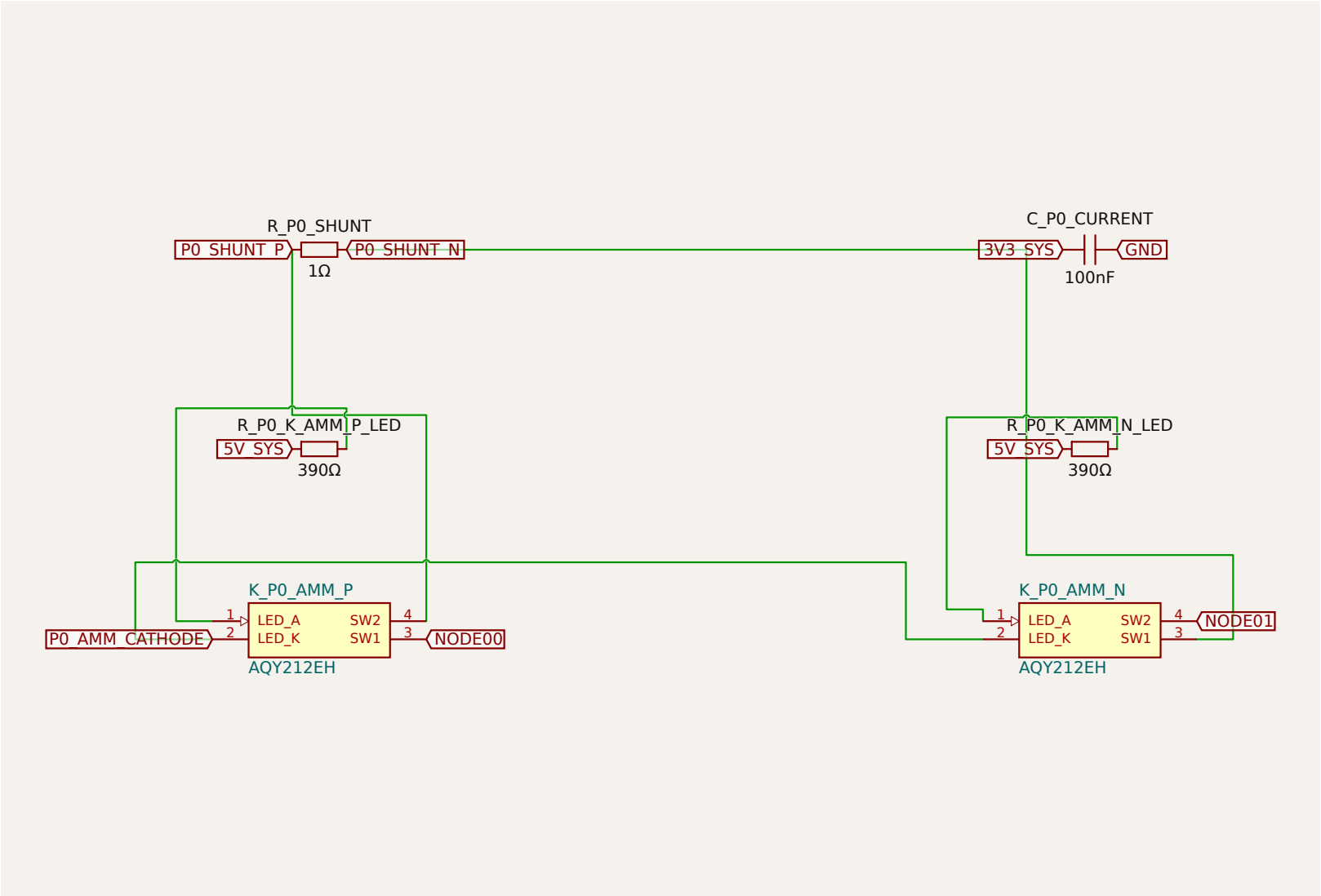
Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

#	Reference	Purpose	Install	Adapter / carrier
1	R_P0_SHUNT	1Ω ±1% external current shunt	CP22	
2	J_P0_CURRENT_MODULE	INA226 carrier: remove onboard shunt; external 1Ω	CP22	INA226 carrier
3	C_P0_CURRENT	100nF	CP22	
4	K_P0_AMM_P	AQY212EH	CP22	
5	R_P0_K_AMM_P_LED	390Ω ±1% PhotoMOS LED drive with ULN2003 saturation allowance	CP22	
6	K_P0_AMM_N	AQY212EH	CP22	
7	R_P0_K_AMM_N_LED	390Ω ±1% PhotoMOS LED drive with ULN2003 saturation allowance	CP22	
8	R_P0_SOURCE_SENSE	10	CP15	
9	R_P0_SINK_SENSE	10	CP15	
10	U_P0_LIMIT	MCP6002 dual source/sink clamp	CP15	
11	C_P0_LIMIT	100nF	CP15	
12	R_P0_HI_GATE	1k	CP15	
13	R_P0_LO_GATE	1k	CP15	
14	C_P0_HI_COMP	1nF	CP15	
15	C_P0_LO_COMP	1nF	CP15	
16	Q_P0_HIGH	BSS84	CP15	SOT23 to numbered 3-pin adapter
17	Q_P0_LOW	2N7000	CP15	
18	K_P0_HIGH	AQY212EH	CP15	
19	R_P0_K_HIGH_LED	390Ω ±1% PhotoMOS LED drive with ULN2003 saturation allowance	CP15	
20	K_P0_LOW	AQY212EH	CP15	
21	R_P0_K_LOW_LED	390Ω ±1% PhotoMOS LED drive with ULN2003 saturation allowance	CP15	
22	U_P0_BUS	TCA9517D	CP6	SOIC8 to DIP8
23	C_P0_BUS_A	100nF	CP6	
24	C_P0_BUS_B	100nF	CP6	
25	R_P0_SDA	4.7k	CP6	
26	R_P0_SCL	4.7k	CP6	
27	J_LINK_1	Harness terminal: SCL_POSITION_0	CP6	
28	J_LINK_2	Harness terminal: SDA_POSITION_0	CP6	

bank-a-position-0 — parts continued

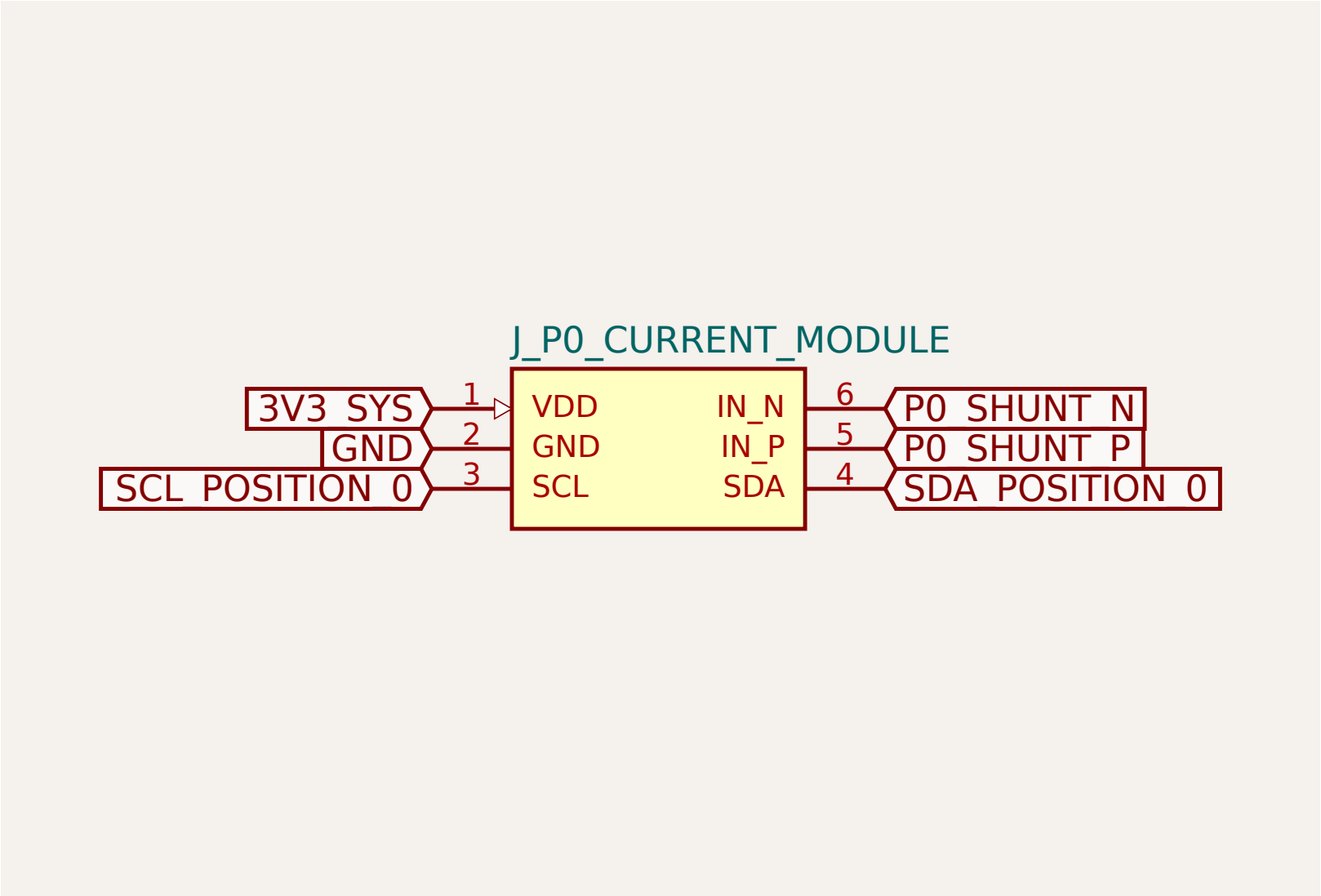
#	Reference	Purpose	Install	Adapter / carrier
29	J_LINK_3	Harness terminal: GND	CP6	
30	J_LINK_4	Harness terminal: 3V3_SYS	CP6	
31	J_LINK_5	Harness terminal: NODE00	CP22	
32	J_LINK_6	Harness terminal: NODE01	CP22	
33	J_LINK_7	Harness terminal: NODE02	CP15	
34	J_LINK_8	Harness terminal: 5V_SYS	CP15	
35	J_LINK_9	Harness terminal: 5V_EDU	CP15	
36	J_LINK_10	Harness terminal: USB_SMART_3V3	CP6	
37	J_LINK_11	Harness terminal: SCL_A	CP6	
38	J_LINK_12	Harness terminal: SDA_0	CP6	
39	J_LINK_13	Harness terminal: P0_K_HIGH_CATHODE	CP15	
40	J_LINK_14	Harness terminal: P0_K_LOW_CATHODE	CP15	
41	J_LINK_15	Harness terminal: P0_AMM_CATHODE	CP22	
42	J_LINK_16	Harness terminal: A_REF_HI	CP15	
43	J_LINK_17	Harness terminal: A_REF_LO	CP15	

bank-a-position-0 — Current



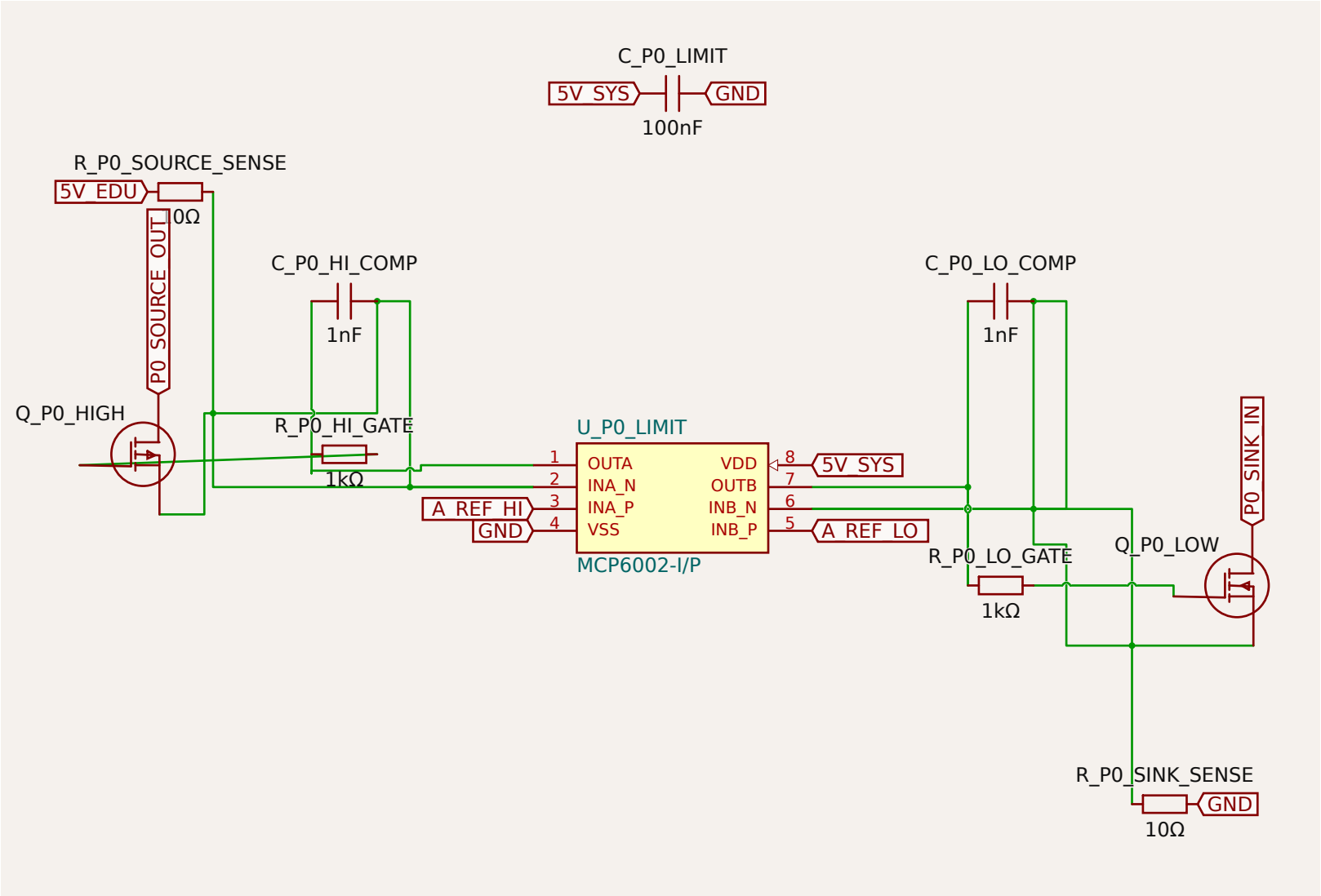
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-a-position-0 — Circuit



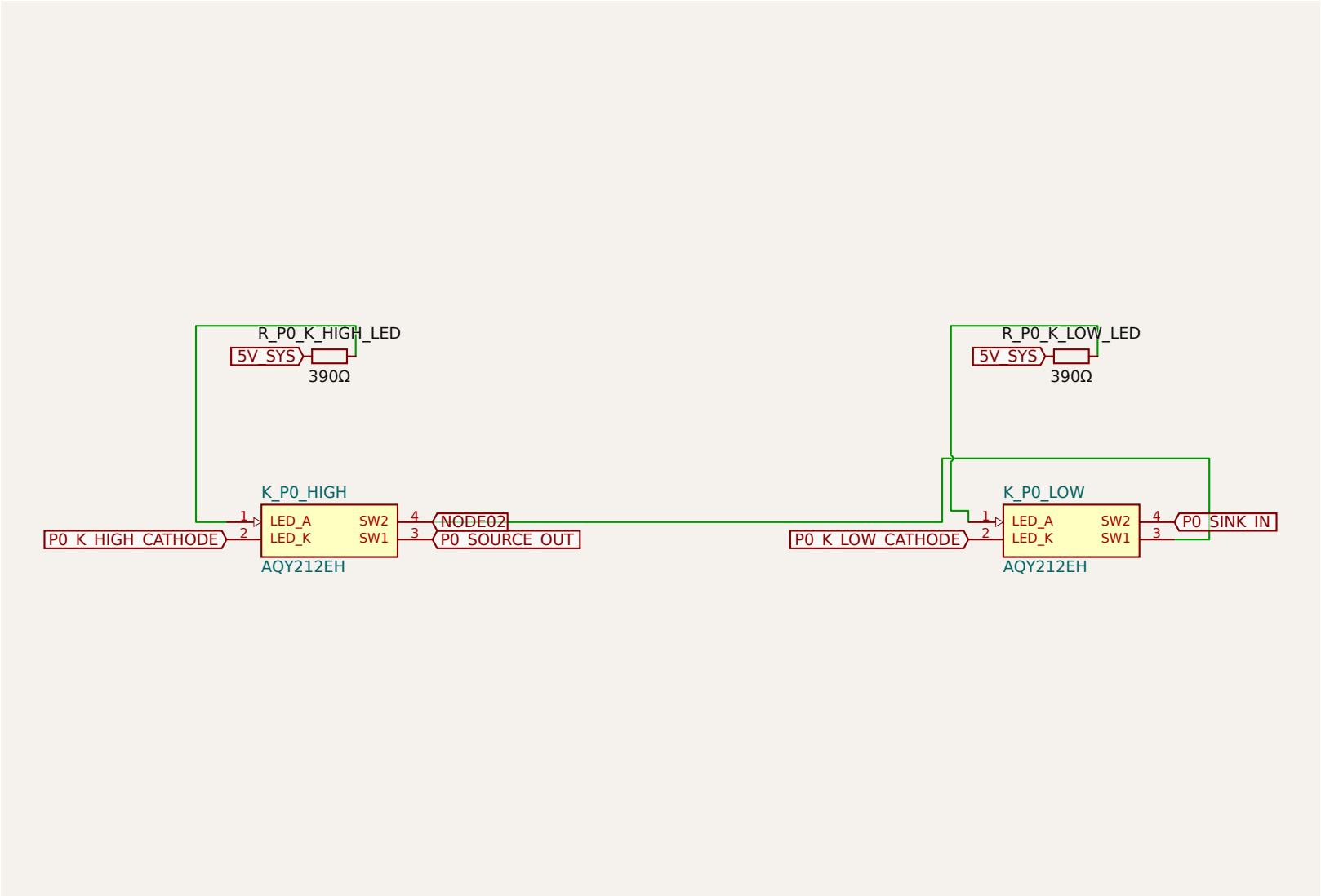
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-a-position-0 — Limits



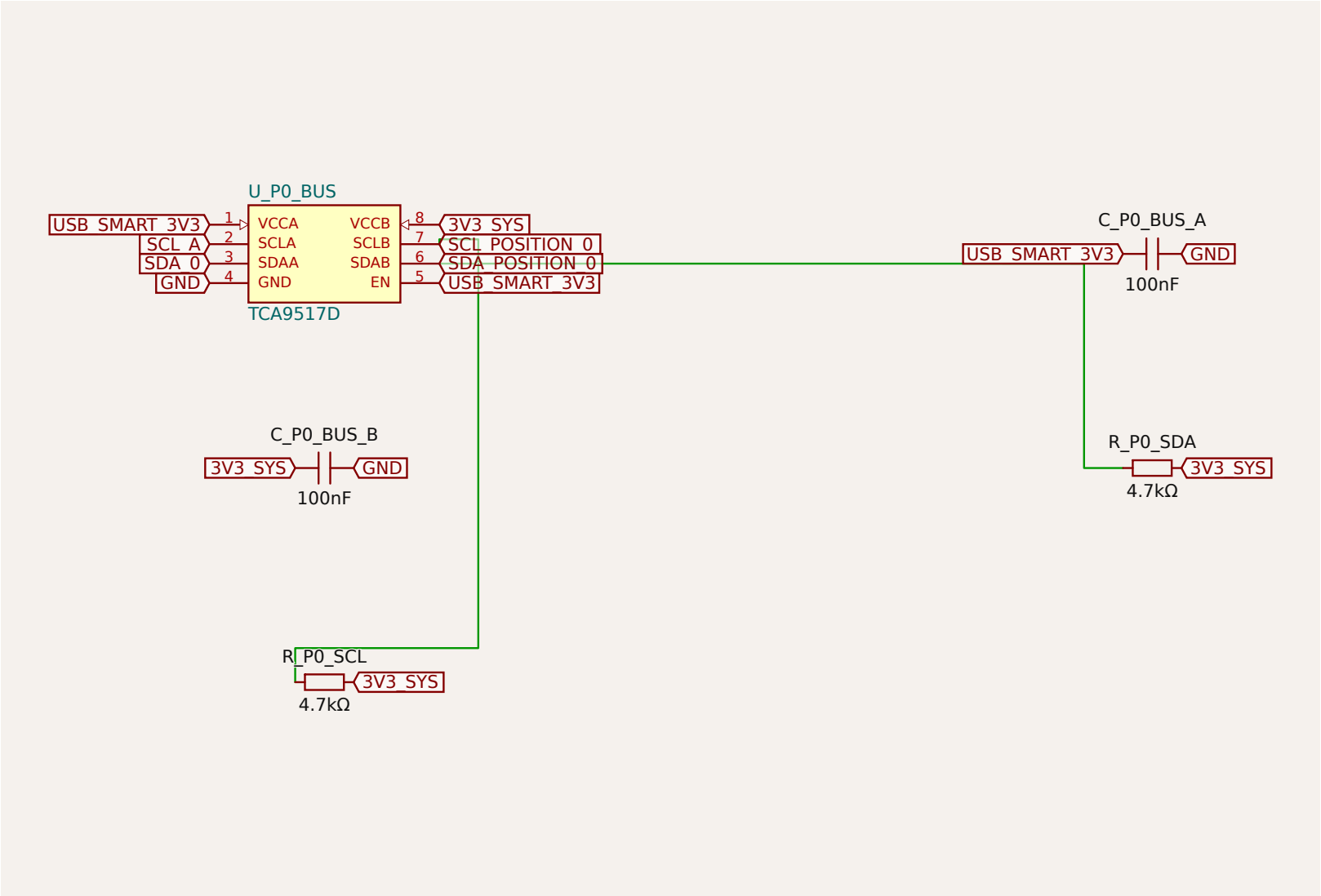
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-a-position-0 — Output



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-a-position-0 — Interface

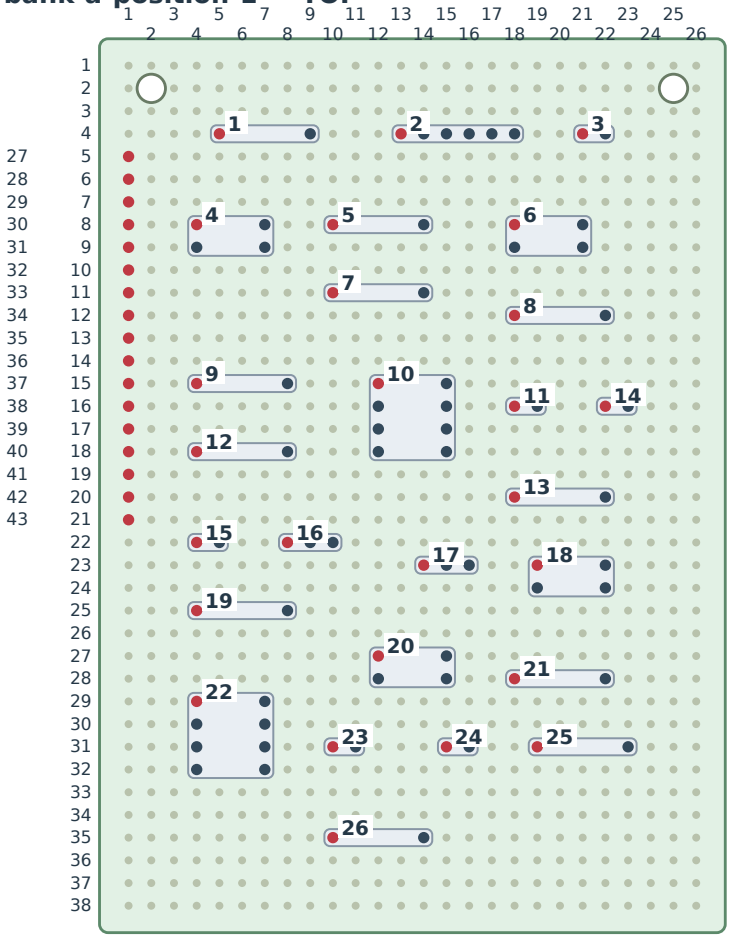


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-a-position-1

Bank A: position 1 complete isolated channel

bank-a-position-1 — TOP

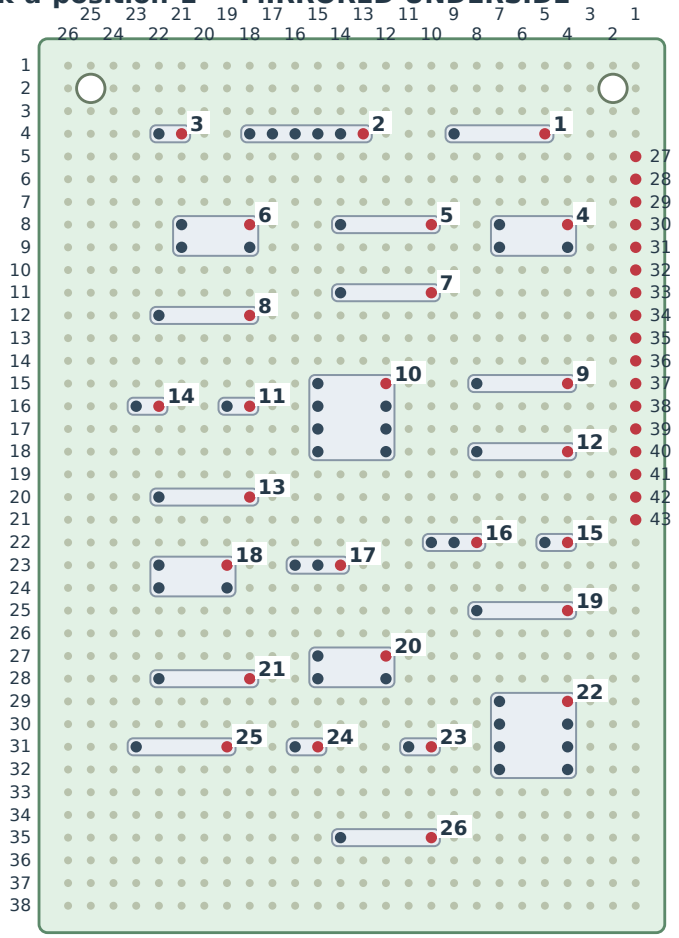


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

bank-a-position-1 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

bank-a-position-1 — numbered parts

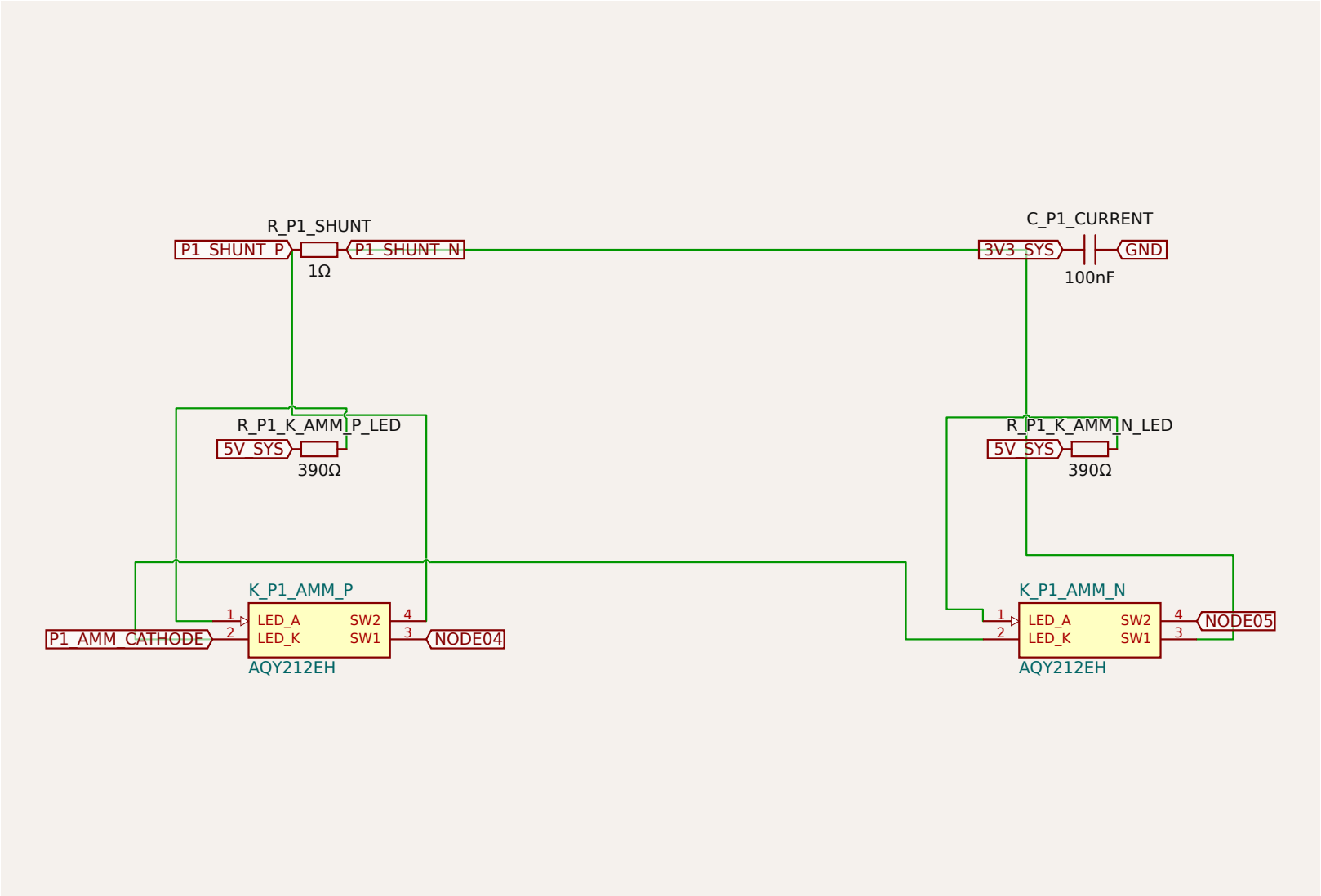
Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

#	Reference	Purpose	Install	Adapter / carrier
1	R_P1_SHUNT	1Ω ±1% external current shunt	CP27	
2	J_P1_CURRENT_MODULE	INA226 carrier: remove onboard shunt; external 1Ω	CP27	INA226 carrier
3	C_P1_CURRENT	100nF	CP27	
4	K_P1_AMM_P	AQY212EH	CP27	
5	R_P1_K_AMM_P_LED	390Ω ±1% PhotoMOS LED drive with ULN2003 saturation allowance	CP27	
6	K_P1_AMM_N	AQY212EH	CP27	
7	R_P1_K_AMM_N_LED	390Ω ±1% PhotoMOS LED drive with ULN2003 saturation allowance	CP27	
8	R_P1_SOURCE_SENSE	10	CP27	
9	R_P1_SINK_SENSE	10	CP27	
10	U_P1_LIMIT	MCP6002 dual source/sink clamp	CP27	
11	C_P1_LIMIT	100nF	CP27	
12	R_P1_HI_GATE	1k	CP27	
13	R_P1_LO_GATE	1k	CP27	
14	C_P1_HI_COMP	1nF	CP27	
15	C_P1_LO_COMP	1nF	CP27	
16	Q_P1_HIGH	BSS84	CP27	SOT23 to numbered 3-pin adapter
17	Q_P1_LOW	2N7000	CP27	
18	K_P1_HIGH	AQY212EH	CP27	
19	R_P1_K_HIGH_LED	390Ω ±1% PhotoMOS LED drive with ULN2003 saturation allowance	CP27	
20	K_P1_LOW	AQY212EH	CP27	
21	R_P1_K_LOW_LED	390Ω ±1% PhotoMOS LED drive with ULN2003 saturation allowance	CP27	
22	U_P1_BUS	TCA9517D	CP27	SOIC8 to DIP8
23	C_P1_BUS_A	100nF	CP27	
24	C_P1_BUS_B	100nF	CP27	
25	R_P1_SDA	4.7k	CP27	
26	R_P1_SCL	4.7k	CP27	
27	J_LINK_1	Harness terminal: GND	CP27	
28	J_LINK_2	Harness terminal: 3V3_SYS	CP27	

bank-a-position-1 — parts continued

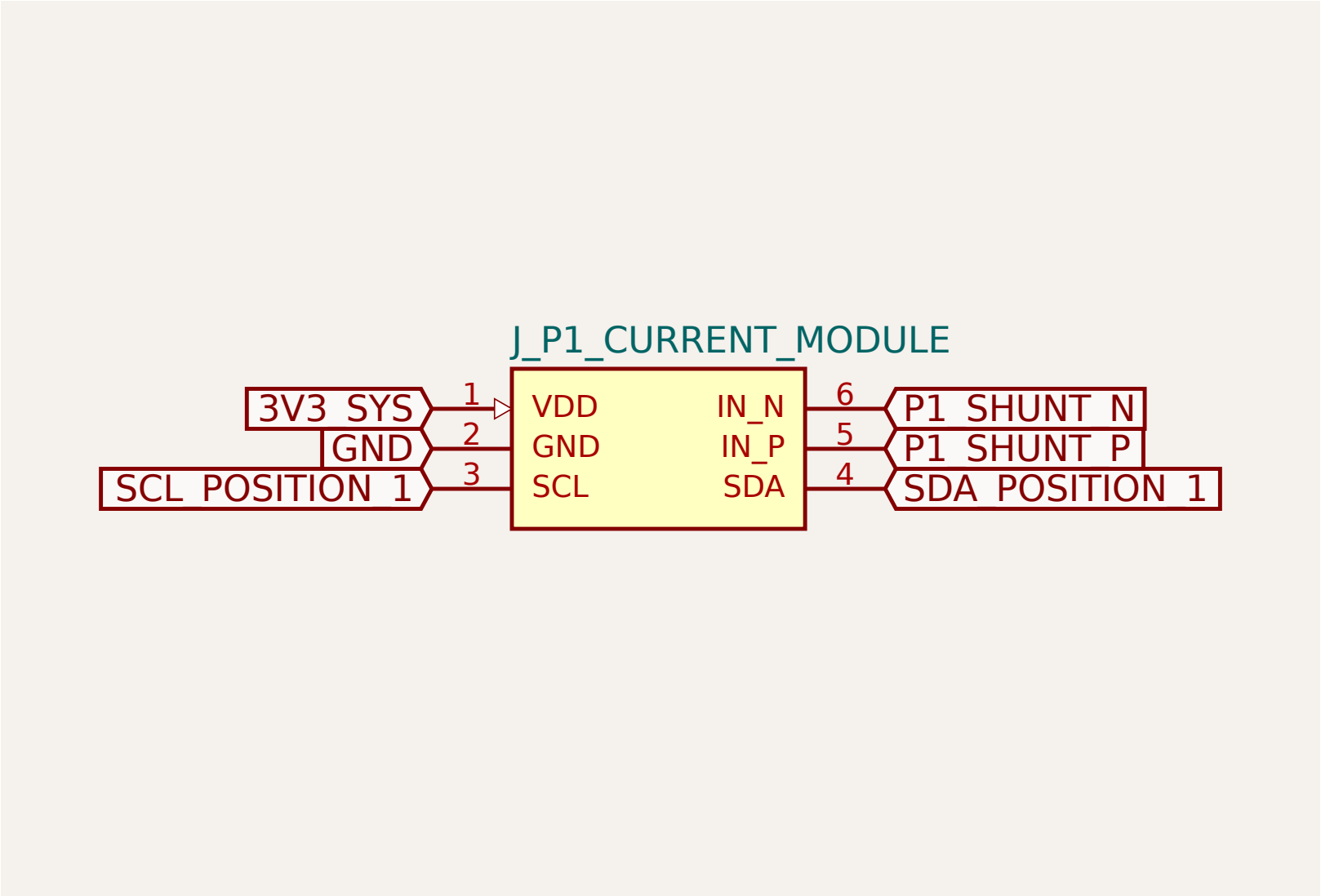
#	Reference	Purpose	Install	Adapter / carrier
29	J_LINK_3	Harness terminal: SCL_POSITION_1	CP27	
30	J_LINK_4	Harness terminal: SDA_POSITION_1	CP27	
31	J_LINK_5	Harness terminal: NODE04	CP27	
32	J_LINK_6	Harness terminal: NODE05	CP27	
33	J_LINK_7	Harness terminal: NODE06	CP27	
34	J_LINK_8	Harness terminal: 5V_SYS	CP27	
35	J_LINK_9	Harness terminal: 5V_EDU	CP27	
36	J_LINK_10	Harness terminal: USB_SMART_3V3	CP27	
37	J_LINK_11	Harness terminal: SCL_A	CP27	
38	J_LINK_12	Harness terminal: SDA_1	CP27	
39	J_LINK_13	Harness terminal: P1_K_HIGH_CATHODE	CP27	
40	J_LINK_14	Harness terminal: P1_K_LOW_CATHODE	CP27	
41	J_LINK_15	Harness terminal: P1_AMM_CATHODE	CP27	
42	J_LINK_16	Harness terminal: A_REF_HI	CP27	
43	J_LINK_17	Harness terminal: A_REF_LO	CP27	

bank-a-position-1 — Current



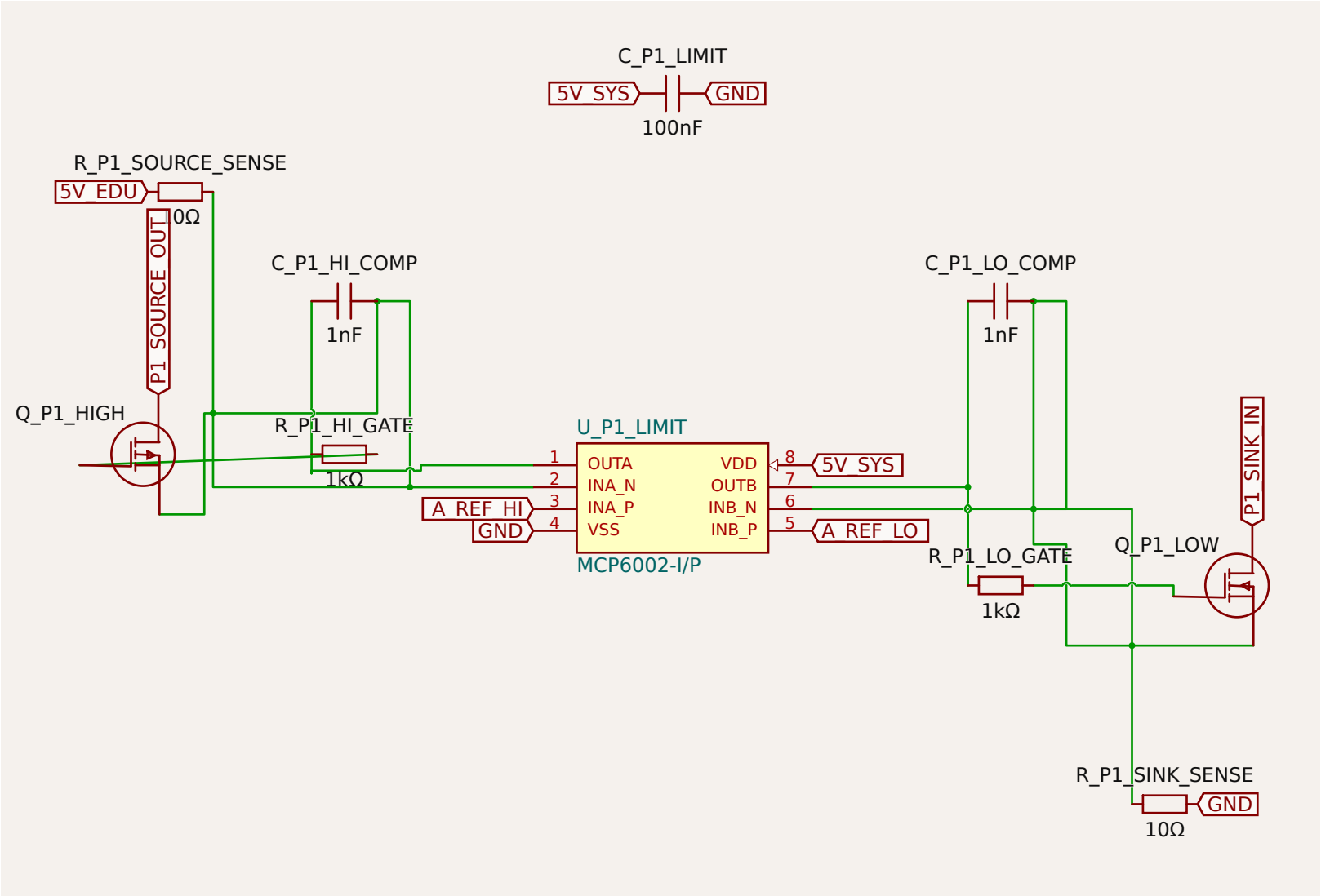
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-a-position-1 — Circuit



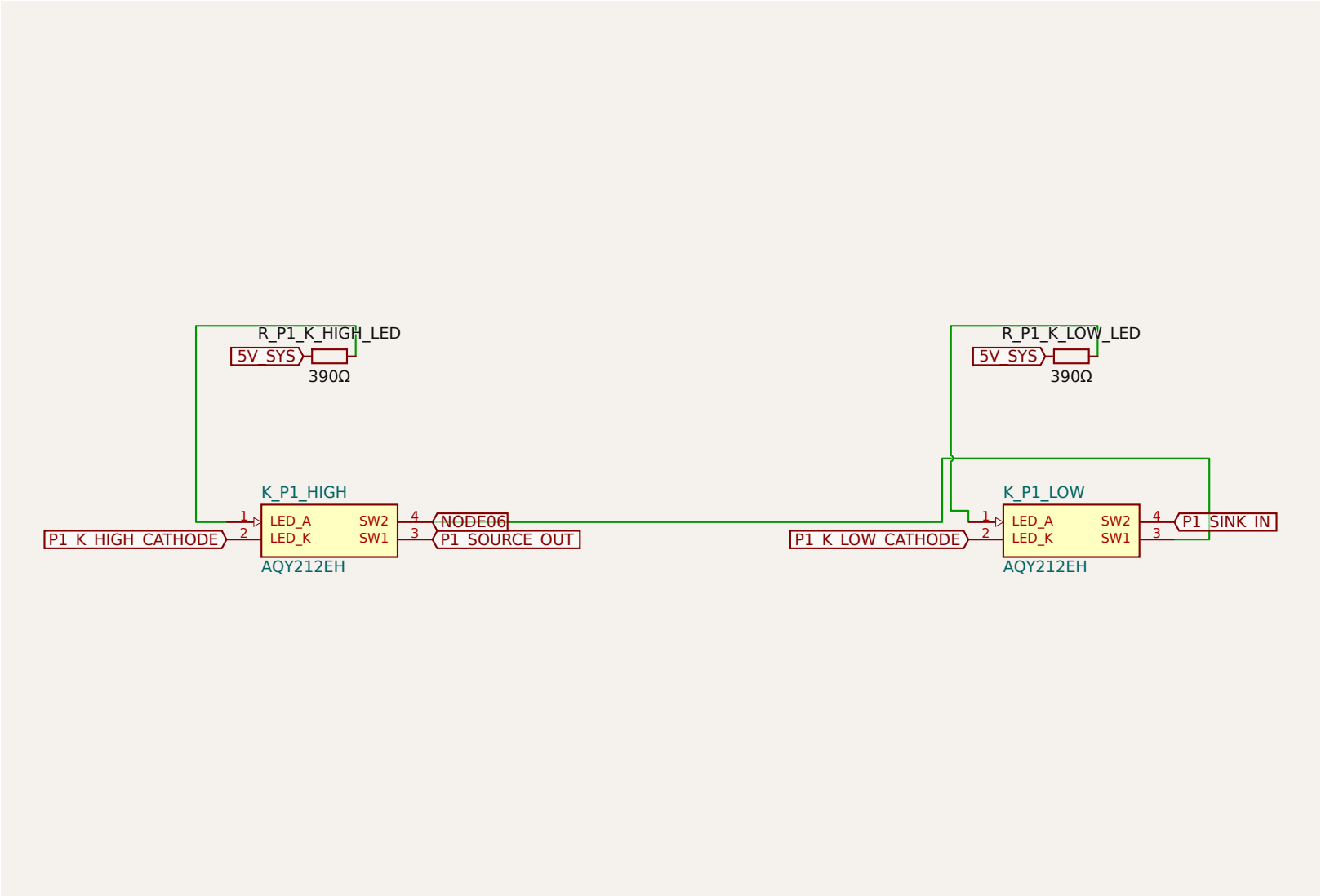
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-a-position-1 — Limits



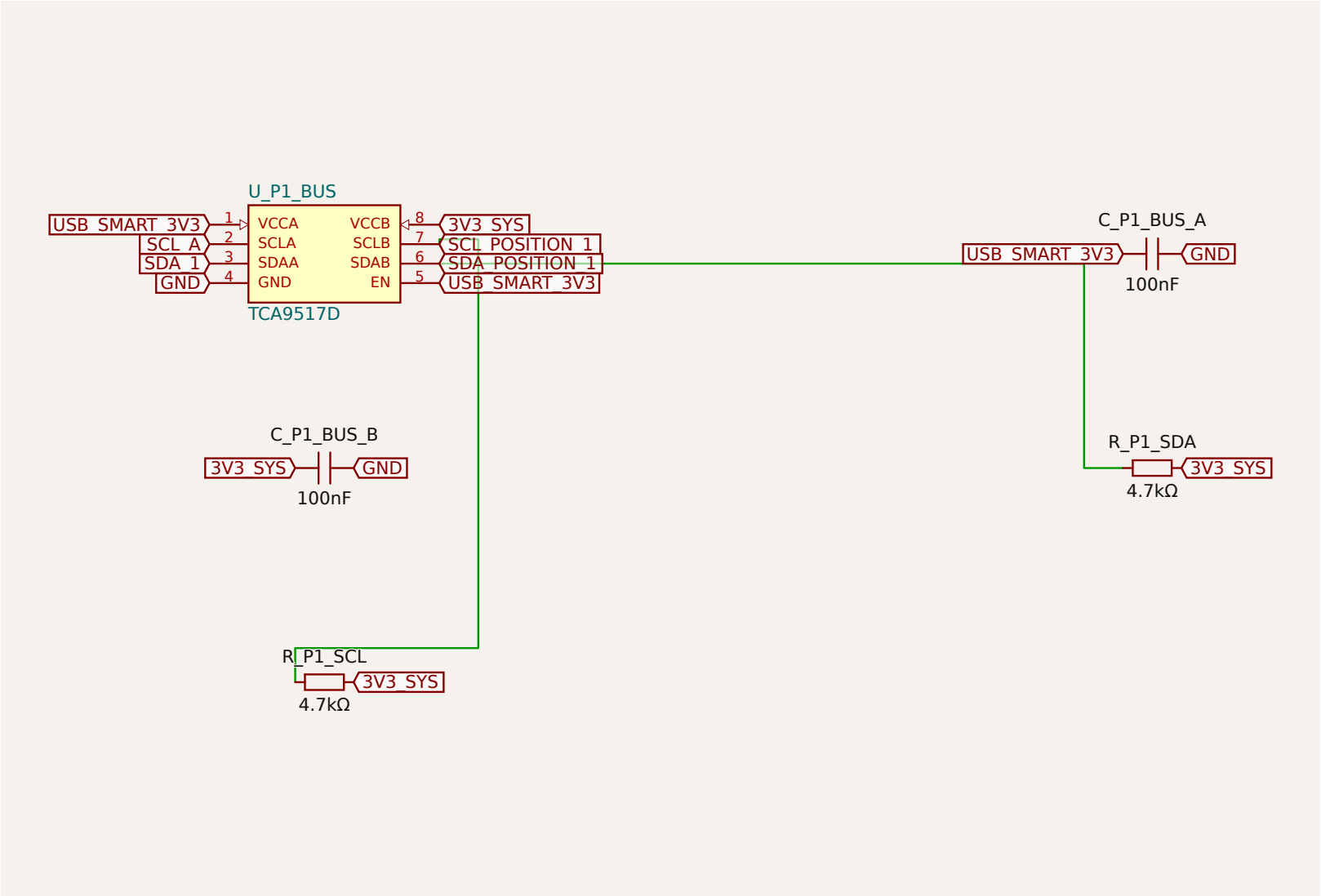
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-a-position-1 — Output



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-a-position-1 — Interface

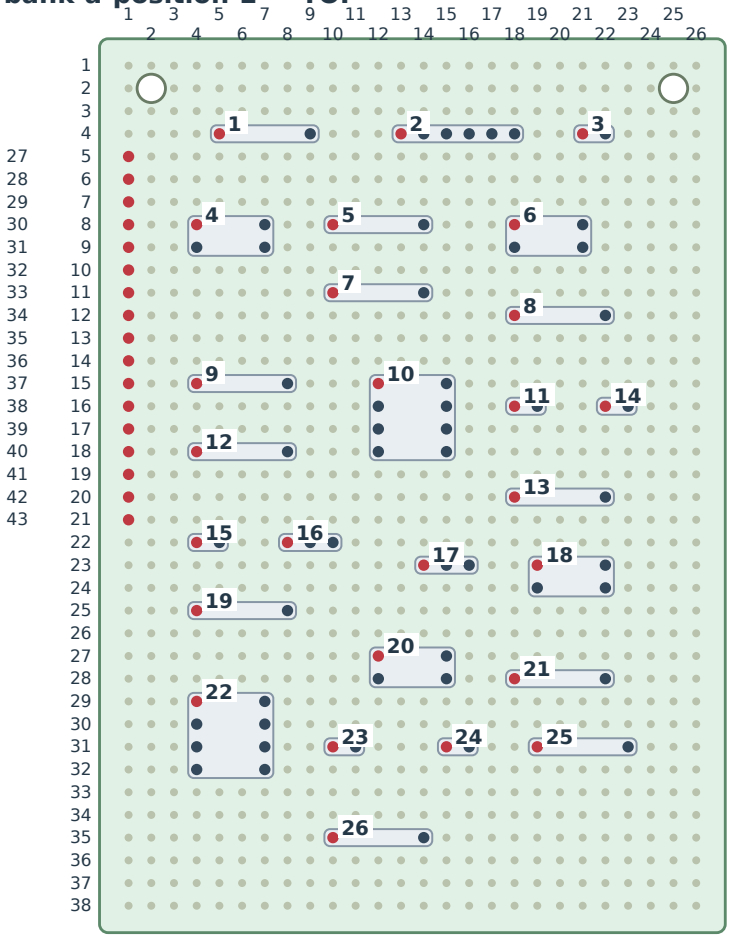


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-a-position-2

Bank A: position 2 complete isolated channel

bank-a-position-2 — TOP

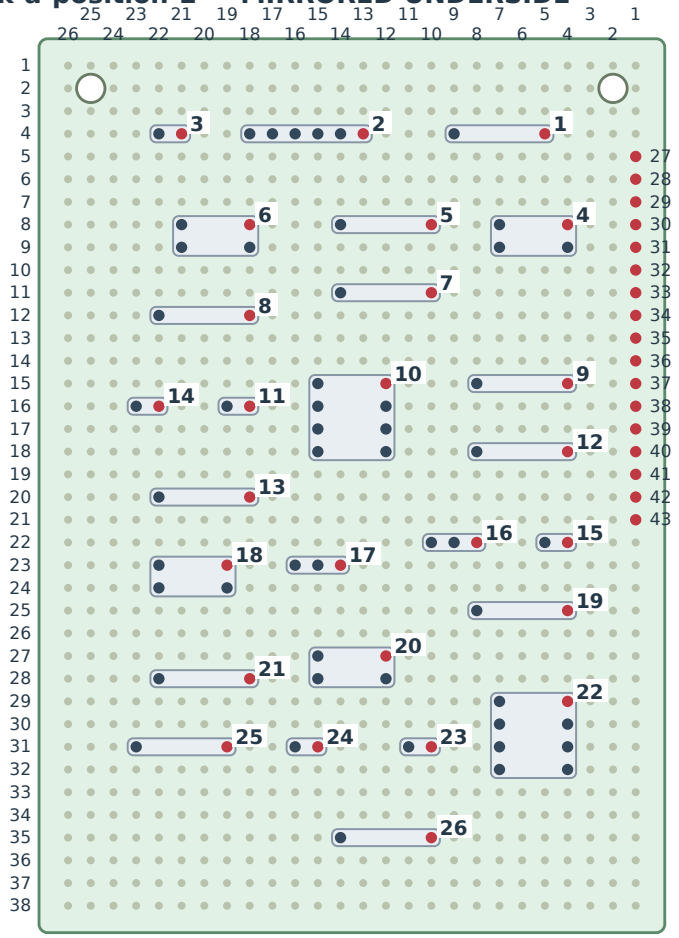


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

bank-a-position-2 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

bank-a-position-2 — numbered parts

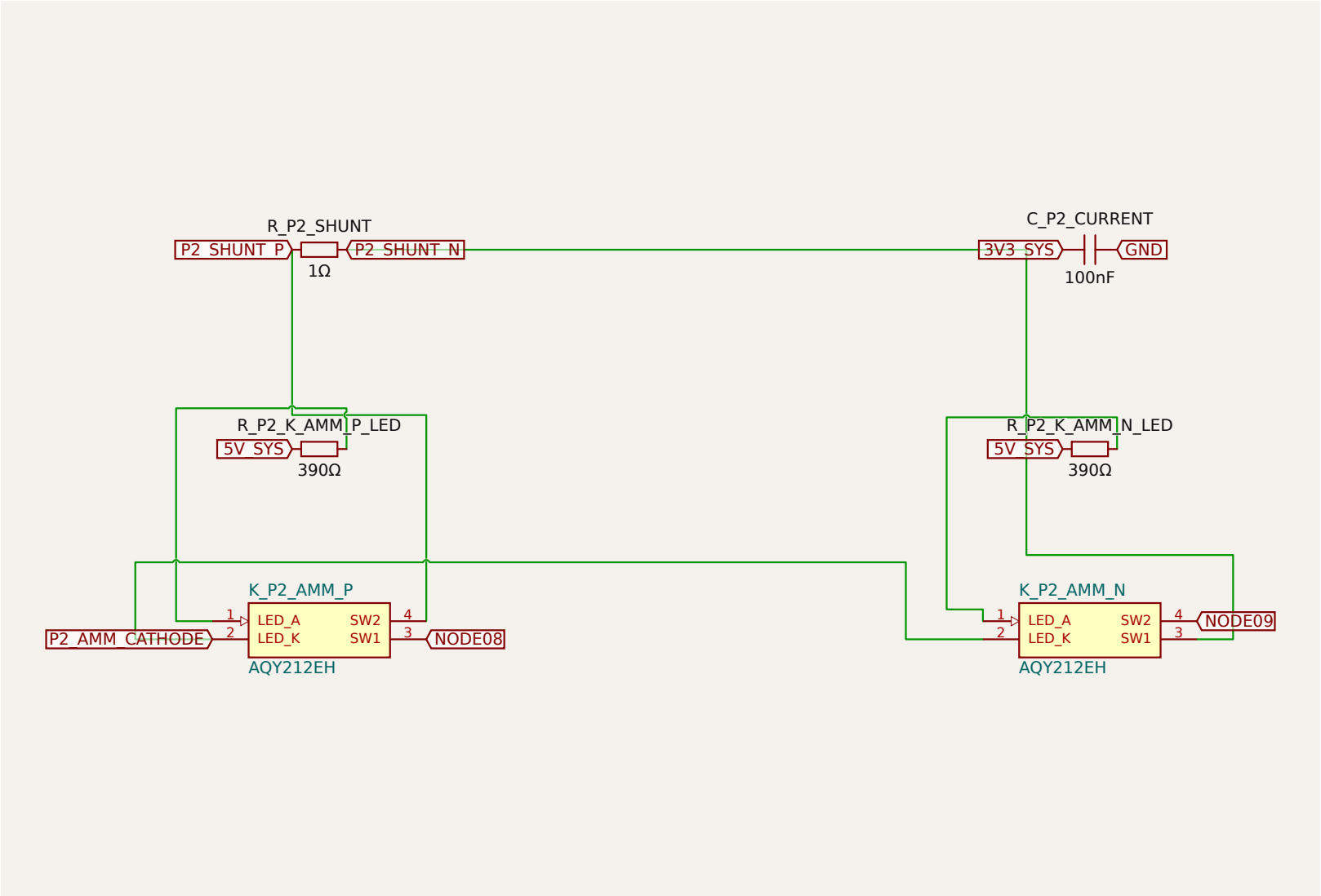
Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

#	Reference	Purpose	Install	Adapter / carrier
1	R_P2_SHUNT	1Ω ±1% external current shunt	CP28	
2	J_P2_CURRENT_MODULE	INA226 carrier: remove onboard shunt; external 1Ω	CP28	INA226 carrier
3	C_P2_CURRENT	100nF	CP28	
4	K_P2_AMM_P	AQY212EH	CP28	
5	R_P2_K_AMM_P_LED	390Ω ±1% PhotoMOS LED drive with ULN2003 saturation allowance	CP28	
6	K_P2_AMM_N	AQY212EH	CP28	
7	R_P2_K_AMM_N_LED	390Ω ±1% PhotoMOS LED drive with ULN2003 saturation allowance	CP28	
8	R_P2_SOURCE_SENSE	10	CP28	
9	R_P2_SINK_SENSE	10	CP28	
10	U_P2_LIMIT	MCP6002 dual source/sink clamp	CP28	
11	C_P2_LIMIT	100nF	CP28	
12	R_P2_HI_GATE	1k	CP28	
13	R_P2_LO_GATE	1k	CP28	
14	C_P2_HI_COMP	1nF	CP28	
15	C_P2_LO_COMP	1nF	CP28	
16	Q_P2_HIGH	BSS84	CP28	SOT23 to numbered 3-pin adapter
17	Q_P2_LOW	2N7000	CP28	
18	K_P2_HIGH	AQY212EH	CP28	
19	R_P2_K_HIGH_LED	390Ω ±1% PhotoMOS LED drive with ULN2003 saturation allowance	CP28	
20	K_P2_LOW	AQY212EH	CP28	
21	R_P2_K_LOW_LED	390Ω ±1% PhotoMOS LED drive with ULN2003 saturation allowance	CP28	
22	U_P2_BUS	TCA9517D	CP28	SOIC8 to DIP8
23	C_P2_BUS_A	100nF	CP28	
24	C_P2_BUS_B	100nF	CP28	
25	R_P2_SDA	4.7k	CP28	
26	R_P2_SCL	4.7k	CP28	
27	J_LINK_1	Harness terminal: GND	CP28	
28	J_LINK_2	Harness terminal: 3V3_SYS	CP28	

bank-a-position-2 — parts continued

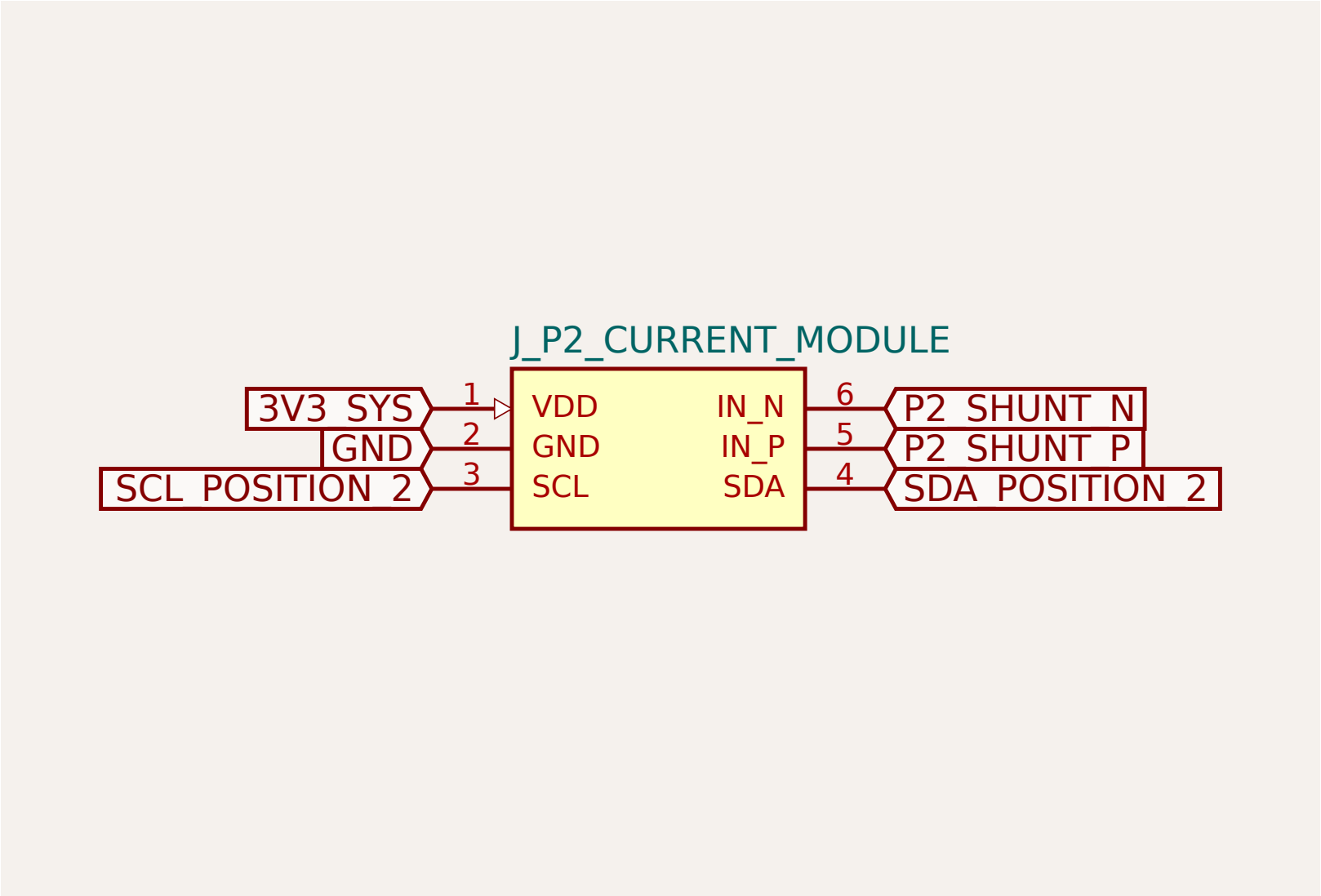
#	Reference	Purpose	Install	Adapter / carrier
29	J_LINK_3	Harness terminal: SCL_POSITION_2	CP28	
30	J_LINK_4	Harness terminal: SDA_POSITION_2	CP28	
31	J_LINK_5	Harness terminal: NODE08	CP28	
32	J_LINK_6	Harness terminal: NODE09	CP28	
33	J_LINK_7	Harness terminal: NODE10	CP28	
34	J_LINK_8	Harness terminal: 5V_SYS	CP28	
35	J_LINK_9	Harness terminal: 5V_EDU	CP28	
36	J_LINK_10	Harness terminal: USB_SMART_3V3	CP28	
37	J_LINK_11	Harness terminal: SCL_A	CP28	
38	J_LINK_12	Harness terminal: SDA_2	CP28	
39	J_LINK_13	Harness terminal: P2_K_HIGH_CATHODE	CP28	
40	J_LINK_14	Harness terminal: P2_K_LOW_CATHODE	CP28	
41	J_LINK_15	Harness terminal: P2_AMM_CATHODE	CP28	
42	J_LINK_16	Harness terminal: A_REF_HI	CP28	
43	J_LINK_17	Harness terminal: A_REF_LO	CP28	

bank-a-position-2 — Current



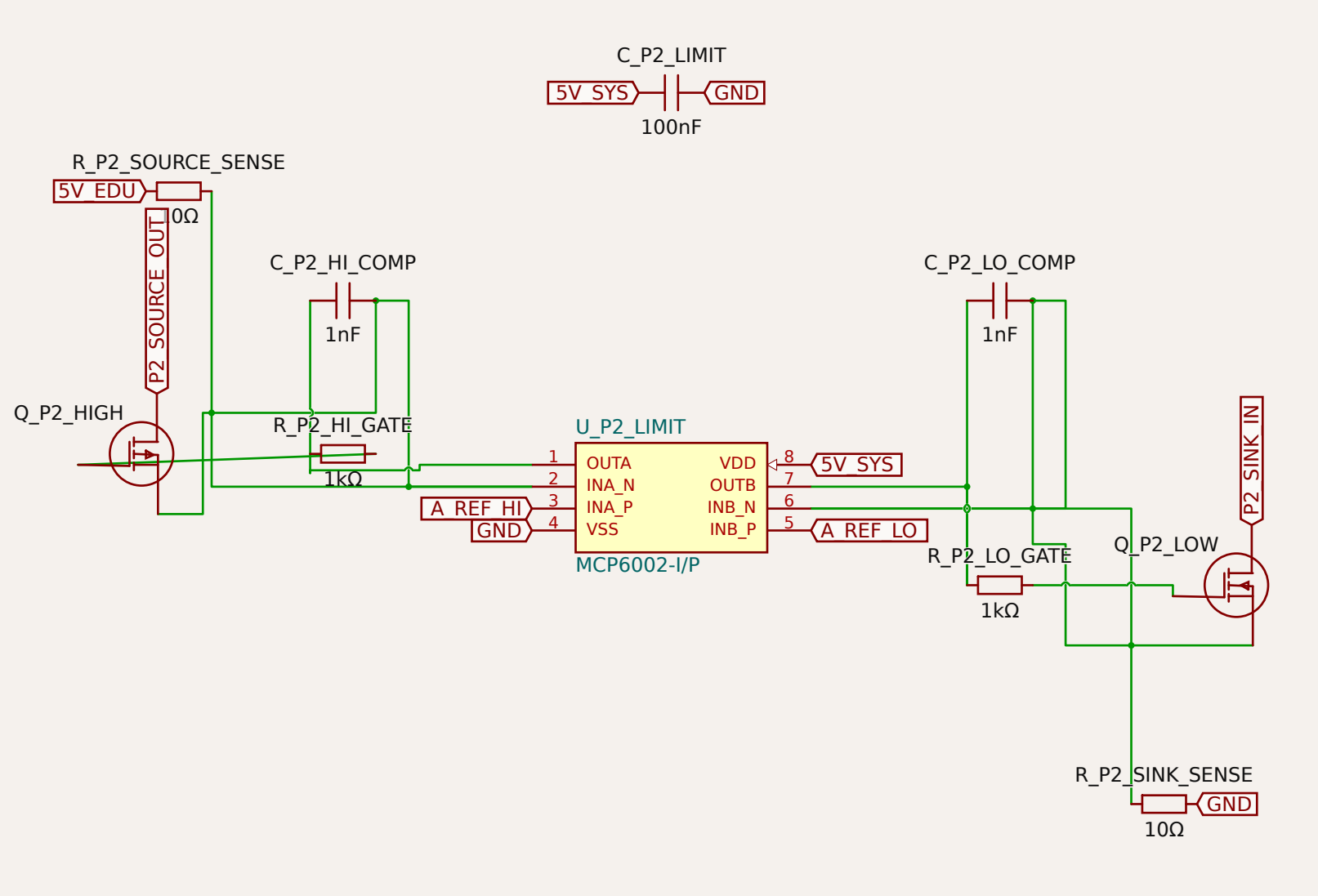
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-a-position-2 — Circuit



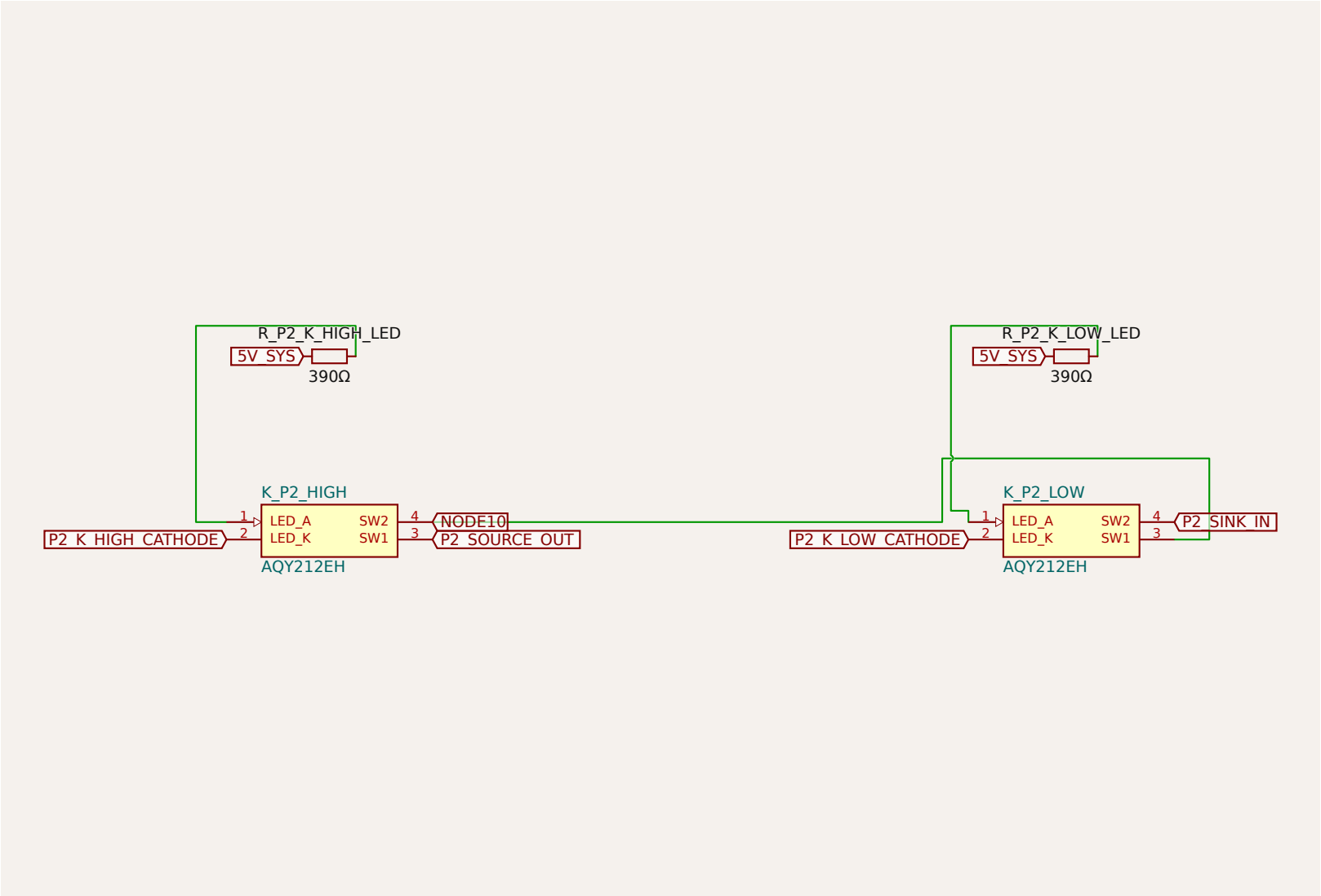
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-a-position-2 — Limits



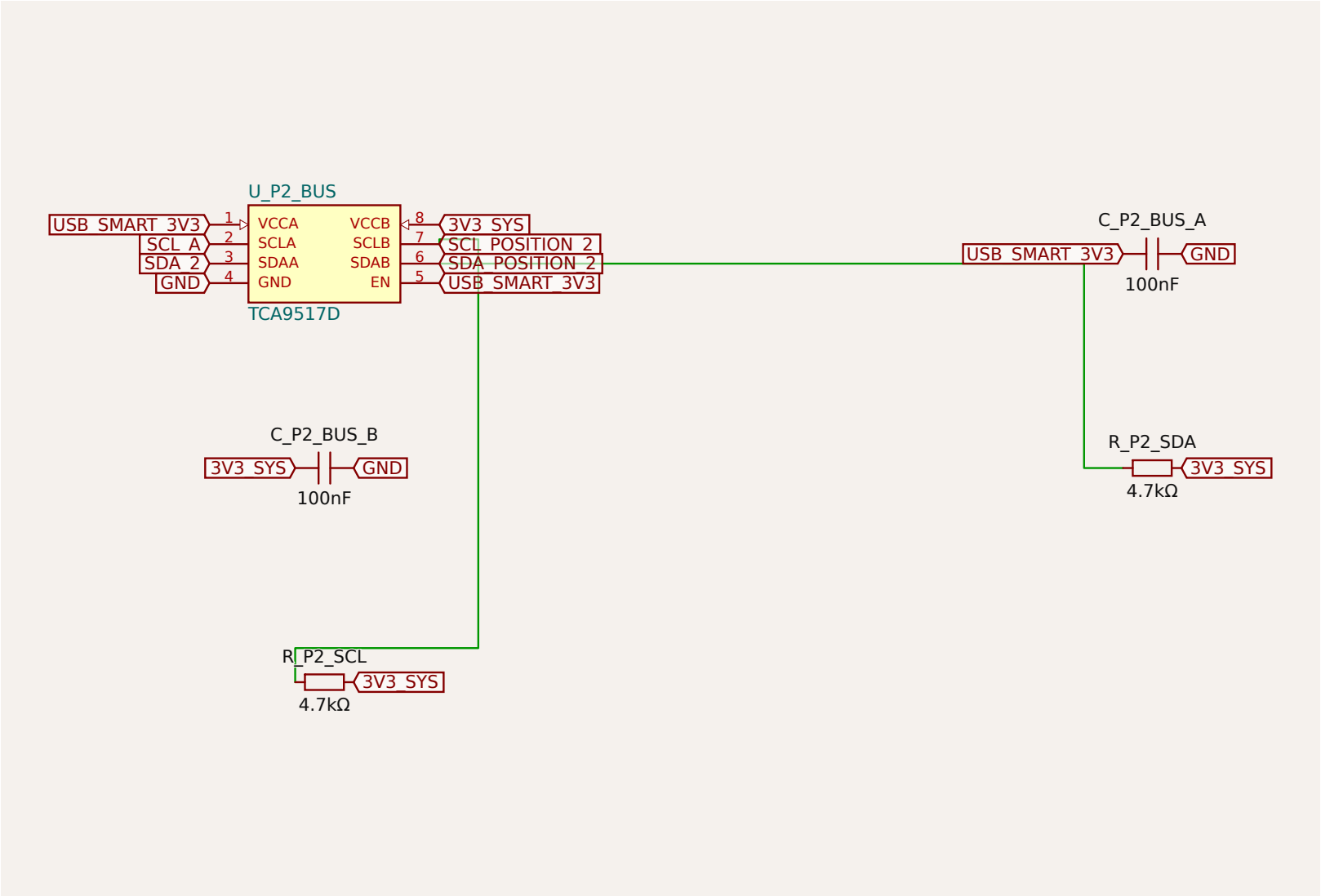
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-a-position-2 — Output



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-a-position-2 — Interface

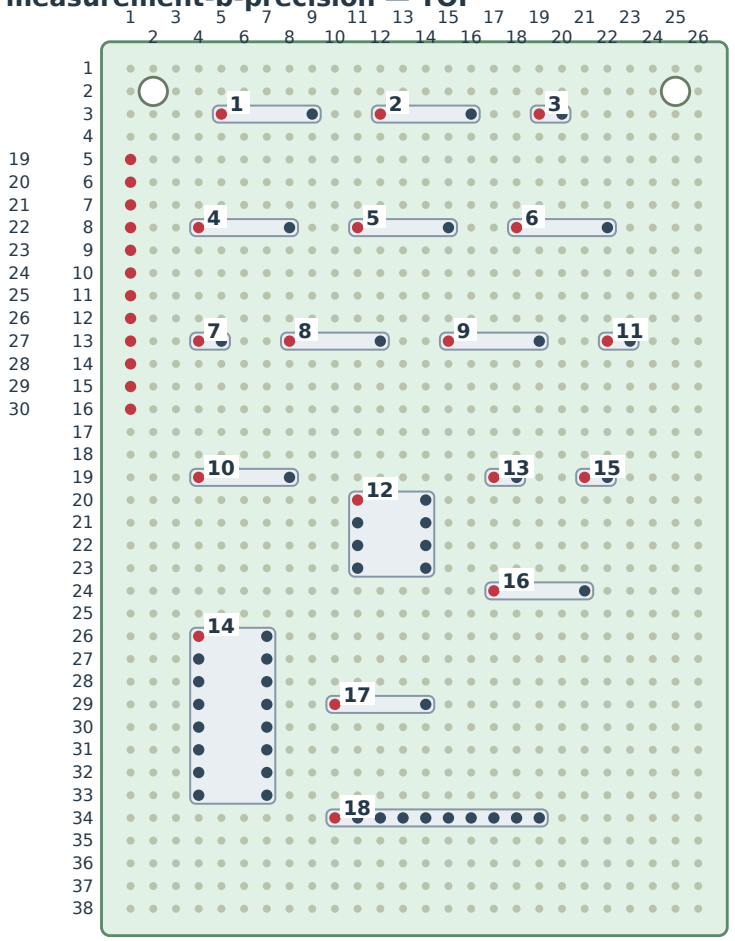


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

measurement-b-precision

Bank B: buffered voltage scanner

measurement-b-precision — TOP

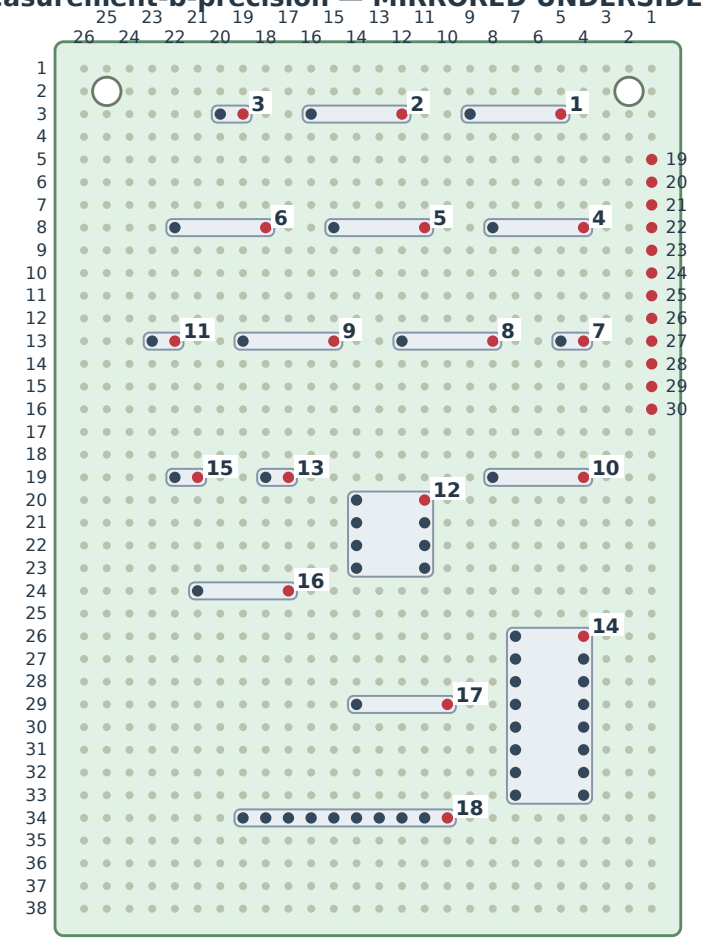


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

measurement-b-precision — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

measurement-b-precision — numbered parts

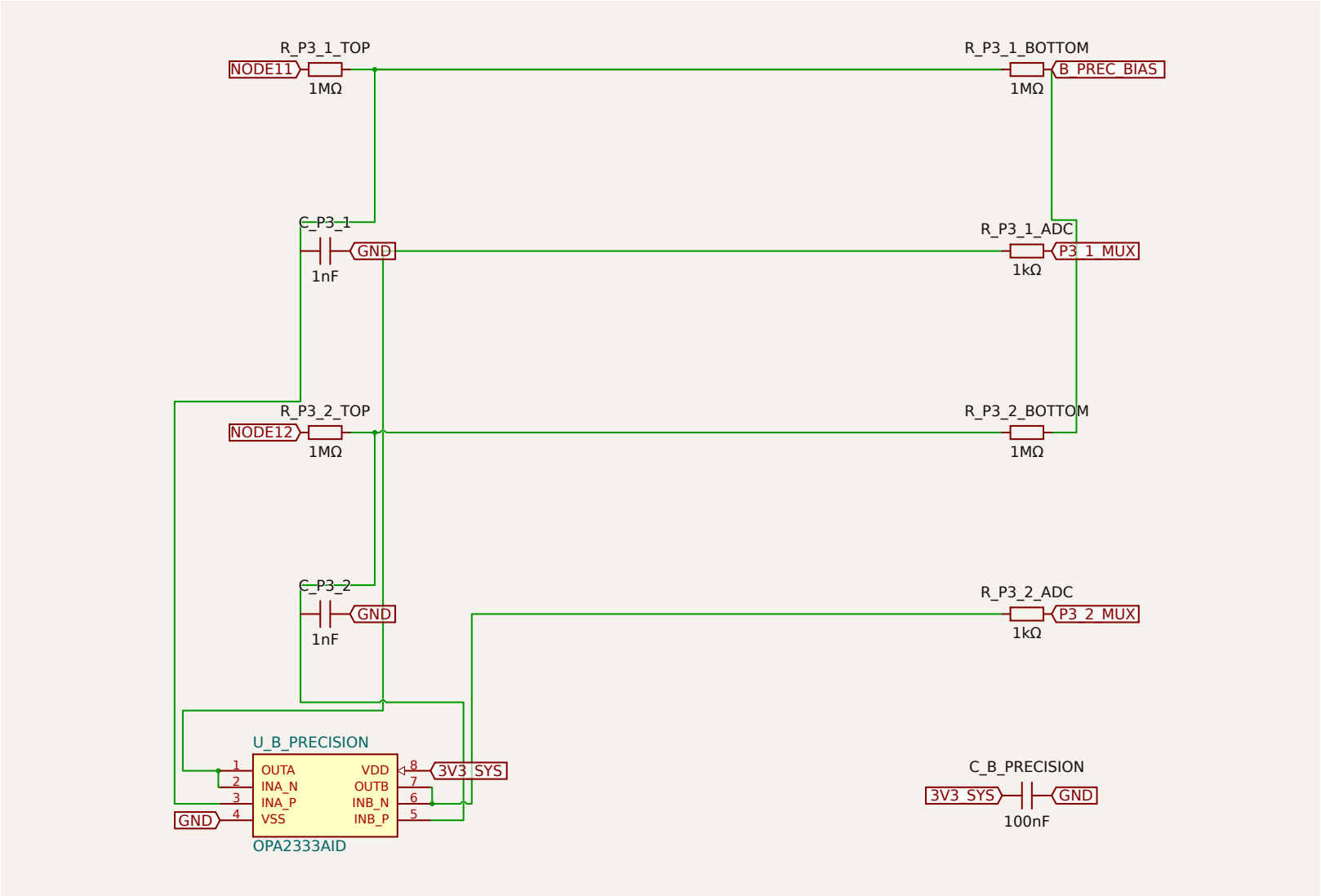
Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

#	Reference	Purpose	Install	Adapter / carrier
1	R_P3_1_TOP	R_P3_1_TOP: 1MΩ	CP29	
2	R_P3_1_BOTTOM	R_P3_1_BOTTOM: 1MΩ	CP29	
3	C_P3_1	C_P3_1: 1nFΩ	CP29	
4	R_P3_1_ADC	R_P3_1_ADC: 1kΩ	CP29	
5	R_P3_2_TOP	R_P3_2_TOP: 1MΩ	CP29	
6	R_P3_2_BOTTOM	R_P3_2_BOTTOM: 1MΩ	CP29	
7	C_P3_2	C_P3_2: 1nFΩ	CP29	
8	R_P3_2_ADC	R_P3_2_ADC: 1kΩ	CP29	
9	R_B_PREC_BIAS_TOP	R_B_PREC_BIAS_TOP: 5.62kΩ	CP29	
10	R_B_PREC_BIAS_BOTTOM	R_B_PREC_BIAS_BOTTOM: 1kΩ	CP29	
11	C_B_PREC_BIAS	C_B_PREC_BIAS: 100nFΩ	CP29	
12	U_B_PRECISION	OPA2333AID precision diagnostic/voltage buffer	CP29	SOIC8 to numbered DIP8
13	C_B_PRECISION	C_B_PRECISION: 100nFΩ	CP29	
14	U_B_MUX	CD4052BE buffered voltage selector	CP29	
15	C_B_MUX	C_B_MUX: 100nFΩ	CP29	
16	R_B_SEL_A	R_B_SEL_A: 100kΩ	CP29	
17	R_B_SEL_B	R_B_SEL_B: 100kΩ	CP29	
18	J_B_ADC_MODULE	ADS1115 carrier interface — verify labelled module pads	CP29	ADS1115 carrier
19	J_LINK_1	Harness terminal: GND	CP29	
20	J_LINK_2	Harness terminal: 3V3_SYS	CP29	
21	J_LINK_3	Harness terminal: SCL_POSITION_3	CP29	
22	J_LINK_4	Harness terminal: SDA_POSITION_3	CP29	
23	J_LINK_5	Harness terminal: NODE11	CP29	
24	J_LINK_6	Harness terminal: NODE12	CP29	
25	J_LINK_7	Harness terminal: MUX_B_A	CP29	
26	J_LINK_8	Harness terminal: MUX_B_B	CP29	
27	J_LINK_9	Harness terminal: P4_1_MUX	CP29	
28	J_LINK_10	Harness terminal: P4_2_MUX	CP29	

measurement-b-precision — parts continued

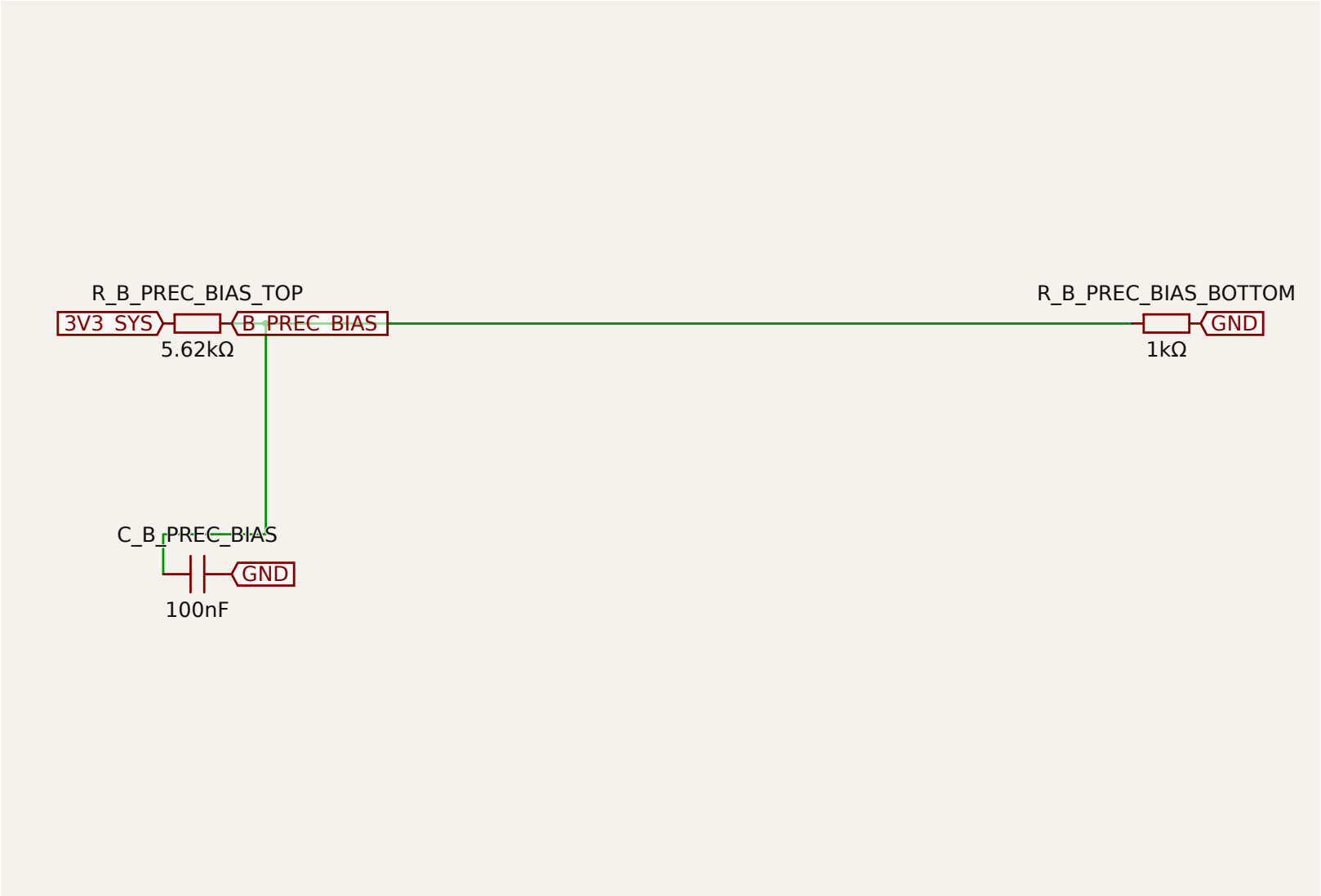
#	Reference	Purpose	Install	Adapter / carrier
29	J_LINK_11	Harness terminal: P5_1_MUX	CP29	
30	J_LINK_12	Harness terminal: P5_2_MUX	CP29	

measurement-b-precision — Voltage inputs



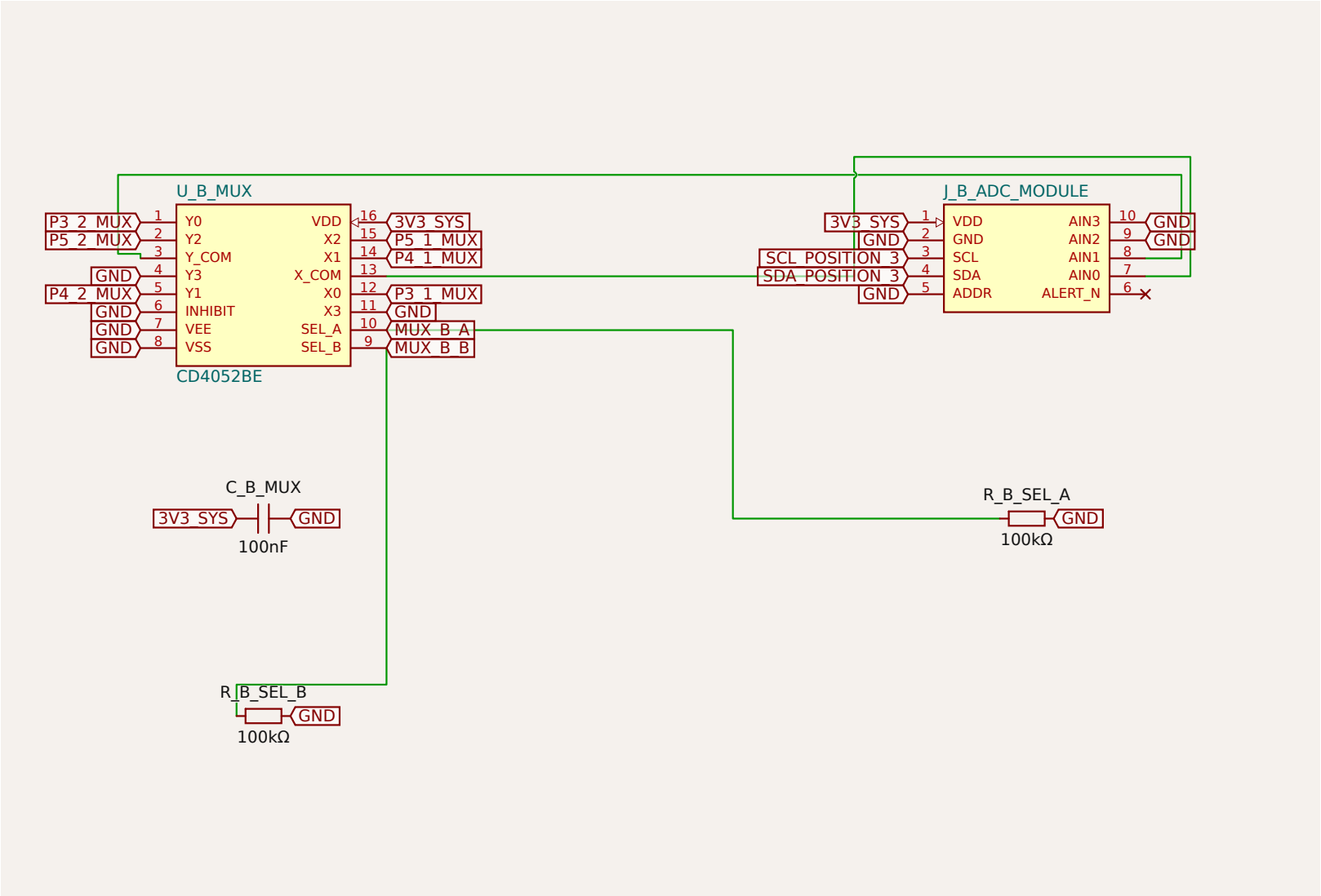
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

measurement-b-precision — Support



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

measurement-b-precision — Buffered selector and ADC

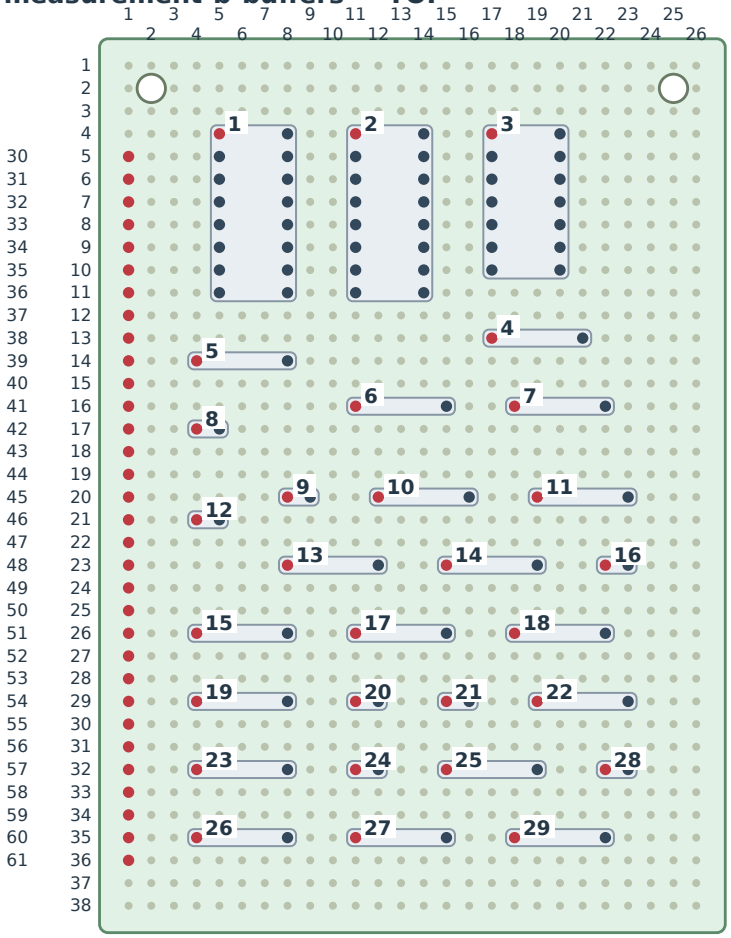


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

measurement-b-buffers

Bank B: buffered voltage scanner

measurement-b-buffers — TOP

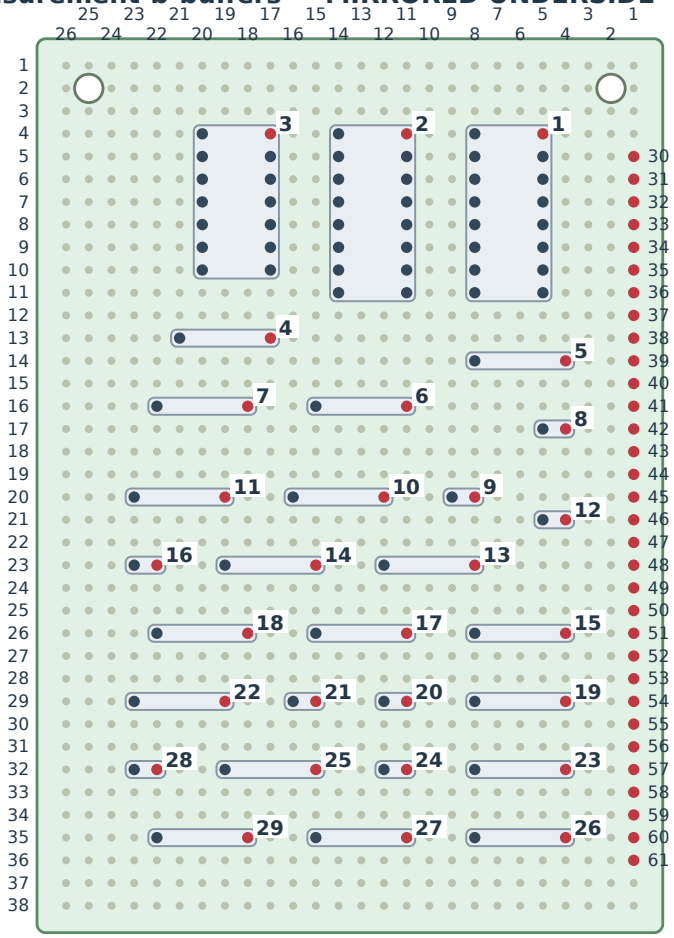


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

measurement-b-buffers — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

measurement-b-buffers — numbered parts

Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

#	Reference	Purpose	Install	Adapter / carrier
1	U_B_LED_DRIVER_0	ULN2003AN PhotoMOS LED driver array	CP29	
2	U_B_LED_DRIVER_1	ULN2003AN PhotoMOS LED driver array	CP29	
3	U_B_BUFFERS	MCP6004-I/P four voltage buffers	CP30	
4	R_B_HI_TOP	1k	CP29	
5	R_B_HI_BOTTOM	49.9k	CP29	
6	R_B_LO_TOP	49.9k	CP29	
7	R_B_LO_BOTTOM	1k	CP29	
8	C_B_REFHI	100nF	CP29	
9	C_B_REFLO	100nF	CP29	
10	R_P4_1_TOP	R_P4_1_TOP: 1MΩ	CP30	
11	R_P4_1_BOTTOM	R_P4_1_BOTTOM: 1MΩ	CP30	
12	C_P4_1	C_P4_1: 1nFΩ	CP30	
13	R_P4_1_ADC	R_P4_1_ADC: 1kΩ	CP30	
14	R_P4_2_TOP	R_P4_2_TOP: 1MΩ	CP30	
15	R_P4_2_BOTTOM	R_P4_2_BOTTOM: 1MΩ	CP30	
16	C_P4_2	C_P4_2: 1nFΩ	CP30	
17	R_P4_2_ADC	R_P4_2_ADC: 1kΩ	CP30	
18	R_B_GENERAL_BIAS_TOP	R_B_GENERAL_BIAS_TOP: 5.62kΩ	CP30	
19	R_B_GENERAL_BIAS_BOTTOM	R_B_GENERAL_BIAS_BOTTOM: 1kΩ	CP30	
20	C_B_GENERAL_BIAS	C_B_GENERAL_BIAS: 100nFΩ	CP30	
21	C_B_BUFFERS	C_B_BUFFERS: 100nFΩ	CP30	
22	R_P5_1_TOP	R_P5_1_TOP: 1MΩ	CP31	
23	R_P5_1_BOTTOM	R_P5_1_BOTTOM: 1MΩ	CP31	
24	C_P5_1	C_P5_1: 1nFΩ	CP31	
25	R_P5_1_ADC	R_P5_1_ADC: 1kΩ	CP31	
26	R_P5_2_TOP	R_P5_2_TOP: 1MΩ	CP31	
27	R_P5_2_BOTTOM	R_P5_2_BOTTOM: 1MΩ	CP31	
28	C_P5_2	C_P5_2: 1nFΩ	CP31	

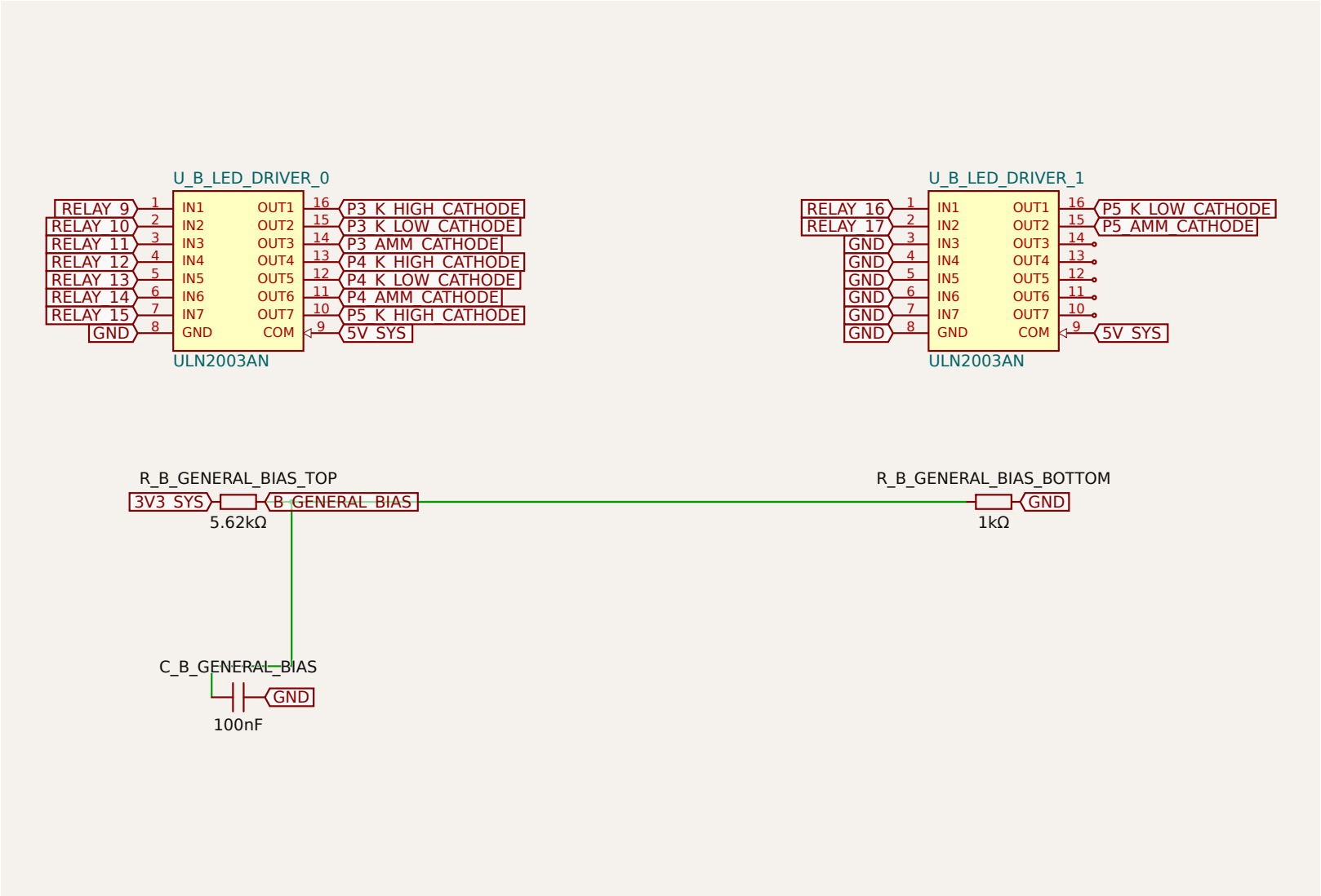
measurement-b-buffers — parts continued

#	Reference	Purpose	Install	Adapter / carrier
29	R_P5_2_ADC	R_P5_2_ADC: 1kΩ	CP31	
30	J_LINK_1	Harness terminal: GND	CP29	
31	J_LINK_2	Harness terminal: 3V3_SYS	CP30	
32	J_LINK_3	Harness terminal: NODE15	CP30	
33	J_LINK_4	Harness terminal: NODE16	CP30	
34	J_LINK_5	Harness terminal: NODE19	CP31	
35	J_LINK_6	Harness terminal: NODE20	CP31	
36	J_LINK_7	Harness terminal: 5V_SYS	CP29	
37	J_LINK_8	Harness terminal: 5V_EDU	CP29	
38	J_LINK_9	Harness terminal: RELAY_9	CP29	
39	J_LINK_10	Harness terminal: RELAY_10	CP29	
40	J_LINK_11	Harness terminal: RELAY_11	CP29	
41	J_LINK_12	Harness terminal: RELAY_12	CP29	
42	J_LINK_13	Harness terminal: RELAY_13	CP29	
43	J_LINK_14	Harness terminal: RELAY_14	CP29	
44	J_LINK_15	Harness terminal: RELAY_15	CP29	
45	J_LINK_16	Harness terminal: RELAY_16	CP29	
46	J_LINK_17	Harness terminal: RELAY_17	CP29	
47	J_LINK_18	Harness terminal: P4_1_MUX	CP30	
48	J_LINK_19	Harness terminal: P4_2_MUX	CP30	
49	J_LINK_20	Harness terminal: P5_1_MUX	CP31	
50	J_LINK_21	Harness terminal: P5_2_MUX	CP31	
51	J_LINK_22	Harness terminal: P3_K_HIGH_CATHODE	CP29	
52	J_LINK_23	Harness terminal: P3_K_LOW_CATHODE	CP29	
53	J_LINK_24	Harness terminal: P3_AMM_CATHODE	CP29	
54	J_LINK_25	Harness terminal: P4_K_HIGH_CATHODE	CP29	
55	J_LINK_26	Harness terminal: P4_K_LOW_CATHODE	CP29	
56	J_LINK_27	Harness terminal: P4_AMM_CATHODE	CP29	

measurement-b-buffers — parts continued

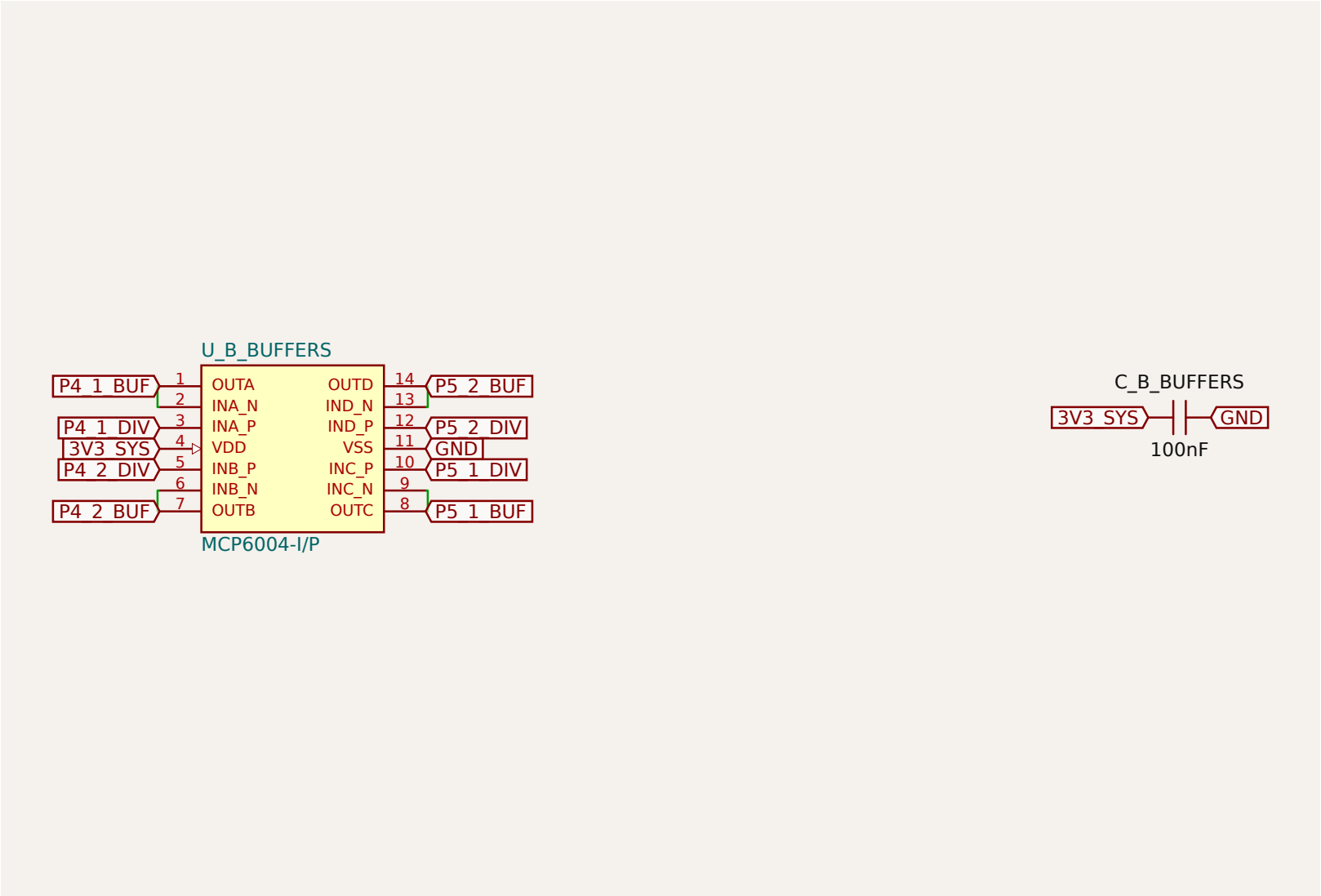
#	Reference	Purpose	Install	Adapter / carrier
57	J_LINK_28	Harness terminal: P5_K_HIGH_CATHODE	CP29	
58	J_LINK_29	Harness terminal: P5_K_LOW_CATHODE	CP29	
59	J_LINK_30	Harness terminal: P5_AMM_CATHODE	CP29	
60	J_LINK_31	Harness terminal: B_REF_HI	CP29	
61	J_LINK_32	Harness terminal: B_REF_LO	CP29	

measurement-b-buffers — Support



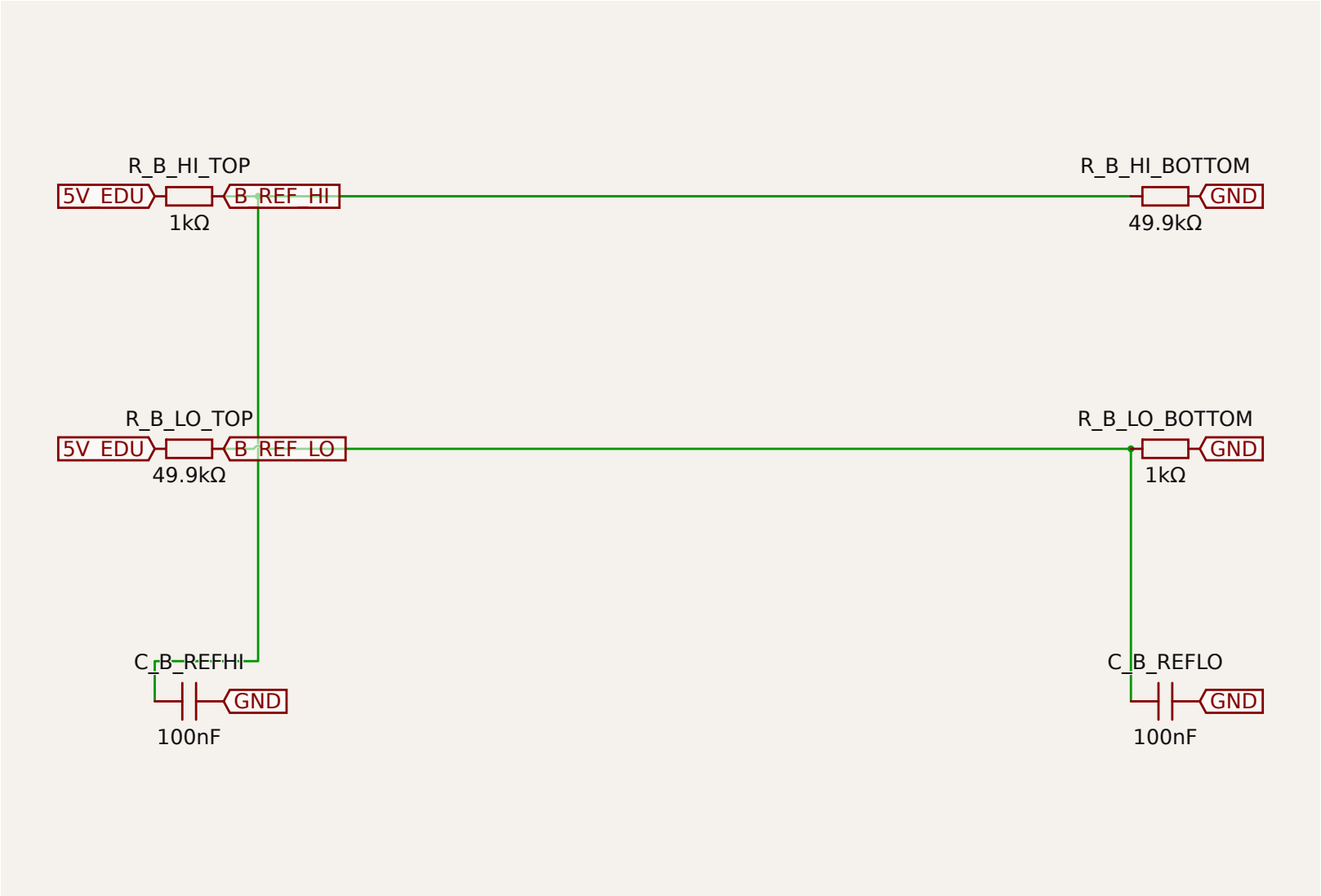
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

measurement-b-buffers — Voltage buffers



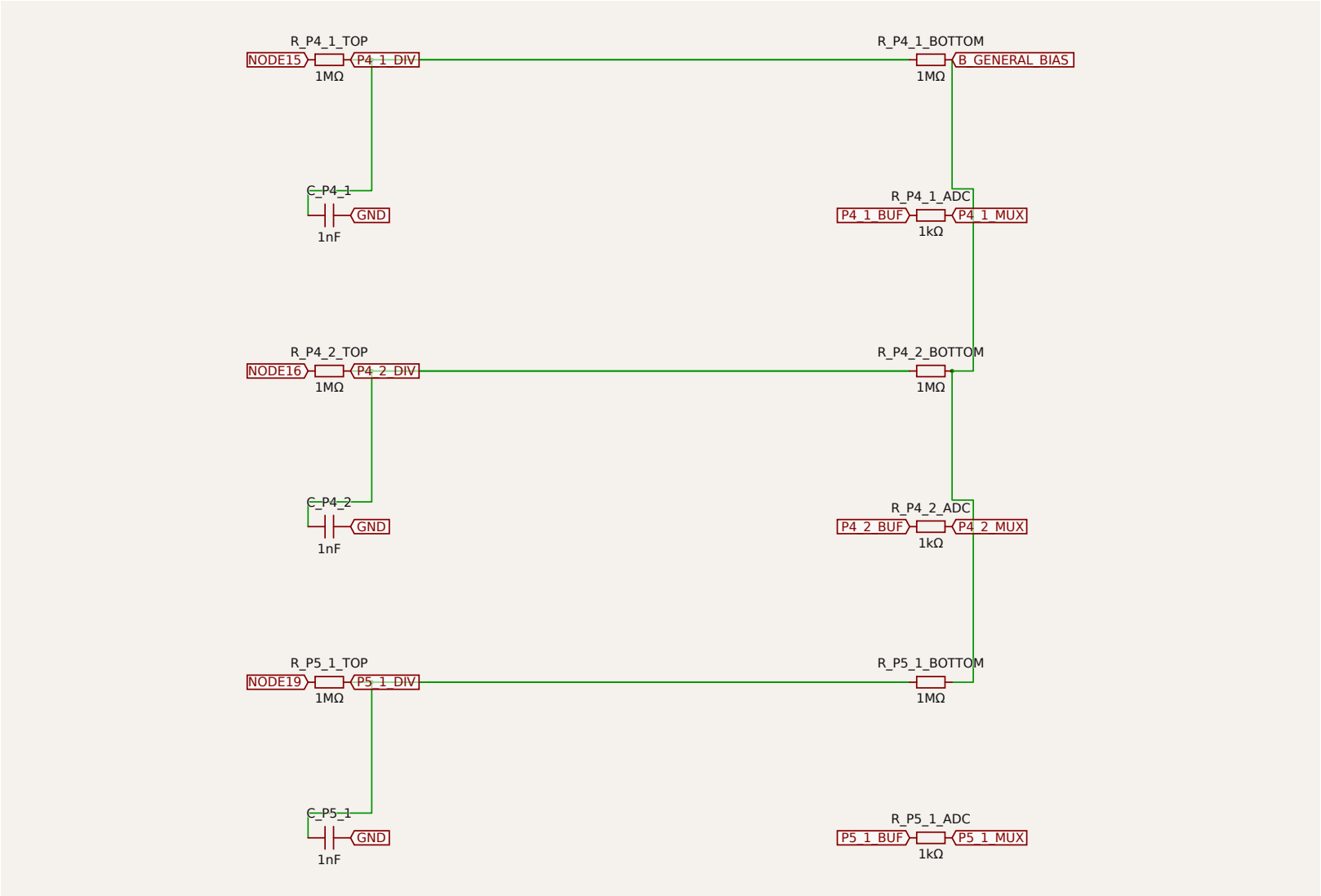
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

measurement-b-buffers — Limits



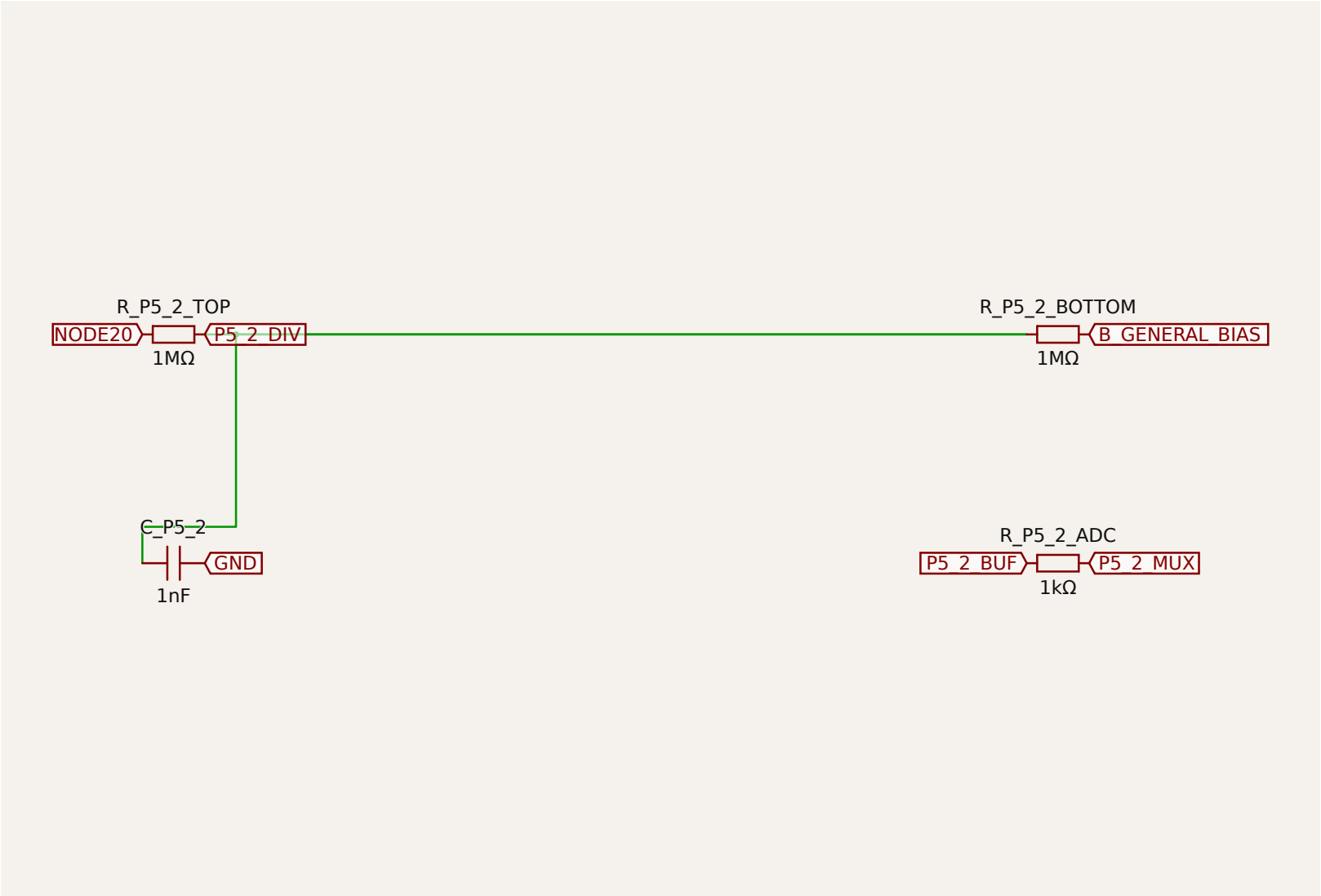
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

measurement-b-buffers — Voltage inputs 1



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

measurement-b-buffers — Voltage inputs 2

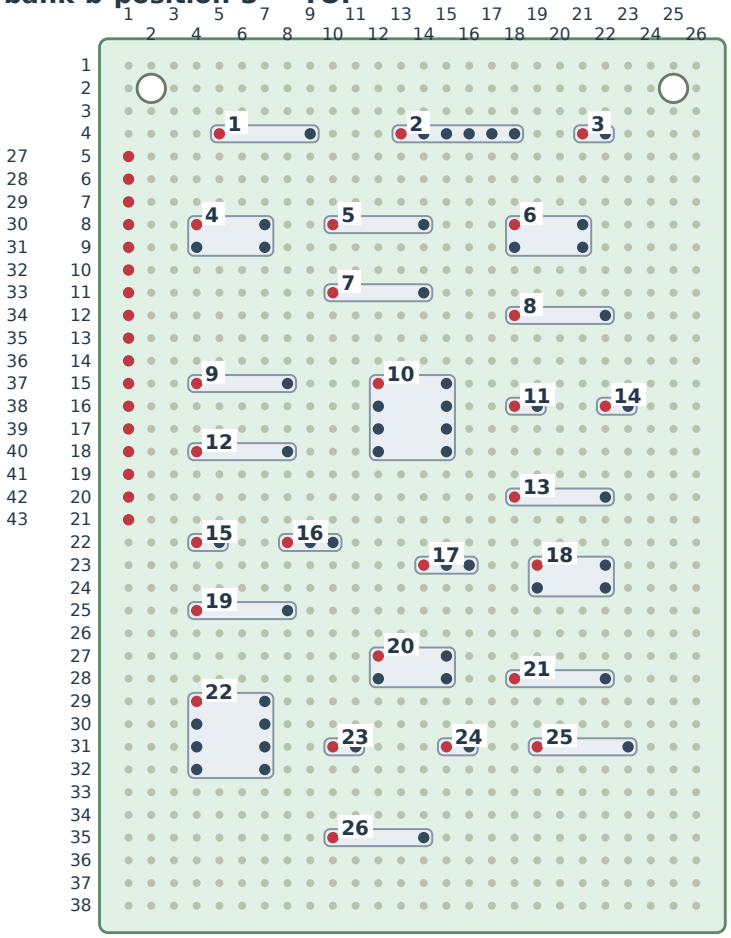


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-b-position-3

Bank B: position 3 complete isolated channel

bank-b-position-3 — TOP

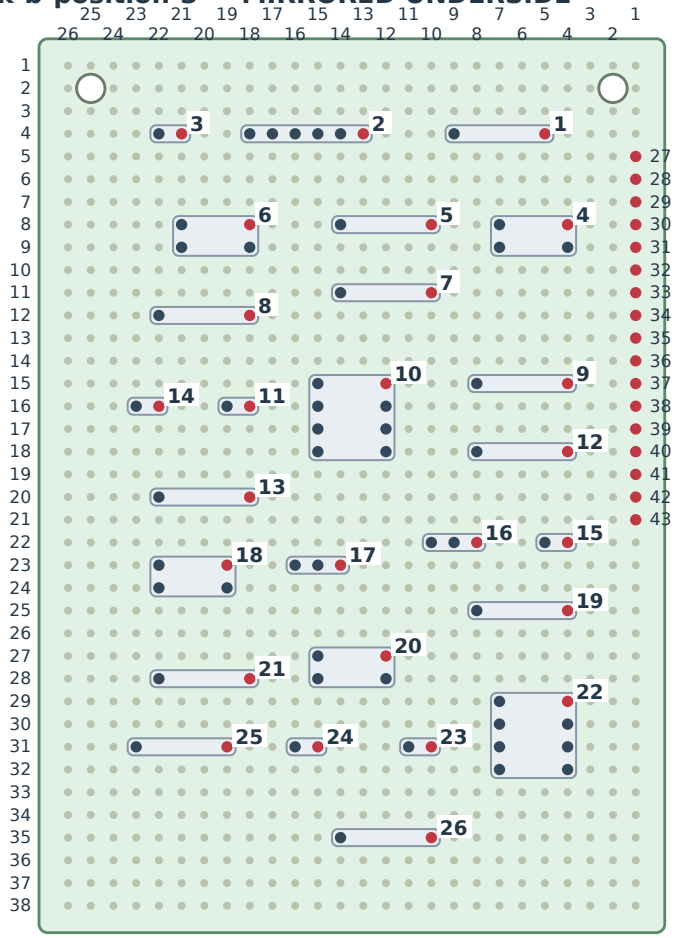


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

bank-b-position-3 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

bank-b-position-3 — numbered parts

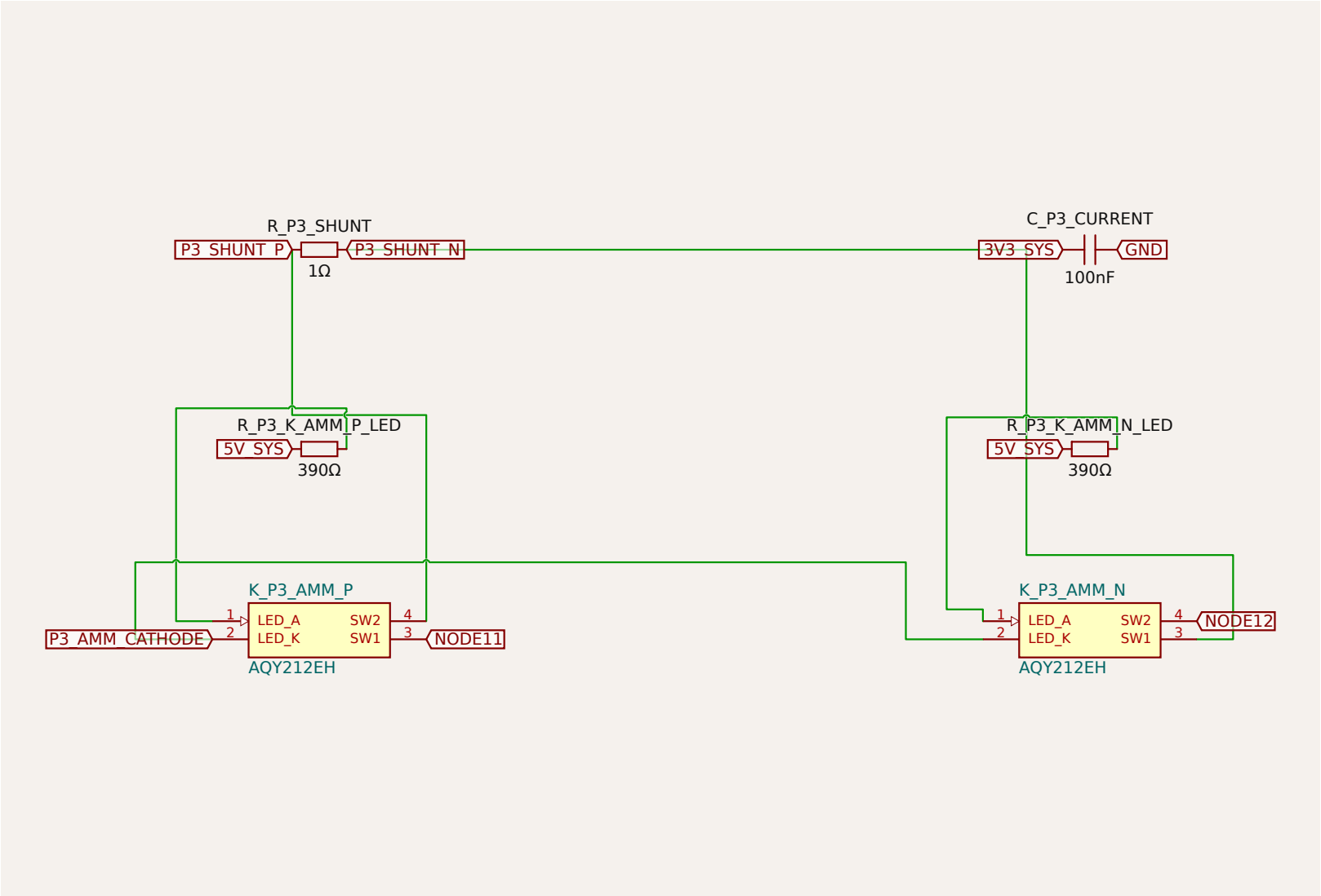
Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

#	Reference	Purpose	Install	Adapter / carrier
1	R_P3_SHUNT	1Ω ±1% external current shunt	CP29	
2	J_P3_CURRENT_MODULE	INA226 carrier: remove onboard shunt; external 1Ω	CP29	INA226 carrier
3	C_P3_CURRENT	100nF	CP29	
4	K_P3_AMM_P	AQY212EH	CP29	
5	R_P3_K_AMM_P_LED	390Ω ±1% PhotoMOS LED drive with ULN2003 saturation allowance	CP29	
6	K_P3_AMM_N	AQY212EH	CP29	
7	R_P3_K_AMM_N_LED	390Ω ±1% PhotoMOS LED drive with ULN2003 saturation allowance	CP29	
8	R_P3_SOURCE_SENSE	10	CP29	
9	R_P3_SINK_SENSE	10	CP29	
10	U_P3_LIMIT	MCP6002 dual source/sink clamp	CP29	
11	C_P3_LIMIT	100nF	CP29	
12	R_P3_HI_GATE	1k	CP29	
13	R_P3_LO_GATE	1k	CP29	
14	C_P3_HI_COMP	1nF	CP29	
15	C_P3_LO_COMP	1nF	CP29	
16	Q_P3_HIGH	BSS84	CP29	SOT23 to numbered 3-pin adapter
17	Q_P3_LOW	2N7000	CP29	
18	K_P3_HIGH	AQY212EH	CP29	
19	R_P3_K_HIGH_LED	390Ω ±1% PhotoMOS LED drive with ULN2003 saturation allowance	CP29	
20	K_P3_LOW	AQY212EH	CP29	
21	R_P3_K_LOW_LED	390Ω ±1% PhotoMOS LED drive with ULN2003 saturation allowance	CP29	
22	U_P3_BUS	TCA9517D	CP29	SOIC8 to DIP8
23	C_P3_BUS_A	100nF	CP29	
24	C_P3_BUS_B	100nF	CP29	
25	R_P3_SDA	4.7k	CP29	
26	R_P3_SCL	4.7k	CP29	
27	J_LINK_1	Harness terminal: GND	CP29	
28	J_LINK_2	Harness terminal: 3V3_SYS	CP29	

bank-b-position-3 — parts continued

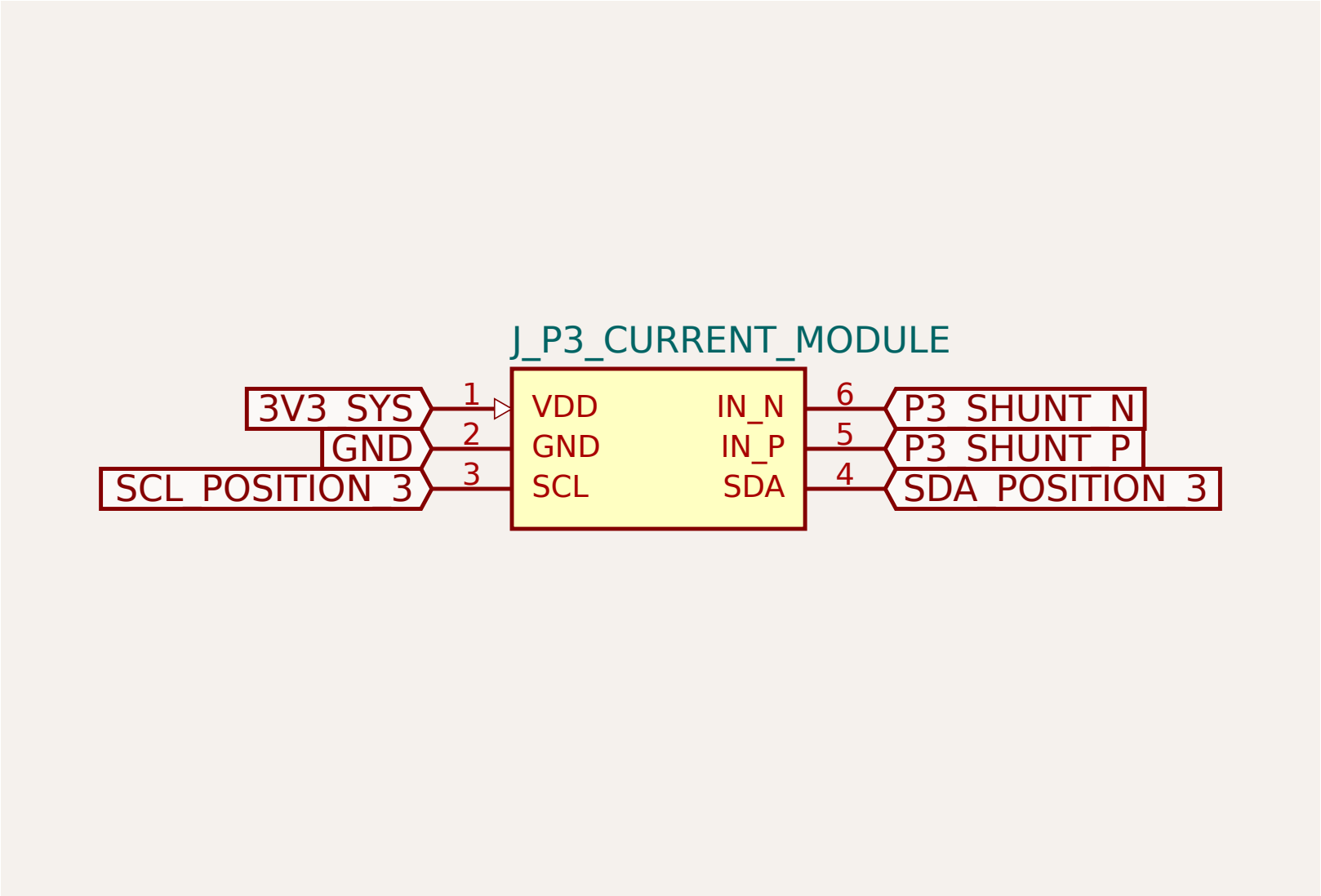
#	Reference	Purpose	Install	Adapter / carrier
29	J_LINK_3	Harness terminal: SCL_POSITION_3	CP29	
30	J_LINK_4	Harness terminal: SDA_POSITION_3	CP29	
31	J_LINK_5	Harness terminal: NODE11	CP29	
32	J_LINK_6	Harness terminal: NODE12	CP29	
33	J_LINK_7	Harness terminal: NODE13	CP29	
34	J_LINK_8	Harness terminal: 5V_SYS	CP29	
35	J_LINK_9	Harness terminal: 5V_EDU	CP29	
36	J_LINK_10	Harness terminal: USB_SMART_3V3	CP29	
37	J_LINK_11	Harness terminal: SCL_A	CP29	
38	J_LINK_12	Harness terminal: SDA_3	CP29	
39	J_LINK_13	Harness terminal: P3_K_HIGH_CATHODE	CP29	
40	J_LINK_14	Harness terminal: P3_K_LOW_CATHODE	CP29	
41	J_LINK_15	Harness terminal: P3_AMM_CATHODE	CP29	
42	J_LINK_16	Harness terminal: B_REF_HI	CP29	
43	J_LINK_17	Harness terminal: B_REF_LO	CP29	

bank-b-position-3 — Current



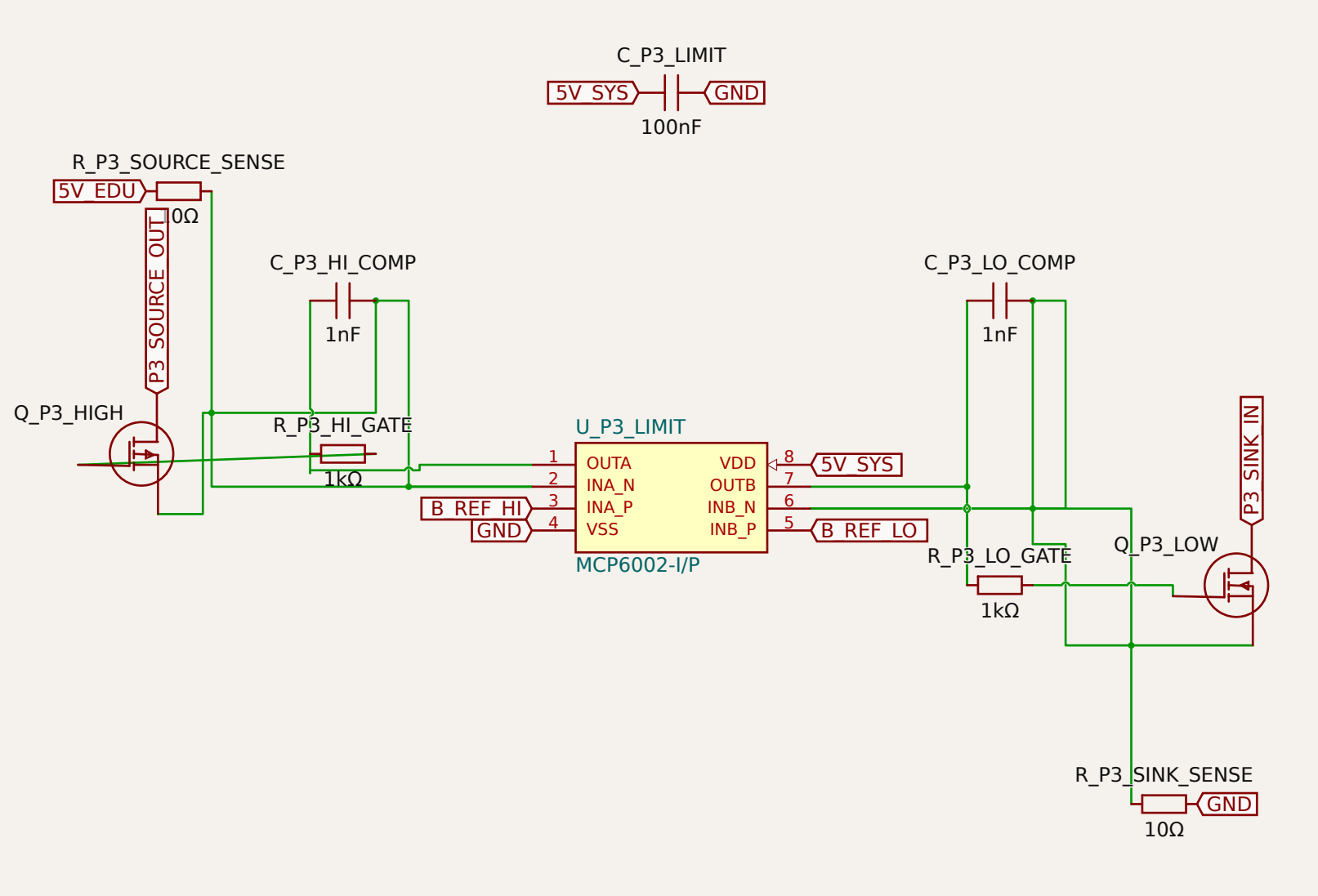
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-b-position-3 — Circuit



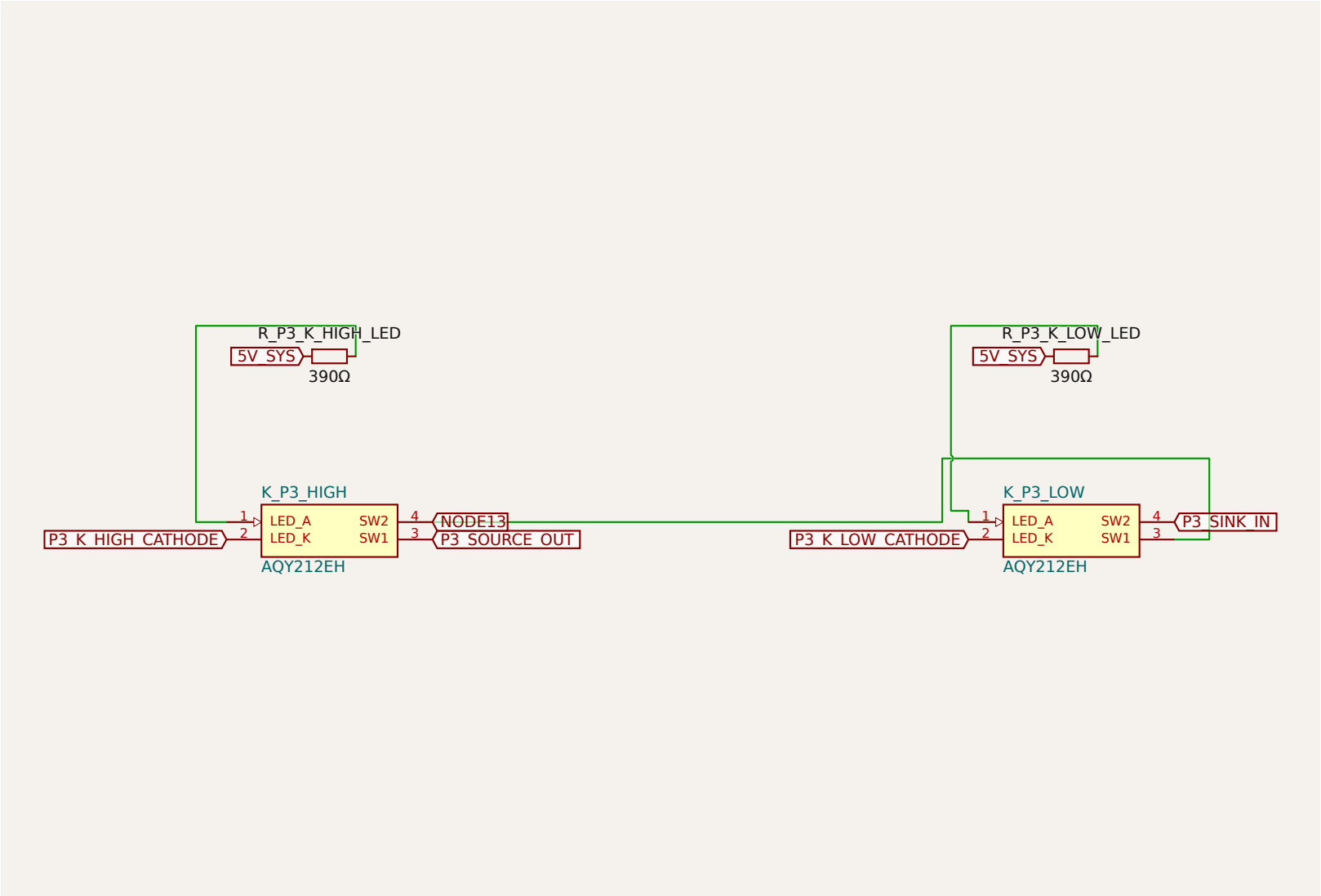
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-b-position-3 — Limits



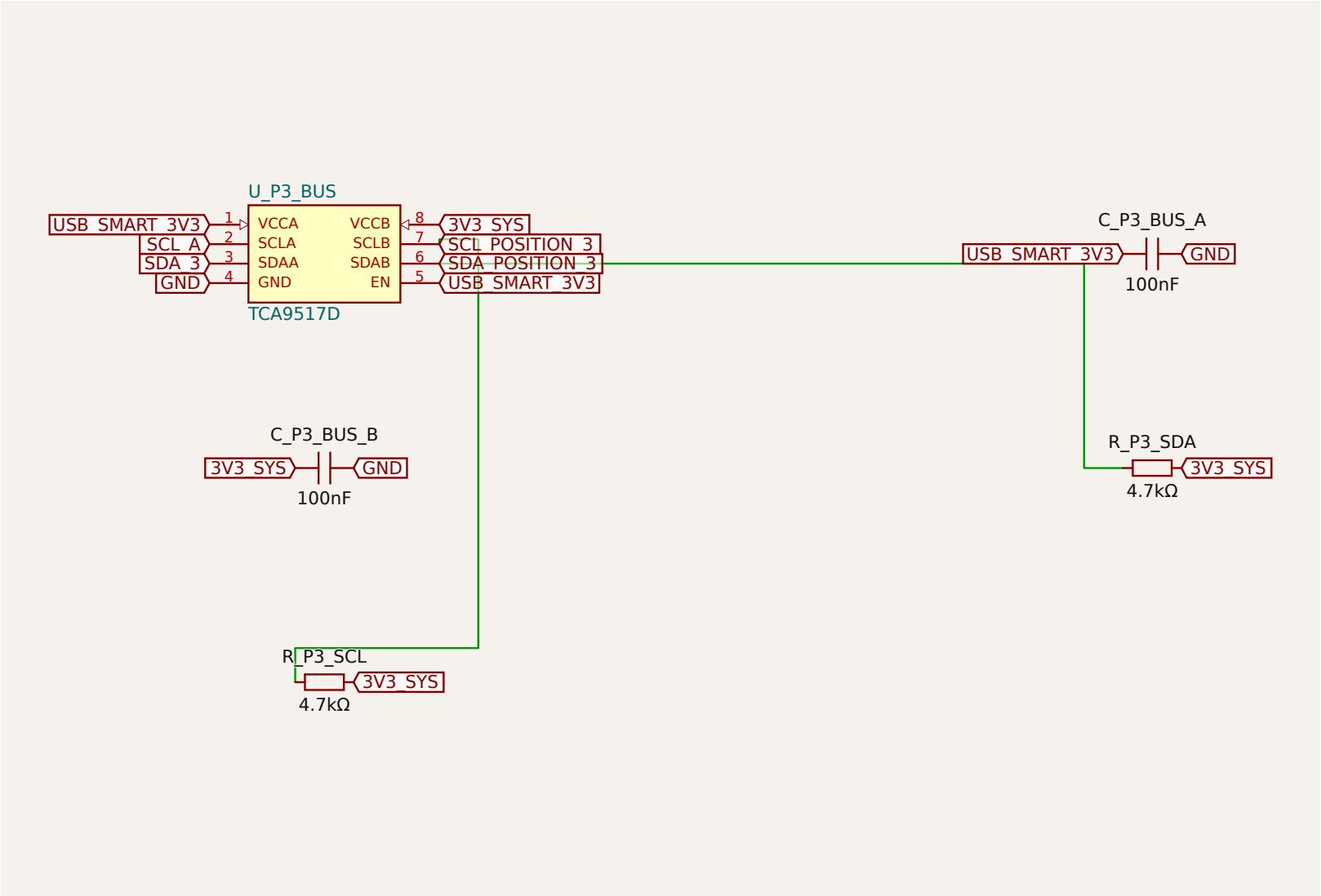
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-b-position-3 — Output



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-b-position-3 — Interface

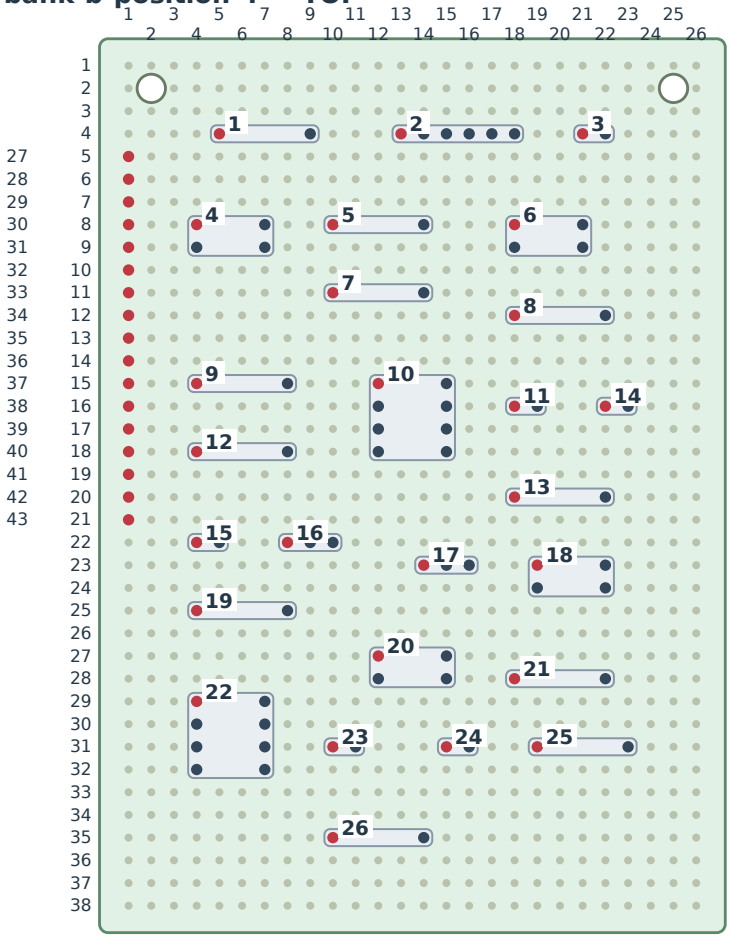


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-b-position-4

Bank B: position 4 complete isolated channel

bank-b-position-4 — TOP

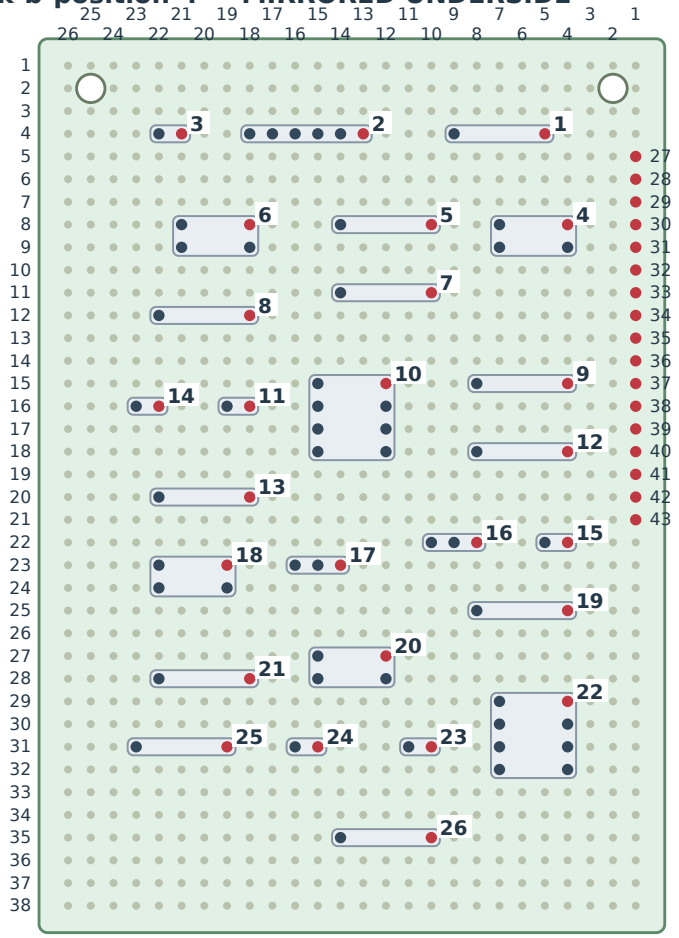


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

bank-b-position-4 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

bank-b-position-4 — numbered parts

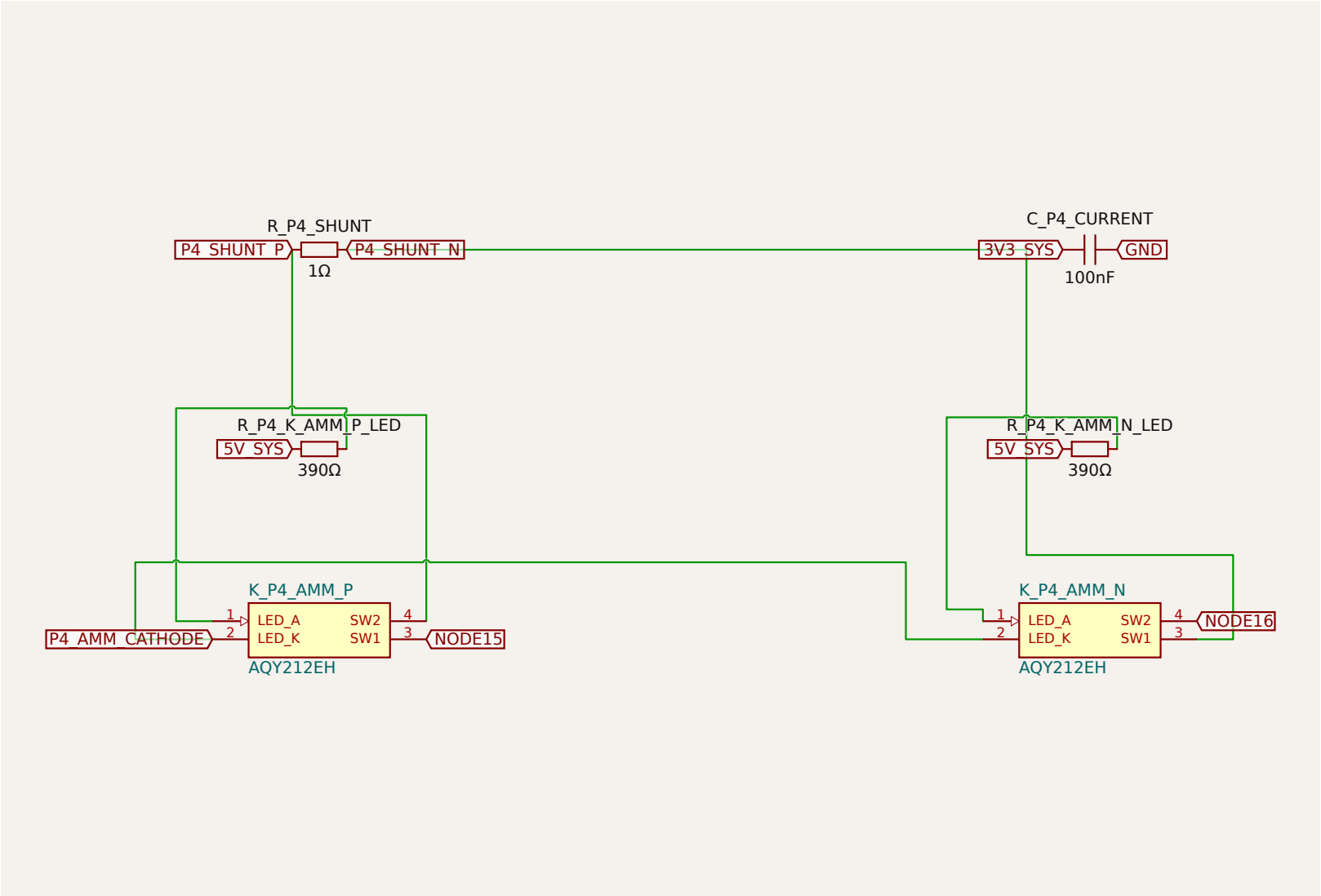
Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

#	Reference	Purpose	Install	Adapter / carrier
1	R_P4_SHUNT	1Ω ±1% external current shunt	CP30	
2	J_P4_CURRENT_MODULE	INA226 carrier: remove onboard shunt; external 1Ω	CP30	INA226 carrier
3	C_P4_CURRENT	100nF	CP30	
4	K_P4_AMM_P	AQY212EH	CP30	
5	R_P4_K_AMM_P_LED	390Ω ±1% PhotoMOS LED drive with ULN2003 saturation allowance	CP30	
6	K_P4_AMM_N	AQY212EH	CP30	
7	R_P4_K_AMM_N_LED	390Ω ±1% PhotoMOS LED drive with ULN2003 saturation allowance	CP30	
8	R_P4_SOURCE_SENSE	10	CP30	
9	R_P4_SINK_SENSE	10	CP30	
10	U_P4_LIMIT	MCP6002 dual source/sink clamp	CP30	
11	C_P4_LIMIT	100nF	CP30	
12	R_P4_HI_GATE	1k	CP30	
13	R_P4_LO_GATE	1k	CP30	
14	C_P4_HI_COMP	1nF	CP30	
15	C_P4_LO_COMP	1nF	CP30	
16	Q_P4_HIGH	BSS84	CP30	SOT23 to numbered 3-pin adapter
17	Q_P4_LOW	2N7000	CP30	
18	K_P4_HIGH	AQY212EH	CP30	
19	R_P4_K_HIGH_LED	390Ω ±1% PhotoMOS LED drive with ULN2003 saturation allowance	CP30	
20	K_P4_LOW	AQY212EH	CP30	
21	R_P4_K_LOW_LED	390Ω ±1% PhotoMOS LED drive with ULN2003 saturation allowance	CP30	
22	U_P4_BUS	TCA9517D	CP30	SOIC8 to DIP8
23	C_P4_BUS_A	100nF	CP30	
24	C_P4_BUS_B	100nF	CP30	
25	R_P4_SDA	4.7k	CP30	
26	R_P4_SCL	4.7k	CP30	
27	J_LINK_1	Harness terminal: GND	CP30	
28	J_LINK_2	Harness terminal: 3V3_SYS	CP30	

bank-b-position-4 — parts continued

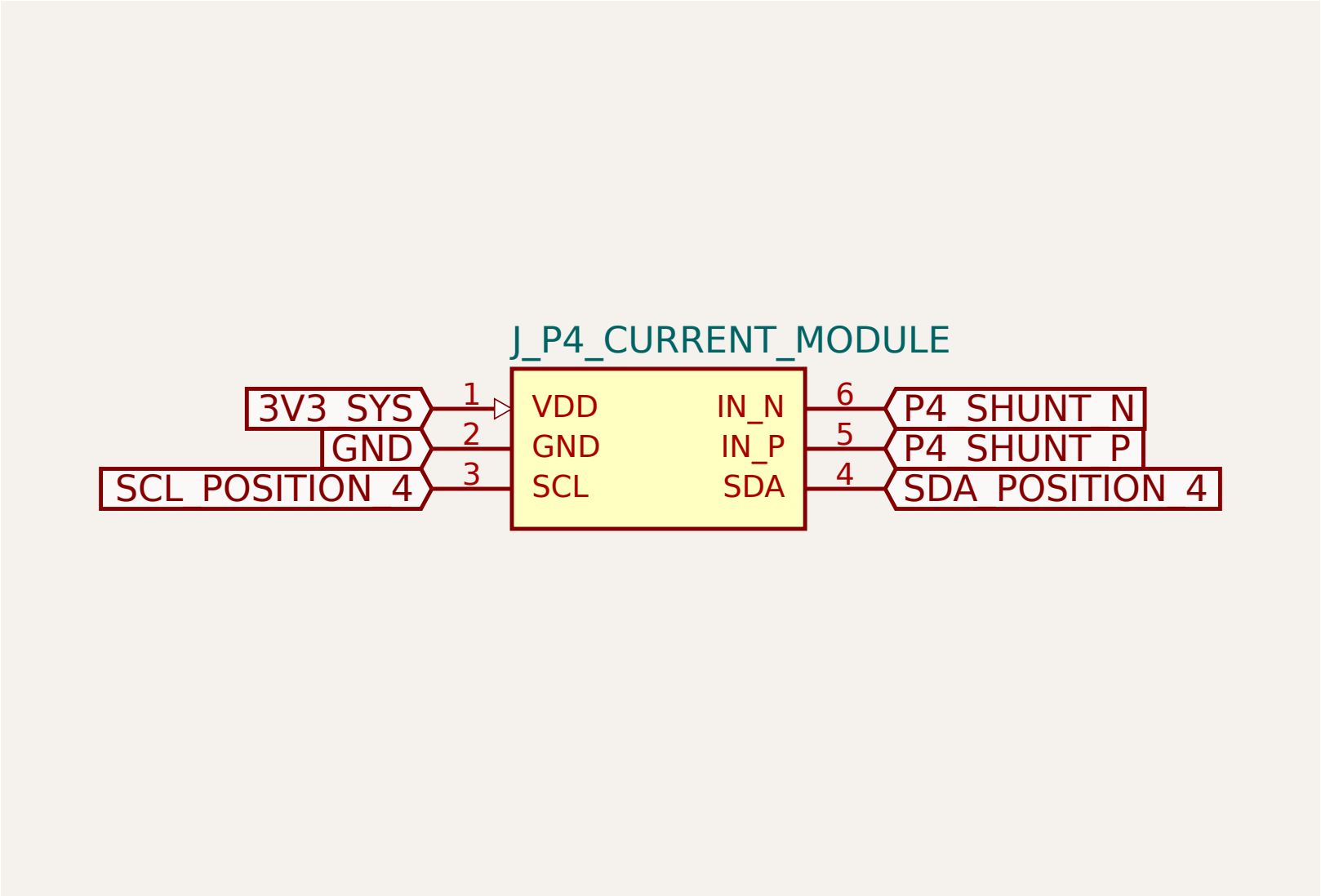
#	Reference	Purpose	Install	Adapter / carrier
29	J_LINK_3	Harness terminal: SCL_POSITION_4	CP30	
30	J_LINK_4	Harness terminal: SDA_POSITION_4	CP30	
31	J_LINK_5	Harness terminal: NODE15	CP30	
32	J_LINK_6	Harness terminal: NODE16	CP30	
33	J_LINK_7	Harness terminal: NODE17	CP30	
34	J_LINK_8	Harness terminal: 5V_SYS	CP30	
35	J_LINK_9	Harness terminal: 5V_EDU	CP30	
36	J_LINK_10	Harness terminal: USB_SMART_3V3	CP30	
37	J_LINK_11	Harness terminal: SCL_A	CP30	
38	J_LINK_12	Harness terminal: SDA_4	CP30	
39	J_LINK_13	Harness terminal: P4_K_HIGH_CATHODE	CP30	
40	J_LINK_14	Harness terminal: P4_K_LOW_CATHODE	CP30	
41	J_LINK_15	Harness terminal: P4_AMM_CATHODE	CP30	
42	J_LINK_16	Harness terminal: B_REF_HI	CP30	
43	J_LINK_17	Harness terminal: B_REF_LO	CP30	

bank-b-position-4 — Current



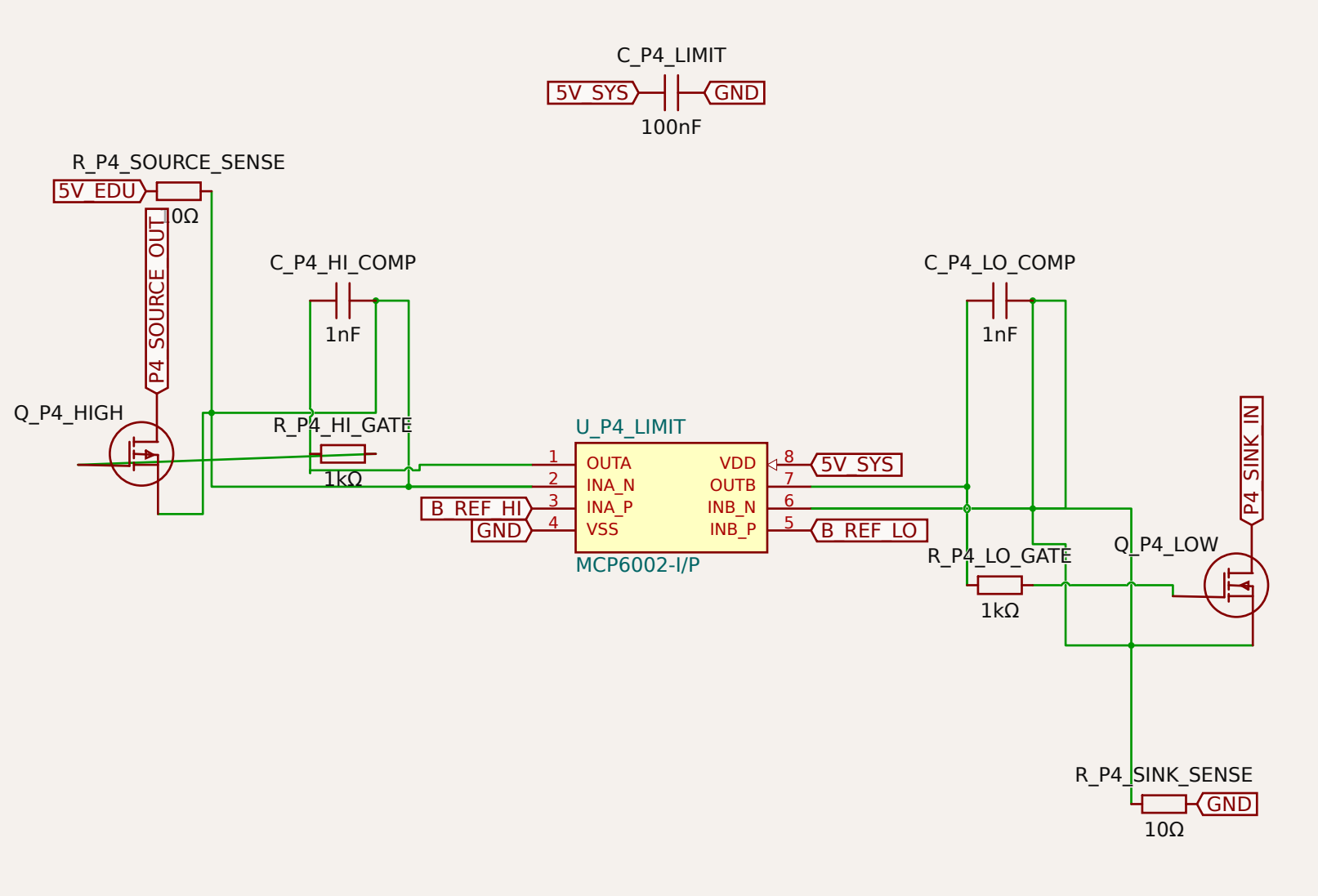
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-b-position-4 — Circuit



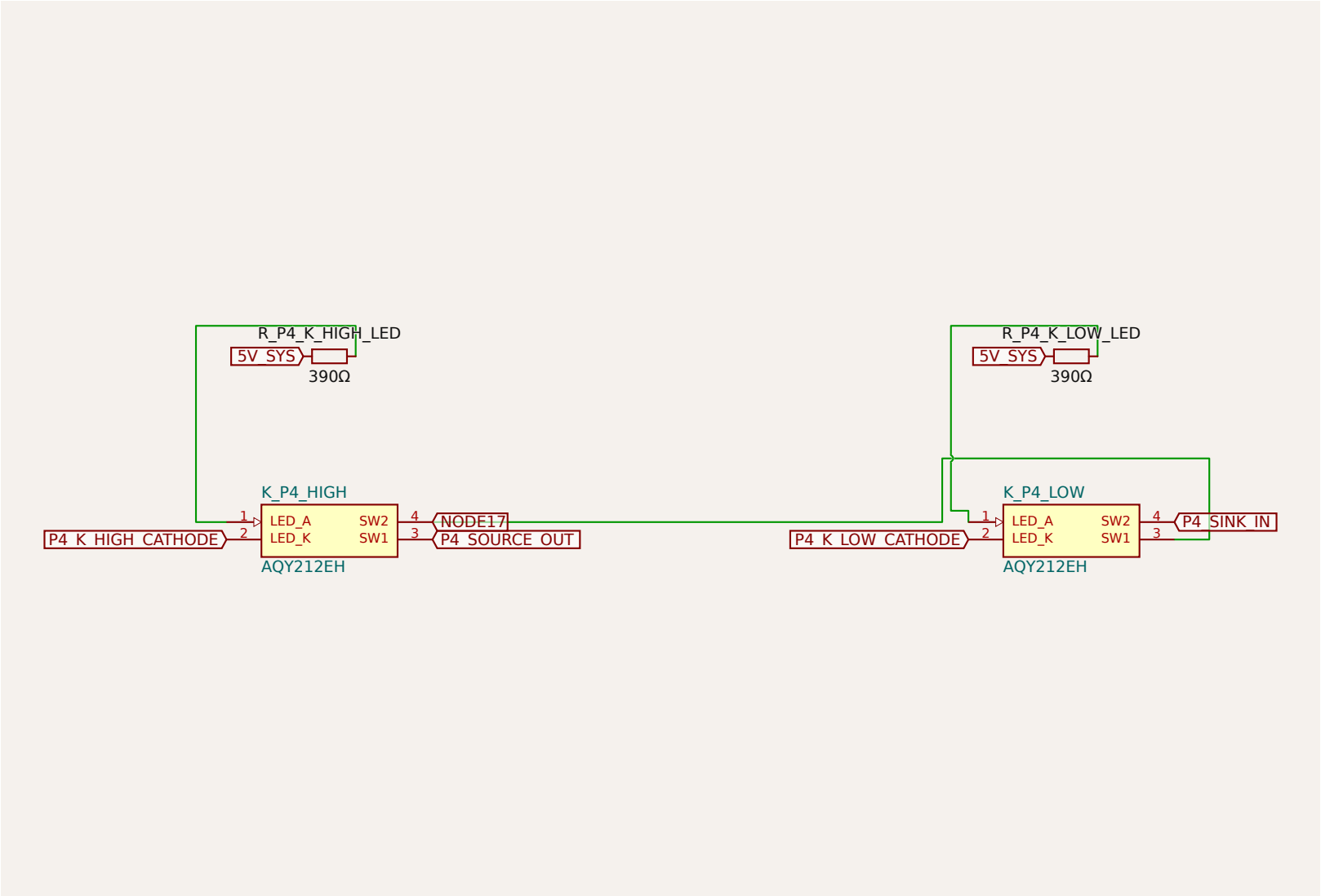
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-b-position-4 — Limits



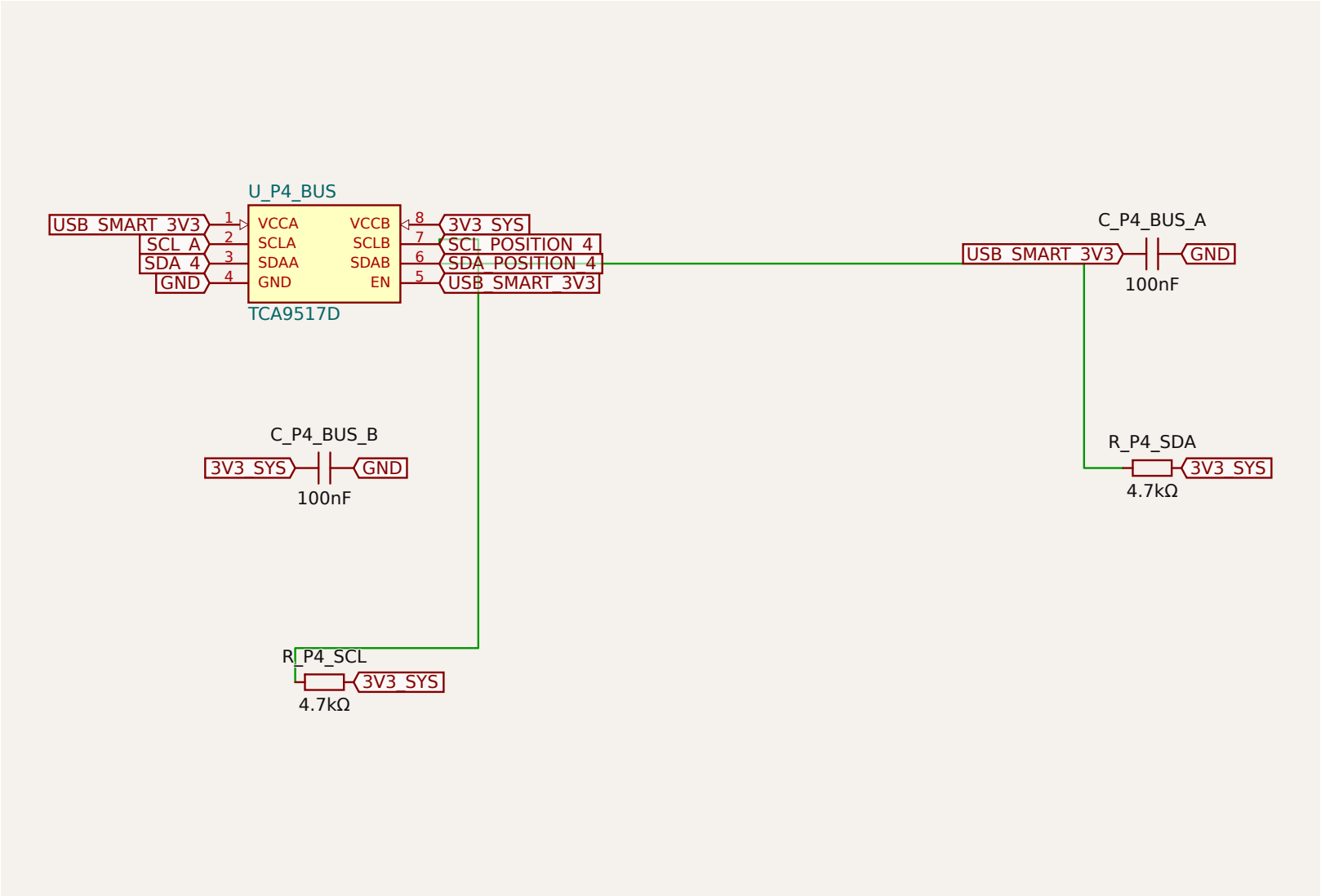
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-b-position-4 — Output



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-b-position-4 — Interface

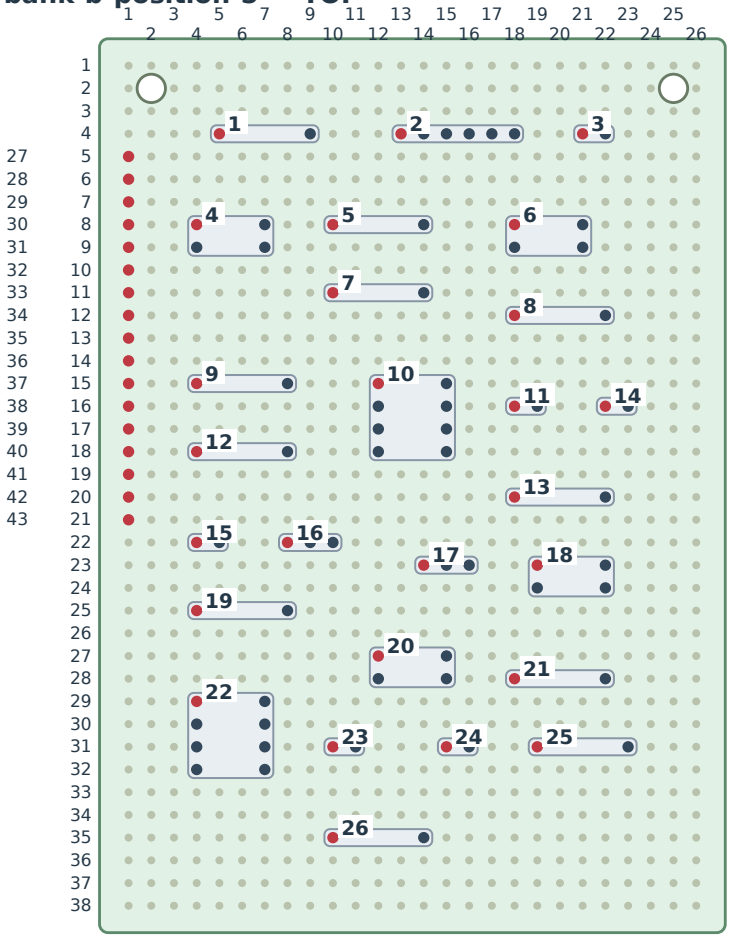


Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-b-position-5

Bank B: position 5 complete isolated channel

bank-b-position-5 — TOP

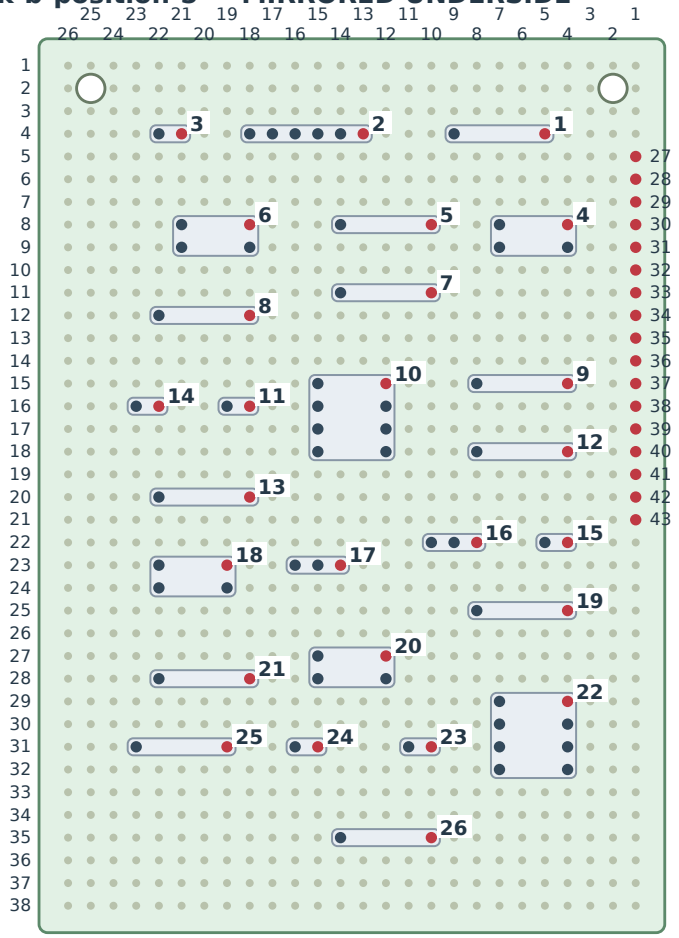


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

bank-b-position-5 — MIRRORED UNDERSIDE



70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

bank-b-position-5 — numbered parts

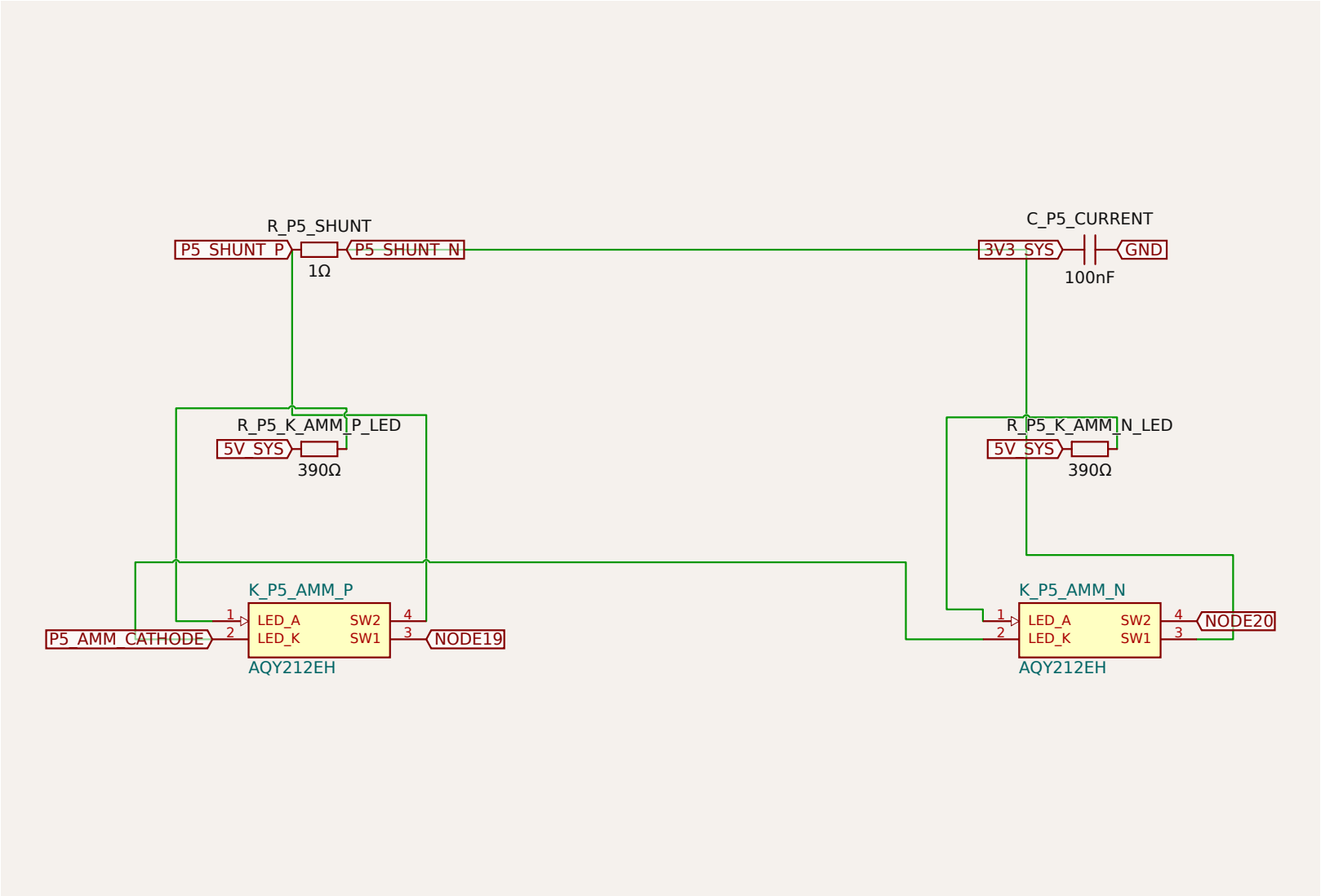
Numbered badges refer to this table. Coordinates and complete pin maps are in the corresponding CSV. Body outlines are provisional; pad coordinates are the design reference.

#	Reference	Purpose	Install	Adapter / carrier
1	R_P5_SHUNT	1Ω ±1% external current shunt	CP31	
2	J_P5_CURRENT_MODULE	INA226 carrier: remove onboard shunt; external 1Ω	CP31	INA226 carrier
3	C_P5_CURRENT	100nF	CP31	
4	K_P5_AMM_P	AQY212EH	CP31	
5	R_P5_K_AMM_P_LED	390Ω ±1% PhotoMOS LED drive with ULN2003 saturation allowance	CP31	
6	K_P5_AMM_N	AQY212EH	CP31	
7	R_P5_K_AMM_N_LED	390Ω ±1% PhotoMOS LED drive with ULN2003 saturation allowance	CP31	
8	R_P5_SOURCE_SENSE	10	CP31	
9	R_P5_SINK_SENSE	10	CP31	
10	U_P5_LIMIT	MCP6002 dual source/sink clamp	CP31	
11	C_P5_LIMIT	100nF	CP31	
12	R_P5_HI_GATE	1k	CP31	
13	R_P5_LO_GATE	1k	CP31	
14	C_P5_HI_COMP	1nF	CP31	
15	C_P5_LO_COMP	1nF	CP31	
16	Q_P5_HIGH	BSS84	CP31	SOT23 to numbered 3-pin adapter
17	Q_P5_LOW	2N7000	CP31	
18	K_P5_HIGH	AQY212EH	CP31	
19	R_P5_K_HIGH_LED	390Ω ±1% PhotoMOS LED drive with ULN2003 saturation allowance	CP31	
20	K_P5_LOW	AQY212EH	CP31	
21	R_P5_K_LOW_LED	390Ω ±1% PhotoMOS LED drive with ULN2003 saturation allowance	CP31	
22	U_P5_BUS	TCA9517D	CP31	SOIC8 to DIP8
23	C_P5_BUS_A	100nF	CP31	
24	C_P5_BUS_B	100nF	CP31	
25	R_P5_SDA	4.7k	CP31	
26	R_P5_SCL	4.7k	CP31	
27	J_LINK_1	Harness terminal: GND	CP31	
28	J_LINK_2	Harness terminal: 3V3_SYS	CP31	

bank-b-position-5 — parts continued

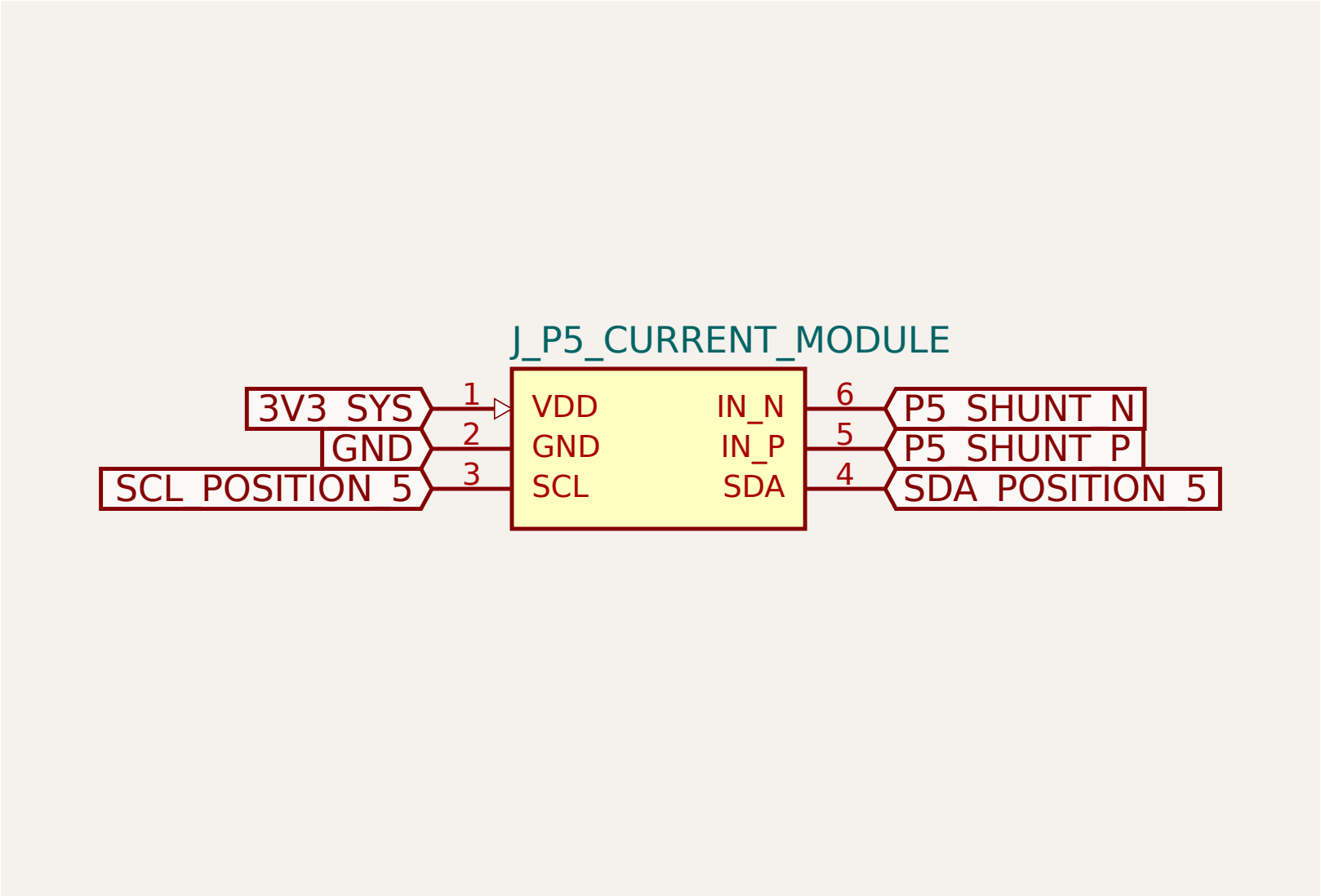
#	Reference	Purpose	Install	Adapter / carrier
29	J_LINK_3	Harness terminal: SCL_POSITION_5	CP31	
30	J_LINK_4	Harness terminal: SDA_POSITION_5	CP31	
31	J_LINK_5	Harness terminal: NODE19	CP31	
32	J_LINK_6	Harness terminal: NODE20	CP31	
33	J_LINK_7	Harness terminal: NODE21	CP31	
34	J_LINK_8	Harness terminal: 5V_SYS	CP31	
35	J_LINK_9	Harness terminal: 5V_EDU	CP31	
36	J_LINK_10	Harness terminal: USB_SMART_3V3	CP31	
37	J_LINK_11	Harness terminal: SCL_A	CP31	
38	J_LINK_12	Harness terminal: SDA_5	CP31	
39	J_LINK_13	Harness terminal: P5_K_HIGH_CATHODE	CP31	
40	J_LINK_14	Harness terminal: P5_K_LOW_CATHODE	CP31	
41	J_LINK_15	Harness terminal: P5_AMM_CATHODE	CP31	
42	J_LINK_16	Harness terminal: B_REF_HI	CP31	
43	J_LINK_17	Harness terminal: B_REF_LO	CP31	

bank-b-position-5 — Current



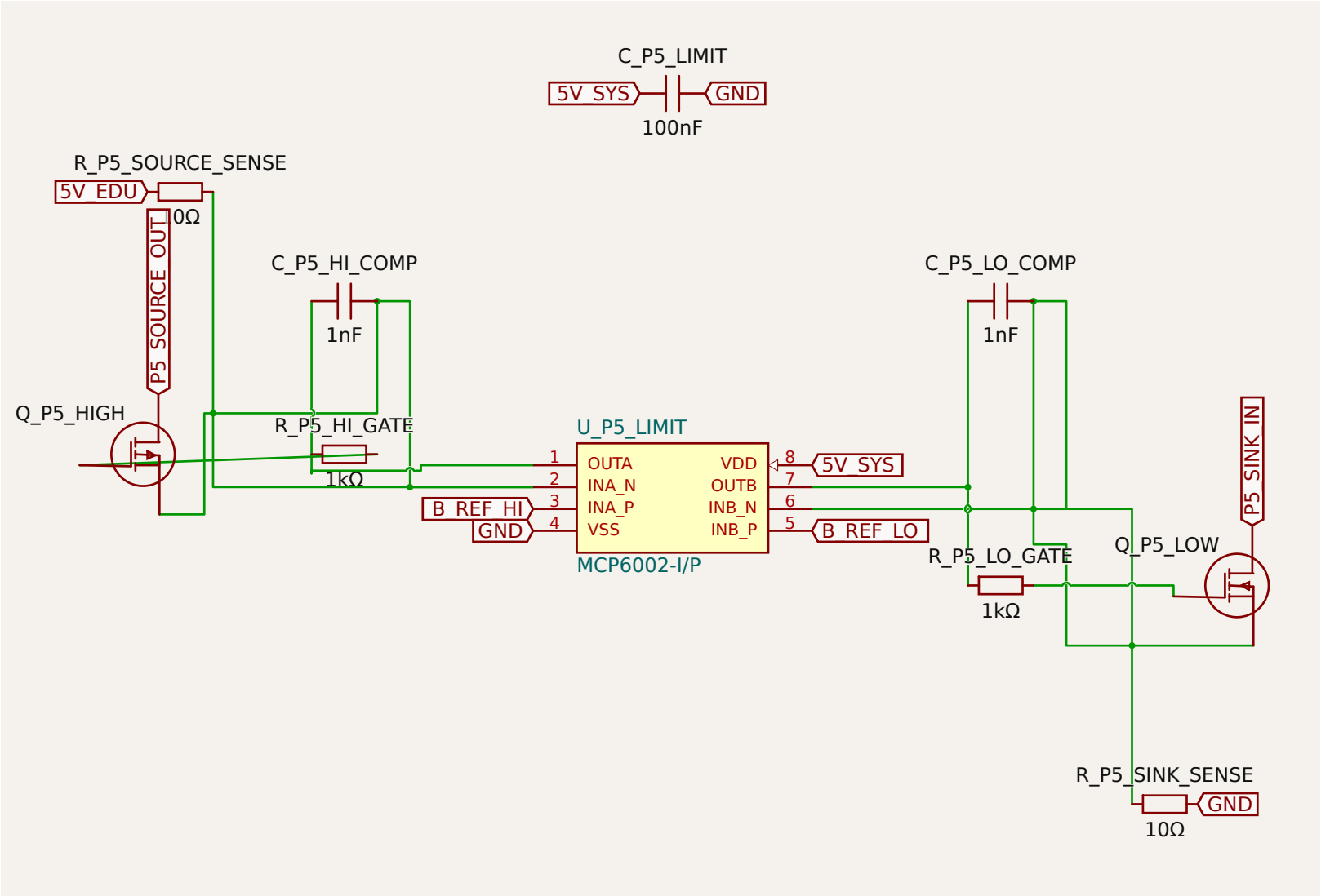
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-b-position-5 — Circuit



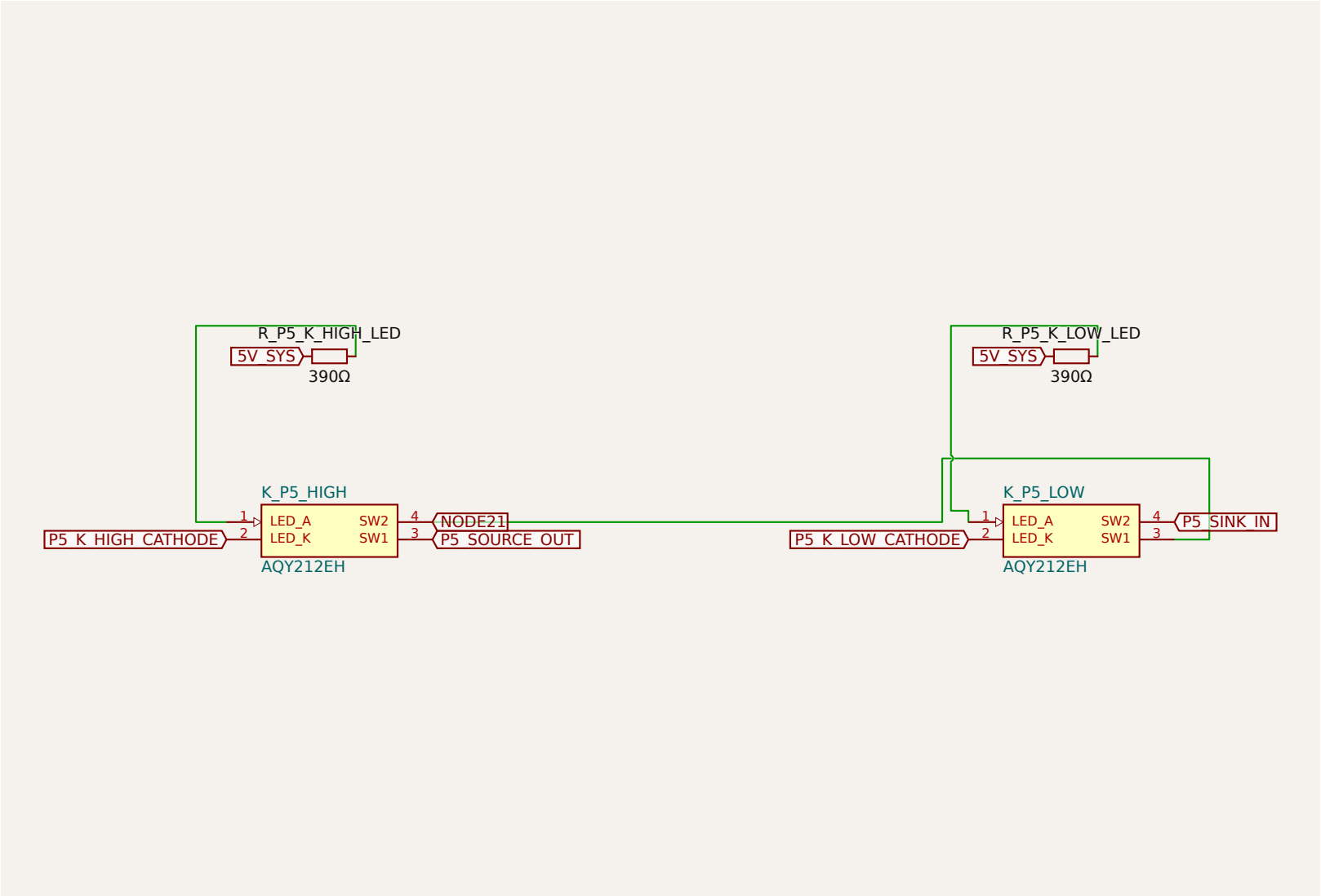
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-b-position-5 — Limits



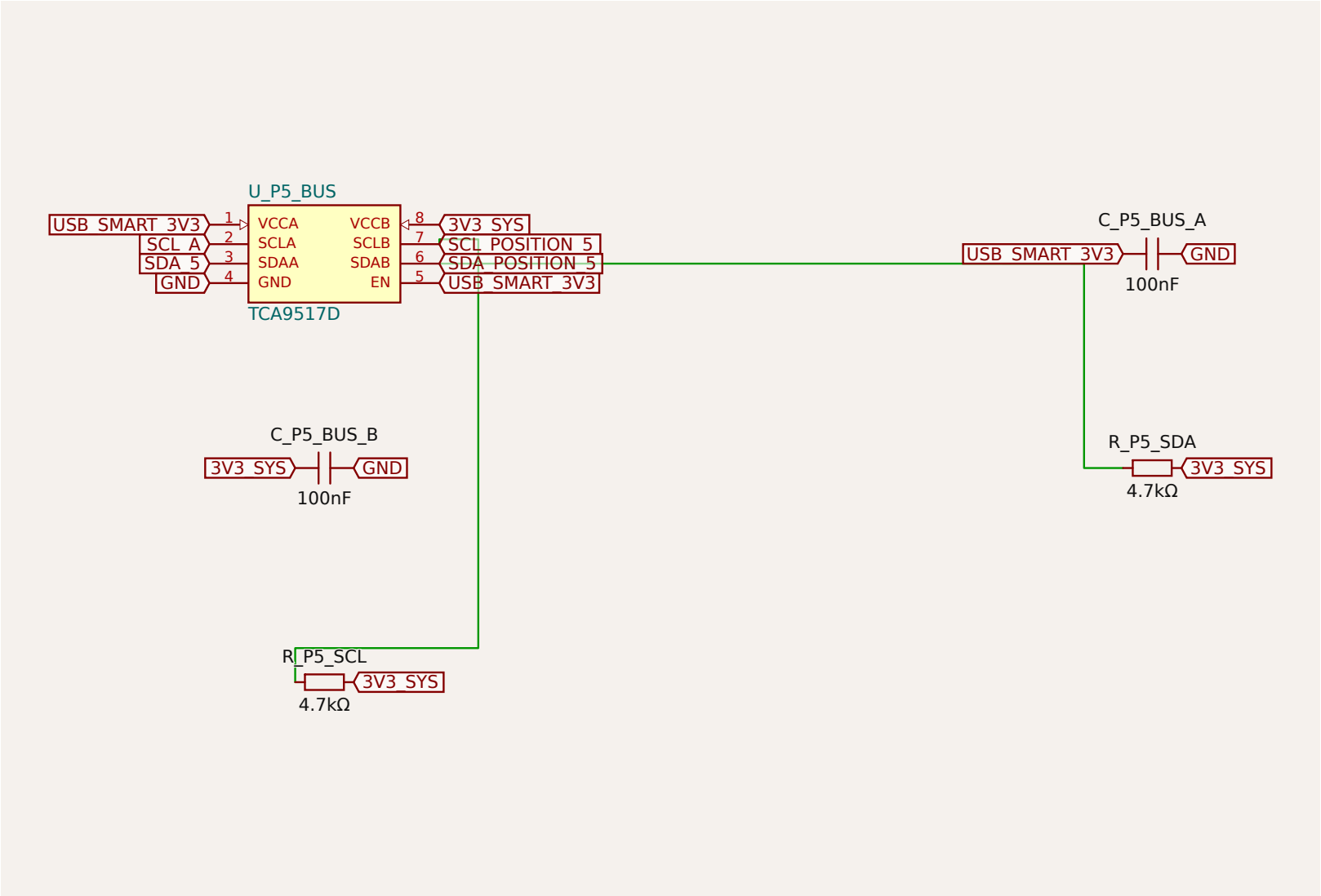
Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-b-position-5 — Output



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.

bank-b-position-5 — Interface



Named boundaries preserve external net membership. See design.json for every boundary destination; no connection is removed from the electrical model.